EXO / GARMENT STUDIO

COMPLETE ENGINEERING PROGRAM

THE REMAINING BUILD

Architecture, native tool contracts, fabric master, physical simulation, manufacturing, system quality and final release gates

2026.10.16 / MASTER EXECUTION SPECIFICATION / EVIDENCE-BOUND / NOT A PARITY CERTIFICATION

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Each indexed requirement is independently reviewable. IDs remain stable across regenerated drafts. Reference annexes are explicitly dated research snapshots. Use PDF bookmarks to navigate source annexes and all requirement entries.

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Executive contract and present-state truth

This dossier specifies the technical work required to bring EXO ClothCraft/EXO Sample 3D toward a reliable CLO-class garment CAD, simulation and production workflow. It is a proposed implementation contract. It does not claim that EXO has achieved exact CLO equivalence, certified apparel fitting or production-approved manufacturing output.

Evidence dimension	Verified repository snapshot	Remaining validation
Native toolbar mapping	70 observed CLO 7.0.242 commands; 11 native / 9 inspect / 50 unavailable	Functional click → model edit → undo → reopen → 2D/3D matching for each slot
Fabric library	400 estimated specs, 405 Rust choices; 385 photo-linked visual variants	400 genuine calibrated tests are NOT present; per fabric optical source matching and drape validation
Physical optics	67 installed CC0 source materials; 44 referenced; 201 map images	Source construction identity, centimeters-per-tile, scene color matching and yarn-scale validation
Preview dynamics	React PBD, Rust VBD, C++ XPBD and Rust PPF paths exist	Single canonical project contract, confidence levels and cloth/body swept collision evidence
Manufacturing	DXF/POM/BOM/marker/factory ZIP generators	Independent third-party import, actual cutting, physical samples and signed package acceptance
Desktop	Signed WebKit app and developing Qt/C++ structures	Full authoring interaction evidence and cross-platform packaging

Priority rule: preserve truthful user-facing capability indicators. A button in the toolbar, a data record or a screenshot does not prove its semantic operation. Each following work item has an acceptance boundary and source pointers. "Future" sections are optional architectural extensions rather than blockers for core CAD.

Source-of-truth audit provenance: repository at /blender-mac-workspace/exo-clothcraft-v2; current working tree contains active uncommitted changes. Evidence labels below are snapshots, not reproducible build green statuses. The PDF/Docx themselves are planning artifacts and do not modify EXO code.

Dependency-critical implementation order

Sequence	Program gate	Blocking evidence
M0	Canonical schema, deterministic builds, versioned operation contracts	Clean native builds, migration and fixture corpus
M1	2D CAD + toolbar/property semantic correctness	70-slot matrix, exact tools, cross-revision undo
M2	Seam topology and 2D↔3D correspondence	Sewing, arrangement, stable mesh/source mapping
M3	Cloth physical solver, avatar, swept contact	Accepted multi-frame tee/pants/layer tests
M4	Calibrated fabrics and truthful PBR	Specimen source and optical/mechanical independent evidence
M5	Fit insights + production measurements	Signed landmarks, explanatory heat maps
M6	Grading, markers, factory interoperability	Third-party CAD DXF roundtrip and cut-test
M7	Signed desktop, QA, scripting, open-source packaging	Clean machine smoke, key control e2e tests
M8	Rendering, asset libraries, photography	Color managed look and artifact provenance
M9+	Community, tech packs, spatial optional products	Authentication, permissions, immutable approved releases

GATE TRANSITION RULE

No issue advances because its file exists or a test passes in only one layer. A gate requires the exact fixture revision, native build hash, automated assertions, adversarial failure test and an inspection report. Each completion ticket must identify a responsible implementation surface and a separate reviewer.

SUGGESTED ISSUE FIELDS

Each ticket: SPEC-ID, feature, source location, acceptance date, current evidence class, blocking prerequisites, exact user workflow, input/output schema, undo state, expected and failure behavior, screenshot/regression path, QA owner and supporting measurement evidence.

System boundaries and canonical dependency graph

Local EXO authoring owns private garment source, edits and physical models. The React/WebKit shell displays 2D/3D state; Rust V2 powers selected live native commands; the C++ CAD library contains a broad but separately verified conversion set; PPF remains a bounded optional solver service. Future community and manufacturer services receive only creator-authorized immutable publish packages, never access to arbitrary local workspace files.

Layer	Owns	Receives	Never assume
UI	tool state, viewport state, temporary selection	stable document snapshots, tool capabilities	UI event alone mutates native model
Project document	IDs, units, 2D pieces, seams, fabric, trim, revisions	atomic commands, migration	mesh vertex indexes are durable IDs
Geometry/CAD	2D curves, topology, seam pairing, meshing, 2D/3D map	authoring graph	cut boundary is equivalent to stitch edge
Physics	material-specific constraints, avatar contact, time trajectories	validated mesh and constitutive calibration	finite run means valid garment fit
Visuals	PBR textures, masks, studio lights, still/video renders	UVs, optical material profiles	visual wash changes fabric mechanics
Factory	grade, POM/BOM, markers, DXF release bundle	approved document revision	export ZIP passes cutter acceptance
Platform	authorized published derivatives, entitlement, proposals	explicit immutable creator release	public preview grants source rights

Recommended data path: user command → precondition and expected revision → document mutation → dependency invalidation → deterministic derived geometry and meshes → selected solver with diagnostics → rendering/inspection → production preflight → authorized immutable export. Persist named transformations, units and hashes at every boundary; record backend identity alongside output.

Do not conflate C++ export-count parity or Rust available commands with finished GUI integration. Errors from a stale build directory or broken test selector are evidence about test setup and should be distinguished from numerical test failures.

Latest known validation gaps and confidence boundaries

The current 16 October worker audit reviewed active Rust V2 and the separate C++ garment CAD and solver modules. These findings supersede optimistic readings of older parity reports but do not erase previously implemented features. They are source-level evidence and targeted test attempts, not blanket statements about future builds.

Boundary	Source-level evidence	Remaining work
Rust V2 authoring	exo-session.rs has edit, undo/redo, set-property, line-spec, pose/pull/grab, step and save	Unify with canonical geometry/project metadata, native command actions and property persistence tests
Fit badge	V2 snapshot explicitly carries fitCertified:false	Only promote per verified trajectory and material/pose fixture
C++ conversion	Dated status reports 175 of 182 JS mappings (96.2%) and previous 58/58 parity scripts	Export mappings ≠ working full desktop/CLO product; retest from current clean checkout
C++ collision parity	Historical collision deviations around 0.074m tee and 0.055m pants RMS	Fix coupled contact and swept checks before any physically validated fit claim
Manufacturing byte agreement	Reported 16/24 tee and 20/24 pants factory ZIP file byte matches	Investigate all divergent metadata and content; independently import exported DXF
Qt native UI	Optional Qt shell and CAD core; native Qt simulation action not fully wired	Integrate actual command path, renderer, property controls and signed app testing
Fresh native suite	58 tests NOT RUN in attempted existing build cache pointing to retired checkout path	Clean out-of-tree CMake configuration, build and new CTest evidence; NOT RUN is not 58 failed assertions

Source references: native_cpp/CONVERSION_STATUS_2026-10-15.md; integrations/clothcraft/exo-session.rs; native_cpp/core/cloth_collision_step.cpp; native_cpp/desktop; native_cpp/tests. Any numerical value in this page must be tied to its dated fixture rather than generalized to every garment, pose, or backend.

A durable program dashboard should attach code revision, hardware, software versions, build flags and fixtures to every result. The same codebase may have a UI capability, an unrelated library function and a separate solver experiment; give each its own verification state.

Canonical garment document

OBJECTIVE / ACCEPTANCE SCOPE

Single revisioned source of truth for 2D pattern, 3D arrangement, seam graph, fabric assignments, avatar pose and exports

ACTUAL SOURCE LOCATIONS

src/core/project.js; src/core/backends/protocol.js; native_cpp/core/garment_document.hpp

IMPLEMENTATION REQUIREMENTS

Define versioned Project/Pattern/Edge/Curve/Seam/Fabric/Avatar/Trim/Manufacturing models; immutable IDs; unit scale; coordinate frames; provenance; explicit migrations. Convert native Rust session snapshots and C++ document snapshots through a schema-preserving boundary.

PROPOSED TYPED CONTRACT

ProjectRevision{id,revision,parents,units,hash,entities,invalidations}; no data-derived element ID may silently shift on edit.

REQUIRED VERIFICATION AND FAILURE CASES

Save/edit/undo/reopen/re-export consistency on tee, pants and jacket; reject stale revision and dangling foreign key without modifying disk

Execution	Dependency and integration	Definition of done
M0/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

The implementation must not claim commercially certified CLO parity simply because a visual analogue exists. Deliver acceptance evidence with numerical physical units and a human-reviewable before/after record. Persist unsupported reasons until tested rather than mapping a reference toolbar action to an unrelated operation.

Typed command engine and undo

OBJECTIVE / ACCEPTANCE SCOPE

Make every toolbar action a deterministic, testable edit with proper cancellation and history

ACTUAL SOURCE LOCATIONS

src/core/workspaceCommandIndex.js; src/core/cloPropertyCommands.js; integrations/clothcraft/exo-session.rs

IMPLEMENTATION REQUIREMENTS

Register typed operations with target preconditions, dry-run diff, effect scopes, inverse snapshot and atomic commit. Choose one transaction owner per change and map browser/native bindings to same command semantics.

PROPOSED TYPED CONTRACT

CommandEnvelope{commandId,targetIds,params,expectedRevision,selection,units}; CommandResult{revision,createdIds,changedIds,invalidated,errors}.

REQUIRED VERIFICATION AND FAILURE CASES

Bad inputs leave project byte-identical; grouped drag is one history step; undo/redo survives native restart and invalidates dependent projections

Execution	Dependency and integration	Definition of done
M0/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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2D planar geometry kernel

OBJECTIVE / ACCEPTANCE SCOPE

Exact point, curve, segment and topology operations without pixel-space drift

ACTUAL SOURCE LOCATIONS

src/core/draftingPrecision.js; src/core/draftingTopology.js; native_cpp/core/drafting_precision.cpp

IMPLEMENTATION REQUIREMENTS

Use stable edge-segment IDs and declared units. Evaluate Bézier derivatives and arc length; robust intersection predicates; exact decimal input; split/reparameterize edges while preserving attachment parameters.

PROPOSED TYPED CONTRACT

CurveEdge{id,from,to,kind,controls,orientation,arcLengthTolerance}; tolerance policy defines mm to internal coordinate conversion.

REQUIRED VERIFICATION AND FAILURE CASES

Golden analytic geometry fixtures at many zoom/DPR levels; no inverted winding or dangling notch after split; no self intersection accepted silently

Execution	Dependency and integration	Definition of done
M1/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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2D drawing and selection

OBJECTIVE / ACCEPTANCE SCOPE

Selection, lasso, snap, node edits, keyboard precision and viewport state

ACTUAL SOURCE LOCATIONS

src/components/PatternEditor.jsx; src/core/draftingSnapping.js

IMPLEMENTATION REQUIREMENTS

Specify pointer hit-test order, snapping priorities, selection cycling, drag preview commit, keyboard nudges, multi-select and focused command routing. Keep screen transforms separate from physical points.

PROPOSED TYPED CONTRACT

SelectionState{kind,ids,active,mode}; DragPreview{baseRevision,proposedGeometry,precisionMm}.

REQUIRED VERIFICATION AND FAILURE CASES

Pointer and keyboard produce equivalent geometry; viewport zoom does not alter final cm measurements

Execution	Dependency and integration	Definition of done
M1/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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2D drafting constructions

OBJECTIVE / ACCEPTANCE SCOPE

Native darts, slash/spread, pleat, fullness, mirror, unfold and linked pieces

ACTUAL SOURCE LOCATIONS

src/core/draftingConstructions.js; src/core/draftingUnfold.js; native_cpp/core/drafting_constructions.cpp

IMPLEMENTATION REQUIREMENTS

Model fold/dart legs, apex, take-up and dependent constraints. Use constructive graph nodes whose control points remain editable; reconstruct boundary stitching, cut offsets and annotations after each operation.

PROPOSED TYPED CONTRACT

Construction{id,type,parentPiece,anchors,parameters,dependentEdges,revision}.

REQUIRED VERIFICATION AND FAILURE CASES

Known dart intake and fold tests, mirrored tangent continuity and seam lengths preserved under undo/export/import

Execution	Dependency and integration	Definition of done
M1/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Notches and seam allowance CAD

OBJECTIVE / ACCEPTANCE SCOPE

Cut-line offsets, variable width, corners, notches and seam line differentiation

ACTUAL SOURCE LOCATIONS

src/core/draftingPrecision.js; src/core/productionPreflight.js

IMPLEMENTATION REQUIREMENTS

Compute offset curves with robust join semantics, corner trimming and collision checks. Distinguish stitch line, cut line, turn/fold line and metadata references. Annotate cut marks with consistent scale.

PROPOSED TYPED CONTRACT

Allowance{edgeId,widthMm,cornerPolicy,lockMode}; Notch{edgeId,t,depthMm,style,gradeMode}.

REQUIRED VERIFICATION AND FAILURE CASES

1:1 paper plot measurements and independently checked DXF layer output; no allowance becomes a sewing edge by accident

Execution	Dependency and integration	Definition of done
M1/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Linked 2D to 3D correspondence

OBJECTIVE / ACCEPTANCE SCOPE

Lossless mapping of every authored piece and semantic line to render/solve mesh

ACTUAL SOURCE LOCATIONS

src/core/clothMeshing.js; src/core/liveDrapeTransfer.js; native_cpp/core/pattern_mesh.cpp

IMPLEMENTATION REQUIREMENTS

Maintain source vertex and edge-param lineage during triangulation and remeshing; define index stable until invalidated. Propagate UVs, grainline, material IDs and seam constraints.

PROPOSED TYPED CONTRACT

MeshMapping{pieceId,meshVersion,vertexSources,edgeRanges,triangleSources,uvFrame}.

REQUIRED VERIFICATION AND FAILURE CASES

Highlighting a source line finds correct 3D surface after subdivision; editing point invalidates only dependent solve caches

Execution	Dependency and integration	Definition of done
M2/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Sewing graph and partial seams

OBJECTIVE / ACCEPTANCE SCOPE

Segment/free sewing, orientation, lengths, easing, seam types and topology diagnostics

ACTUAL SOURCE LOCATIONS

src/core/sewingConstruction.js; src/core/backends/seamAssembly.js; native_cpp/core/native_seam_assembly.cpp

IMPLEMENTATION REQUIREMENTS

Treat seam as a pair of oriented parameter intervals on persistent boundary edges. Build stitch samples by arc-length correspondence, reconcile different discretizations and detect gaps or duplicates.

PROPOSED TYPED CONTRACT

Seam{id,left:{edge,t0,t1},right:{edge,t0,t1},direction,ease,stitch,allowance}.

REQUIRED VERIFICATION AND FAILURE CASES

Full/partial seam fixtures, reverse-edge direction, oversized ease ratio rejection, seam walk report and undo/redo reopening

Execution	Dependency and integration	Definition of done
M2/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Complex construction and hardware

OBJECTIVE / ACCEPTANCE SCOPE

Zippers, buttons, buttonholes, binding, piping, trims, quilting, lining and multi-layer assembly

ACTUAL SOURCE LOCATIONS

src/core/trims.js; src/core/sewingConstruction.js; native_cpp/core/cloth_seam_constraints.cpp

IMPLEMENTATION REQUIREMENTS

Store separately authored trim geometry and pattern-space attachment constraints. Separate visual paths from physical/production semantics. Add collision proxies only after robust primitive support.

PROPOSED TYPED CONTRACT

TrimInstance{id,type,material,anchors,fastenerState,geometryVersion,manufacturingCode}.

REQUIRED VERIFICATION AND FAILURE CASES

A jacket zipper closes/reopens with no orphan edges; hardware BOM is traceable to rendered attachment and version

Execution	Dependency and integration	Definition of done
M2/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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3D garment editing tools

OBJECTIVE / ACCEPTANCE SCOPE

Mesh selection, stylelines, pins, transforms and surface-bound strokes

ACTUAL SOURCE LOCATIONS

src/core/meshSelection.js; src/components/GarmentViewport.jsx

IMPLEMENTATION REQUIREMENTS

Add area/brush/lasso selection on triangle IDs with source map, editable styleline contours and persistent projected control points. Preserve source panel ownership after remesh.

PROPOSED TYPED CONTRACT

SurfaceStroke{id,meshRevision,barycentricAnchors,parentPieceIds,projectionTolerance}.

REQUIRED VERIFICATION AND FAILURE CASES

Click/drag modifies correct authored entity, undo/redo, 3D→2D continuity and reopen; unsupported tools remain labelled

Execution	Dependency and integration	Definition of done
M2/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Avatar geometry and measurement

OBJECTIVE / ACCEPTANCE SCOPE

Body collision mesh, pose rig, standardized landmarks, shape and anthropometric control

ACTUAL SOURCE LOCATIONS

src/core/meshAvatarContact.js; public/avatars; blender_bridge

IMPLEMENTATION REQUIREMENTS

Record avatar license, units, rig hierarchy, landmark definitions, skinning and signed mesh normals. Validate open scans, custom morphology, poses and robust collision proxy generation.

PROPOSED TYPED CONTRACT

Avatar{id,meshHash,unit,rig,bonePoses,measurements,landmarks,collisionProxyVersion}.

REQUIRED VERIFICATION AND FAILURE CASES

Independent tape measurement within tolerance, neutral/arms-up poses, malformed scan rejection, accurate visual and collision alignment

Execution	Dependency and integration	Definition of done
M3/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

The implementation must not claim commercially certified CLO parity simply because a visual analogue exists. Deliver acceptance evidence with numerical physical units and a human-reviewable before/after record. Persist unsupported reasons until tested rather than mapping a reference toolbar action to an unrelated operation.

Animation timeline

OBJECTIVE / ACCEPTANCE SCOPE

Pose, clip, keyframe editing and motion-driven drape

ACTUAL SOURCE LOCATIONS

src/components/GarmentViewport.jsx; src/core/physics/timestep.js

IMPLEMENTATION REQUIREMENTS

Build typed animation tracks for avatar bones, camera and garment states, keyframe interpolation, timeline editing, scrub/cancel, cache invalidation and export.

PROPOSED TYPED CONTRACT

AnimationTrack{targetId,property,keyframes,interp}; MotionSample{time,poseHash}.

REQUIRED VERIFICATION AND FAILURE CASES

Repeatable pose reconstruction and defined contact checks between frames; time scrubs do not corrupt source geometry

Execution	Dependency and integration	Definition of done
M3/P2	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Interactive VBD and XPBD preview

OBJECTIVE / ACCEPTANCE SCOPE

Fast cloth response with explicit accuracy tier and selectable backend

ACTUAL SOURCE LOCATIONS

integrations/clothcraft/exo-session.rs; native_backend/server.mjs; src/core/physics.js

IMPLEMENTATION REQUIREMENTS

Specify substep/unit conventions and initial assembly, anisotropic stretch/shear/bend/damping, panel pins and restart. Draw boundaries among Rust VBD, browser PBD and C++ XPBD; report runtime and constraints.

PROPOSED TYPED CONTRACT

SimulationRequest{documentRevision,backend,materials,avatar,config,seed}; Frame{positions,velocities,diagnostics}.

REQUIRED VERIFICATION AND FAILURE CASES

Tee/pants/jacket quality curves, finite values, reproducibility, controlled pause/reset and recorded unsupported contact conditions

Execution	Dependency and integration	Definition of done
M3/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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PPF contact batch acceptance

OBJECTIVE / ACCEPTANCE SCOPE

Full-frame and swept-contact validation rather than solver-completed status

ACTUAL SOURCE LOCATIONS

ppf_backend/server.mjs; ppf_backend/fitAcceptance.mjs; ppf_backend/assemblyAcceptance.mjs

IMPLEMENTATION REQUIREMENTS

Solve with initialized seam graph and validated boundaries; run cloth/body and cloth/cloth swept intersection analysis, seam-gap metrics, friction, strain and temporal jitter reports.

PROPOSED TYPED CONTRACT

AcceptanceReport{fixture,solverHash,frames,maxPenetrationMm,sweptHits,seamGapMm,strain,passed,failures}.

REQUIRED VERIFICATION AND FAILURE CASES

Known failing tee/arm motion must be rejected; acceptance only when independent metrics meet stated fixture thresholds

Execution	Dependency and integration	Definition of done
M3/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Simulation numerical reliability

OBJECTIVE / ACCEPTANCE SCOPE

Collision tolerances, convergence, diagnostics and safe restart

ACTUAL SOURCE LOCATIONS

native_cpp/core/cloth_collision_step.cpp; src/core/physics/qualityGate.js

IMPLEMENTATION REQUIREMENTS

Test stretch/shear energy, non-inverted triangles, mass conservation, energy drift, self-contact and avatar-contact residuals under variable timestep and remeshing.

PROPOSED TYPED CONTRACT

SolverDiagnostics{iteration,energy,residual,contactCount,violations,stable,runId}.

REQUIRED VERIFICATION AND FAILURE CASES

NaN, teleport, tunneling, seam bursting and stale frame tests; bounded cancellation; reproducible seed and environment

Execution	Dependency and integration	Definition of done
M3/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Physical fabric model

OBJECTIVE / ACCEPTANCE SCOPE

Traceable anisotropic sheet mechanics separate from optics

ACTUAL SOURCE LOCATIONS

src/core/fabricCalibration.js; src/core/fabricEvidenceCalibration.js; src/core/clothcraftFabrics.js

IMPLEMENTATION REQUIREMENTS

Use GSM, pressure-qualified thickness, directional force-extension curves, shear, bending hysteresis, friction pair and damping. Map measurements to each solver explicitly with uncertainty and benchmark fits.

PROPOSED TYPED CONTRACT

PhysicalFabricSpec{specimenId,conditioning,tests,SI,uncertainty,calibration,solverMappings}.

REQUIRED VERIFICATION AND FAILURE CASES

Real swatch and solver model measured against drape/force data; show estimated flag without calibration; 400-count not certification

Execution	Dependency and integration	Definition of done
M4/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Physical fabric acquisition

OBJECTIVE / ACCEPTANCE SCOPE

Lab specimen intake, unit normalization, calibration and provenance

ACTUAL SOURCE LOCATIONS

docs/FABRIC_MEASUREMENT_INGESTION.md; src/core/fabricEvidenceCalibration.js

IMPLEMENTATION REQUIREMENTS

Define machine calibration, specimen orientation, conditioning, repeat runs and raw curve storage; automate outlier/units checks and fit report. Keep unverifiable claims explicitly unverified.

PROPOSED TYPED CONTRACT

FabricTest{sampleId,instrument,method,orientation,replicate,rawSeries,operator,environment}.

REQUIRED VERIFICATION AND FAILURE CASES

Reject inconsistent load units, missing repeats and bad sampling; independent examiner can reproduce one fitted stretch response

Execution	Dependency and integration	Definition of done
M4/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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PBR textile optical pipeline

OBJECTIVE / ACCEPTANCE SCOPE

Measured/estimated PBR maps, yarn-scale repeat, panel UVs and color management

ACTUAL SOURCE LOCATIONS

src/core/textileProcedurals.js; src/core/fabricPbrGenerator.js; src/core/fabricSurfaceModel.js

IMPLEMENTATION REQUIREMENTS

Separate source albedo, normal, roughness, opacity, height, metallic, sheen, weave anisotropy and color-space tagging. Follow grainline in 2D UVs and preserve map provenance/hashes.

PROPOSED TYPED CONTRACT

VisualFabricProfile{id,textureSets,licenses,texelScaleCm,normalConvention,uvRotation,renderTransform}.

REQUIRED VERIFICATION AND FAILURE CASES

Neutral-light swatch + oblique-light comparisons, no flipped normals, consistent repeat size across garment sizes

Execution	Dependency and integration	Definition of done
M4/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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400 fabric catalog engineering

OBJECTIVE / ACCEPTANCE SCOPE

Preserve all 400 stable IDs, add specimen fidelity, swatch and construction truth

ACTUAL SOURCE LOCATIONS

src/data/fabricCatalog400.js; src/data/defaultProject.js; docs/reports/EXO_400_FABRICS_COVERAGE.json

IMPLEMENTATION REQUIREMENTS

Audit every current estimate and unique scan assignment, create priorities by cotton/French terry/denim/wool/knit/leather/nylon, measure high-volume archetypes, then populate calibrated variants only when tested.

PROPOSED TYPED CONTRACT

FabricCatalogEntry{id,parentConstructionId,appearanceVariant,evidenceLevel,specimenId,license,QA}.

REQUIRED VERIFICATION AND FAILURE CASES

400 stable entries selectable and saved; separate coverage measures for 400 parameter presets, 44 used image sources and 0 measured mechanics

Execution	Dependency and integration	Definition of done
M4/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Fabric optical finishes and washes

OBJECTIVE / ACCEPTANCE SCOPE

Non-destructive denim/garment washing, distress, colorway layers and masks

ACTUAL SOURCE LOCATIONS

src/core/clothWashProcedurals.js; src/core/clothWashImage.js

IMPLEMENTATION REQUIREMENTS

Create versioned finish stack with seeded masks, material-aware choices, physical-scale edge distances, seam masks, layered roughness effects and source-preserving compositing.

PROPOSED TYPED CONTRACT

FinishLayer{kind,strength,seed,blend,uvTransform,seamWear,masks,sourceHash}.

REQUIRED VERIFICATION AND FAILURE CASES

Undo/reopen deterministic colorways; mask aligned after grading; textile physics does not change for optical wash

Execution	Dependency and integration	Definition of done
M4/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Fabric browser and property editing

OBJECTIVE / ACCEPTANCE SCOPE

Context-sensitive metadata, measured status, face materials and revisioned edits

ACTUAL SOURCE LOCATIONS

src/components/ClothCraftFabricLibrary.jsx; src/components/CloPropertyEditorAccordion.jsx; src/core/cloPropertyCommands.js

IMPLEMENTATION REQUIREMENTS

Add editable Name/Item No./Construction/Supplier/Owner with revision validation, front/back/side appearance fields, physical properties, texture transforms, assignment and bulk edits.

PROPOSED TYPED CONTRACT

PropertyMutation{entityId,field,value,expectedRevision}; field registry defines type/units/read-only rules.

REQUIRED VERIFICATION AND FAILURE CASES

Selected fabric edits appear instantly, persist after native restart and never silently edit the selected panel

Execution	Dependency and integration	Definition of done
M4/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Avatar/garment fit interpretation

OBJECTIVE / ACCEPTANCE SCOPE

Clearance maps, strain maps, body measure POM, comfort limitations

ACTUAL SOURCE LOCATIONS

src/core/meshAvatarContact.js; docs/PPF_ACTUAL_GARMENT_FIT_DEEP_AUDIT_2026-10-14.md

IMPLEMENTATION REQUIREMENTS

Compute fit visualizations from tested geometry and solver diagnostics. Distinguish contact, high tensile strain, pressure proxy, nonconvergence and actual physical comfort.

PROPOSED TYPED CONTRACT

FitLayer{metric,units,min,max,confidence,fixtureId,sourceRevision}.

REQUIRED VERIFICATION AND FAILURE CASES

Known 2mm penetration appears as failure; legend and dataset distinguish measured and estimated metrics

Execution	Dependency and integration	Definition of done
M5/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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POM and measurement tools

OBJECTIVE / ACCEPTANCE SCOPE

Permanent dimension anchors, garment points of measure, grading and report tables

ACTUAL SOURCE LOCATIONS

src/core/measurements.js; src/core/clothcraftSelectionMetrics.js

IMPLEMENTATION REQUIREMENTS

Create authored POM paths anchored to pattern topology/2D/3D body landmarks; store measurement type, protocol, tolerance, placement and sizes.

PROPOSED TYPED CONTRACT

POM{id,measureKind,anchors,method,units,tolerance,sizeValues,revision}.

REQUIRED VERIFICATION AND FAILURE CASES

POM persists/grades; edit a pattern then recalculate; temporary native tape not mistaken for saved POM

Execution	Dependency and integration	Definition of done
M5/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Grading and multi-size

OBJECTIVE / ACCEPTANCE SCOPE

Point grade rules, size tables, curve propagation and notch/graphic sizes

ACTUAL SOURCE LOCATIONS

src/core/grading.js; native_cpp/core/grading.cpp

IMPLEMENTATION REQUIREMENTS

Model base-size control points with per-size deltas; interpolate graded curves, sew compatibility and item IDs; import/export AAMA/ASTM rules.

PROPOSED TYPED CONTRACT

GradeRule{pointId,sizeGroup,baseSize,deltaBySize,notchBehavior}.

REQUIRED VERIFICATION AND FAILURE CASES

Measurements for XS–XXL independently checked; seam match after grading; roundtrip through external CAD

Execution	Dependency and integration	Definition of done
M6/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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DXF and industrial CAD interchange

OBJECTIVE / ACCEPTANCE SCOPE

Bidirectional AAMA/ASTM/standard DXF with units, layers, curve and notch fidelity

ACTUAL SOURCE LOCATIONS

src/core/dxf.js; native_cpp/core/dxf_import.cpp; native_cpp/core/dxf_export.cpp

IMPLEMENTATION REQUIREMENTS

Explicitly map polylines/curves, grainlines, cut/stitch lines, notches, seam allowances, grade layers, units and import warnings.

PROPOSED TYPED CONTRACT

ImportReport{sourceFileHash,format,unitScale,mappedEntities,losses,warnings}.

REQUIRED VERIFICATION AND FAILURE CASES

External CAD roundtrips + measured millimeter tolerances on test tee and trouser markers

Execution	Dependency and integration	Definition of done
M6/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Markers and cutting layout

OBJECTIVE / ACCEPTANCE SCOPE

Fabric-width, nap, print direction, grain, multi-size nesting and yardage

ACTUAL SOURCE LOCATIONS

src/core/marker.js; native_cpp/core/marker_layout.cpp

IMPLEMENTATION REQUIREMENTS

Represent constrained polygon packing, repeated patterns, seam allowances, shrink, fabric flaws, stripes/plaids and lay plans with independently measured efficiency.

PROPOSED TYPED CONTRACT

MarkerJob{rollWidth,repeat,grainTolerance,nap,patternVersions,quantity};
MarkerReport{placements,utilization,violations}.

REQUIRED VERIFICATION AND FAILURE CASES

Verify no overlaps, actual length on plotted layout, stripe alignment and factory-signed sample marker

Execution	Dependency and integration	Definition of done
M6/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Factory BOM and cost

OBJECTIVE / ACCEPTANCE SCOPE

Traceable fabrics, hardware, seams, shrink allowances, suppliers, consumption and costing

ACTUAL SOURCE LOCATIONS

src/core/factoryPackage.js; src/core/bomXml.js; native_cpp/core/bom_export.cpp

IMPLEMENTATION REQUIREMENTS

Build line items from authored assignments and actual marker usage; retain vendor costs private; track alternates, unit conversions, minimum order and revision history.

PROPOSED TYPED CONTRACT

BOMLine{itemId,type,supplier,sku,qty,unit,usageBasis,sourceRevision}.

REQUIRED VERIFICATION AND FAILURE CASES

Per-size usage reconciles with marker; 3D hardware count equals BOM; private prices excluded from public previews

Execution	Dependency and integration	Definition of done
M6/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Technical pack production

OBJECTIVE / ACCEPTANCE SCOPE

Editable and immutable approved releases of POM, BOM, 2D, images and notes

ACTUAL SOURCE LOCATIONS

src/core/factoryPackage.js; docs/EXO_INTERACTIVE_WEB_TECHPACK_PLANNING_2026-10-16.md

IMPLEMENTATION REQUIREMENTS

Define source-linked tables, component callouts, sewing illustrations, grade measurements, tolerance sign-off, sample rounds and export manifest with reviewer approvals.

PROPOSED TYPED CONTRACT

TechPackRelease{id,sourceHash,sections,approvals,assets,visibility,signature}.

REQUIRED VERIFICATION AND FAILURE CASES

Factory receives complete selected release, revisions compare field by field; prior approved release cannot be silently overwritten

Execution	Dependency and integration	Definition of done
M6/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Project save, migration and recovery

OBJECTIVE / ACCEPTANCE SCOPE

Transactional atomic persistence, autosave and damaged-file recovery

ACTUAL SOURCE LOCATIONS

src/core/project.js; integrations/clothcraft/exo-session.rs

IMPLEMENTATION REQUIREMENTS

Use complete version headers, checked sums, forward-only migration, atomic temp/rename and off-site optional backup. Keep project source separate from generated caches.

PROPOSED TYPED CONTRACT

ProjectFile{schemaVersion,sourceRevision,authoringGraph,resourceManifest,historyWatermark}.

REQUIRED VERIFICATION AND FAILURE CASES

Crash during save cannot corrupt last good revision; 1-year-old fixture migrates with no loss of topology

Execution	Dependency and integration	Definition of done
M0/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Desktop app shell and Qt strategy

OBJECTIVE / ACCEPTANCE SCOPE

Consolidate signed macOS WebKit/Rust and native C++ Qt without dual sources of truth

ACTUAL SOURCE LOCATIONS

integrations/clothcraft/macos; native_app; native_cpp/desktop

IMPLEMENTATION REQUIREMENTS

Select one supported desktop product path per release; secure localhost RPC, lifecycle/IPC, signed sandbox, file dialogs, native accessibility, workspace state and crash logs.

PROPOSED TYPED CONTRACT

DesktopCapability{runtime,buildHash,supportedTools,backendEndpoints,securityScope}.

REQUIRED VERIFICATION AND FAILURE CASES

Installed signed app is tested on clean machine; no browser-only toolbar mistaken for native implementation

Execution	Dependency and integration	Definition of done
M7/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Typed automation and scripting

OBJECTIVE / ACCEPTANCE SCOPE

Stable operation coverage for human and AI tools

ACTUAL SOURCE LOCATIONS

docs/EXO_V2_AGENT_API_FOUNDATION.md; src/core/workspaceCommandIndex.js

IMPLEMENTATION REQUIREMENTS

Generate tool registry from exact command schemas; strict read/write scopes, previews, revision guards, atomic multi-step transaction, deterministic replay and logs.

PROPOSED TYPED CONTRACT

AgentOperation{command,targets,params,expectedRevision,allowSideEffects,previewOnly}.

REQUIRED VERIFICATION AND FAILURE CASES

Agent creates tee seam, edits fabric, exports on test fixture while all actions remain reversible

Execution	Dependency and integration	Definition of done
M7/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Rendering and visual studio

OBJECTIVE / ACCEPTANCE SCOPE

Film-grade textile optics, neutral reference lighting, still/movie output

ACTUAL SOURCE LOCATIONS

src/components/GarmentViewport.jsx; docs/EXO_UI_RENDERING_AND_AVATARS.md

IMPLEMENTATION REQUIREMENTS

Design physically plausible light presets, studio HDRI licenses, camera composition, tone maps, microtexture LOD, output color profiles and queued exports.

PROPOSED TYPED CONTRACT

RenderJob{sceneHash,materialHashes,camera,lights,resolution,colorSpace,output}.

REQUIRED VERIFICATION AND FAILURE CASES

Repeat captures compare visual refs under neutral light; no proprietary third-party HDRI bundled without license

Execution	Dependency and integration	Definition of done
M8/P2	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Asset library and local ownership

OBJECTIVE / ACCEPTANCE SCOPE

Searchable editable library of patterns, garments, avatars, fabrics, trims and studios

ACTUAL SOURCE LOCATIONS

docs/EXO_DATA_MODEL.md; public/avatars; docs/EXO_FABRIC_MATERIALS_AND_TRIMS.md

IMPLEMENTATION REQUIREMENTS

Implement content-addressed assets, preview indexes, local folders and reversible relinking, tags, duplicate detection, license metadata, offline access.

PROPOSED TYPED CONTRACT

LibraryAsset{id,type,contentHash,sourceUri,license,dependencies,previewHashes}.

REQUIRED VERIFICATION AND FAILURE CASES

Renaming a folder and reopening an outfit rebinds by hash without losing authored fabric IDs

Execution	Dependency and integration	Definition of done
M8/P2	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Open-source packaging and CI

OBJECTIVE / ACCEPTANCE SCOPE

Portable builds, dependency licenses, reproducible assets and support policy

ACTUAL SOURCE LOCATIONS

package.json; .github/workflows; docs/LICENSING.md

IMPLEMENTATION REQUIREMENTS

Pin native toolchains, generate SBOM, static checks, test fixtures, binary signing/notarization, contributor templates and regression dashboards.

PROPOSED TYPED CONTRACT

ReleaseManifest{commit,buildEnv,licenseInventory,testResults,knownGaps,packageHashes}.

REQUIRED VERIFICATION AND FAILURE CASES

Clean machine install/build passes supported command set; distributed binary dependencies and licenses match manifest

Execution	Dependency and integration	Definition of done
M7/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

The implementation must not claim commercially certified CLO parity simply because a visual analogue exists. Deliver acceptance evidence with numerical physical units and a human-reviewable before/after record. Persist unsupported reasons until tested rather than mapping a reference toolbar action to an unrelated operation.

Community and pattern marketplace

OBJECTIVE / ACCEPTANCE SCOPE

Independent future service with creator profiles, assets, rights and pricing

ACTUAL SOURCE LOCATIONS

docs/EXO_COMMUNITY_MARKETPLACE_SOCIAL_COLLABORATION_ARCHITECTURE_2026-10-16.md

IMPLEMENTATION REQUIREMENTS

Do not expose local source by default. Implement authorized publish previews, versioned entitlement, sanctioned remixes, refund and moderation events with no private CAD leakage.

PROPOSED TYPED CONTRACT

PublishedAsset{id,version,license,rights,publicDerivatives,privateSourceRef}.

REQUIRED VERIFICATION AND FAILURE CASES

An anonymous follower cannot fetch editable source; a purchased entitlement receives only licensed immutable revision

Execution	Dependency and integration	Definition of done
M9/P2	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Interactive web tech packs

OBJECTIVE / ACCEPTANCE SCOPE

Team collaboration, comments, proposals, immutable manufacturer release

ACTUAL SOURCE LOCATIONS

docs/EXO_INTERACTIVE_WEB_TECHPACK_PLANNING_2026-10-16.md

IMPLEMENTATION REQUIREMENTS

Design separate tenancy, section-based ACL, comment anchors on stable CAD IDs, proposal merges with revision preconditions, factory acknowledgments and diff receipts.

PROPOSED TYPED CONTRACT

Proposal{id,baseRevision,patch,author,reviewStatus}; Approval{releaseHash,role,time}.

REQUIRED VERIFICATION AND FAILURE CASES

Factory B cannot read Factory A quote; approved R01 stays downloadable after new R02 proposal

Execution	Dependency and integration	Definition of done
M9/P2	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Spatial showroom and projection

OBJECTIVE / ACCEPTANCE SCOPE

Optional walking wardrobe, projection staging, AR and video production

ACTUAL SOURCE LOCATIONS

docs/EXO_SPATIAL_STUDIO_EXECUTION_BACKLOG_2026-10-16.md

IMPLEMENTATION REQUIREMENTS

Prototype after CAD reliability; isolate projectable display meshes and 3D room masters, maintain stable looks and explicit asset rights. Avoid requiring PPF on web page load.

PROPOSED TYPED CONTRACT

Look{id,garmentRevision,avatarPose,studio,lights,cameras,derivativeManifest}.

REQUIRED VERIFICATION AND FAILURE CASES

Walkable 10-outfit scene runs on target hardware and never requests private unpublished geometry

Execution	Dependency and integration	Definition of done
M10/P3	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Security/privacy and permissions

OBJECTIVE / ACCEPTANCE SCOPE

Project privacy, import sandboxing, safe local bridge and public publishing

ACTUAL SOURCE LOCATIONS

integrations/clothcraft/server.mjs; docs/EXO_RELEASE_STATUS.md

IMPLEMENTATION REQUIREMENTS

Bind local services to loopback, validate origins/input size, guard destructive mutations by version; block path traversal, zip bombs, malformed GLB/DXF and untrusted texture payloads.

PROPOSED TYPED CONTRACT

CapabilityGrant{principal,scope,resource,operation,expiry}; SecurityEvent{id,reason,source}.

REQUIRED VERIFICATION AND FAILURE CASES

Untrusted project import is quarantined, no extra disk paths are readable, serialized source is never leaked to previews

Execution	Dependency and integration	Definition of done
M0/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Performance budgets and telemetry

OBJECTIVE / ACCEPTANCE SCOPE

Large pattern sets, 400 catalog, textures, simulation, import and signed app responsiveness

ACTUAL SOURCE LOCATIONS

src/components/GarmentViewport.jsx; src/core/backends/registry.js

IMPLEMENTATION REQUIREMENTS

Define workload fixtures per device, memory/latency budgets, shader/texture loading and worker cancellation. Cache by exact revision and avoid nested full rebuild on visual-only edit.

PROPOSED TYPED CONTRACT

PerfTrace{operation,fixture,cpuMs,gpuMs,memoryPeak,outputBytes,revision}.

REQUIRED VERIFICATION AND FAILURE CASES

Measure 100-piece scene interactions and worst-case 400 swatch browsing without UI deadlock or stale state

Execution	Dependency and integration	Definition of done
M7/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Accessibility and localization

OBJECTIVE / ACCEPTANCE SCOPE

Keyboard complete editing, focus rules, units and descriptive commands

ACTUAL SOURCE LOCATIONS

src/components/CloUI.jsx; src/components/RightToolPalette.jsx

IMPLEMENTATION REQUIREMENTS

Specify accessible role/description of controls, hover+keyboard parity, radial tool groups, screen-reader property labels and regional numeric/unit conversion.

PROPOSED TYPED CONTRACT

UserPreferences{unitSystem,locale,shortcutMap,workspaceLayout}; UICommand{ariaLabel,shortcut,disableReason}.

REQUIRED VERIFICATION AND FAILURE CASES

Every toolbar button has exact accessible name, keyboard path, valid focus and meaningful disabled explanation

Execution	Dependency and integration	Definition of done
M7/P1	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Tool-bar and property parity reference

OBJECTIVE / ACCEPTANCE SCOPE

70 verified CLO 7 toolbar positions as reference; precise supported/inspect/disabled EXO states

ACTUAL SOURCE LOCATIONS

src/data/clo7ToolbarOrder.js; src/core/clo7NativeToolCapabilities.js

IMPLEMENTATION REQUIREMENTS

Preserve observed order separately for 3D and 2D independent A/B columns, live semantics, context switches, icons and flyouts. Implement actual Rust commands before updating capability registry.

PROPOSED TYPED CONTRACT

ToolCapability{id,dimension,column,position,mode,nativeAction,reason,testEvidence}.

REQUIRED VERIFICATION AND FAILURE CASES

70/70 controls visible in documented order; 11 native / 9 inspect / 50 unavailable treated as present snapshot, never all functional

Execution	Dependency and integration	Definition of done
M1/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Context-aware Property Editor

OBJECTIVE / ACCEPTANCE SCOPE

Pattern/fabric/seam/avatar/trim selection and editable sections with validation

ACTUAL SOURCE LOCATIONS

src/components/CloPropertyEditorAccordion.jsx; src/components/CloObjectPropertyFields.jsx

IMPLEMENTATION REQUIREMENTS

Normalize selection context and field registry. Each field has type, range, units, disable reason, localization, dependency invalidations and conflict messaging.

PROPOSED TYPED CONTRACT

PropertyDescriptor{entityKinds,key,type,units,min,max,nullable,editable,backendCapability}.

REQUIRED VERIFICATION AND FAILURE CASES

Selecting a fabric exposes Fabric1 metadata/PBR; selecting a pattern exposes construction/particle distance without cross-writing

Execution	Dependency and integration	Definition of done
M1/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Material calibration regression

OBJECTIVE / ACCEPTANCE SCOPE

Cross-solver physical mapping, optical maps and objective lab-data acceptance

ACTUAL SOURCE LOCATIONS

docs/FABRIC_MEASUREMENT_INGESTION.md; src/core/fabricCalibration.js

IMPLEMENTATION REQUIREMENTS

Create calibrated material cards with instrument source, unit normalization, error bars, differential patch tests, numerical integration and hold-out material tests.

PROPOSED TYPED CONTRACT

CalibrationReport{specimenId,solverBuild,parameterMapping,errors,heldOutCurves,approvedBy}.

REQUIRED VERIFICATION AND FAILURE CASES

Exactly zero measured presets until reviewed dataset; every promoted measured state links raw series and reproducible fitting script

Execution	Dependency and integration	Definition of done
M4/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Cross-backend numerical parity

OBJECTIVE / ACCEPTANCE SCOPE

JS, C++ and Rust share semantics without conflating matching with physical accuracy

ACTUAL SOURCE LOCATIONS

native_cpp/CONVERSION_STATUS_2026-10-15.md; docs/CLOTHCRAFT_CPP_RUST_INTEROP_2026-10-15.md

IMPLEMENTATION REQUIREMENTS

Compare data model/operation outputs by type: geometry in mm, solver samples in SI, frame arrays, UV and factory payloads. Keep solver maturity separate from exact JS parity.

PROPOSED TYPED CONTRACT

ParityResult{fixture,engines,version,field,expected,actual,tolerance,difference,classification}.

REQUIRED VERIFICATION AND FAILURE CASES

Replicate 58 tests from clean native build; reject >tolerance collision or style metadata drift, with machine-recorded reports

Execution	Dependency and integration	Definition of done
M0/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

The implementation must not claim commercially certified CLO parity simply because a visual analogue exists. Deliver acceptance evidence with numerical physical units and a human-reviewable before/after record. Persist unsupported reasons until tested rather than mapping a reference toolbar action to an unrelated operation.

Engineering release gating

OBJECTIVE / ACCEPTANCE SCOPE

Evidence-weighted readiness by workflow, never decorative percentage complete

ACTUAL SOURCE LOCATIONS

docs/EXO_VALIDATION_TEST_PLAN.md; docs/EXO_MILESTONE_ISSUE_BOARD.md

IMPLEMENTATION REQUIREMENTS

Every issue has traceable fixture, failure record, explicit acceptance, code owner, dependency, risk and command. Require independent visual and numerical regression.

PROPOSED TYPED CONTRACT

ReleaseGate{id,feature,owner,fixture,evidence,passed,blockers,risk,reviewedAt}.

REQUIRED VERIFICATION AND FAILURE CASES

No factory-ready/fit-certified claim without external reviews; warn on unverified native controls and PPF crossings

Execution	Dependency and integration	Definition of done
M0/P0	Source-model revision guard, source IDs and explicit invalidation of dependent caches	Named QA evidence, reproducible fixture, snapshot, proper error state
Integration ownership	Browser/React view, signed Rust V2 action, separate C++/PPF service as applicable	Do not declare implemented until actual active workspace path uses new semantics
Rollback and compatibility	Migration route for old native/browser documents, inverse command or snapshot	Save/reopen/undo and alternate-backend non-regression

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Interchange format matrix

FUNCTIONAL BOUNDARY

DXF AAMA/ASTM, SVG, OBJ, GLB, FBX, USD, ZPRJ/ZFAB import/export intent and unsupported boundaries

REFERENCE SURFACE

src/core/dxf.js; integrations/clo_sdk/audit_zprj.py

IMPLEMENTATION DETAIL

Per-format semantic matrix of geometry, sizes, materials, seams, animations, textures, versioning and license; import warnings rather than silent loss.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Regression fixture repository

FUNCTIONAL BOUNDARY

Reference garments across multiple body shapes and garments

REFERENCE SURFACE

docs/EXO_VALIDATION_TEST_PLAN.md; native_cpp/tests

IMPLEMENTATION DETAIL

Fixtures for sleeveless tee, raglan hoodie, 5-pocket jeans, structured jacket and layered garments in smallest and largest supported sizes.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Cloth solver benchmark protocol

FUNCTIONAL BOUNDARY

Timed, seeded and measured acceptances across device classes

REFERENCE SURFACE

src/core/physics/qualityGate.js; ppf_backend/fitAcceptance.mjs

IMPLEMENTATION DETAIL

Pin timestep and tolerances and report absolute seam gaps, body penetration, triangle inversion, strain, collision and normal tangency.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Deployment update architecture

FUNCTIONAL BOUNDARY

Signed macOS bundle, upgrades, rollback and notarization

REFERENCE SURFACE

integrations/clothcraft/macos; native_app

IMPLEMENTATION DETAIL

Versioned signed app and native service builds; checksum install and reversible migration; protect project documents from forced overwrite.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Typography and design quality

FUNCTIONAL BOUNDARY

Consistent editor layout, docking, icons, scale and workflows

REFERENCE SURFACE

docs/CLO7_LIVE_TOOLBAR_AUDIT_20261016.md; src/clo-interface.css

IMPLEMENTATION DETAIL

Design-system tokens with per-platform high DPI, accessible commands and standard interaction rules; audit real app screenshots.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Manufacturing traceability

FUNCTIONAL BOUNDARY

Cutter-ready package provenance and release instructions

REFERENCE SURFACE

docs/EXO_PRODUCTION_PIPELINE.md; src/core/factoryPackage.js

IMPLEMENTATION DETAIL

Every exported drawing, POM, BOM and marker points back to an immutable garment revision and measured shrinkage/width data.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Fabric inspection laboratory

FUNCTIONAL BOUNDARY

Light booth view and transparent physical/visual evidence

REFERENCE SURFACE

docs/FABRIC_PHYSICS_AND_CLO_RESEARCH_2026-10-16.md

IMPLEMENTATION DETAIL

Enforce standardized lighting/color targets and physical swatch sample card, calibrated repeat scale and known instrument readings.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Finish masking studio

FUNCTIONAL BOUNDARY

Sew and fold aware fade/abrasion authoring

REFERENCE SURFACE

src/core/clothWashProcedurals.js; src/core/clothWashImage.js

IMPLEMENTATION DETAIL

Editable UV-space procedural masks, painting layer revision and stable seam landmark coordinates with nondestructive export.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

CLO version change monitoring

FUNCTIONAL BOUNDARY

Maintain exact reference audit by version without copying proprietary assets

REFERENCE SURFACE

src/data/clo7ToolbarOrder.js; docs/CLO7_LIVE_TOOLBAR_AUDIT_20261016.md

IMPLEMENTATION DETAIL

Pin reference version 7.0.242; changes in 2026 releases are separate additions, not silently appended into 70-slot order.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Backend isolation and diagnostics

FUNCTIONAL BOUNDARY

Reliable watchdog and bounded local process orchestration

REFERENCE SURFACE

native_backend/server.mjs; ppf_backend/server.mjs

IMPLEMENTATION DETAIL

Deadline, job cancellation, preview artifact manifests, child PID cleanup and errors without blocking UI process.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Asset licensing and provenance

FUNCTIONAL BOUNDARY

Verify rights for photos, scans, avatar meshes and external source

REFERENCE SURFACE

docs/LICENSING.md; docs/reports/EXO_400_LICENSED_SCAN_REGISTRY.json

IMPLEMENTATION DETAIL

Keep CC0 source evidence, user-uploaded restrictions, per-map hash and dependencies; never embed CLO proprietary resources.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Automated tool functional matrix

FUNCTIONAL BOUNDARY

Every native/legacy UI control from click to persistence

REFERENCE SURFACE

integrations/clothcraft/check-clo7-native-capabilities.mjs; scripts/qa-clo-interface.mjs

IMPLEMENTATION DETAIL

Test click, command dispatch, state change, selection, undo, save/load, screenshot and error case for each toolbar tool.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Context property exhaustive matrix

FUNCTIONAL BOUNDARY

Property Editor all selected object kinds and multi-select

REFERENCE SURFACE

docs/CLO3D_PROPERTY_EDITOR_DEEP_AUDIT_20261016.md;
docs/CLO7_LIVE_PROPERTY_EDITOR_AUDIT_20261016.md

IMPLEMENTATION DETAIL

Exact names, types, units and default state for pattern, fabric, seam, trims, avatar, simulation, graphic, grading and POM; test null/mixed values.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Look and rendering ownership

FUNCTIONAL BOUNDARY

Final presentation without replacing CAD sources

REFERENCE SURFACE

docs/EXO_SPATIAL_STUDIO_TEN_PAGE_IMPLEMENTATION_BLUEPRINT_2026-10-16.md

IMPLEMENTATION DETAIL

Store immutable looks and render settings with source hashes; public visual derivatives contain no private pattern coordinates.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Development operation and observability

FUNCTIONAL BOUNDARY

Build health, persistent regression dashboards, failure triage

REFERENCE SURFACE

package.json; docs/EXO_DEVELOPER_GUIDE.md

IMPLEMENTATION DETAIL

Continuous CI on clean environment, runtime capability report, checksums, bug repro archive and automated artifact index.

Work stage	Required output
Model	Typed versioned entities with explicit units, IDs and reference hashes
User workflow	Inputs, click/keyboard selection, live response, contextual inspector and cancel path
Failure mode	Invalid file, selection, missing capability, concurrency conflict and stale cache
Persistence	Undo/redo, save/load/migrate, chosen export and provenance
Verification	Positive/negative fixture, UI interaction, machine metrics and reviewer sign-off

Design the interaction against the currently connected native operation registry. A presentational equivalent and an operational equivalent require separate status flags and evidence. Record the relevant fixture and exact reproduction command when marking each of the above outputs accepted.

Simulate

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A01	Workspace / simulation	NATIVE	simulation.toggle

OBSERVED PURPOSE AND TARGET USER ACTION

Simulate is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for workspace / simulation; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Validate current state and command preconditions, expose real operation only when backend implements it and return typed state change with revision and diagnostics.

DATA AND TRANSACTION CONTRACT

CommandEnvelope and ProjectRevision. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: native. Native step changes simulated cloth state; pause stops scheduled steps. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Click with valid state, run without selection, undo/restart and assert no fake execution or mislabeled support. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Select/Move

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A02	3D editing / avatar	NATIVE	viewport3d.selectMove

OBSERVED PURPOSE AND TARGET USER ACTION

Select/Move is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: native. Native Rust Place/pose-panel changes selected garment placement; no CLO gizmo parity claim. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Select Mesh (Brush)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A03	3D editing / avatar	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Select Mesh (Brush) is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. No equivalent verified Rust V2 operation; keep this reference tool disabled. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Styleline

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A04	3D editing / avatar	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Styleline is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs persistent 3D projected pen/stroke geometry with editable anchors, surface attachment, undo and export. Existing 2D construction lines are not a substitute. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Sewing

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A05	Stitch topology	INSPECT	sewing.inspectSeams

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Sewing is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for stitch topology; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Selection must find boundary IDs and parameter ranges; pair orientation/ease; update sewing graph, constraint buffer and diagnostics.

DATA AND TRANSACTION CONTRACT

SeamRelation and edge intervals. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: inspect. Related native information is viewable, but this CLO editing operation is not implemented. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Sew across one segment, reverse it, undo, reopen, measure residual seam gap, then intentionally select invalid edges. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Segment Sewing

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A06	Stitch topology	NATIVE	sewing.freeSew

OBSERVED PURPOSE AND TARGET USER ACTION

Segment Sewing is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for stitch topology; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Selection must find boundary IDs and parameter ranges; pair orientation/ease; update sewing graph, constraint buffer and diagnostics.

DATA AND TRANSACTION CONTRACT

SeamRelation and edge intervals. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: native. Select two garment edge ranges in the native 3D viewport to author a segment sewing relationship; requires actual valid edges. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Sew across one segment, reverse it, undo, reopen, measure residual seam gap, then intentionally select invalid edges. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Free Sewing

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A07	Stitch topology	NATIVE	sewing.freeSew

OBSERVED PURPOSE AND TARGET USER ACTION

Free Sewing is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for stitch topology; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Selection must find boundary IDs and parameter ranges; pair orientation/ease; update sewing graph, constraint buffer and diagnostics.

DATA AND TRANSACTION CONTRACT

SeamRelation and edge intervals. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: native. Select two garment edges in the native 3D viewport to sew them with the existing Rust seam operation. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Sew across one segment, reverse it, undo, reopen, measure residual seam gap, then intentionally select invalid edges. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Auto Sewing

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A08	Stitch topology	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Auto Sewing is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for stitch topology; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Selection must find boundary IDs and parameter ranges; pair orientation/ease; update sewing graph, constraint buffer and diagnostics.

DATA AND TRANSACTION CONTRACT

SeamRelation and edge intervals. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires first-class seam/tack selection and editing with valid boundary pair relationships, attachment constraints and reversible native operations; only selected simpler sewing modes work so far. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Sew across one segment, reverse it, undo, reopen, measure residual seam gap, then intentionally select invalid edges. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Tack

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A09	Stitch topology	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Tack is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for stitch topology; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Selection must find boundary IDs and parameter ranges; pair orientation/ease; update sewing graph, constraint buffer and diagnostics.

DATA AND TRANSACTION CONTRACT

SeamRelation and edge intervals. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires first-class seam/tack selection and editing with valid boundary pair relationships, attachment constraints and reversible native operations; only selected simpler sewing modes work so far. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Sew across one segment, reverse it, undo, reopen, measure residual seam gap, then intentionally select invalid edges. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Tack

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A10	Stitch topology	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Tack is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for stitch topology; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Selection must find boundary IDs and parameter ranges; pair orientation/ease; update sewing graph, constraint buffer and diagnostics.

DATA AND TRANSACTION CONTRACT

SeamRelation and edge intervals. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires first-class seam/tack selection and editing with valid boundary pair relationships, attachment constraints and reversible native operations; only selected simpler sewing modes work so far. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Sew across one segment, reverse it, undo, reopen, measure residual seam gap, then intentionally select invalid edges. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Fold Arrangement

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A11	3D editing / avatar	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Fold Arrangement is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a dedicated arrangement, fold or remeshing operation that preserves authored patterns and measures resulting cloth state. Use supported native move/reset/simulation controls until verified. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Fold 3D Garment

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A12	3D editing / avatar	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Fold 3D Garment is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a dedicated arrangement, fold or remeshing operation that preserves authored patterns and measures resulting cloth state. Use supported native move/reset/simulation controls until verified. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Reset 2D Arrangement (All)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A13	3D editing / avatar	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Reset 2D Arrangement (All) is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a dedicated arrangement, fold or remeshing operation that preserves authored patterns and measures resulting cloth state. Use supported native move/reset/simulation controls until verified. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Reset 3D Arrangement (All)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A14	3D editing / avatar	NATIVE	arrangement.reset3d

OBSERVED PURPOSE AND TARGET USER ACTION

Reset 3D Arrangement (All) is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: native. Native Rust reset restores the 3D initial off-body pose while preserving authored 2D patterns, material and sewing relations. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Re-drape 3D Arrangement

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A15	3D editing / avatar	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Re-drape 3D Arrangement is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a dedicated arrangement, fold or remeshing operation that preserves authored patterns and measures resulting cloth state. Use supported native move/reset/simulation controls until verified. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Hi-Res Garment

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A16	Workspace / simulation	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Hi-Res Garment is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for workspace / simulation; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Validate current state and command preconditions, expose real operation only when backend implements it and return typed state change with revision and diagnostics.

DATA AND TRANSACTION CONTRACT

CommandEnvelope and ProjectRevision. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a dedicated arrangement, fold or remeshing operation that preserves authored patterns and measures resulting cloth state. Use supported native move/reset/simulation controls until verified. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Click with valid state, run without selection, undo/restart and assert no fake execution or mislabeled support. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Measure (Avatar)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A17	Measurement / drafting	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Measure (Avatar) is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for measurement / drafting; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select referenced geometry, preserve object IDs through grading/point edits, and report exact units and tolerances in active inspector.

DATA AND TRANSACTION CONTRACT

MeasurementAnchor and GradeRule. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires typed avatar measurements, animation tracks or garment measurement objects that remain attached to native 3D geometry; no matching authoring command is currently available. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Author, edit, measure, save/reopen, grade across sizes and export selected dimension with unchanged units. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Basic Tape Measure (Avatar)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A18	Measurement / drafting	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Basic Tape Measure (Avatar) is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for measurement / drafting; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select referenced geometry, preserve object IDs through grading/point edits, and report exact units and tolerances in active inspector.

DATA AND TRANSACTION CONTRACT

MeasurementAnchor and GradeRule. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires typed avatar measurements, animation tracks or garment measurement objects that remain attached to native 3D geometry; no matching authoring command is currently available. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Author, edit, measure, save/reopen, grade across sizes and export selected dimension with unchanged units. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Measure (Garment)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A19	Measurement / drafting	INSPECT	dimensions.show

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Measure (Garment) is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for measurement / drafting; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select referenced geometry, preserve object IDs through grading/point edits, and report exact units and tolerances in active inspector.

DATA AND TRANSACTION CONTRACT

MeasurementAnchor and GradeRule. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: inspect. Related native information is viewable, but this CLO editing operation is not implemented. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Author, edit, measure, save/reopen, grade across sizes and export selected dimension with unchanged units. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Linear Measure (Garment)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · A · A20	Measurement / drafting	INSPECT	dimensions.show

OBSERVED PURPOSE AND TARGET USER ACTION

Linear Measure (Garment) is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for measurement / drafting; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select referenced geometry, preserve object IDs through grading/point edits, and report exact units and tolerances in active inspector.

DATA AND TRANSACTION CONTRACT

MeasurementAnchor and GradeRule. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: inspect. Related native information is viewable, but this CLO editing operation is not implemented. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Author, edit, measure, save/reopen, grade across sizes and export selected dimension with unchanged units. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Play Motion

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B01	3D editing / avatar	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Play Motion is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires typed avatar measurements, animation tracks or garment measurement objects that remain attached to native 3D geometry; no matching authoring command is currently available. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Line (3D Pattern)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B02	3D editing / avatar	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Line (3D Pattern) is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs persistent 3D projected pen/stroke geometry with editable anchors, surface attachment, undo and export. Existing 2D construction lines are not a substitute. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Line (3D Pattern)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B03	3D editing / avatar	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Line (3D Pattern) is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs persistent 3D projected pen/stroke geometry with editable anchors, surface attachment, undo and export. Existing 2D construction lines are not a substitute. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

3D Base Pen

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B04	3D editing / avatar	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

3D Base Pen is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs persistent 3D projected pen/stroke geometry with editable anchors, surface attachment, undo and export. Existing 2D construction lines are not a substitute. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit 3D Pen (Avatar)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B05	3D editing / avatar	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Edit 3D Pen (Avatar) is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs persistent 3D projected pen/stroke geometry with editable anchors, surface attachment, undo and export. Existing 2D construction lines are not a substitute. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

3D Pen (Avatar)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B06	3D editing / avatar	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

3D Pen (Avatar) is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 3d editing / avatar; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Map hit-tested 3D triangle and barycentric coordinates to source pattern, update surface or arrangement state, retain pose and stable ID lineage under remeshing.

DATA AND TRANSACTION CONTRACT

SurfaceAttachment, Pose and GarmentFrame. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs persistent 3D projected pen/stroke geometry with editable anchors, surface attachment, undo and export. Existing 2D construction lines are not a substitute. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Orbit, pick near an occluded edge, edit in native 3D, inspect 2D mapping, undo and reopen across a remesh. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Texture (3D)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B07	Material / finish	INSPECT	fabric.inspectProperties

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Texture (3D) is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for material / finish; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select owning surface/pattern and bind texture space to grainline; typed layer operations must invalidate visual maps, and physically meaningful changes require a separate calibrated solver operation.

DATA AND TRANSACTION CONTRACT

MaterialAssignment and FinishLayer. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: inspect. Related native information is viewable, but this CLO editing operation is not implemented. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Apply to one panel, rotate grainline, undo, export swatch and check no accidental change to physical coefficients. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Transform Graphic

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B08	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Transform Graphic is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs native object selection/placement and a typed instance/style model linked to the cloth surface. Related EXO draft annotations exist, but are not yet native CLO-equivalent geometry. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Graphic (3D Pattern)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B09	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Graphic (3D Pattern) is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs native object selection/placement and a typed instance/style model linked to the cloth surface. Related EXO draft annotations exist, but are not yet native CLO-equivalent geometry. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Select/Move Button

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B10	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Select/Move Button is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs native object selection/placement and a typed instance/style model linked to the cloth surface. Related EXO draft annotations exist, but are not yet native CLO-equivalent geometry. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Button

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B11	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Button is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs native object selection/placement and a typed instance/style model linked to the cloth surface. Related EXO draft annotations exist, but are not yet native CLO-equivalent geometry. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Fasten Button

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B12	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Fasten Button is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires typed 3D trim geometry, attachment/sewing topology, rigid or deformable contact, material/style assets, and undo. EXO draft trim annotations alone do not satisfy this. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Zipper

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B13	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Zipper is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires typed 3D trim geometry, attachment/sewing topology, rigid or deformable contact, material/style assets, and undo. EXO draft trim annotations alone do not satisfy this. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Piping

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B14	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Piping is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires typed 3D trim geometry, attachment/sewing topology, rigid or deformable contact, material/style assets, and undo. EXO draft trim annotations alone do not satisfy this. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Piping

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B15	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Piping is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires typed 3D trim geometry, attachment/sewing topology, rigid or deformable contact, material/style assets, and undo. EXO draft trim annotations alone do not satisfy this. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Select Binding

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B16	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Select Binding is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires typed 3D trim geometry, attachment/sewing topology, rigid or deformable contact, material/style assets, and undo. EXO draft trim annotations alone do not satisfy this. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Binding

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B17	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Binding is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires typed 3D trim geometry, attachment/sewing topology, rigid or deformable contact, material/style assets, and undo. EXO draft trim annotations alone do not satisfy this. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Press

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
3D · B · B18	Material / finish	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Press is a separately named control in the 3D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for material / finish; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select owning surface/pattern and bind texture space to grainline; typed layer operations must invalidate visual maps, and physically meaningful changes require a separate calibrated solver operation.

DATA AND TRANSACTION CONTRACT

MaterialAssignment and FinishLayer. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a solver-backed textile finishing or reinforcement constraint, with material/temperature parameters and measurable cloth response. EXO does not yet simulate this manufacturing process. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Apply to one panel, rotate grainline, undo, export swatch and check no accidental change to physical coefficients. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Transform Pattern

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A01	2D pattern CAD	NATIVE	pattern2d.selectTransform

OBSERVED PURPOSE AND TARGET USER ACTION

Transform Pattern is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 2d pattern cad; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Edit in physical units on the canonical planar source using robust snapping and exact point/curve transforms. Rebuild dependent seam and mesh mappings deterministically.

DATA AND TRANSACTION CONTRACT

PatternPiece, BoundaryEdge and CurveAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: native. Native BoardTransform changes 2D panel placement. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Use exact numeric distances at two zooms, verify geometry/topology, undo, reopen and DXF export with mm tolerances. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Pattern

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A02	2D pattern CAD	NATIVE	pattern2d.editPoints

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Pattern is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 2d pattern cad; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Edit in physical units on the canonical planar source using robust snapping and exact point/curve transforms. Rebuild dependent seam and mesh mappings deterministically.

DATA AND TRANSACTION CONTRACT

PatternPiece, BoundaryEdge and CurveAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: native. Native MovePoints/CurvePoint edit existing pattern points. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Use exact numeric distances at two zooms, verify geometry/topology, undo, reopen and DXF export with mm tolerances. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Fullness (Point)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A03	2D pattern CAD	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Fullness (Point) is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 2d pattern cad; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Edit in physical units on the canonical planar source using robust snapping and exact point/curve transforms. Rebuild dependent seam and mesh mappings deterministically.

DATA AND TRANSACTION CONTRACT

PatternPiece, BoundaryEdge and CurveAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a native parametric drafting operation that updates real pattern points/segments and survives undo, reopen and production export. Current EXO draft tools are not native Rust equivalents. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Use exact numeric distances at two zooms, verify geometry/topology, undo, reopen and DXF export with mm tolerances. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Polygon

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A04	2D pattern CAD	NATIVE	pattern2d.drawPolygon

OBSERVED PURPOSE AND TARGET USER ACTION

Polygon is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 2d pattern cad; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Edit in physical units on the canonical planar source using robust snapping and exact point/curve transforms. Rebuild dependent seam and mesh mappings deterministically.

DATA AND TRANSACTION CONTRACT

PatternPiece, BoundaryEdge and CurveAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: native. Native Polygon creates authored pattern geometry. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Use exact numeric distances at two zooms, verify geometry/topology, undo, reopen and DXF export with mm tolerances. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Internal Polygon/Line

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A05	2D pattern CAD	NATIVE	pattern2d.drawInternalLine

OBSERVED PURPOSE AND TARGET USER ACTION

Internal Polygon/Line is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 2d pattern cad; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Edit in physical units on the canonical planar source using robust snapping and exact point/curve transforms. Rebuild dependent seam and mesh mappings deterministically.

DATA AND TRANSACTION CONTRACT

PatternPiece, BoundaryEdge and CurveAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: native. Native InternalLine authors panel interior lines. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Use exact numeric distances at two zooms, verify geometry/topology, undo, reopen and DXF export with mm tolerances. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Base Polygon

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A06	2D pattern CAD	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Base Polygon is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 2d pattern cad; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Edit in physical units on the canonical planar source using robust snapping and exact point/curve transforms. Rebuild dependent seam and mesh mappings deterministically.

DATA AND TRANSACTION CONTRACT

PatternPiece, BoundaryEdge and CurveAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a native parametric drafting operation that updates real pattern points/segments and survives undo, reopen and production export. Current EXO draft tools are not native Rust equivalents. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Use exact numeric distances at two zooms, verify geometry/topology, undo, reopen and DXF export with mm tolerances. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Trace

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A07	2D pattern CAD	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Trace is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 2d pattern cad; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Edit in physical units on the canonical planar source using robust snapping and exact point/curve transforms. Rebuild dependent seam and mesh mappings deterministically.

DATA AND TRANSACTION CONTRACT

PatternPiece, BoundaryEdge and CurveAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a native parametric drafting operation that updates real pattern points/segments and survives undo, reopen and production export. Current EXO draft tools are not native Rust equivalents. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Use exact numeric distances at two zooms, verify geometry/topology, undo, reopen and DXF export with mm tolerances. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Notch

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A08	Measurement / drafting	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Notch is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for measurement / drafting; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select referenced geometry, preserve object IDs through grading/point edits, and report exact units and tolerances in active inspector.

DATA AND TRANSACTION CONTRACT

MeasurementAnchor and GradeRule. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a native parametric drafting operation that updates real pattern points/segments and survives undo, reopen and production export. Current EXO draft tools are not native Rust equivalents. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Author, edit, measure, save/reopen, grade across sizes and export selected dimension with unchanged units. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Seam Allowance

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A09	2D pattern CAD	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Seam Allowance is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 2d pattern cad; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Edit in physical units on the canonical planar source using robust snapping and exact point/curve transforms. Rebuild dependent seam and mesh mappings deterministically.

DATA AND TRANSACTION CONTRACT

PatternPiece, BoundaryEdge and CurveAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a native parametric drafting operation that updates real pattern points/segments and survives undo, reopen and production export. Current EXO draft tools are not native Rust equivalents. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Use exact numeric distances at two zooms, verify geometry/topology, undo, reopen and DXF export with mm tolerances. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Walk Pattern

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A10	2D pattern CAD	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Walk Pattern is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for 2d pattern cad; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Edit in physical units on the canonical planar source using robust snapping and exact point/curve transforms. Rebuild dependent seam and mesh mappings deterministically.

DATA AND TRANSACTION CONTRACT

PatternPiece, BoundaryEdge and CurveAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a native parametric drafting operation that updates real pattern points/segments and survives undo, reopen and production export. Current EXO draft tools are not native Rust equivalents. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Use exact numeric distances at two zooms, verify geometry/topology, undo, reopen and DXF export with mm tolerances. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Point of Measure

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A11	Measurement / drafting	INSPECT	dimensions.measureTemporary

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Point of Measure is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for measurement / drafting; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select referenced geometry, preserve object IDs through grading/point edits, and report exact units and tolerances in active inspector.

DATA AND TRANSACTION CONTRACT

MeasurementAnchor and GradeRule. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: inspect. Click two positions on the native 2D pattern to measure exact metric distance. This is a temporary tape overlay, not an authored production POM. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Author, edit, measure, save/reopen, grade across sizes and export selected dimension with unchanged units. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Point of Measure

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A12	Measurement / drafting	INSPECT	dimensions.measureTemporary

OBSERVED PURPOSE AND TARGET USER ACTION

Point of Measure is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for measurement / drafting; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select referenced geometry, preserve object IDs through grading/point edits, and report exact units and tolerances in active inspector.

DATA AND TRANSACTION CONTRACT

MeasurementAnchor and GradeRule. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: inspect. Temporary two-click tape measure is available on the native 2D board. Saved and graded CLO POM construction remains unimplemented. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Author, edit, measure, save/reopen, grade across sizes and export selected dimension with unchanged units. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Annotation

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A13	Measurement / drafting	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Annotation is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for measurement / drafting; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select referenced geometry, preserve object IDs through grading/point edits, and report exact units and tolerances in active inspector.

DATA AND TRANSACTION CONTRACT

MeasurementAnchor and GradeRule. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. No equivalent verified Rust V2 operation; keep this reference tool disabled. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Author, edit, measure, save/reopen, grade across sizes and export selected dimension with unchanged units. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Pattern Annotation

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A14	Measurement / drafting	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Pattern Annotation is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for measurement / drafting; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select referenced geometry, preserve object IDs through grading/point edits, and report exact units and tolerances in active inspector.

DATA AND TRANSACTION CONTRACT

MeasurementAnchor and GradeRule. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. No equivalent verified Rust V2 operation; keep this reference tool disabled. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Author, edit, measure, save/reopen, grade across sizes and export selected dimension with unchanged units. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Pleats

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A15	Measurement / drafting	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Pleats is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for measurement / drafting; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select referenced geometry, preserve object IDs through grading/point edits, and report exact units and tolerances in active inspector.

DATA AND TRANSACTION CONTRACT

MeasurementAnchor and GradeRule. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a native parametric drafting operation that updates real pattern points/segments and survives undo, reopen and production export. Current EXO draft tools are not native Rust equivalents. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Author, edit, measure, save/reopen, grade across sizes and export selected dimension with unchanged units. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Grading

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A16	Measurement / drafting	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Grading is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for measurement / drafting; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select referenced geometry, preserve object IDs through grading/point edits, and report exact units and tolerances in active inspector.

DATA AND TRANSACTION CONTRACT

MeasurementAnchor and GradeRule. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a native parametric drafting operation that updates real pattern points/segments and survives undo, reopen and production export. Current EXO draft tools are not native Rust equivalents. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Author, edit, measure, save/reopen, grade across sizes and export selected dimension with unchanged units. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Auto Grading

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · A · A17	Measurement / drafting	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Auto Grading is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for measurement / drafting; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select referenced geometry, preserve object IDs through grading/point edits, and report exact units and tolerances in active inspector.

DATA AND TRANSACTION CONTRACT

MeasurementAnchor and GradeRule. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a native parametric drafting operation that updates real pattern points/segments and survives undo, reopen and production export. Current EXO draft tools are not native Rust equivalents. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Author, edit, measure, save/reopen, grade across sizes and export selected dimension with unchanged units. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Sewing

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B01	Stitch topology	INSPECT	sewing.inspectSeams

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Sewing is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for stitch topology; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Selection must find boundary IDs and parameter ranges; pair orientation/ease; update sewing graph, constraint buffer and diagnostics.

DATA AND TRANSACTION CONTRACT

SeamRelation and edge intervals. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: inspect. Related native information is viewable, but this CLO editing operation is not implemented. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Sew across one segment, reverse it, undo, reopen, measure residual seam gap, then intentionally select invalid edges. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Segment Sewing

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B02	Stitch topology	NATIVE	sewing.freeSew

OBSERVED PURPOSE AND TARGET USER ACTION

Segment Sewing is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for stitch topology; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Selection must find boundary IDs and parameter ranges; pair orientation/ease; update sewing graph, constraint buffer and diagnostics.

DATA AND TRANSACTION CONTRACT

SeamRelation and edge intervals. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: native. Native Rust Sew pairs selected 2D boundary ranges, with undo and edge validation. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Sew across one segment, reverse it, undo, reopen, measure residual seam gap, then intentionally select invalid edges. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Free Sewing

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B03	Stitch topology	NATIVE	sewing.freeSew

OBSERVED PURPOSE AND TARGET USER ACTION

Free Sewing is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for stitch topology; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Selection must find boundary IDs and parameter ranges; pair orientation/ease; update sewing graph, constraint buffer and diagnostics.

DATA AND TRANSACTION CONTRACT

SeamRelation and edge intervals. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: native. Native Sew joins selected boundary ranges; requires two valid boundary selections. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Sew across one segment, reverse it, undo, reopen, measure residual seam gap, then intentionally select invalid edges. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Check Sewing Length

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B04	Stitch topology	INSPECT	sewing.inspectSeams

OBSERVED PURPOSE AND TARGET USER ACTION

Check Sewing Length is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for stitch topology; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Selection must find boundary IDs and parameter ranges; pair orientation/ease; update sewing graph, constraint buffer and diagnostics.

DATA AND TRANSACTION CONTRACT

SeamRelation and edge intervals. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: inspect. Select a native sewing relationship to inspect both connected 2D/3D boundary lengths and simulated gap; automatic sewing-length correction is not implemented. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Sew across one segment, reverse it, undo, reopen, measure residual seam gap, then intentionally select invalid edges. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Steam

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B05	Material / finish	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Steam is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for material / finish; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select owning surface/pattern and bind texture space to grainline; typed layer operations must invalidate visual maps, and physically meaningful changes require a separate calibrated solver operation.

DATA AND TRANSACTION CONTRACT

MaterialAssignment and FinishLayer. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a solver-backed textile finishing or reinforcement constraint, with material/temperature parameters and measurable cloth response. EXO does not yet simulate this manufacturing process. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Apply to one panel, rotate grainline, undo, export swatch and check no accidental change to physical coefficients. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Seam Taping

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B06	Workspace / simulation	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Seam Taping is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for workspace / simulation; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Validate current state and command preconditions, expose real operation only when backend implements it and return typed state change with revision and diagnostics.

DATA AND TRANSACTION CONTRACT

CommandEnvelope and ProjectRevision. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a solver-backed textile finishing or reinforcement constraint, with material/temperature parameters and measurable cloth response. EXO does not yet simulate this manufacturing process. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Click with valid state, run without selection, undo/restart and assert no fake execution or mislabeled support. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Texture (2D)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B07	Material / finish	INSPECT	fabric.inspectProperties

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Texture (2D) is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for material / finish; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select owning surface/pattern and bind texture space to grainline; typed layer operations must invalidate visual maps, and physically meaningful changes require a separate calibrated solver operation.

DATA AND TRANSACTION CONTRACT

MaterialAssignment and FinishLayer. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: inspect. Related native information is viewable, but this CLO editing operation is not implemented. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Apply to one panel, rotate grainline, undo, export swatch and check no accidental change to physical coefficients. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Transform Graphic

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B08	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Transform Graphic is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs native object selection/placement and a typed instance/style model linked to the cloth surface. Related EXO draft annotations exist, but are not yet native CLO-equivalent geometry. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Graphic (2D Pattern)

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B09	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Graphic (2D Pattern) is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs native object selection/placement and a typed instance/style model linked to the cloth surface. Related EXO draft annotations exist, but are not yet native CLO-equivalent geometry. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Topstitch

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B10	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Topstitch is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs native object selection/placement and a typed instance/style model linked to the cloth surface. Related EXO draft annotations exist, but are not yet native CLO-equivalent geometry. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Segment Topstitch

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B11	Attachment / trim	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Segment Topstitch is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for attachment / trim; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Place an authored component against a persistent edge/surface anchor with dimensions, material, orientation, assembly metadata and source panel linkage.

DATA AND TRANSACTION CONTRACT

TrimInstance and PatternSurfaceAnchor. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Needs native object selection/placement and a typed instance/style model linked to the cloth surface. Related EXO draft annotations exist, but are not yet native CLO-equivalent geometry. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Place at measured seam point, change geometry, verify tracking, undo, save/reopen and ensure BOM reflects the item. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Edit Puckering

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B12	Material / finish	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Edit Puckering is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for material / finish; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select owning surface/pattern and bind texture space to grainline; typed layer operations must invalidate visual maps, and physically meaningful changes require a separate calibrated solver operation.

DATA AND TRANSACTION CONTRACT

MaterialAssignment and FinishLayer. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a solver-backed textile finishing or reinforcement constraint, with material/temperature parameters and measurable cloth response. EXO does not yet simulate this manufacturing process. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Apply to one panel, rotate grainline, undo, export swatch and check no accidental change to physical coefficients. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Segment Puckering

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B13	Material / finish	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Segment Puckering is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for material / finish; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select owning surface/pattern and bind texture space to grainline; typed layer operations must invalidate visual maps, and physically meaningful changes require a separate calibrated solver operation.

DATA AND TRANSACTION CONTRACT

MaterialAssignment and FinishLayer. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a solver-backed textile finishing or reinforcement constraint, with material/temperature parameters and measurable cloth response. EXO does not yet simulate this manufacturing process. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Apply to one panel, rotate grainline, undo, export swatch and check no accidental change to physical coefficients. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Set Sublayer

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B14	Workspace / simulation	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Set Sublayer is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for workspace / simulation; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Validate current state and command preconditions, expose real operation only when backend implements it and return typed state change with revision and diagnostics.

DATA AND TRANSACTION CONTRACT

CommandEnvelope and ProjectRevision. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a native pattern-side paint/layer/UV tool with persistent material masks, per-face maps and reversible edits. The separate Fabric workspace handles existing material properties. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Click with valid state, run without selection, undo/restart and assert no fake execution or mislabeled support. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Fill

Observed CLO7 slot	Category	EXO Rust V2 status	Current native route
2D · B · B15	Material / finish	UNAVAILABLE	not yet routed

OBSERVED PURPOSE AND TARGET USER ACTION

Fill is a separately named control in the 2D reference viewport. Exact interaction contracts, flyout variants and selection preconditions must be confirmed against the saved CLO 7.0.242 reference session. In EXO, selection must resolve to the expected authored object kind for material / finish; an active icon or cursor alone is insufficient.

REQUIRED NATIVE IMPLEMENTATION

Select owning surface/pattern and bind texture space to grainline; typed layer operations must invalidate visual maps, and physically meaningful changes require a separate calibrated solver operation.

DATA AND TRANSACTION CONTRACT

MaterialAssignment and FinishLayer. Use stable object identifiers, physical millimeter/centimeter units where geometry is involved, explicit typed parameters and expectedRevision. The command must return affected IDs, history inverse and which 2D/3D, optical, simulation or manufacturing caches need regeneration. Selection tools may return a typed read-only selection result; they must not report an edit revision unless source data changed.

PRESENT EVIDENCE / HONEST CAPABILITY STATE

Observed registry mode: unavailable. Requires a native pattern-side paint/layer/UV tool with persistent material masks, per-face maps and reversible edits. The separate Fabric workspace handles existing material properties. This describes the Rust V2 implementation snapshot, not a guarantee that an analogous browser V1 tool is absent. A complete implementation needs a real native GUI and service exercise and must not inherit success from a placeholder control.

ACCEPTANCE WORKFLOW

Apply to one panel, rotate grainline, undo, export swatch and check no accidental change to physical coefficients. Also verify keyboard/flyout focus, tooltip truth, no-op handling, cancellation, deterministic error messages and visible selection feedback at 390px viewport.

Scenario	Pass criterion
Nominal	Native tool executes its named action on a valid target; expected geometry/property delta confirmed
Undo/reopen	Undo restores full source state; redo and restart reconstruct identical entity IDs and topology
Bad selection	No silent wrong-object mutation; clear disabled/error reason; source revision does not change
2D ↔ 3D	Changes highlight and transfer to paired source/mesh, or capability accurately marked read-only

Fabric catalog / specimens 001–010

The table lists the exact next 10 stable catalog IDs (1–10 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
jersey	Cotton Jersey	220	2.0	0.12	0.18	0.68	0.34
fleece	Heavy French Terry	460	4.0	0.06	0.09	0.82	0.46
denim	Rigid Denim	390	3.2	0.015	0.02	0.91	0.42
silk	Silk Twill	95	1.2	0.035	0.045	0.42	0.22
interfacing	Fusible Interfacing	52	0.7	0.012	0.018	0.88	0.52
cottonPoplin	Cotton Poplin	135	1.1	0.014	0.025	0.77	0.36
cottonCanvas	Cotton Canvas	320	2.7	0.008	0.014	0.94	0.48
woolSuiting	Wool Suiting	250	2.1	0.035	0.04	0.69	0.43
woolMelton	Wool Melton	580	5.0	0.015	0.018	0.94	0.56
softLeather	Soft Garment Leather	650	1.8	0.02	0.035	0.89	0.51

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 011–020

The table lists the exact next 10 stable catalog IDs (11–20 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
suede	Suede / Napped Leather	580	2.2	0.018	0.024	0.86	0.64
lightNylon	Lightweight Nylon	80	0.7	0.012	0.016	0.3	0.29
nylonRipstop	Nylon Ripstop	110	0.9	0.011	0.018	0.47	0.3
linen	Plain Linen	180	1.6	0.013	0.017	0.8	0.39
silkSatin	Silk Satin	85	0.9	0.035	0.045	0.32	0.19
ribKnit	Stretch Rib Knit	320	2.6	0.22	0.45	0.59	0.45
stretchDenim	Stretch Denim	345	2.9	0.025	0.14	0.79	0.42
velvet	Cotton Velvet	310	2.0	0.025	0.04	0.66	0.55
silkChiffon	Silk Chiffon	52	0.5	0.02	0.035	0.2	0.22
silkOrganza	Silk Organza	38	0.38	0.008	0.012	0.5	0.21

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 021–030

The table lists the exact next 10 stable catalog IDs (21–30 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
cottonCorduroy	Cotton Corduroy	370	3.0	0.015	0.028	0.84	0.51
scubaKnit	Scuba Double Knit	360	3.5	0.22	0.31	0.74	0.31
technicalSoftshell	Technical Softshell	290	2.6	0.045	0.095	0.76	0.28
cottonLightJersey	Light Cotton Jersey	160	1.05	0.11	0.19	0.58	0.34
cottonHeavyJersey	Heavyweight Cotton Jersey	285	1.7	0.08	0.15	0.75	0.34
cottonInterlock	Cotton Interlock	235	1.8	0.14	0.22	0.65	0.34
cottonPique	Cotton Piqué	225	1.65	0.08	0.14	0.73	0.34
waffleKnit	Cotton Waffle Knit	290	2.5	0.16	0.29	0.66	0.34
cottonRib1x1	Cotton 1×1 Rib	245	1.65	0.2	0.4	0.58	0.45
cottonRib2x2	Cotton 2×2 Rib	320	2.4	0.2	0.46	0.67	0.45

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 031–040

The table lists the exact next 10 stable catalog IDs (31–40 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
frenchTerryMid	Midweight French Terry	330	2.35	0.08	0.13	0.76	0.46
frenchTerryHeavy	Heavyweight French Terry	520	4.1	0.05	0.1	0.87	0.46
brushedFleece	Brushed Back Sweatshirt Fleece	385	3.5	0.07	0.12	0.78	0.46
polarFleece	Polyester Polar Fleece	285	4.5	0.08	0.13	0.51	0.46
sherpa	Polyester Sherpa	390	6.2	0.09	0.14	0.66	0.46
cottonFlannel	Brushed Cotton Flannel	195	1.5	0.02	0.04	0.7	0.36
cottonOxford	Cotton Oxford Shirting	170	1.1	0.02	0.04	0.76	0.36
cottonBroadcloth	Cotton Broadcloth	130	0.9	0.015	0.025	0.73	0.36
cottonChambray	Cotton Chambray	150	1.1	0.015	0.03	0.7	0.36
cottonTwill	Cotton Workwear Twill	290	1.75	0.015	0.025	0.87	0.48

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 041–050

The table lists the exact next 10 stable catalog IDs (41–50 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
cottonDuck	Cotton Duck Canvas	405	3.1	0.008	0.013999999999999999	0.96	0.48
waxedCanvas	Waxed Cotton Canvas	360	2.5	0.01	0.015	0.9	0.48
linenShirting	Lightweight Linen Shirting	145	0.85	0.012	0.018000000000000002	0.66	0.39
linenHeavy	Heavyweight Linen	285	1.95	0.01	0.02	0.81	0.39
hempCanvas	Hemp Canvas	365	2.55	0.009000000000000001	0.017	0.92	0.48
rawDenim	Raw Indigo Denim 14oz	475	1.45	0.008	0.015	0.95	0.42
lightDenim	Lightweight Denim	270	0.95	0.015	0.027999999999999997	0.77	0.42
heavyDenim	Heavyweight Denim 18oz	610	2.3	0.006	0.01	0.97	0.42
denimCottonElastane	Cotton Elastane Denim	350	1.15	0.03	0.18	0.77	0.42
woolFlannel	Wool Flannel	275	1.9	0.03	0.05	0.68	0.43

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 051–060

The table lists the exact next 10 stable catalog IDs (51–60 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
woolGabardine	Wool Gabardine	280	1.45	0.023	0.032	0.82	0.43
woolTweed	Wool Tweed	410	2.85	0.02	0.035	0.83	0.56
boucle	Bouclé Wool Blend	460	4.2	0.06	0.08	0.68	0.56
cashmereKnit	Cashmere Sweater Knit	225	2.1	0.12	0.21	0.46	0.45
viscoseChallis	Viscose Challis	125	0.75	0.04	0.06	0.33	0.22
tencelTwill	Lyocell Twill	195	0.95	0.038	0.057999999999999996	0.42	0.22
rayonCrepe	Rayon Crêpe	145	1.0	0.05	0.07	0.36	0.22
silkCharmeuse	Silk Charmeuse	90	0.5	0.03	0.05	0.3	0.22
silkGeorgette	Silk Georgette	70	0.42	0.04	0.06	0.25	0.22
polySatin	Polyester Satin	155	0.85	0.025	0.04	0.48	0.19

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 061–070

The table lists the exact next 10 stable catalog IDs (61–70 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
polyTaffeta	Polyester Taffeta	100	0.58	0.015	0.025	0.7	0.29
polyMesh	Athletic Polyester Mesh	130	1.25	0.15	0.27	0.41	0.29
powerMesh	Nylon Spandex Power Mesh	180	0.9	0.28	0.48	0.45	0.45
spandexTricot	Stretch Nylon Spandex Tricot	240	0.7	0.4	0.55	0.39	0.45
neoprene	Neoprene Laminated Foam	590	3.8	0.13	0.23	0.91	0.31
shellNylon	Coated Nylon Jacket Shell	95	0.18	0.01	0.015	0.47	0.29
ripstopUltralight	Ultralight Nylon Ripstop	55	0.16	0.01	0.02	0.35	0.3
polyTulle	Polyester Tulle Net	48	0.4	0.1	0.17	0.35	0.21
veganLeather	Polyurethane Coated Textile	540	1.2	0.03	0.06	0.86	0.51
lambskin	Lambskin Garment Leather	460	0.8	0.04	0.07	0.64	0.51

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 071–080

The table lists the exact next 10 stable catalog IDs (71–80 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
nappaLeather	Nappa Cowhide Garment Leather	640	1.25	0.02	0.04	0.86	0.51
pebbledLeather	Pebbled Cowhide Leather	730	1.65	0.02	0.04	0.89	0.51
vegetableLeather	Vegetable-Tanned Cowhide	840	2.2	0.015	0.03	0.97	0.51
patentLeather	Patent Leather	660	1.2	0.025	0.04	0.88	0.51
nubuck	Nubuck Leather	650	1.35	0.02	0.032	0.83	0.64
suedeLambskin	Lambskin Suede	505	1.1	0.03	0.05	0.66	0.64
fauxSuede	Microfiber Faux Suede	320	1.25	0.04	0.08	0.62	0.64
velour	Cotton Velour Knit	295	2.1	0.14	0.22	0.55	0.55
fauxFur	Short Pile Faux Fur	530	8.0	0.06	0.11	0.72	0.46
quiltedNylon	Quilted Nylon with Batting	310	6.0	0.03	0.05	0.81	0.28

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 081–090

The table lists the exact next 10 stable catalog IDs (81–90 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
jersey__fine	Cotton Jersey · Fine / Light Construction · 163 GSM	162.8	1.62	0.1356	0.2034	0.595	0.3264
jersey__dense	Cotton Jersey · Dense / Firm Construction · 271 GSM	270.6	2.32	0.1032	0.1548	0.75	0.3672
jersey__softened	Cotton Jersey · Soft-Hand / Broken-In Finish · 212 GSM	212.3	2.06	0.1344	0.2016	0.575	0.3162
jersey__crisp	Cotton Jersey · Crisp / Structured Finish · 238 GSM	237.6	2.06	0.1056	0.1584	0.765	0.357
fleece__fine	French Terry · Fine / Light Construction · 340 GSM	340.4	3.24	0.0678	0.1017	0.735	0.4416
fleece__dense	French Terry · Dense / Firm Construction · 566 GSM	565.8	4.64	0.0516	0.0774	0.89	0.4968
fleece__softened	French Terry · Soft-Hand / Broken-In Finish · 444 GSM	443.9	4.12	0.0672	0.1008	0.715	0.4278
fleece__crisp	French Terry · Crisp / Structured Finish · 497 GSM	496.8	4.12	0.0528	0.0792	0.905	0.483
denim__fine	Rigid Denim · Fine / Light Construction · 289 GSM	288.6	2.592	0.01695	0.0226	0.825	0.4032
denim__dense	Rigid Denim · Dense / Firm Construction · 480 GSM	479.7	3.712	0.0129	0.0172	0.98	0.4536

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 091–100

The table lists the exact next 10 stable catalog IDs (91–100 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
denim__softened	Rigid Denim · Soft-Hand / Broken-In Finish · 376 GSM	376.3	3.296	0.0168	0.0224	0.805	0.3906
denim__crisp	Rigid Denim · Crisp / Structured Finish · 421 GSM	421.2	3.296	0.0132	0.0176	0.985	0.441
silk__fine	Silk Twill · Fine / Light Construction · 70 GSM	70.3	0.972	0.03955	0.05085	0.335	0.2112
silk__dense	Silk Twill · Dense / Firm Construction · 117 GSM	116.8	1.392	0.0301	0.0387	0.49	0.2376
silk__softened	Silk Twill · Soft-Hand / Broken-In Finish · 92 GSM	91.7	1.236	0.0392	0.0504	0.315	0.2046
silk__crisp	Silk Twill · Crisp / Structured Finish · 103 GSM	102.6	1.236	0.0308	0.0396	0.505	0.231
interfacing__fine	Fusible Interfacing · Fine / Light Construction · 38 GSM	38.5	0.567	0.01356	0.02034	0.795	0.4992
interfacing__dense	Fusible Interfacing · Dense / Firm Construction · 64 GSM	64	0.812	0.01032	0.01548	0.95	0.5616
interfacing__softened	Fusible Interfacing · Soft-Hand / Broken-In Finish · 50 GSM	50.2	0.721	0.01344	0.02016	0.775	0.4836
interfacing__crisp	Fusible Interfacing · Crisp / Structured Finish · 56 GSM	56.2	0.721	0.01056	0.01584	0.965	0.546

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 101–110

The table lists the exact next 10 stable catalog IDs (101–110 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
cottonPoplin__fine	Cotton Poplin · Fine / Light Construction · 100 GSM	99.9	0.891	0.01582	0.02825	0.685	0.3456
cottonPoplin__dense	Cotton Poplin · Dense / Firm Construction · 166 GSM	166.1	1.276	0.01204	0.0215	0.84	0.3888
cottonPoplin__softened	Cotton Poplin · Soft-Hand / Broken-In Finish · 130 GSM	130.3	1.133	0.01568	0.028	0.665	0.3348
cottonPoplin__crisp	Cotton Poplin · Crisp / Structured Finish · 146 GSM	145.8	1.133	0.01232	0.022	0.855	0.378
cottonCanvas__fine	Cotton Canvas · Fine / Light Construction · 237 GSM	236.8	2.187	0.00904	0.01582	0.855	0.4608
cottonCanvas__dense	Cotton Canvas · Dense / Firm Construction · 394 GSM	393.6	3.132	0.00688	0.01204	0.985	0.5184
cottonCanvas__softened	Cotton Canvas · Soft-Hand / Broken-In Finish · 309 GSM	308.8	2.781	0.00896	0.01568	0.835	0.4464
cottonCanvas__crisp	Cotton Canvas · Crisp / Structured Finish · 346 GSM	345.6	2.781	0.00704	0.01232	0.985	0.504
woolSuiting__fine	Wool Suiting · Fine / Light Construction · 185 GSM	185	1.701	0.03955	0.0452	0.605	0.4128
woolSuiting__dense	Wool Suiting · Dense / Firm Construction · 308 GSM	307.5	2.436	0.0301	0.0344	0.76	0.4644

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 111–120

The table lists the exact next 10 stable catalog IDs (111–120 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
woolSuiting__softened	Wool Suiting · Soft-Hand / Broken-In Finish · 241 GSM	241.3	2.163	0.0392	0.0448	0.585	0.3999
woolSuiting__crisp	Wool Suiting · Crisp / Structured Finish · 270 GSM	270	2.163	0.0308	0.0352	0.775	0.4515
woolMelton__fine	Wool Melton · Fine / Light Construction · 429 GSM	429.2	4.05	0.01695	0.02034	0.855	0.5376
woolMelton__dense	Wool Melton · Dense / Firm Construction · 713 GSM	713.4	5.8	0.0129	0.01548	0.985	0.6048
woolMelton__softened	Wool Melton · Soft-Hand / Broken-In Finish · 560 GSM	559.7	5.15	0.0168	0.02016	0.835	0.5208
woolMelton__crisp	Wool Melton · Crisp / Structured Finish · 626 GSM	626.4	5.15	0.0132	0.01584	0.985	0.588
softLeather__fine	Garment Leather · Fine / Light Construction · 481 GSM	481	1.458	0.0226	0.03955	0.805	0.4896
softLeather__dense	Garment Leather · Dense / Firm Construction · 800 GSM	799.5	2.088	0.0172	0.0301	0.96	0.5508
softLeather__softened	Garment Leather · Soft-Hand / Broken-In Finish · 627 GSM	627.3	1.854	0.0224	0.0392	0.785	0.4743
softLeather__crisp	Garment Leather · Crisp / Structured Finish · 702 GSM	702	1.854	0.0176	0.0308	0.975	0.5355

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 121–130

The table lists the exact next 10 stable catalog IDs (121–130 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
suede__fine	Suede / Napped Leather · Fine / Light Construction · 429 GSM	429.2	1.782	0.02034	0.02712	0.775	0.6144
suede__dense	Suede / Napped Leather · Dense / Firm Construction · 713 GSM	713.4	2.552	0.01548	0.02064	0.93	0.6912
suede__softened	Suede / Napped Leather · Soft-Hand / Broken-In Finish · 560 GSM	559.7	2.266	0.02016	0.02688	0.755	0.5952
suede__crisp	Suede / Napped Leather · Crisp / Structured Finish · 626 GSM	626.4	2.266	0.01584	0.02112	0.945	0.672
lightNylon__fine	Nylon · Fine / Light Construction · 59 GSM	59.2	0.567	0.01356	0.01808	0.215	0.2784
lightNylon__dense	Nylon · Dense / Firm Construction · 98 GSM	98.4	0.812	0.01032	0.01376	0.37	0.3132
lightNylon__softened	Nylon · Soft-Hand / Broken-In Finish · 77 GSM	77.2	0.721	0.01344	0.01792	0.195	0.2697
lightNylon__crisp	Nylon · Crisp / Structured Finish · 86 GSM	86.4	0.721	0.01056	0.01408	0.385	0.3045
nylonRipstop__fine	Nylon Ripstop · Fine / Light Construction · 81 GSM	81.4	0.729	0.01243	0.02034	0.385	0.288
nylonRipstop__dense	Nylon Ripstop · Dense / Firm Construction · 135 GSM	135.3	1.044	0.00946	0.01548	0.54	0.324

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 131–140

The table lists the exact next 10 stable catalog IDs (131–140 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
nylonRipstop__softened	Nylon Ripstop · Soft-Hand / Broken-In Finish · 106 GSM	106.1	0.927	0.01232	0.02016	0.365	0.279
nylonRipstop__crisp	Nylon Ripstop · Crisp / Structured Finish · 119 GSM	118.8	0.927	0.00968	0.01584	0.555	0.315
linen__fine	Plain Linen · Fine / Light Construction · 133 GSM	133.2	1.296	0.01469	0.01921	0.715	0.3744
linen__dense	Plain Linen · Dense / Firm Construction · 221 GSM	221.4	1.856	0.01118	0.01462	0.87	0.4212
linen__softened	Plain Linen · Soft-Hand / Broken-In Finish · 174 GSM	173.7	1.648	0.01456	0.01904	0.695	0.3627
linen__crisp	Plain Linen · Crisp / Structured Finish · 194 GSM	194.4	1.648	0.01144	0.01496	0.885	0.4095
silkSatin__fine	Silk Satin · Fine / Light Construction · 63 GSM	62.9	0.729	0.03955	0.05085	0.235	0.1824
silkSatin__dense	Silk Satin · Dense / Firm Construction · 105 GSM	104.5	1.044	0.0301	0.0387	0.39	0.2052
silkSatin__softened	Silk Satin · Soft-Hand / Broken-In Finish · 82 GSM	82	0.927	0.0392	0.0504	0.215	0.1767
silkSatin__crisp	Silk Satin · Crisp / Structured Finish · 92 GSM	91.8	0.927	0.0308	0.0396	0.405	0.1995

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 141–150

The table lists the exact next 10 stable catalog IDs (141–150 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
ribKnit__fine	Stretch Rib Knit · Fine / Light Construction · 237 GSM	236.8	2.106	0.2486	0.5085	0.505	0.432
ribKnit__dense	Stretch Rib Knit · Dense / Firm Construction · 394 GSM	393.6	3.016	0.1892	0.387	0.66	0.486
ribKnit__softened	Stretch Rib Knit · Soft-Hand / Broken-In Finish · 309 GSM	308.8	2.678	0.2464	0.504	0.485	0.4185
ribKnit__crisp	Stretch Rib Knit · Crisp / Structured Finish · 346 GSM	345.6	2.678	0.1936	0.396	0.675	0.4725
stretchDenim__fine	Stretch Denim · Fine / Light Construction · 255 GSM	255.3	2.349	0.02825	0.1582	0.705	0.4032
stretchDenim__dense	Stretch Denim · Dense / Firm Construction · 424 GSM	424.3	3.364	0.0215	0.1204	0.86	0.4536
stretchDenim__softened	Stretch Denim · Soft-Hand / Broken-In Finish · 333 GSM	332.9	2.987	0.028	0.1568	0.685	0.3906
stretchDenim__crisp	Stretch Denim · Crisp / Structured Finish · 373 GSM	372.6	2.987	0.022	0.1232	0.875	0.441
velvet__fine	Cotton Velvet · Fine / Light Construction · 229 GSM	229.4	1.62	0.02825	0.0452	0.575	0.528
velvet__dense	Cotton Velvet · Dense / Firm Construction · 381 GSM	381.3	2.32	0.0215	0.0344	0.73	0.594

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 151–160

The table lists the exact next 10 stable catalog IDs (151–160 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
velvet__softened	Cotton Velvet · Soft-Hand / Broken-In Finish · 299 GSM	299.1	2.06	0.028	0.0448	0.555	0.5115
velvet__crisp	Cotton Velvet · Crisp / Structured Finish · 335 GSM	334.8	2.06	0.022	0.0352	0.745	0.5775
silkChiffon__fine	Silk Chiffon · Fine / Light Construction · 38 GSM	38.5	0.405	0.0226	0.03955	0.12	0.2112
silkChiffon__dense	Silk Chiffon · Dense / Firm Construction · 64 GSM	64	0.58	0.0172	0.0301	0.27	0.2376
silkChiffon__softened	Silk Chiffon · Soft-Hand / Broken-In Finish · 50 GSM	50.2	0.515	0.0224	0.0392	0.12	0.2046
silkChiffon__crisp	Silk Chiffon · Crisp / Structured Finish · 56 GSM	56.2	0.515	0.0176	0.0308	0.285	0.231
silkOrganza__fine	Silk Organza · Fine / Light Construction · 28 GSM	28.1	0.308	0.00904	0.01356	0.415	0.2016
silkOrganza__dense	Silk Organza · Dense / Firm Construction · 47 GSM	46.7	0.441	0.00688	0.01032	0.57	0.2268
silkOrganza__softened	Silk Organza · Soft-Hand / Broken-In Finish · 37 GSM	36.7	0.391	0.00896	0.01344	0.395	0.1953
silkOrganza__crisp	Silk Organza · Crisp / Structured Finish · 41 GSM	41	0.391	0.00704	0.01056	0.585	0.2205

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 161–170

The table lists the exact next 10 stable catalog IDs (161–170 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
cottonCorduroy__fine	Cotton Corduroy · Fine / Light Construction · 274 GSM	273.8	2.43	0.01695	0.03164	0.755	0.4896
cottonCorduroy__dense	Cotton Corduroy · Dense / Firm Construction · 455 GSM	455.1	3.48	0.0129	0.02408	0.91	0.5508
cottonCorduroy__softened	Cotton Corduroy · Soft-Hand / Broken-In Finish · 357 GSM	357.1	3.09	0.0168	0.03136	0.735	0.4743
cottonCorduroy__crisp	Cotton Corduroy · Crisp / Structured Finish · 400 GSM	399.6	3.09	0.0132	0.02464	0.925	0.5355
scubaKnit__fine	Scuba Double Knit · Fine / Light Construction · 266 GSM	266.4	2.835	0.2486	0.3503	0.655	0.2976
scubaKnit__dense	Scuba Double Knit · Dense / Firm Construction · 443 GSM	442.8	4.06	0.1892	0.2666	0.81	0.3348
scubaKnit__softened	Scuba Double Knit · Soft-Hand / Broken-In Finish · 347 GSM	347.4	3.605	0.2464	0.3472	0.635	0.2883
scubaKnit__crisp	Scuba Double Knit · Crisp / Structured Finish · 389 GSM	388.8	3.605	0.1936	0.2728	0.825	0.3255
technicalSoftshell__fine	Technical Softshell · Fine / Light Construction · 215 GSM	214.6	2.106	0.05085	0.10735	0.675	0.2688
technicalSoftshell__dense	Technical Softshell · Dense / Firm Construction · 357 GSM	356.7	3.016	0.0387	0.0817	0.83	0.3024

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 171–180

The table lists the exact next 10 stable catalog IDs (171–180 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
technicalSoftshell__softened	Technical Softshell · Soft-Hand / Broken-In Finish · 280 GSM	279.8	2.678	0.0504	0.1064	0.655	0.2604
technicalSoftshell__crisp	Technical Softshell · Crisp / Structured Finish · 313 GSM	313.2	2.678	0.0396	0.0836	0.845	0.294
cottonLightJersey__fine	Cotton Jersey · Fine / Light Construction · 118 GSM	118.4	0.851	0.1243	0.2147	0.495	0.3264
cottonLightJersey__dense	Cotton Jersey · Dense / Firm Construction · 197 GSM	196.8	1.218	0.0946	0.1634	0.65	0.3672
cottonLightJersey__softened	Cotton Jersey · Soft-Hand / Broken-In Finish · 154 GSM	154.4	1.081	0.1232	0.2128	0.475	0.3162
cottonLightJersey__crisp	Cotton Jersey · Crisp / Structured Finish · 173 GSM	172.8	1.081	0.0968	0.1672	0.665	0.357
cottonHeavyJersey__fine	Cotton Jersey · Fine / Light Construction · 211 GSM	210.9	1.377	0.0904	0.1695	0.665	0.3264
cottonHeavyJersey__dense	Cotton Jersey · Dense / Firm Construction · 351 GSM	350.6	1.972	0.0688	0.129	0.82	0.3672
cottonHeavyJersey__softened	Cotton Jersey · Soft-Hand / Broken-In Finish · 275 GSM	275	1.751	0.0896	0.168	0.645	0.3162
cottonHeavyJersey__crisp	Cotton Jersey · Crisp / Structured Finish · 308 GSM	307.8	1.751	0.0704	0.132	0.835	0.357

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 181–190

The table lists the exact next 10 stable catalog IDs (181–190 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
cottonInterlock__fine	Cotton Interlock · Fine / Light Construction · 174 GSM	173.9	1.458	0.1582	0.2486	0.565	0.3264
cottonInterlock__dense	Cotton Interlock · Dense / Firm Construction · 289 GSM	289.1	2.088	0.1204	0.1892	0.72	0.3672
cottonInterlock__softened	Cotton Interlock · Soft-Hand / Broken-In Finish · 227 GSM	226.8	1.854	0.1568	0.2464	0.545	0.3162
cottonInterlock__crisp	Cotton Interlock · Crisp / Structured Finish · 254 GSM	253.8	1.854	0.1232	0.1936	0.735	0.357
cottonPique__fine	Cotton Piqué · Fine / Light Construction · 167 GSM	166.5	1.337	0.0904	0.1582	0.645	0.3264
cottonPique__dense	Cotton Piqué · Dense / Firm Construction · 277 GSM	276.8	1.914	0.0688	0.1204	0.8	0.3672
cottonPique__softened	Cotton Piqué · Soft-Hand / Broken-In Finish · 217 GSM	217.1	1.7	0.0896	0.1568	0.625	0.3162
cottonPique__crisp	Cotton Piqué · Crisp / Structured Finish · 243 GSM	243	1.7	0.0704	0.1232	0.815	0.357
waffleKnit__fine	Cotton Waffle Knit · Fine / Light Construction · 215 GSM	214.6	2.025	0.1808	0.3277	0.575	0.3264
waffleKnit__dense	Cotton Waffle Knit · Dense / Firm Construction · 357 GSM	356.7	2.9	0.1376	0.2494	0.73	0.3672

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 191–200

The table lists the exact next 10 stable catalog IDs (191–200 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
waffleKnit__softened	Cotton Waffle Knit · Soft-Hand / Broken-In Finish · 280 GSM	279.8	2.575	0.1792	0.3248	0.555	0.3162
waffleKnit__crisp	Cotton Waffle Knit · Crisp / Structured Finish · 313 GSM	313.2	2.575	0.1408	0.2552	0.745	0.357
cottonRib1x1__fine	Cotton 1×1 Rib · Fine / Light Construction · 181 GSM	181.3	1.337	0.226	0.452	0.495	0.432
cottonRib1x1__dense	Cotton 1×1 Rib · Dense / Firm Construction · 301 GSM	301.4	1.914	0.172	0.344	0.65	0.486
cottonRib1x1__softened	Cotton 1×1 Rib · Soft-Hand / Broken-In Finish · 236 GSM	236.4	1.7	0.224	0.448	0.475	0.4185
cottonRib1x1__crisp	Cotton 1×1 Rib · Crisp / Structured Finish · 265 GSM	264.6	1.7	0.176	0.352	0.665	0.4725
cottonRib2x2__fine	Cotton 2×2 Rib · Fine / Light Construction · 237 GSM	236.8	1.944	0.226	0.5198	0.585	0.432
cottonRib2x2__dense	Cotton 2×2 Rib · Dense / Firm Construction · 394 GSM	393.6	2.784	0.172	0.3956	0.74	0.486
cottonRib2x2__softened	Cotton 2×2 Rib · Soft-Hand / Broken-In Finish · 309 GSM	308.8	2.472	0.224	0.5152	0.565	0.4185
cottonRib2x2__crisp	Cotton 2×2 Rib · Crisp / Structured Finish · 346 GSM	345.6	2.472	0.176	0.4048	0.755	0.4725

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 201–210

The table lists the exact next 10 stable catalog IDs (201–210 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
frenchTerryMid__fine	French Terry · Fine / Light Construction · 244 GSM	244.2	1.903	0.0904	0.1469	0.675	0.4416
frenchTerryMid__dense	French Terry · Dense / Firm Construction · 406 GSM	405.9	2.726	0.0688	0.1118	0.83	0.4968
frenchTerryMid__softened	French Terry · Soft-Hand / Broken-In Finish · 318 GSM	318.4	2.421	0.0896	0.1456	0.655	0.4278
frenchTerryMid__crisp	French Terry · Crisp / Structured Finish · 356 GSM	356.4	2.421	0.0704	0.1144	0.845	0.483
frenchTerryHeavy__fine	French Terry · Fine / Light Construction · 385 GSM	384.8	3.321	0.0565	0.113	0.785	0.4416
frenchTerryHeavy__dense	French Terry · Dense / Firm Construction · 640 GSM	639.6	4.756	0.043	0.086	0.94	0.4968
frenchTerryHeavy__softened	French Terry · Soft-Hand / Broken-In Finish · 502 GSM	501.8	4.223	0.056	0.112	0.765	0.4278
frenchTerryHeavy__crisp	French Terry · Crisp / Structured Finish · 562 GSM	561.6	4.223	0.044	0.088	0.955	0.483
brushedFleece__fine	Brushed Back Sweatshirt Fleece · Fine / Light Construction · 285 GSM	284.9	2.835	0.0791	0.1356	0.695	0.4416
brushedFleece__dense	Brushed Back Sweatshirt Fleece · Dense / Firm Construction · 474 GSM	473.6	4.06	0.0602	0.1032	0.85	0.4968

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 211–220

The table lists the exact next 10 stable catalog IDs (211–220 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
brushedFleece__softened	Brushed Back Sweatshirt Fleece · Soft-Hand / Broken-In Finish · 372 GSM	371.5	3.605	0.0784	0.1344	0.675	0.4278
brushedFleece__crisp	Brushed Back Sweatshirt Fleece · Crisp / Structured Finish · 416 GSM	415.8	3.605	0.0616	0.1056	0.865	0.483
polarFleece__fine	Polyester Polar Fleece · Fine / Light Construction · 211 GSM	210.9	3.645	0.0904	0.1469	0.425	0.4416
polarFleece__dense	Polyester Polar Fleece · Dense / Firm Construction · 351 GSM	350.6	5.22	0.0688	0.1118	0.58	0.4968
polarFleece__softened	Polyester Polar Fleece · Soft-Hand / Broken-In Finish · 275 GSM	275	4.635	0.0896	0.1456	0.405	0.4278
polarFleece__crisp	Polyester Polar Fleece · Crisp / Structured Finish · 308 GSM	307.8	4.635	0.0704	0.1144	0.595	0.483
sherpa__fine	Polyester Sherpa · Fine / Light Construction · 289 GSM	288.6	5.022	0.1017	0.1582	0.575	0.4416
sherpa__dense	Polyester Sherpa · Dense / Firm Construction · 480 GSM	479.7	7.192	0.0774	0.1204	0.73	0.4968
sherpa__softened	Polyester Sherpa · Soft-Hand / Broken-In Finish · 376 GSM	376.3	6.386	0.1008	0.1568	0.555	0.4278
sherpa__crisp	Polyester Sherpa · Crisp / Structured Finish · 421 GSM	421.2	6.386	0.0792	0.1232	0.745	0.483

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 221–230

The table lists the exact next 10 stable catalog IDs (221–230 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
cottonFlannel__fine	Brushed Cotton Flannel · Fine / Light Construction · 144 GSM	144.3	1.215	0.0226	0.0452	0.615	0.3456
cottonFlannel__dense	Brushed Cotton Flannel · Dense / Firm Construction · 240 GSM	239.8	1.74	0.0172	0.0344	0.77	0.3888
cottonFlannel__softened	Brushed Cotton Flannel · Soft-Hand / Broken-In Finish · 188 GSM	188.2	1.545	0.0224	0.0448	0.595	0.3348
cottonFlannel__crisp	Brushed Cotton Flannel · Crisp / Structured Finish · 211 GSM	210.6	1.545	0.0176	0.0352	0.785	0.378
cottonOxford__fine	Cotton Oxford Shirting · Fine / Light Construction · 126 GSM	125.8	0.891	0.0226	0.0452	0.675	0.3456
cottonOxford__dense	Cotton Oxford Shirting · Dense / Firm Construction · 209 GSM	209.1	1.276	0.0172	0.0344	0.83	0.3888
cottonOxford__softened	Cotton Oxford Shirting · Soft-Hand / Broken-In Finish · 164 GSM	164	1.133	0.0224	0.0448	0.655	0.3348
cottonOxford__crisp	Cotton Oxford Shirting · Crisp / Structured Finish · 184 GSM	183.6	1.133	0.0176	0.0352	0.845	0.378
cottonBroadcloth__fine	Cotton Broadcloth · Fine / Light Construction · 96 GSM	96.2	0.729	0.016955	0.028255	0.645	0.3456
cottonBroadcloth__dense	Cotton Broadcloth · Dense / Firm Construction · 160 GSM	159.9	1.044	0.0129	0.0215	0.8	0.3888

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 231–240

The table lists the exact next 10 stable catalog IDs (231–240 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
cottonBroadcloth__softened	Cotton Broadcloth · Soft-Hand / Broken-In Finish · 125 GSM	125.5	0.927	0.0168	0.028	0.625	0.3348
cottonBroadcloth__crisp	Cotton Broadcloth · Crisp / Structured Finish · 140 GSM	140.4	0.927	0.0132	0.022	0.815	0.378
cottonChambray__fine	Cotton Chambray · Fine / Light Construction · 111 GSM	111	0.891	0.01695	0.0339	0.615	0.3456
cottonChambray__dense	Cotton Chambray · Dense / Firm Construction · 185 GSM	184.5	1.276	0.0129	0.0258	0.77	0.3888
cottonChambray__softened	Cotton Chambray · Soft-Hand / Broken-In Finish · 145 GSM	144.8	1.133	0.0168	0.0336	0.595	0.3348
cottonChambray__crisp	Cotton Chambray · Crisp / Structured Finish · 162 GSM	162	1.133	0.0132	0.0264	0.785	0.378
cottonTwill__fine	Cotton Workwear Twill · Fine / Light Construction · 215 GSM	214.6	1.417	0.01695	0.02825	0.785	0.4608
cottonTwill__dense	Cotton Workwear Twill · Dense / Firm Construction · 357 GSM	356.7	2.03	0.0129	0.0215	0.94	0.5184
cottonTwill__softened	Cotton Workwear Twill · Soft-Hand / Broken-In Finish · 280 GSM	279.8	1.802	0.0168	0.028	0.765	0.4464
cottonTwill__crisp	Cotton Workwear Twill · Crisp / Structured Finish · 313 GSM	313.2	1.802	0.0132	0.022	0.955	0.504

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 241–250

The table lists the exact next 10 stable catalog IDs (241–250 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
cottonDuck__fine	Cotton Duck Canvas · Fine / Light Construction · 300 GSM	299.7	2.511	0.00904	0.01582	0.875	0.4608
cottonDuck__dense	Cotton Duck Canvas · Dense / Firm Construction · 498 GSM	498.1	3.596	0.00688	0.01204	0.985	0.5184
cottonDuck__softened	Cotton Duck Canvas · Soft-Hand / Broken-In Finish · 391 GSM	390.8	3.193	0.00896	0.01568	0.855	0.4464
cottonDuck__crisp	Cotton Duck Canvas · Crisp / Structured Finish · 437 GSM	437.4	3.193	0.00704	0.01232	0.985	0.504
waxedCanvas__fine	Waxed Cotton Canvas · Fine / Light Construction · 266 GSM	266.4	2.025	0.0113	0.01695	0.815	0.4608
waxedCanvas__dense	Waxed Cotton Canvas · Dense / Firm Construction · 443 GSM	442.8	2.9	0.0086	0.0129	0.97	0.5184
waxedCanvas__softened	Waxed Cotton Canvas · Soft-Hand / Broken-In Finish · 347 GSM	347.4	2.575	0.0112	0.0168	0.795	0.4464
waxedCanvas__crisp	Waxed Cotton Canvas · Crisp / Structured Finish · 389 GSM	388.8	2.575	0.0088	0.0132	0.985	0.504
linenShirting__fine	Linen Shirting · Fine / Light Construction · 107 GSM	107.3	0.689	0.01356	0.02034	0.575	0.3744
linenShirting__dense	Linen Shirting · Dense / Firm Construction · 178 GSM	178.3	0.986	0.01032	0.01548	0.73	0.4212

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 251–260

The table lists the exact next 10 stable catalog IDs (251–260 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
linenShirting__softened	Linen Shirting · Soft-Hand / Broken-In Finish · 140 GSM	139.9	0.875	0.01344	0.02016	0.555	0.3627
linenShirting__crisp	Linen Shirting · Crisp / Structured Finish · 157 GSM	156.6	0.875	0.01056	0.01584	0.745	0.4095
linenHeavy__fine	Linen · Fine / Light Construction · 211 GSM	210.9	1.579	0.0113	0.0226	0.725	0.3744
linenHeavy__dense	Linen · Dense / Firm Construction · 351 GSM	350.6	2.262	0.0086	0.0172	0.88	0.4212
linenHeavy__softened	Linen · Soft-Hand / Broken-In Finish · 275 GSM	275	2.008	0.0112	0.0224	0.705	0.3627
linenHeavy__crisp	Linen · Crisp / Structured Finish · 308 GSM	307.8	2.008	0.0088	0.0176	0.895	0.4095
hempCanvas__fine	Hemp Canvas · Fine / Light Construction · 270 GSM	270.1	2.066	0.01017	0.01921	0.835	0.4608
hempCanvas__dense	Hemp Canvas · Dense / Firm Construction · 449 GSM	448.9	2.958	0.00774	0.01462	0.985	0.5184
hempCanvas__softened	Hemp Canvas · Soft-Hand / Broken-In Finish · 352 GSM	352.2	2.627	0.01008	0.01904	0.815	0.4464
hempCanvas__crisp	Hemp Canvas · Crisp / Structured Finish · 394 GSM	394.2	2.627	0.00792	0.01496	0.985	0.504

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 261–270

The table lists the exact next 10 stable catalog IDs (261–270 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
rawDenim__fine	Raw Indigo Denim · Fine / Light Construction · 352 GSM	351.5	1.174	0.00904	0.01695	0.865	0.4032
rawDenim__dense	Raw Indigo Denim · Dense / Firm Construction · 584 GSM	584.3	1.682	0.00688	0.0129	0.985	0.4536
rawDenim__softened	Raw Indigo Denim · Soft-Hand / Broken-In Finish · 458 GSM	458.4	1.494	0.00896	0.0168	0.845	0.3906
rawDenim__crisp	Raw Indigo Denim · Crisp / Structured Finish · 513 GSM	513	1.494	0.00704	0.0132	0.985	0.441
lightDenim__fine	Denim · Fine / Light Construction · 200 GSM	199.8	0.769	0.01695	0.03164	0.685	0.4032
lightDenim__dense	Denim · Dense / Firm Construction · 332 GSM	332.1	1.102	0.0129	0.02408	0.84	0.4536
lightDenim__softened	Denim · Soft-Hand / Broken-In Finish · 261 GSM	260.6	0.979	0.0168	0.03136	0.665	0.3906
lightDenim__crisp	Denim · Crisp / Structured Finish · 292 GSM	291.6	0.979	0.0132	0.02464	0.855	0.441
heavyDenim__fine	Denim · Fine / Light Construction · 451 GSM	451.4	1.863	0.00678	0.0113	0.885	0.4032
heavyDenim__dense	Denim · Dense / Firm Construction · 750 GSM	750.3	2.668	0.00516	0.0086	0.985	0.4536

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 271–280

The table lists the exact next 10 stable catalog IDs (271–280 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
heavyDenim__softened	Denim · Soft-Hand / Broken-In Finish · 589 GSM	588.6	2.369	0.00672	0.0112	0.865	0.3906
heavyDenim__crisp	Denim · Crisp / Structured Finish · 659 GSM	658.8	2.369	0.00528	0.0088	0.985	0.441
denimCottonElastane__fine	Cotton Elastane Denim · Fine / Light Construction · 259 GSM	259	0.931	0.0339	0.2034	0.685	0.4032
denimCottonElastane__dense	Cotton Elastane Denim · Dense / Firm Construction · 431 GSM	430.5	1.334	0.0258	0.1548	0.84	0.4536
denimCottonElastane__softened	Cotton Elastane Denim · Soft-Hand / Broken-In Finish · 338 GSM	337.8	1.185	0.0336	0.2016	0.665	0.3906
denimCottonElastane__crisp	Cotton Elastane Denim · Crisp / Structured Finish · 378 GSM	378	1.185	0.0264	0.1584	0.855	0.441
woolFlannel__fine	Wool Flannel · Fine / Light Construction · 204 GSM	203.5	1.539	0.0339	0.0565	0.595	0.4128
woolFlannel__dense	Wool Flannel · Dense / Firm Construction · 338 GSM	338.3	2.204	0.0258	0.043	0.75	0.4644
woolFlannel__softened	Wool Flannel · Soft-Hand / Broken-In Finish · 265 GSM	265.4	1.957	0.0336	0.056	0.575	0.3999
woolFlannel__crisp	Wool Flannel · Crisp / Structured Finish · 297 GSM	297	1.957	0.0264	0.044	0.765	0.4515

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 281–290

The table lists the exact next 10 stable catalog IDs (281–290 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
woolGabardine__fine	Wool Gabardine · Fine / Light Construction · 207 GSM	207.2	1.174	0.02599	0.03616	0.735	0.4128
woolGabardine__dense	Wool Gabardine · Dense / Firm Construction · 344 GSM	344.4	1.682	0.01978	0.02752	0.89	0.4644
woolGabardine__softened	Wool Gabardine · Soft-Hand / Broken-In Finish · 270 GSM	270.2	1.494	0.02576	0.03584	0.715	0.3999
woolGabardine__crisp	Wool Gabardine · Crisp / Structured Finish · 302 GSM	302.4	1.494	0.02024	0.02816	0.905	0.4515
woolTweed__fine	Wool Tweed · Fine / Light Construction · 303 GSM	303.4	2.308	0.0226	0.03955	0.745	0.5376
woolTweed__dense	Wool Tweed · Dense / Firm Construction · 504 GSM	504.3	3.306	0.0172	0.0301	0.9	0.6048
woolTweed__softened	Wool Tweed · Soft-Hand / Broken-In Finish · 396 GSM	395.6	2.936	0.0224	0.0392	0.725	0.5208
woolTweed__crisp	Wool Tweed · Crisp / Structured Finish · 443 GSM	442.8	2.936	0.0176	0.0308	0.915	0.588
boucle__fine	Bouclé Wool Blend · Fine / Light Construction · 340 GSM	340.4	3.402	0.0678	0.0904	0.595	0.5376
boucle__dense	Bouclé Wool Blend · Dense / Firm Construction · 566 GSM	565.8	4.872	0.0516	0.0688	0.75	0.6048

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 291–300

The table lists the exact next 10 stable catalog IDs (291–300 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
boucle__softened	Bouclé Wool Blend · Soft-Hand / Broken-In Finish · 444 GSM	443.9	4.326	0.0672	0.0896	0.575	0.5208
boucle__crisp	Bouclé Wool Blend · Crisp / Structured Finish · 497 GSM	496.8	4.326	0.0528	0.0704	0.765	0.588
cashmereKnit__fine	Cashmere Sweater Knit · Fine / Light Construction · 167 GSM	166.5	1.701	0.1356	0.2373	0.375	0.432
cashmereKnit__dense	Cashmere Sweater Knit · Dense / Firm Construction · 277 GSM	276.8	2.436	0.1032	0.1806	0.53	0.486
cashmereKnit__softened	Cashmere Sweater Knit · Soft-Hand / Broken-In Finish · 217 GSM	217.1	2.163	0.1344	0.2352	0.355	0.4185
cashmereKnit__crisp	Cashmere Sweater Knit · Crisp / Structured Finish · 243 GSM	243	2.163	0.1056	0.1848	0.545	0.4725
viscoseChallis__fine	Viscose Challis · Fine / Light Construction · 93 GSM	92.5	0.607	0.0452	0.0678	0.245	0.2112
viscoseChallis__dense	Viscose Challis · Dense / Firm Construction · 154 GSM	153.8	0.87	0.0344	0.0516	0.4	0.2376
viscoseChallis__softened	Viscose Challis · Soft-Hand / Broken-In Finish · 121 GSM	120.6	0.772	0.0448	0.0672	0.225	0.2046
viscoseChallis__crisp	Viscose Challis · Crisp / Structured Finish · 135 GSM	135	0.772	0.0352	0.0528	0.415	0.231

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 301–310

The table lists the exact next 10 stable catalog IDs (301–310 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
tencelTwill__fine	Lyocell Twill · Fine / Light Construction · 144 GSM	144.3	0.769	0.04294	0.06554	0.335	0.2112
tencelTwill__dense	Lyocell Twill · Dense / Firm Construction · 240 GSM	239.8	1.102	0.03268	0.04988	0.49	0.2376
tencelTwill__softened	Lyocell Twill · Soft-Hand / Broken-In Finish · 188 GSM	188.2	0.979	0.04256	0.06496	0.315	0.2046
tencelTwill__crisp	Lyocell Twill · Crisp / Structured Finish · 211 GSM	210.6	0.979	0.03344	0.05104	0.505	0.231
rayonCrepe__fine	Rayon Crêpe · Fine / Light Construction · 107 GSM	107.3	0.81	0.0565	0.0791	0.275	0.2112
rayonCrepe__dense	Rayon Crêpe · Dense / Firm Construction · 178 GSM	178.3	1.16	0.043	0.0602	0.43	0.2376
rayonCrepe__softened	Rayon Crêpe · Soft-Hand / Broken-In Finish · 140 GSM	139.9	1.03	0.056	0.0784	0.255	0.2046
rayonCrepe__crisp	Rayon Crêpe · Crisp / Structured Finish · 157 GSM	156.6	1.03	0.044	0.0616	0.445	0.231
silkCharmeuse__fine	Silk Charmeuse · Fine / Light Construction · 67 GSM	66.6	0.405	0.0339	0.0565	0.215	0.2112
silkCharmeuse__dense	Silk Charmeuse · Dense / Firm Construction · 111 GSM	110.7	0.58	0.0258	0.043	0.37	0.2376

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 311–320

The table lists the exact next 10 stable catalog IDs (311–320 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
silkCharmeuse__softened	Silk Charmeuse · Soft-Hand / Broken-In Finish · 87 GSM	86.8	0.515	0.0336	0.056	0.195	0.2046
silkCharmeuse__crisp	Silk Charmeuse · Crisp / Structured Finish · 97 GSM	97.2	0.515	0.0264	0.044	0.385	0.231
silkGeorgette__fine	Silk Georgette · Fine / Light Construction · 52 GSM	51.8	0.34	0.0452	0.0678	0.165	0.2112
silkGeorgette__dense	Silk Georgette · Dense / Firm Construction · 86 GSM	86.1	0.487	0.0344	0.0516	0.32	0.2376
silkGeorgette__softened	Silk Georgette · Soft-Hand / Broken-In Finish · 68 GSM	67.5	0.433	0.0448	0.0672	0.145	0.2046
silkGeorgette__crisp	Silk Georgette · Crisp / Structured Finish · 76 GSM	75.6	0.433	0.0352	0.0528	0.335	0.231
polySatin__fine	Polyester Satin · Fine / Light Construction · 115 GSM	114.7	0.689	0.02825	0.0452	0.395	0.1824
polySatin__dense	Polyester Satin · Dense / Firm Construction · 191 GSM	190.7	0.986	0.0215	0.0344	0.55	0.2052
polySatin__softened	Polyester Satin · Soft-Hand / Broken-In Finish · 150 GSM	149.6	0.875	0.028	0.0448	0.375	0.1767
polySatin__crisp	Polyester Satin · Crisp / Structured Finish · 167 GSM	167.4	0.875	0.022	0.0352	0.565	0.1995

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 321–330

The table lists the exact next 10 stable catalog IDs (321–330 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
polyTaffeta__fine	Polyester Taffeta · Fine / Light Construction · 74 GSM	74	0.47	0.01695	0.02825	0.615	0.2784
polyTaffeta__dense	Polyester Taffeta · Dense / Firm Construction · 123 GSM	123	0.673	0.0129	0.0215	0.77	0.3132
polyTaffeta__softened	Polyester Taffeta · Soft-Hand / Broken-In Finish · 97 GSM	96.5	0.597	0.0168	0.028	0.595	0.2697
polyTaffeta__crisp	Polyester Taffeta · Crisp / Structured Finish · 108 GSM	108	0.597	0.0132	0.022	0.785	0.3045
polyMesh__fine	Athletic Polyester Mesh · Fine / Light Construction · 96 GSM	96.2	1.012	0.1695	0.3051	0.325	0.2784
polyMesh__dense	Athletic Polyester Mesh · Dense / Firm Construction · 160 GSM	159.9	1.45	0.129	0.2322	0.48	0.3132
polyMesh__softened	Athletic Polyester Mesh · Soft-Hand / Broken-In Finish · 125 GSM	125.5	1.288	0.168	0.3024	0.305	0.2697
polyMesh__crisp	Athletic Polyester Mesh · Crisp / Structured Finish · 140 GSM	140.4	1.288	0.132	0.2376	0.495	0.3045
powerMesh__fine	Nylon Spandex Power Mesh · Fine / Light Construction · 133 GSM	133.2	0.729	0.3164	0.5424	0.365	0.432
powerMesh__dense	Nylon Spandex Power Mesh · Dense / Firm Construction · 221 GSM	221.4	1.044	0.2408	0.4128	0.52	0.486

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 331–340

The table lists the exact next 10 stable catalog IDs (331–340 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
powerMesh__softened	Nylon Spandex Power Mesh · Soft-Hand / Broken-In Finish · 174 GSM	173.7	0.927	0.3136	0.5376	0.345	0.4185
powerMesh__crisp	Nylon Spandex Power Mesh · Crisp / Structured Finish · 194 GSM	194.4	0.927	0.2464	0.4224	0.535	0.4725
spandexTricot__fine	Stretch Nylon Spandex Tricot · Fine / Light Construction · 178 GSM	177.6	0.567	0.452	0.6215	0.305	0.432
spandexTricot__dense	Stretch Nylon Spandex Tricot · Dense / Firm Construction · 295 GSM	295.2	0.812	0.344	0.473	0.46	0.486
spandexTricot__softened	Stretch Nylon Spandex Tricot · Soft-Hand / Broken-In Finish · 232 GSM	231.6	0.721	0.448	0.616	0.285	0.4185
spandexTricot__crisp	Stretch Nylon Spandex Tricot · Crisp / Structured Finish · 259 GSM	259.2	0.721	0.352	0.484	0.475	0.4725
neoprene__fine	Neoprene Laminated Foam · Fine / Light Construction · 437 GSM	436.6	3.078	0.1469	0.2599	0.825	0.2976
neoprene__dense	Neoprene Laminated Foam · Dense / Firm Construction · 726 GSM	725.7	4.408	0.1118	0.1978	0.98	0.3348
neoprene__softened	Neoprene Laminated Foam · Soft-Hand / Broken-In Finish · 569 GSM	569.4	3.914	0.1456	0.2576	0.805	0.2883
neoprene__crisp	Neoprene Laminated Foam · Crisp / Structured Finish · 637 GSM	637.2	3.914	0.1144	0.2024	0.985	0.3255

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 341–350

The table lists the exact next 10 stable catalog IDs (341–350 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
shellNylon__fine	Coated Nylon Jacket Shell · Fine / Light Construction · 70 GSM	70.3	0.146	0.0113	0.01695	0.385	0.2784
shellNylon__dense	Coated Nylon Jacket Shell · Dense / Firm Construction · 117 GSM	116.8	0.209	0.0086	0.0129	0.54	0.3132
shellNylon__softened	Coated Nylon Jacket Shell · Soft-Hand / Broken-In Finish · 92 GSM	91.7	0.185	0.0112	0.0168	0.365	0.2697
shellNylon__crisp	Coated Nylon Jacket Shell · Crisp / Structured Finish · 103 GSM	102.6	0.185	0.0088	0.0132	0.555	0.3045
ripstopUltralight__fine	Nylon Ripstop · Fine / Light Construction · 41 GSM	40.7	0.13	0.0113	0.0226	0.265	0.288
ripstopUltralight__dense	Nylon Ripstop · Dense / Firm Construction · 68 GSM	67.7	0.186	0.0086	0.0172	0.42	0.324
ripstopUltralight__softened	Nylon Ripstop · Soft-Hand / Broken-In Finish · 53 GSM	53.1	0.165	0.0112	0.0224	0.245	0.279
ripstopUltralight__crisp	Nylon Ripstop · Crisp / Structured Finish · 59 GSM	59.4	0.165	0.0088	0.0176	0.435	0.315
polyTulle__fine	Polyester Tulle Net · Fine / Light Construction · 36 GSM	35.5	0.324	0.113	0.1921	0.265	0.2016
polyTulle__dense	Polyester Tulle Net · Dense / Firm Construction · 59 GSM	59	0.464	0.086	0.1462	0.42	0.2268

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 351–360

The table lists the exact next 10 stable catalog IDs (351–360 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
polyTulle__softened	Polyester Tulle Net · Soft-Hand / Broken-In Finish · 46 GSM	46.3	0.412	0.112	0.1904	0.245	0.1953
polyTulle__crisp	Polyester Tulle Net · Crisp / Structured Finish · 52 GSM	51.8	0.412	0.088	0.1496	0.435	0.2205
veganLeather__fine	Polyurethane Coated Textile · Fine / Light Construction · 400 GSM	399.6	0.972	0.0339	0.0678	0.775	0.4896
veganLeather__dense	Polyurethane Coated Textile · Dense / Firm Construction · 664 GSM	664.2	1.392	0.0258	0.0516	0.93	0.5508
veganLeather__softened	Polyurethane Coated Textile · Soft-Hand / Broken-In Finish · 521 GSM	521.1	1.236	0.0336	0.0672	0.755	0.4743
veganLeather__crisp	Polyurethane Coated Textile · Crisp / Structured Finish · 583 GSM	583.2	1.236	0.0264	0.0528	0.945	0.5355
lambskin__fine	Lambskin Garment Leather · Fine / Light Construction · 340 GSM	340.4	0.648	0.0452	0.0791	0.555	0.4896
lambskin__dense	Lambskin Garment Leather · Dense / Firm Construction · 566 GSM	565.8	0.928	0.0344	0.0602	0.71	0.5508
lambskin__softened	Lambskin Garment Leather · Soft-Hand / Broken-In Finish · 444 GSM	443.9	0.824	0.0448	0.0784	0.535	0.4743
lambskin__crisp	Lambskin Garment Leather · Crisp / Structured Finish · 497 GSM	496.8	0.824	0.0352	0.0616	0.725	0.5355

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 361–370

The table lists the exact next 10 stable catalog IDs (361–370 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
nappaLeather__fine	Nappa Cowhide Garment Leather · Fine / Light Construction · 474 GSM	473.6	1.012	0.0226	0.0452	0.775	0.4896
nappaLeather__dense	Nappa Cowhide Garment Leather · Dense / Firm Construction · 787 GSM	787.2	1.45	0.0172	0.0344	0.93	0.5508
nappaLeather__softened	Nappa Cowhide Garment Leather · Soft-Hand / Broken-In Finish · 618 GSM	617.6	1.288	0.0224	0.0448	0.755	0.4743
nappaLeather__crisp	Nappa Cowhide Garment Leather · Crisp / Structured Finish · 691 GSM	691.2	1.288	0.0176	0.0352	0.945	0.5355
pebbledLeather__fine	Pebbled Cowhide Leather · Fine / Light Construction · 540 GSM	540.2	1.337	0.0226	0.0452	0.805	0.4896
pebbledLeather__dense	Pebbled Cowhide Leather · Dense / Firm Construction · 898 GSM	897.9	1.914	0.0172	0.0344	0.96	0.5508
pebbledLeather__softened	Pebbled Cowhide Leather · Soft-Hand / Broken-In Finish · 704 GSM	704.4	1.7	0.0224	0.0448	0.785	0.4743
pebbledLeather__crisp	Pebbled Cowhide Leather · Crisp / Structured Finish · 788 GSM	788.4	1.7	0.0176	0.0352	0.975	0.5355
vegetableLeather__fine	Vegetable-Tanned Cowhide · Fine / Light Construction · 622 GSM	621.6	1.782	0.01695	0.0339	0.885	0.4896
vegetableLeather__dense	Vegetable-Tanned Cowhide · Dense / Firm Construction · 1033 GSM	1033.2	2.552	0.0129	0.0258	0.985	0.5508

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 371–380

The table lists the exact next 10 stable catalog IDs (371–380 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
vegetableLeather__softened	Vegetable-Tanned Cowhide · Soft-Hand / Broken-In Finish · 811 GSM	810.6	2.266	0.0168	0.0336	0.865	0.4743
vegetableLeather__crisp	Vegetable-Tanned Cowhide · Crisp / Structured Finish · 907 GSM	907.2	2.266	0.0132	0.0264	0.985	0.5355
patentLeather__fine	Patent Leather · Fine / Light Construction · 488 GSM	488.4	0.972	0.02825	0.0452	0.795	0.4896
patentLeather__dense	Patent Leather · Dense / Firm Construction · 812 GSM	811.8	1.392	0.0215	0.0344	0.95	0.5508
patentLeather__softened	Patent Leather · Soft-Hand / Broken-In Finish · 637 GSM	636.9	1.236	0.028	0.0448	0.775	0.4743
patentLeather__crisp	Patent Leather · Crisp / Structured Finish · 713 GSM	712.8	1.236	0.022	0.0352	0.965	0.5355
nubuck__fine	Nubuck Leather · Fine / Light Construction · 481 GSM	481	1.094	0.0226	0.03616	0.745	0.6144
nubuck__dense	Nubuck Leather · Dense / Firm Construction · 800 GSM	799.5	1.566	0.0172	0.02752	0.9	0.6912
nubuck__softened	Nubuck Leather · Soft-Hand / Broken-In Finish · 627 GSM	627.3	1.391	0.0224	0.03584	0.725	0.5952
nubuck__crisp	Nubuck Leather · Crisp / Structured Finish · 702 GSM	702	1.391	0.0176	0.02816	0.915	0.672

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 381–390

The table lists the exact next 10 stable catalog IDs (381–390 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
suedeLambskin__fine	Lambskin Suede · Fine / Light Construction · 374 GSM	373.7	0.891	0.0339	0.0565	0.575	0.6144
suedeLambskin__dense	Lambskin Suede · Dense / Firm Construction · 621 GSM	621.1	1.276	0.0258	0.043	0.73	0.6912
suedeLambskin__softened	Lambskin Suede · Soft-Hand / Broken-In Finish · 487 GSM	487.3	1.133	0.0336	0.056	0.555	0.5952
suedeLambskin__crisp	Lambskin Suede · Crisp / Structured Finish · 545 GSM	545.4	1.133	0.0264	0.044	0.745	0.672
fauxSuede__fine	Microfiber Faux Suede · Fine / Light Construction · 237 GSM	236.8	1.012	0.0452	0.0904	0.535	0.6144
fauxSuede__dense	Microfiber Faux Suede · Dense / Firm Construction · 394 GSM	393.6	1.45	0.0344	0.0688	0.69	0.6912
fauxSuede__softened	Microfiber Faux Suede · Soft-Hand / Broken-In Finish · 309 GSM	308.8	1.288	0.0448	0.0896	0.515	0.5952
fauxSuede__crisp	Microfiber Faux Suede · Crisp / Structured Finish · 346 GSM	345.6	1.288	0.0352	0.0704	0.705	0.672
velour__fine	Cotton Velour Knit · Fine / Light Construction · 218 GSM	218.3	1.701	0.1582	0.2486	0.465	0.528
velour__dense	Cotton Velour Knit · Dense / Firm Construction · 363 GSM	362.9	2.436	0.1204	0.1892	0.62	0.594

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Fabric catalog / specimens 391–400

The table lists the exact next 10 stable catalog IDs (391–400 of 400). These are current EXO records and parameter estimates, not 10 independently lab-measured specimen materials. Missing values and source/visual proxy mapping require investigation before any stronger physical or optical quality label is published.

ID	Construction / saved display name	GSM	mm	warp	weft	bend	friction
velour__softened	Cotton Velour Knit · Soft-Hand / Broken-In Finish · 285 GSM	284.7	2.163	0.1568	0.2464	0.445	0.5115
velour__crisp	Cotton Velour Knit · Crisp / Structured Finish · 319 GSM	318.6	2.163	0.1232	0.1936	0.635	0.5775
fauxFur__fine	Short Pile Faux Fur · Fine / Light Construction · 392 GSM	392.2	6.48	0.0678	0.1243	0.635	0.4416
fauxFur__dense	Short Pile Faux Fur · Dense / Firm Construction · 652 GSM	651.9	9.28	0.0516	0.0946	0.79	0.4968
fauxFur__softened	Short Pile Faux Fur · Soft-Hand / Broken-In Finish · 511 GSM	511.4	8.24	0.0672	0.1232	0.615	0.4278
fauxFur__crisp	Short Pile Faux Fur · Crisp / Structured Finish · 572 GSM	572.4	8.24	0.0528	0.0968	0.805	0.483
quiltedNylon__fine	Quilted Nylon with Batting · Fine / Light Construction · 229 GSM	229.4	4.86	0.0339	0.0565	0.725	0.2688
quiltedNylon__dense	Quilted Nylon with Batting · Dense / Firm Construction · 381 GSM	381.3	6.96	0.0258	0.043	0.88	0.3024
quiltedNylon__softened	Quilted Nylon with Batting · Soft-Hand / Broken-In Finish · 299 GSM	299.1	6.18	0.0336	0.056	0.705	0.2604
quiltedNylon__crisp	Quilted Nylon with Batting · Crisp / Structured Finish · 335 GSM	334.8	6.18	0.0264	0.044	0.895	0.294

IMPLEMENTATION PER ROW

For each stable ID: preserve loaded project compatibility; attach verified manufacturing composition, construction/weave/knit, unique parent specimen, conditioning history and proposed weight variants. Record machine-tested warp and weft force curves, bias/shear behavior, bending with hysteresis and friction pairs in standard SI units. Convert values through solver-specific constitutive calibration rather than using visual roughness as a stiffness surrogate.

OPTICAL/FINISH REQUIREMENTS

Classify whether the assigned licensed source imagery is construction-exact, related-fabric proxy or fully procedural. Confirm image map checksums, linear vs sRGB channels, real centimeters per repeat, normal direction, mipmaps, face-specific use and grayscale optical wear masks. Maintain an independent, reversible colorway/finish stack for denim, French terry, cotton and other suitable groups.

ACCEPTANCE GATE FOR THESE EXACT RECORDS

Check all ten IDs through selection, assignment to panel, material swatch, variant/reopen, native 2D/3D rendering, 1k scan asset loading, texture scale and optical wash undo. Separately compare a physical swatch and calibrated solver output for each promoted specimen. Catalog existence is not material validation.

Actual evidence summary for whole catalog	Status
Physically measured EXO catalog entries	0
Fabric presets with photo source	385
Direct construction photo matches	220
Related-family source proxies	165
Unmatched/procedural	15

Execution matrix and hard numerical release gates

A completed feature proves named behavior through more than a compile or screenshot. For every tool: dispatch, preconditions, model mutation, history, reopen, 2D/3D mapping, alternate runtime behavior and negative fixtures. For each fabric: stable ID, swatch, units, optical file hash, solver assignment and actual calibration state. For each manufacturer export: independent CAD consumption and dimensional verification.

Gate	Mandatory evidence	Failure prevents
Source hygiene	Clean source clone, locked dependencies, no private external assets	release build
Native bridge	Signed application launch, command response, revision persistence	native-ready toolbar label
Geometry	Unit and tolerance report on curved boundaries, darts, seam allowances	CAD-complete claim
Solver	Whole-trajectory contact, seam, strain and drift metrics	fit accepted status
Materials	Specimen readings, optical map sources, unit mapping	measured/calibrated badge
Manufacturing	Third-party DXF roundtrip, factory sample sign-off	factory-approved release
Security	Local service scoping, import validation, rights isolation	community publication

When a fixture fails, preserve its revision and numeric maximum error, avoid replacing the baseline screenshot and record the exact test binary paths. Historical 58/58 native CTest claims are not the same as a freshly configured build; reconfigure out of tree where stale CMake caches refer to old checkout paths.

Official CLO research and compatibility boundaries

The following links are public documented behavior to guide independent EXO workflows. CLO's current web articles describe multiple newer 2026 features, while the live observed toolbar positions are from a pinned installed 7.0.242 session. Keep these as two separate baseline versions. EXO implementation should use original assets, original code and licensed media.

Resource	Engineering reason
CLO live 7.0.242 inventory	Project live app inspector; four columns of 20+18+17+15 controls.
CLO physical fabrication documentation	Fabric emulator measures directional physical sample responses; not automatically EXO data.
CLO material settings	Front/Back/Side optical property model and PBR settings.
CLO DXF interoperability	Industrial AAMA/ASTM/standard DXF import/export options and losses to test.
CLO grading	Size groups, rules and grade editing behaviors.
CLO pattern information	Pattern metadata, revision and shrinkage requirements.
CLO particle distance	Numerical cloth mesh resolution property in mm.
CLO sewing	Sewing selection/mutation behavior.
CLO POM	Permanent measurement geometry editing behavior.
CLO finishing	Optical map generation options.
CLO 2D seams	Custom seam allowance editing and selection.
CLO recent grading review	2026 grading side-by-side review; future parity scope.
Licensed optical scan registry	CC0 third-party scan record and checksums.

Do not infer CLO internal solver constants, unpublished Photoshop-like wash masks, 400 exclusive CLO shipped fabrics, or proprietary project file completeness from these pages. Vendor descriptions show reference user behavior, not an authorization to copy intellectual property.

Reference URLs are enumerated in the accompanying Markdown source and HTML. For audit, fetch a dated copy or retain article identification and update dates; mark unconfirmed behavior explicitly. The design goal is comparable usefulness with original EXO implementation.

GARMENT_STUDIO_COMPLETE_CLO3D_PARITY_MASTER_PLAN_2026-10-13.md

Audit interpretation: This annex preserves earlier engineering research for traceability. Read reported pass counts and statuses as of its original date only. The current 70-slot and 400-fabric inventories above supersede any older source totals.

GARMENT STUDIO — EXHAUSTIVE CLO-CLASS MASTER ENGINEERING PLAN AND CHECKLIST

Created 2026-10-13. Scope: entire application, 2D CAD, 3D tools, fitting/simulation, native app, manufacture, automation and QA. This is a **living inventory**, not proof of CLO3D feature parity or a claim of having checked every feature of every CLO release.

> **Safety of current work:** This roadmap is documentation only. Existing uncommitted source edits, garments, simulation fixtures, personal avatars and local builds were not reset, replaced or deleted.

1. Executive interpretation, status and priority vocabulary

- **Baseline:** prior user-facing estimate ~30–35% feature parity was rough opinion, not a measured benchmark; do not convert into a burndown percentage without actual scored acceptance data.
- **Current source evidence:** React/Vite + Three.js UI (`src/App.jsx`, `src/components/*`); core CAD/production modules (`src/core/*`); JS PBD, C++ native XPBD and Metal/PPF bridge; Blender bridge; native macOS bundle; many engineering check commands in `package.json`.
- **Verified historically:** a production build, the consolidated non-simulation engineering checks, and a signed/opened native bundle in an earlier pass. **No new builds/tests are claimed by this planning document.**
- **Known important defects (from previous audits):** floating garment and unacceptable strain; cloth-body and cloth-self intersections in dynamic fitting; incomplete CAD constraints and styleline operations; incomplete industrial interchange; shallow hierarchy/editor actions; UI automation selector ambiguity.
- **P0 = correctness or foundation blocker; P1 = significant workflow parity; P2 = production polish / advanced breadth.** All unchecked boxes below are *work/verification required*, not a declaration that the code does not yet exist. A box can be marked done only with direct evidence linked to a test and artifact.

1A. Traceability rules and honest scope boundaries

Each GS entry is an **atomic capability criterion**, not yet a resolved task ticket. To execute it, split into independently estimated tickets for design, implementation, unit tests, integration tests, native acceptance and docs. Before claiming parity, compare against a specified CLO3D product version, platform and documented command behavior using original assets. The expression “every possible feature” is open-ended; this plan aims for broad engineering coverage and must grow as verified gaps are discovered.

Technical-debt and defect baseline: preserve all uncommitted local changes; do not infer absence from an unsearched module; quantify solver regressions only under the same baseline; treat disabled/placeholder controls as incomplete but avoid calling a test timeout a product defect. Each backlog entry stays open until evidence shows it passes the universal acceptance gates. A source file existing is not acceptance evidence.

2. Universal Definition of Done for every feature

For every feature below, the assignee must complete all of these gates before marking it done:

- [ ] G-01 User workflow and precision requirements written down.
- [ ] G-02 Existing implementation inspected; duplicate capability avoided.
- [ ] G-03 Data model and units defined; stable IDs/provenance preserved.
- [ ] G-04 Behavior works with real project documents, not a UI-only mock.
- [ ] G-05 Undo and redo both restore exact state.
- [ ] G-06 Save, close, reopen preserves feature state and required assets.
- [ ] G-07 2D↔3D linkage is correct where applicable.
- [ ] G-08 Import/export and manufacturing consequences tested where applicable.
- [ ] G-09 Numeric edge cases, units, precision, degenerate geometry tested.

- [ ] G-10 Native Mac and browser/preview interaction checked where applicable.
- [ ] G-11 Automated regression plus representative manual quality review.
- [ ] G-12 Performance, cancellation and error feedback checked.
- [ ] G-13 No regression in other apparel samples and avatar poses.
- [ ] G-14 Docs, feature matrix, provenance and acceptance evidence updated.

3. Detailed system-by-system backlog

Checkbox IDs are permanent roadmap references. Use them as issue IDs in future agent passes. Each line is a discrete testable target, sometimes requiring several engineering subtasks. Long-term acceptance must be grounded in verified samples and not just a screenshot.

00 — Architecture, technical debt, source of truth (P0)

00.01 Document model

- [ ] **GS-00-01-01** Versioned canonical project schema.
- [ ] **GS-00-01-02** Stable IDs.
- [ ] **GS-00-01-03** Units and axes.
- [ ] **GS-00-01-04** Authored vs derived data.
- [ ] **GS-00-01-05** Constraints and dependencies.
- [ ] **GS-00-01-06** Migration and compatibility.
- [ ] **GS-00-01-07** Project diff.
- [ ] **GS-00-01-08** Undo records.
- [ ] **GS-00-01-09** Forward compatibility.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

00.02 Geometry engine

- [ ] **GS-00-02-01** Double precision coordinates.
- [ ] **GS-00-02-02** Robust predicates.
- [ ] **GS-00-02-03** Deterministic polygon booleans.
- [ ] **GS-00-02-04** Exact curve intersections.
- [ ] **GS-00-02-05** Offsets.
- [ ] **GS-00-02-06** No sliver artifacts.
- [ ] **GS-00-02-07** Controlled tolerances.
- [ ] **GS-00-02-08** Precision reporting.
- [ ] **GS-00-02-09** Stable shape topology.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

00.03 2D/3D pipeline

- [ ] **GS-00-03-01** Source-to-mesh provenance.
- [ ] **GS-00-03-02** Incremental invalidation.
- [ ] **GS-00-03-03** Remesh map retention.
- [ ] **GS-00-03-04** Garment state machine.
- [ ] **GS-00-03-05** Immutable snapshots for async jobs.
- [ ] **GS-00-03-06** Event delivery ordering.
- [ ] **GS-00-03-07** Rendering separation.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

00.04 Backend strategy

- [ ] **GS-00-04-01** PBD/native XPBD/PPF capability matrix.
- [ ] **GS-00-04-02** Shared payload schema.
- [ ] **GS-00-04-03** Job cancellation.
- [ ] **GS-00-04-04** Progress/status.
- [ ] **GS-00-04-05** Failure modes.
- [ ] **GS-00-04-06** Deterministic seed.
- [ ] **GS-00-04-07** Solver version tracking.
- [ ] **GS-00-04-08** Fallback clearly labeled.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

00.05 Native direction

- [ ] **GS-00-05-01** Assess Qt/C++20 host vs current wrapper.
- [ ] **GS-00-05-02** Module API boundaries.
- [ ] **GS-00-05-03** Metal rendering feasibility.
- [ ] **GS-00-05-04** Native file dialogs.
- [ ] **GS-00-05-05** Codesign/notarization.
- [ ] **GS-00-05-06** Upgrade plan without disrupting React app.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

00.06 Code health

- [ ] **GS-00-06-01** Break up monolithic App.jsx.
- [ ] **GS-00-06-02** Isolate document actions.
- [ ] **GS-00-06-03** Typed validation.
- [ ] **GS-00-06-04** Lint/typecheck.
- [ ] **GS-00-06-05** Dependency risk.
- [ ] **GS-00-06-06** Feature flags.
- [ ] **GS-00-06-07** Error boundaries.
- [ ] **GS-00-06-08** Logs.
- [ ] **GS-00-06-09** Profiling.
- [ ] **GS-00-06-10** License inventory.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

01 — Native desktop application and project lifecycle (P0)

01.01 Documents

- [ ] **GS-01-01-01** New/open/save/save as.
- [ ] **GS-01-01-02** Recent projects.
- [ ] **GS-01-01-03** Unsaved-changes prompt.
- [ ] **GS-01-01-04** Autosave.
- [ ] **GS-01-01-05** Backup.

- [ ] **GS-01-01-06** Recovery.
- [ ] **GS-01-01-07** External media.
- [ ] **GS-01-01-08** Changed-file conflict.
- [ ] **GS-01-01-09** Atomic writes.
- [ ] **GS-01-01-10** Path remapping.
- [ ] **GS-01-01-11** Version migration.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

01.02 App packaging

- [ ] **GS-01-02-01** Native arm64 build.
- [ ] **GS-01-02-02** Bundle resources.
- [ ] **GS-01-02-03** Signed binaries.
- [ ] **GS-01-02-04** Installer/updater.
- [ ] **GS-01-02-05** Quarantine behavior.
- [ ] **GS-01-02-06** App permission prompts.
- [ ] **GS-01-02-07** Offline operation.
- [ ] **GS-01-02-08** Graceful service launch/shutdown.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

01.03 Undo and history

- [ ] **GS-01-03-01** Every tool reversible.
- [ ] **GS-01-03-02** Grouping/transactions.
- [ ] **GS-01-03-03** Redo invalidation.
- [ ] **GS-01-03-04** Branch management.
- [ ] **GS-01-03-05** Persistent history design.
- [ ] **GS-01-03-06** Predictable large-simulation snapshots.
- [ ] **GS-01-03-07** Crash-state restore.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

01.04 Reliability

- [ ] **GS-01-04-01** Memory bounds.
- [ ] **GS-01-04-02** Background processing.
- [ ] **GS-01-04-03** App reopen.
- [ ] **GS-01-04-04** Resource cleanup.
- [ ] **GS-01-04-05** Deactivation/reactivation.
- [ ] **GS-01-04-06** Sleep/wake.
- [ ] **GS-01-04-07** Missing GPU.
- [ ] **GS-01-04-08** Corrupted file alerts.
- [ ] **GS-01-04-09** Safe default recovery.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

01.05 Preferences

- [ ] **GS-01-05-01** Units.
- [ ] **GS-01-05-02** Precision.
- [ ] **GS-01-05-03** Mouse/trackpad.
- [ ] **GS-01-05-04** Localization.
- [ ] **GS-01-05-05** UI density.
- [ ] **GS-01-05-06** Shortcuts.
- [ ] **GS-01-05-07** Theme.
- [ ] **GS-01-05-08** Autosave interval.
- [ ] **GS-01-05-09** Workspace restore.
- [ ] **GS-01-05-10** Cache locations.
- [ ] **GS-01-05-11** Telemetry opt-in.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

02 — Workspace, UI and interaction parity (P1)

02.01 Shell layout

- [ ] **GS-02-01-01** Resizable/dockable panels.
- [ ] **GS-02-01-02** Detachable 2D/3D views.
- [ ] **GS-02-01-03** Saved layouts.
- [ ] **GS-02-01-04** Multi-monitor.
- [ ] **GS-02-01-05** Collapsed sidebars.
- [ ] **GS-02-01-06** Toolbar placement.
- [ ] **GS-02-01-07** Responsive minimum sizes.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

02.02 Tool discoverability

- [ ] **GS-02-02-01** Complete tool inventory.
- [ ] **GS-02-02-02** Non-placeholder commands.
- [ ] **GS-02-02-03** Flyouts.
- [ ] **GS-02-02-04** Names and icons.
- [ ] **GS-02-02-05** Tooltip examples.
- [ ] **GS-02-02-06** Keyboard hints.
- [ ] **GS-02-02-07** Context-aware disabled reasons.
- [ ] **GS-02-02-08** Command palette.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

02.03 Object Browser

- [ ] **GS-02-03-01** Hierarchical scene graph.
- [ ] **GS-02-03-02** 2D and 3D visibility independently.
- [ ] **GS-02-03-03** Lock.
- [ ] **GS-02-03-04** Folders.
- [ ] **GS-02-03-05** Parent/child.
- [ ] **GS-02-03-06** Reordering.
- [ ] **GS-02-03-07** Drag/drop.

- [ ] **GS-02-03-08** Multi-selection.
- [ ] **GS-02-03-09** Search.
- [ ] **GS-02-03-10** Filters.
- [ ] **GS-02-03-11** Isolate/solo.
- [ ] **GS-02-03-12** Groups.
- [ ] **GS-02-03-13** Batch edit.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

02.04 Property Editor

- [ ] **GS-02-04-01** Selection-based inspectors.
- [ ] **GS-02-04-02** Batch mixed values.
- [ ] **GS-02-04-03** Undo-aware numeric fields.
- [ ] **GS-02-04-04** Units.
- [ ] **GS-02-04-05** Inline errors.
- [ ] **GS-02-04-06** Linked properties.
- [ ] **GS-02-04-07** Presets.
- [ ] **GS-02-04-08** Searchable sections.
- [ ] **GS-02-04-09** Contextual help.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

02.05 Selection model

- [ ] **GS-02-05-01** Select/extend/toggle.
- [ ] **GS-02-05-02** Lasso/box.
- [ ] **GS-02-05-03** Target priority.
- [ ] **GS-02-05-04** Locked-item protection.
- [ ] **GS-02-05-05** Hidden-item selection rules.
- [ ] **GS-02-05-06** Sync 2D-to-3D.
- [ ] **GS-02-05-07** Outliner-to-canvas.
- [ ] **GS-02-05-08** Hover focus.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

02.06 Input and accessibility

- [ ] **GS-02-06-01** Keyboard-only operation.
- [ ] **GS-02-06-02** Focus order.
- [ ] **GS-02-06-03** Screen reader labels.
- [ ] **GS-02-06-04** Pointer capture.
- [ ] **GS-02-06-05** Retina scaling.
- [ ] **GS-02-06-06** Touchpad pinch.
- [ ] **GS-02-06-07** Zoom center behavior.
- [ ] **GS-02-06-08** High contrast.
- [ ] **GS-02-06-09** Localization.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

02.07 UI testing

- [ ] **GS-02-07-01** Golden screenshots in native and browser.
- [ ] **GS-02-07-02** Actual click-to-change assertions.
- [ ] **GS-02-07-03** Duplicate selectors fixed.
- [ ] **GS-02-07-04** Panel overlap and clipping.
- [ ] **GS-02-07-05** Workspace persistence.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

03 — Precision 2D pattern CAD (P0)

03.01 Primitive geometry

- [ ] **GS-03-01-01** Exact points.
- [ ] **GS-03-01-02** Lines.
- [ ] **GS-03-01-03** Ray/segment.
- [ ] **GS-03-01-04** Circular arcs.
- [ ] **GS-03-01-05** Splines.
- [ ] **GS-03-01-06** Cubic Bézier.
- [ ] **GS-03-01-07** Composite curves.
- [ ] **GS-03-01-08** Tangents.
- [ ] **GS-03-01-09** Normals.
- [ ] **GS-03-01-10** Intersections.
- [ ] **GS-03-01-11** Offset.
- [ ] **GS-03-01-12** Trim.
- [ ] **GS-03-01-13** Join.
- [ ] **GS-03-01-14** Fillet.
- [ ] **GS-03-01-15** Chamfer.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

03.02 Drafting interactions

- [ ] **GS-03-02-01** Pen/point insertion.
- [ ] **GS-03-02-02** Curve edit.
- [ ] **GS-03-02-03** Split.
- [ ] **GS-03-02-04** Merge.
- [ ] **GS-03-02-05** Trace.
- [ ] **GS-03-02-06** Clone.
- [ ] **GS-03-02-07** Rotate.
- [ ] **GS-03-02-08** Translate.
- [ ] **GS-03-02-09** Scale.
- [ ] **GS-03-02-10** Reflect.
- [ ] **GS-03-02-11** Pivot.
- [ ] **GS-03-02-12** Pivot-preserving dimension.
- [ ] **GS-03-02-13** Multi-point transform.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

03.03 Constraint solver

- [ ] **GS-03-03-01** Horizontal/vertical.
- [ ] **GS-03-03-02** Coincident.
- [ ] **GS-03-03-03** Parallel.
- [ ] **GS-03-03-04** Perpendicular.
- [ ] **GS-03-03-05** Tangent.
- [ ] **GS-03-03-06** Symmetric.
- [ ] **GS-03-03-07** Collinear.
- [ ] **GS-03-03-08** Concentric.
- [ ] **GS-03-03-09** Equal length/radius.
- [ ] **GS-03-03-10** Dimensions.
- [ ] **GS-03-03-11** Over/underconstrained diagnosis.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

03.04 Numeric input

- [ ] **GS-03-04-01** Precise length.
- [ ] **GS-03-04-02** Angle.
- [ ] **GS-03-04-03** Coordinate.
- [ ] **GS-03-04-04** Radius.
- [ ] **GS-03-04-05** Curve handles.
- [ ] **GS-03-04-06** Units mm/cm/in.
- [ ] **GS-03-04-07** Expression fields.
- [ ] **GS-03-04-08** Fractional inches.
- [ ] **GS-03-04-09** Keyboard confirm/cancel.
- [ ] **GS-03-04-10** Snapping tolerances.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

03.05 Snapping/guides

- [ ] **GS-03-05-01** Grid.
- [ ] **GS-03-05-02** Vertex.
- [ ] **GS-03-05-03** Midpoint.
- [ ] **GS-03-05-04** Tangent.
- [ ] **GS-03-05-05** Intersection.
- [ ] **GS-03-05-06** Projection.
- [ ] **GS-03-05-07** Grainline.
- [ ] **GS-03-05-08** Construction guide.
- [ ] **GS-03-05-09** Smart alignment.
- [ ] **GS-03-05-10** Weighted snapping.
- [ ] **GS-03-05-11** Temporary overrides.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

03.06 Topology

- [ ] **GS-03-06-01** Holes.

- [ ] **GS-03-06-02** Cut lines.
- [ ] **GS-03-06-03** Multiple contours.
- [ ] **GS-03-06-04** Nested pattern regions.
- [ ] **GS-03-06-05** Closed/open validation.
- [ ] **GS-03-06-06** Nonmanifold diagnosis.
- [ ] **GS-03-06-07** Winding.
- [ ] **GS-03-06-08** Seam-to-edge stable IDs.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

03.07 Pattern transformations

- [ ] **GS-03-07-01** Slash/spread.
- [ ] **GS-03-07-02** Dart pivot/rotate/convert/close.
- [ ] **GS-03-07-03** Pleat.
- [ ] **GS-03-07-04** Flare.
- [ ] **GS-03-07-05** Slash-and-overlap.
- [ ] **GS-03-07-06** Flatten.
- [ ] **GS-03-07-07** True.
- [ ] **GS-03-07-08** Blend.
- [ ] **GS-03-07-09** Reshape.
- [ ] **GS-03-07-10** Curved hem.
- [ ] **GS-03-07-11** Notch transfer.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

03.08 Symmetry

- [ ] **GS-03-08-01** Linked half pattern.
- [ ] **GS-03-08-02** Unfold.
- [ ] **GS-03-08-03** Paired edits.
- [ ] **GS-03-08-04** Mirrored darts.
- [ ] **GS-03-08-05** Independent break-link.
- [ ] **GS-03-08-06** Symmetry around chosen axis.
- [ ] **GS-03-08-07** Exact reproducible transformations.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

03.09 Internal construction

- [ ] **GS-03-09-01** Internal curves.
- [ ] **GS-03-09-02** Cut/fold lines.
- [ ] **GS-03-09-03** Drill holes.
- [ ] **GS-03-09-04** Placement marks.
- [ ] **GS-03-09-05** Button/buttonhole paths.
- [ ] **GS-03-09-06** Alignment.
- [ ] **GS-03-09-07** Grain and nap arrows.
- [ ] **GS-03-09-08** Annotation properties.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

03.10 Pattern labeling

- [ ] **GS-03-10-01** Piece names.
- [ ] **GS-03-10-02** Cut quantities.
- [ ] **GS-03-10-03** Material assignment.
- [ ] **GS-03-10-04** Sizes.
- [ ] **GS-03-10-05** Fold status.
- [ ] **GS-03-10-06** Grainline direction.
- [ ] **GS-03-10-07** Labels in print/plot exports.
- [ ] **GS-03-10-08** Placement and collision avoidance.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

03.11 Seam allowances

- [ ] **GS-03-11-01** Per-edge widths.
- [ ] **GS-03-11-02** Corner miters.
- [ ] **GS-03-11-03** Notch retention.
- [ ] **GS-03-11-04** Variable width.
- [ ] **GS-03-11-05** Inset/outset.
- [ ] **GS-03-11-06** Cut/sew contour separation.
- [ ] **GS-03-11-07** Join topology.
- [ ] **GS-03-11-08** Corner error detection.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

03.12 Pattern verification

- [ ] **GS-03-12-01** Area/perimeter.
- [ ] **GS-03-12-02** Edge length.
- [ ] **GS-03-12-03** Angle.
- [ ] **GS-03-12-04** Curvature.
- [ ] **GS-03-12-05** Dimensions.
- [ ] **GS-03-12-06** Sewing length.
- [ ] **GS-03-12-07** Baseline geometry regression.
- [ ] **GS-03-12-08** Reference manufacturer-ready blocks.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

03.13 Drafting QA

- [ ] **GS-03-13-01** Constraint solve residual.
- [ ] **GS-03-13-02** Sub-mm deterministic test.
- [ ] **GS-03-13-03** Random topology fuzzing.
- [ ] **GS-03-13-04** Undo/redo.
- [ ] **GS-03-13-05** Save/reload.
- [ ] **GS-03-13-06** Large piece count.

- [ ] **GS-03-13-07** Export/import geometry.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

04 — Measurement, avatars as drafting references, grading (P0)

04.01 Measurement system

- [ ] **GS-04-01-01** Named POM.
- [ ] **GS-04-01-02** Landmarks.
- [ ] **GS-04-01-03** Horizontal/vertical/curved distances.
- [ ] **GS-04-01-04** Circumference.
- [ ] **GS-04-01-05** Ease.
- [ ] **GS-04-01-06** Tolerance.
- [ ] **GS-04-01-07** Chart.
- [ ] **GS-04-01-08** Cross-size measurement interpolation.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

04.02 Parametric patterns

- [ ] **GS-04-02-01** Formula measurements.
- [ ] **GS-04-02-02** Dependent geometry.
- [ ] **GS-04-02-03** Parameter graph.
- [ ] **GS-04-02-04** Sizing rules.
- [ ] **GS-04-02-05** Adjustable silhouette.
- [ ] **GS-04-02-06** Ease distribution.
- [ ] **GS-04-02-07** Validated update propagation.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

04.03 Pattern block library

- [ ] **GS-04-03-01** Tees.
- [ ] **GS-04-03-02** Hoodies.
- [ ] **GS-04-03-03** Shirts.
- [ ] **GS-04-03-04** Pants/jeans.
- [ ] **GS-04-03-05** Skirts.
- [ ] **GS-04-03-06** Dresses.
- [ ] **GS-04-03-07** Jackets.
- [ ] **GS-04-03-08** Linings.
- [ ] **GS-04-03-09** Collars.
- [ ] **GS-04-03-10** Cuffs.
- [ ] **GS-04-03-11** Pockets.
- [ ] **GS-04-03-12** Hoods.
- [ ] **GS-04-03-13** Variants.
- [ ] **GS-04-03-14** Tested fit baseline.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

04.04 Grading authoring

- [ ] **GS-04-04-01** Point grading.
- [ ] **GS-04-04-02** X/y rules.
- [ ] **GS-04-04-03** Grading tables.
- [ ] **GS-04-04-04** Base size.
- [ ] **GS-04-04-05** Asymmetric rules.
- [ ] **GS-04-04-06** Piece-specific rules.
- [ ] **GS-04-04-07** Cross-size edits.
- [ ] **GS-04-04-08** Nested sizes.
- [ ] **GS-04-04-09** Rule reuse.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

04.05 Grading validation

- [ ] **GS-04-05-01** Seam lengths across sizes.
- [ ] **GS-04-05-02** No inverted curves.
- [ ] **GS-04-05-03** Allowance preservation.
- [ ] **GS-04-05-04** Balance notches.
- [ ] **GS-04-05-05** Measurement tolerances.
- [ ] **GS-04-05-06** Size run roundtrip.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

04.06 Measurement export

- [ ] **GS-04-06-01** POM diagrams.
- [ ] **GS-04-06-02** Size charts.
- [ ] **GS-04-06-03** Factory tolerance tables.
- [ ] **GS-04-06-04** Inspection point IDs.
- [ ] **GS-04-06-05** Printable measure diagrams.
- [ ] **GS-04-06-06** Change history.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

05 — Sewing graph, construction and 3D arrangement (P0)

05.01 Sew links

- [ ] **GS-05-01-01** Edge interval selection.
- [ ] **GS-05-01-02** Segment sewing.
- [ ] **GS-05-01-03** Free sewing.
- [ ] **GS-05-01-04** Orientation arrows.
- [ ] **GS-05-01-05** Seam IDs.
- [ ] **GS-05-01-06** Multiple seam relationships.
- [ ] **GS-05-01-07** Arc-length remapping.
- [ ] **GS-05-01-08** Mismatch warnings.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

05.02 Stitch logic

- [ ] **GS-05-02-01** Seam line vs cut edge.
- [ ] **GS-05-02-02** Seam allowances.
- [ ] **GS-05-02-03** Notches.
- [ ] **GS-05-02-04** Ease/gather ratio.
- [ ] **GS-05-02-05** Differential feed.
- [ ] **GS-05-02-06** Stitch spacing.
- [ ] **GS-05-02-07** Sew order.
- [ ] **GS-05-02-08** Offset/unequal edges.
- [ ] **GS-05-02-09** Fold construction.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

05.03 Seam diagnostics

- [ ] **GS-05-03-01** Seam walk.
- [ ] **GS-05-03-02** Mismatch histogram.
- [ ] **GS-05-03-03** Reversed links.
- [ ] **GS-05-03-04** Self-links.
- [ ] **GS-05-03-05** T-junctions.
- [ ] **GS-05-03-06** Intersecting stitching.
- [ ] **GS-05-03-07** Dangling ends.
- [ ] **GS-05-03-08** Graph completeness.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

05.04 3D arrangement

- [ ] **GS-05-04-01** Panel transform.
- [ ] **GS-05-04-02** Arrangement points.
- [ ] **GS-05-04-03** Around-body shell.
- [ ] **GS-05-04-04** Orientation and flip.
- [ ] **GS-05-04-05** Flat pre-sew arrangement.
- [ ] **GS-05-04-06** No overlapping initial shell.
- [ ] **GS-05-04-07** Reversible staging.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

05.05 Construction features

- [ ] **GS-05-05-01** Facings.
- [ ] **GS-05-05-02** Collars.
- [ ] **GS-05-05-03** Cuffs.
- [ ] **GS-05-05-04** Vents.
- [ ] **GS-05-05-05** Lapels.
- [ ] **GS-05-05-06** Waistbands.
- [ ] **GS-05-05-07** Pockets.
- [ ] **GS-05-05-08** Darts.
- [ ] **GS-05-05-09** Pleats.

- [ ] **GS-05-05-10** Gussets.
- [ ] **GS-05-05-11** Multiple material panels.
- [ ] **GS-05-05-12** Lining.
- [ ] **GS-05-05-13** Fusible layers.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

05.06 Sewing UX

- [ ] **GS-05-06-01** Linked 2D/3D highlighting.
- [ ] **GS-05-06-02** Drag to sew.
- [ ] **GS-05-06-03** Sew/unsew.
- [ ] **GS-05-06-04** Seam inspector.
- [ ] **GS-05-06-05** Temporary tack.
- [ ] **GS-05-06-06** Seam edit after pattern deformation.
- [ ] **GS-05-06-07** Guided correction.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

05.07 Acceptance

- [ ] **GS-05-07-01** Manufacturer-style tee front/back/2 sleeves sewn by explicit oriented seams.
- [ ] **GS-05-07-02** Validate vertex correspondence.
- [ ] **GS-05-07-03** No orphan stitch endpoints across updates.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

06 — Mesh generation, correspondence and remeshing (P0)

06.01 Mesher

- [ ] **GS-06-01-01** Metric target edge length.
- [ ] **GS-06-01-02** Constrained triangulation.
- [ ] **GS-06-01-03** Curve discretization.
- [ ] **GS-06-01-04** Seam boundary synchronization.
- [ ] **GS-06-01-05** Stable orientation.
- [ ] **GS-06-01-06** Aspect ratio limits.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

06.02 Topology mapping

- [ ] **GS-06-02-01** Source edge/point provenance.
- [ ] **GS-06-02-02** Barycentric face map.
- [ ] **GS-06-02-03** UVs.
- [ ] **GS-06-02-04** Grain direction.
- [ ] **GS-06-02-05** Remesh transferring pins, trims, seams, annotations and graphics.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

06.03 Adaptive remesh

- [ ] **GS-06-03-01** Refine at high curvature.
- [ ] **GS-06-03-02** Stress regions.
- [ ] **GS-06-03-03** Intersections and stitches.
- [ ] **GS-06-03-04** Coarsen stable zones.
- [ ] **GS-06-03-05** Preserve boundaries.
- [ ] **GS-06-03-06** Avoid invalid triangles.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

06.04 Mesh QA

- [ ] **GS-06-04-01** Inverted faces.
- [ ] **GS-06-04-02** Triangles with zero area.
- [ ] **GS-06-04-03** Nonmanifold edges.
- [ ] **GS-06-04-04** Duplicate vertices.
- [ ] **GS-06-04-05** Inconsistent normals.
- [ ] **GS-06-04-06** Slivers.
- [ ] **GS-06-04-07** Degenerate UV.
- [ ] **GS-06-04-08** Mesh convergence.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

06.05 Mesh authoring

- [ ] **GS-06-05-01** 2D-driven vs direct 3D mesh edits.
- [ ] **GS-06-05-02** Mesh selection.
- [ ] **GS-06-05-03** Brush/lasso.
- [ ] **GS-06-05-04** Garment stylelines.
- [ ] **GS-06-05-05** Pin topology.
- [ ] **GS-06-05-06** Maintain pattern ownership and save/load.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

07 — Body/garment contact, cloth physics and solver correctness (P0)

07.01 Unified collider

- [ ] **GS-07-01-01** Same skinned posed avatar for display and collision.
- [ ] **GS-07-01-02** Body-only mesh separation.
- [ ] **GS-07-01-03** Transforms.
- [ ] **GS-07-01-04** BVH/SDF.
- [ ] **GS-07-01-05** Normals.
- [ ] **GS-07-01-06** Scale.
- [ ] **GS-07-01-07** Thickness.
- [ ] **GS-07-01-08** Joints.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

07.02 Body collision

- [ ] **GS-07-02-01** Continuous collision detection.
- [ ] **GS-07-02-02** Vertex-face and edge-edge sweeps.
- [ ] **GS-07-02-03** Signed penetration depth.
- [ ] **GS-07-02-04** Stable collision normals.
- [ ] **GS-07-02-05** Friction.
- [ ] **GS-07-02-06** Tunneling rejection.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

07.03 Self-contact

- [ ] **GS-07-03-01** Cloth-cloth triangle collision.
- [ ] **GS-07-03-02** Edge-edge.
- [ ] **GS-07-03-03** Layered collision.
- [ ] **GS-07-03-04** Near-fold precision.
- [ ] **GS-07-03-05** Robust broadphase.
- [ ] **GS-07-03-06** Thin-shell thickness.
- [ ] **GS-07-03-07** Nonlocal self-contact.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

07.04 Constraint coupling

- [ ] **GS-07-04-01** Sewing/contact/shape/bending solve coupled.
- [ ] **GS-07-04-02** Coherent compliance.
- [ ] **GS-07-04-03** No aggressive sequential projection.
- [ ] **GS-07-04-04** Convergence checks.
- [ ] **GS-07-04-05** Adaptive stepping.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

07.05 Material mechanics

- [ ] **GS-07-05-01** Measured anisotropic warp/weft.
- [ ] **GS-07-05-02** Bias and shear.
- [ ] **GS-07-05-03** Bending.
- [ ] **GS-07-05-04** Buckling.
- [ ] **GS-07-05-05** Damping.
- [ ] **GS-07-05-06** Friction.
- [ ] **GS-07-05-07** Thickness.
- [ ] **GS-07-05-08** Stretch recovery.
- [ ] **GS-07-05-09** Weight.
- [ ] **GS-07-05-10** Shrinkage.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

07.06 Gravity equilibrium

- [ ] **GS-07-06-01** Natural collar/shoulder contact.
- [ ] **GS-07-06-02** No floating.
- [ ] **GS-07-06-03** No upward bias.
- [ ] **GS-07-06-04** Stable settled kinetic energy.
- [ ] **GS-07-06-05** Hem hangs realistically.
- [ ] **GS-07-06-06** Drape under motion.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

07.07 Interactive controls

- [ ] **GS-07-07-01** Start/pause/reset.
- [ ] **GS-07-07-02** Staged sew.
- [ ] **GS-07-07-03** Drag/pin.
- [ ] **GS-07-07-04** Strain freeze.
- [ ] **GS-07-07-05** Wind and gravity.
- [ ] **GS-07-07-06** Deterministic pause/restart.
- [ ] **GS-07-07-07** Resimulate after edits.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

07.08 Long-run numerical stability

- [ ] **GS-07-08-01** Finite positions.
- [ ] **GS-07-08-02** Bounded kinetic/strain.
- [ ] **GS-07-08-03** 10k-step stress.
- [ ] **GS-07-08-04** Timestep/substep convergence.
- [ ] **GS-07-08-05** Mesh-size convergence.
- [ ] **GS-07-08-06** Fail closed on divergence.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

07.09 PPF/Metal bridge

- [ ] **GS-07-09-01** Physical stitch/contact integration.
- [ ] **GS-07-09-02** Quality-bake adapter.
- [ ] **GS-07-09-03** Exact input avatar/pattern snapshot.
- [ ] **GS-07-09-04** Solver job status.
- [ ] **GS-07-09-05** Output provenance.
- [ ] **GS-07-09-06** Failures not auto-accepted.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

07.10 Validation metrics

- [ ] **GS-07-10-01** Peak/p95 strain per region.
- [ ] **GS-07-10-02** Mean/p95/max stitch gap.
- [ ] **GS-07-10-03** Body and self intersections per frame.
- [ ] **GS-07-10-04** Depth in mm.
- [ ] **GS-07-10-05** Contact clearance.

- [ ] **GS-07-10-06** Silhouette.
- [ ] **GS-07-10-07** Runtime.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

07.11 Golden scenarios

- [ ] **GS-07-11-01** Tee at 160/180/200 cm.
- [ ] **GS-07-11-02** Arms down/out.
- [ ] **GS-07-11-03** Sleeves correct orientation.
- [ ] **GS-07-11-04** Hoodie.
- [ ] **GS-07-11-05** Fitted jeans.
- [ ] **GS-07-11-06** Pants.
- [ ] **GS-07-11-07** Multi-layer garment.
- [ ] **GS-07-11-08** Manipulations.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

07.12 Known urgent defects

- [ ] **GS-07-12-01** Floating/standing-off shirt.
- [ ] **GS-07-12-02** Seam vs shoulder constraint conflict.
- [ ] **GS-07-12-03** Prior massive strain.
- [ ] **GS-07-12-04** Final-frame PPF body and self intersections.
- [ ] **GS-07-12-05** Solver timeouts.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

07.13 Release gate

- [ ] **GS-07-13-01** No accepted fit unless complete dynamic trajectory passes contact, seam, strain, settling and geometry checks.
- [ ] **GS-07-13-02** Report failed criteria.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

08 — Auto fitting, pattern optimization and AI assembly (P1)

08.01 Placement inference

- [ ] **GS-08-01-01** Infer panel-to-body anatomical region.
- [ ] **GS-08-01-02** Sleeve handness.
- [ ] **GS-08-01-03** Up direction.
- [ ] **GS-08-01-04** Front/back.
- [ ] **GS-08-01-05** Mirror errors.
- [ ] **GS-08-01-06** Clearance.
- [ ] **GS-08-01-07** Reversible staging.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

08.02 Auto seam recognition

- [ ] **GS-08-02-01** Candidate seam pairs ranked by geometry.
- [ ] **GS-08-02-02** Notches.
- [ ] **GS-08-02-03** Naming.
- [ ] **GS-08-02-04** Edge lengths.
- [ ] **GS-08-02-05** Sewing direction.
- [ ] **GS-08-02-06** Human confirmation and editable previews.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

08.03 Auto fit pipeline

- [ ] **GS-08-03-01** Pattern->staging->sew->simulate->metrics->candidate adjustments->retest.
- [ ] **GS-08-03-02** No silent source changes.
- [ ] **GS-08-03-03** Retain seed and snapshots.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

08.04 Fit adjustment

- [ ] **GS-08-04-01** Pattern length/width.
- [ ] **GS-08-04-02** Shoulder slope.
- [ ] **GS-08-04-03** Armhole.
- [ ] **GS-08-04-04** Sleeve cap.
- [ ] **GS-08-04-05** Ease.
- [ ] **GS-08-04-06** Hem.
- [ ] **GS-08-04-07** Rise.
- [ ] **GS-08-04-08** Inseam.
- [ ] **GS-08-04-09** Waistband.
- [ ] **GS-08-04-10** Localized objective functions.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

08.05 AI command system

- [ ] **GS-08-05-01** Typed editing API.
- [ ] **GS-08-05-02** Complete command coverage.
- [ ] **GS-08-05-03** Dry-run.
- [ ] **GS-08-05-04** Proposals.
- [ ] **GS-08-05-05** Preview.
- [ ] **GS-08-05-06** Undo/rollback.
- [ ] **GS-08-05-07** Schema validation.
- [ ] **GS-08-05-08** Permission scopes.
- [ ] **GS-08-05-09** Audit trail.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

08.06 Optimization QA

- [ ] **GS-08-06-01** Test fit score against withheld avatars.
- [ ] **GS-08-06-02** Do not trade penetration for seam closure.
- [ ] **GS-08-06-03** No overfitting one pose.
- [ ] **GS-08-06-04** Preserve garment identity.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

09 — Fabrics, trims, seam finishes and graphical decoration (P1)

09.01 Fabric library

- [ ] **GS-09-01-01** Presets with measurement provenance.
- [ ] **GS-09-01-02** Fabric weight.
- [ ] **GS-09-01-03** Thickness.
- [ ] **GS-09-01-04** Stretch.
- [ ] **GS-09-01-05** Bending.
- [ ] **GS-09-01-06** Shear.
- [ ] **GS-09-01-07** Damping.
- [ ] **GS-09-01-08** Friction.
- [ ] **GS-09-01-09** Calibrated simulation conversion.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

09.02 Rendering maps

- [ ] **GS-09-02-01** Base color.
- [ ] **GS-09-02-02** Normal.
- [ ] **GS-09-02-03** Roughness.
- [ ] **GS-09-02-04** Metallic where applicable.
- [ ] **GS-09-02-05** Weave microdetail.
- [ ] **GS-09-02-06** Anisotropy.
- [ ] **GS-09-02-07** Translucency.
- [ ] **GS-09-02-08** Fuzz.
- [ ] **GS-09-02-09** Sheen.
- [ ] **GS-09-02-10** Displacement.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

09.03 Texture workflows

- [ ] **GS-09-03-01** UV.
- [ ] **GS-09-03-02** Tile/repeat.
- [ ] **GS-09-03-03** Grain alignment.
- [ ] **GS-09-03-04** Offset.
- [ ] **GS-09-03-05** Rotation.
- [ ] **GS-09-03-06** Fabric face and back.
- [ ] **GS-09-03-07** Plaids/stripes.
- [ ] **GS-09-03-08** Scale reference.
- [ ] **GS-09-03-09** Graphics mask and color space.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

09.04 Seam finishes

- [ ] **GS-09-04-01** Topstitch multi-line.
- [ ] **GS-09-04-02** Overlock.
- [ ] **GS-09-04-03** Zigzag.
- [ ] **GS-09-04-04** Coverstitch.
- [ ] **GS-09-04-05** Chain.
- [ ] **GS-09-04-06** Bartack.
- [ ] **GS-09-04-07** Thread properties.
- [ ] **GS-09-04-08** Stitch distance and density.
- [ ] **GS-09-04-09** Separate print vs sim detail.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

09.05 Trims

- [ ] **GS-09-05-01** Buttons.
- [ ] **GS-09-05-02** Buttonholes.
- [ ] **GS-09-05-03** Snaps.
- [ ] **GS-09-05-04** Zippers.
- [ ] **GS-09-05-05** Drawstrings.
- [ ] **GS-09-05-06** Buckles.
- [ ] **GS-09-05-07** Eyelets.
- [ ] **GS-09-05-08** Labels.
- [ ] **GS-09-05-09** Elastic.
- [ ] **GS-09-05-10** Grommets.
- [ ] **GS-09-05-11** 2D placement and 3D attachment.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

09.06 Structural components

- [ ] **GS-09-06-01** Interfacing.
- [ ] **GS-09-06-02** Fusing.
- [ ] **GS-09-06-03** Padding.
- [ ] **GS-09-06-04** Quilt/trapunto.
- [ ] **GS-09-06-05** Inner/outer layers.
- [ ] **GS-09-06-06** Collision ordering.
- [ ] **GS-09-06-07** Stiffness layers.
- [ ] **GS-09-06-08** Welt/pocket construction.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

09.07 Decorations

- [ ] **GS-09-07-01** Decals.
- [ ] **GS-09-07-02** Placement prints.

- [ ] **GS-09-07-03** Embroidery preview.
- [ ] **GS-09-07-04** Patches.
- [ ] **GS-09-07-05** Applique.
- [ ] **GS-09-07-06** PBR graphic material.
- [ ] **GS-09-07-07** Print/export positions.
- [ ] **GS-09-07-08** Asset anchoring.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

09.08 Material acceptance

- [ ] **GS-09-08-01** Calibration swatches.
- [ ] **GS-09-08-02** Stable saving.
- [ ] **GS-09-08-03** Physical parameters validated separately from appearance.
- [ ] **GS-09-08-04** Consistent UV and print sizes.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

10 — 3D editing, viewport and fit inspection (P1)

10.01 Camera navigation

- [ ] **GS-10-01-01** Orbit/pan/zoom.
- [ ] **GS-10-01-02** Focal-point tracking.
- [ ] **GS-10-01-03** Front/back/side.
- [ ] **GS-10-01-04** Orthographic/perspective.
- [ ] **GS-10-01-05** Bookmarks.
- [ ] **GS-10-01-06** Camera reset.
- [ ] **GS-10-01-07** Focus selection.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

10.02 3D selection

- [ ] **GS-10-02-01** Piece.
- [ ] **GS-10-02-02** Face.
- [ ] **GS-10-02-03** Vertex.
- [ ] **GS-10-02-04** Edge.
- [ ] **GS-10-02-05** Box/lasso/brush.
- [ ] **GS-10-02-06** Additive/subtractive selection.
- [ ] **GS-10-02-07** Hidden/locked logic.
- [ ] **GS-10-02-08** Accuracy near occlusion.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

10.03 Manipulators

- [ ] **GS-10-03-01** Translate/rotate/scale.
- [ ] **GS-10-03-02** Axis/plane.
- [ ] **GS-10-03-03** Numeric transform.

- [ ] **GS-10-03-04** Snapping.
- [ ] **GS-10-03-05** Local/world.
- [ ] **GS-10-03-06** Pivot.
- [ ] **GS-10-03-07** Multi-select.
- [ ] **GS-10-03-08** Undo.
- [ ] **GS-10-03-09** Mapping from 3D to 2D.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

10.04 Styleline editing

- [ ] **GS-10-04-01** Draw 3D lines.
- [ ] **GS-10-04-02** Wrap/snap on garment surface.
- [ ] **GS-10-04-03** Surface cut.
- [ ] **GS-10-04-04** Propagate splits to 2D.
- [ ] **GS-10-04-05** Re-sew.
- [ ] **GS-10-04-06** Remesh.
- [ ] **GS-10-04-07** Preserve fabric assignments.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

10.05 Pins and attachments

- [ ] **GS-10-05-01** Pin to avatar or world.
- [ ] **GS-10-05-02** Sticky point.
- [ ] **GS-10-05-03** Position constraints.
- [ ] **GS-10-05-04** Delete and edit.
- [ ] **GS-10-05-05** Stable save/load.
- [ ] **GS-10-05-06** Collision during direct garment drag.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

10.06 Fit diagnosis

- [ ] **GS-10-06-01** Strain/pressure/contact heatmaps.
- [ ] **GS-10-06-02** Ease map.
- [ ] **GS-10-06-03** Tension arrows.
- [ ] **GS-10-06-04** Length comparisons.
- [ ] **GS-10-06-05** Panel overlaps.
- [ ] **GS-10-06-06** Visual seam/collider wireframes.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

10.07 Viewport QA

- [ ] **GS-10-07-01** Screen-space picking.
- [ ] **GS-10-07-02** High DPI.
- [ ] **GS-10-07-03** 2D/3D focus sync.
- [ ] **GS-10-07-04** Z-fighting.
- [ ] **GS-10-07-05** Culling.

- [ ] **GS-10-07-06** Transparent fabric sorting.
- [ ] **GS-10-07-07** Outline stability.
- [ ] **GS-10-07-08** GPU disposal.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

11 — Avatars, body measurement, poses and animation (P1)

11.01 Avatar library

- [ ] **GS-11-01-01** Male/female body presets.
- [ ] **GS-11-01-02** Thomas and custom models.
- [ ] **GS-11-01-03** Height/shape variety.
- [ ] **GS-11-01-04** Realistic silhouettes.
- [ ] **GS-11-01-05** Anatomically grounded dimensions.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

11.02 Import and rig

- [ ] **GS-11-02-01** GLB/FBX/USD import feasibility.
- [ ] **GS-11-02-02** Skeleton validation.
- [ ] **GS-11-02-03** Skinning weights.
- [ ] **GS-11-02-04** Scaling.
- [ ] **GS-11-02-05** Body-region tags.
- [ ] **GS-11-02-06** Clothing/skin separation.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

11.03 Shape editor

- [ ] **GS-11-03-01** Body circumferences.
- [ ] **GS-11-03-02** Proportions.
- [ ] **GS-11-03-03** Anatomical ranges.
- [ ] **GS-11-03-04** Symmetry.
- [ ] **GS-11-03-05** Landmarks.
- [ ] **GS-11-03-06** Pose-safe modifications.
- [ ] **GS-11-03-07** Body scan import path.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

11.04 Pose system

- [ ] **GS-11-04-01** Standing.
- [ ] **GS-11-04-02** T/A pose.
- [ ] **GS-11-04-03** Arms raised.
- [ ] **GS-11-04-04** Walking.
- [ ] **GS-11-04-05** Seated.
- [ ] **GS-11-04-06** Joint limits.
- [ ] **GS-11-04-07** Collision mesh updates from posed rig.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

11.05 Motion and animation

- [ ] **GS-11-05-01** Timeline.
- [ ] **GS-11-05-02** Keyframes.
- [ ] **GS-11-05-03** Rig clips.
- [ ] **GS-11-05-04** Retargeting.
- [ ] **GS-11-05-05** Blend.
- [ ] **GS-11-05-06** Physical cloth playback.
- [ ] **GS-11-05-07** Baked cache.
- [ ] **GS-11-05-08** Collision safe movement.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

11.06 Export

- [ ] **GS-11-06-01** Avatar mesh/rig.
- [ ] **GS-11-06-02** Clothing sim sequence.
- [ ] **GS-11-06-03** UV and materials.
- [ ] **GS-11-06-04** Blender bridge.
- [ ] **GS-11-06-05** Consistent scale and frame rate.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

12 — Rendering, lighting and presentation (P2)

12.01 PBR

- [ ] **GS-12-01-01** Correct material layers.
- [ ] **GS-12-01-02** Physically plausible microtexture.
- [ ] **GS-12-01-03** Fabric sheen and fuzz.
- [ ] **GS-12-01-04** Thickness.
- [ ] **GS-12-01-05** Transmission.
- [ ] **GS-12-01-06** Consistent normal maps.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

12.02 Lights and environments

- [ ] **GS-12-02-01** HDRI.
- [ ] **GS-12-02-02** Area/key/rim lights.
- [ ] **GS-12-02-03** Floor and backdrop.
- [ ] **GS-12-02-04** Shadows.
- [ ] **GS-12-02-05** Ambient occlusion.
- [ ] **GS-12-02-06** Exposure.
- [ ] **GS-12-02-07** Tone mapping.
- [ ] **GS-12-02-08** Color management.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an

automated assertion.

12.03 Camera and output

- [ ] **GS-12-03-01** Camera presets.
- [ ] **GS-12-03-02** Saved views.
- [ ] **GS-12-03-03** Depth of field.
- [ ] **GS-12-03-04** Alpha background.
- [ ] **GS-12-03-05** High-res still.
- [ ] **GS-12-03-06** Turntable.
- [ ] **GS-12-03-07** Animation.
- [ ] **GS-12-03-08** Render queue and presets.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

12.04 Production handoff

- [ ] **GS-12-04-01** Blender import round-trip.
- [ ] **GS-12-04-02** UV validation.
- [ ] **GS-12-04-03** Material parity.
- [ ] **GS-12-04-04** Stable authored seam/trim positions.
- [ ] **GS-12-04-05** File names and metadata.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

12.05 Rendering QA

- [ ] **GS-12-05-01** Golden material scenes.
- [ ] **GS-12-05-02** Texture scale.
- [ ] **GS-12-05-03** Tone color consistency.
- [ ] **GS-12-05-04** GPU memory.
- [ ] **GS-12-05-05** Lighting differences tracked across engines.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

13 — Manufacturing, tech packs and factory-ready data (P1)

13.01 Tech packs

- [ ] **GS-13-01-01** Cover sheet.
- [ ] **GS-13-01-02** Flats.
- [ ] **GS-13-01-03** BOM.
- [ ] **GS-13-01-04** POM.
- [ ] **GS-13-01-05** Specs.
- [ ] **GS-13-01-06** Graded measurements.
- [ ] **GS-13-01-07** Sketches.
- [ ] **GS-13-01-08** Construction notes.
- [ ] **GS-13-01-09** Colorways.
- [ ] **GS-13-01-10** Fit notes.
- [ ] **GS-13-01-11** Care labels.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

13.02 BOM

- [ ] **GS-13-02-01** Fabric and trim supplier metadata.
- [ ] **GS-13-02-02** Quantities.
- [ ] **GS-13-02-03** Unit of measure.
- [ ] **GS-13-02-04** Wastage.
- [ ] **GS-13-02-05** SKUs.
- [ ] **GS-13-02-06** Versions.
- [ ] **GS-13-02-07** Size/color variants.
- [ ] **GS-13-02-08** Costing hooks.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

13.03 Grading and sizes

- [ ] **GS-13-03-01** Nested cut outlines.
- [ ] **GS-13-03-02** Per-size allowances and notches.
- [ ] **GS-13-03-03** Growth equations.
- [ ] **GS-13-03-04** No invalid patterns.
- [ ] **GS-13-03-05** Factory style size tables.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

13.04 Marker making

- [ ] **GS-13-04-01** Width/roll constraints.
- [ ] **GS-13-04-02** Grain and nap.
- [ ] **GS-13-04-03** Ply.
- [ ] **GS-13-04-04** Multi-size.
- [ ] **GS-13-04-05** Defect avoidance.
- [ ] **GS-13-04-06** Nesting.
- [ ] **GS-13-04-07** Stripe/plaid matching.
- [ ] **GS-13-04-08** Yield/utilization report.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

13.05 Construction annotations

- [ ] **GS-13-05-01** Seam type.
- [ ] **GS-13-05-02** Topstitch.
- [ ] **GS-13-05-03** Stitch density.
- [ ] **GS-13-05-04** Notches.
- [ ] **GS-13-05-05** Drill points.
- [ ] **GS-13-05-06** Cut-on-fold.
- [ ] **GS-13-05-07** Facing.
- [ ] **GS-13-05-08** Fusing.
- [ ] **GS-13-05-09** Operation instructions.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

13.06 Validation and approvals

- [ ] **GS-13-06-01** Tolerances.
- [ ] **GS-13-06-02** Measured sample vs target.
- [ ] **GS-13-06-03** Issue log.
- [ ] **GS-13-06-04** Comments.
- [ ] **GS-13-06-05** Approval revisions.
- [ ] **GS-13-06-06** Export manifest.
- [ ] **GS-13-06-07** Audit history.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

13.07 Exports

- [ ] **GS-13-07-01** Print PDF.
- [ ] **GS-13-07-02** SVG.
- [ ] **GS-13-07-03** DXF.
- [ ] **GS-13-07-04** AAMA/ASTM compatibility feasibility.
- [ ] **GS-13-07-05** Units.
- [ ] **GS-13-07-06** Layers.
- [ ] **GS-13-07-07** Rotations.
- [ ] **GS-13-07-08** Graded marker.
- [ ] **GS-13-07-09** Factory ZIP.
- [ ] **GS-13-07-10** Exact plot scale.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

13.08 Factory acceptance

- [ ] **GS-13-08-01** Vendor sample fixtures.
- [ ] **GS-13-08-02** Preserve notches/internal lines/grainline/quantity/angles.
- [ ] **GS-13-08-03** Round-trip report.
- [ ] **GS-13-08-04** No loss without warning.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

14 — File formats, asset management and interoperability (P1)

14.01 Native format

- [ ] **GS-14-01-01** Versioned project.
- [ ] **GS-14-01-02** Deterministic serialization.
- [ ] **GS-14-01-03** Round-trip identical geometry.
- [ ] **GS-14-01-04** Link external assets.
- [ ] **GS-14-01-05** Inline vs reference policy.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

14.02 2D interchange

- [ ] **GS-14-02-01** DXF open/save.
- [ ] **GS-14-02-02** Standards compatibility.
- [ ] **GS-14-02-03** Curves and construction lines.
- [ ] **GS-14-02-04** Mm/in.
- [ ] **GS-14-02-05** Rotation.
- [ ] **GS-14-02-06** Line layers.
- [ ] **GS-14-02-07** Notches.
- [ ] **GS-14-02-08** Grading.
- [ ] **GS-14-02-09** Warnings.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

14.03 3D interchange

- [ ] **GS-14-03-01** OBJ.
- [ ] **GS-14-03-02** GLB.
- [ ] **GS-14-03-03** GlTF.
- [ ] **GS-14-03-04** FBX where practical.
- [ ] **GS-14-03-05** USD.
- [ ] **GS-14-03-06** Morph/rig.
- [ ] **GS-14-03-07** UVs.
- [ ] **GS-14-03-08** Textures.
- [ ] **GS-14-03-09** Animation.
- [ ] **GS-14-03-10** Blender and DCC consistency.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

14.04 CLO compatibility

- [ ] **GS-14-04-01** Document unsupported proprietary formats.
- [ ] **GS-14-04-02** Investigate lawful supported import/export paths.
- [ ] **GS-14-04-03** No claim of native .zprj round-trip without tests.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

14.05 Asset catalogue

- [ ] **GS-14-05-01** Thumbnails.
- [ ] **GS-14-05-02** Tagging.
- [ ] **GS-14-05-03** Search.
- [ ] **GS-14-05-04** Import folders.
- [ ] **GS-14-05-05** Deduplicate.
- [ ] **GS-14-05-06** Relocate/relink.
- [ ] **GS-14-05-07** Categories.
- [ ] **GS-14-05-08** Versions.
- [ ] **GS-14-05-09** Permissions.
- [ ] **GS-14-05-10** Starter content.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

14.06 External workflow

- [ ] **GS-14-06-01** Seamly2D import.
- [ ] **GS-14-06-02** GarmentCode procedural apparel.
- [ ] **GS-14-06-03** Blender exports.
- [ ] **GS-14-06-04** Documented transform and naming contracts.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

15 — Quality assurance, benchmarks, observability and release (P0)

15.01 Coverage matrix

- [ ] **GS-15-01-01** Every toolbar/command linked to implementation, model mutation, undo/redo, save/reload, exports and screenshots.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

15.02 Unit tests

- [ ] **GS-15-02-01** Geometry precision.
- [ ] **GS-15-02-02** Constraints.
- [ ] **GS-15-02-03** Grading.
- [ ] **GS-15-02-04** Measure.
- [ ] **GS-15-02-05** Sew ranges.
- [ ] **GS-15-02-06** Seam allowances.
- [ ] **GS-15-02-07** Math edge cases.
- [ ] **GS-15-02-08** Zero/NaN guards.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

15.03 Integration tests

- [ ] **GS-15-03-01** 2D edit->3D->save->load->export.
- [ ] **GS-15-03-02** Fabric->renderer/sim.
- [ ] **GS-15-03-03** Avatar pose->collision.
- [ ] **GS-15-03-04** AI API->undo.
- [ ] **GS-15-03-05** Services restart.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

15.04 UI E2E

- [ ] **GS-15-04-01** Native click paths.
- [ ] **GS-15-04-02** Dropdown/flyout.
- [ ] **GS-15-04-03** Right click.
- [ ] **GS-15-04-04** Drag.
- [ ] **GS-15-04-05** Context menus.

- [ ] **GS-15-04-06** Automated fixture screenshot.
- [ ] **GS-15-04-07** Stable selectors.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

15.05 Physics regressions

- [ ] **GS-15-05-01** Continuous-frame contact/strain/closure/energy metrics.
- [ ] **GS-15-05-02** Deterministic runs.
- [ ] **GS-15-05-03** No fake PASS from initial/last-frame-only checks.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

15.06 Performance

- [ ] **GS-15-06-01** Cold start.
- [ ] **GS-15-06-02** Load 100 pattern pieces.
- [ ] **GS-15-06-03** Memory growth.
- [ ] **GS-15-06-04** Target 2D latency.
- [ ] **GS-15-06-05** Frame rate.
- [ ] **GS-15-06-06** 3D select latency.
- [ ] **GS-15-06-07** Simulation throughput.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

15.07 Crash/error handling

- [ ] **GS-15-07-01** Malformed project.
- [ ] **GS-15-07-02** Failed assets.
- [ ] **GS-15-07-03** Missing native service.
- [ ] **GS-15-07-04** Unexpected EOF.
- [ ] **GS-15-07-05** Cancel job.
- [ ] **GS-15-07-06** Disk full.
- [ ] **GS-15-07-07** Sleep/resume.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

15.08 Benchmark comparison

- [ ] **GS-15-08-01** Same avatar/pattern/material/pose/time against licensed CLO build.
- [ ] **GS-15-08-02** Record version.
- [ ] **GS-15-08-03** Source assets.
- [ ] **GS-15-08-04** Silhouette metrics.
- [ ] **GS-15-08-05** Manual expert review.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

15.09 Automation and CI

- [ ] **GS-15-09-01** Unit/visual/nightly solver jobs.

- [ ] **GS-15-09-02** CI artifact reports.
- [ ] **GS-15-09-03** Reproducible dependencies.
- [ ] **GS-15-09-04** Release smoke test.
- [ ] **GS-15-09-05** Regression triage.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

15.10 Security/licensing

- [ ] **GS-15-10-01** Local service binding.
- [ ] **GS-15-10-02** File path validation.
- [ ] **GS-15-10-03** Resource limits.
- [ ] **GS-15-10-04** No arbitrary script injection.
- [ ] **GS-15-10-05** SBOM.
- [ ] **GS-15-10-06** GPL/LGPL compliance.
- [ ] **GS-15-10-07** Import provenance.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

15.11 Release gates

- [ ] **GS-15-11-01** P0 zero known data loss.
- [ ] **GS-15-11-02** No fake supported feature.
- [ ] **GS-15-11-03** Native app smoke.
- [ ] **GS-15-11-04** Saved sample opens.
- [ ] **GS-15-11-05** Published known limitations.
- [ ] **GS-15-11-06** Rollback plan.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

16 — Documentation, examples, onboarding and productization (P2)

16.01 User journeys

- [ ] **GS-16-01-01** First tee.
- [ ] **GS-16-01-02** Hoodie.
- [ ] **GS-16-01-03** Jeans.
- [ ] **GS-16-01-04** Custom pattern.
- [ ] **GS-16-01-05** Import CAD.
- [ ] **GS-16-01-06** Auto fit.
- [ ] **GS-16-01-07** Garment edit.
- [ ] **GS-16-01-08** Factory tech pack.
- [ ] **GS-16-01-09** Render demo.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

16.02 Training

- [ ] **GS-16-02-01** Step-by-step tool tips.
- [ ] **GS-16-02-02** Videos.

- [ ] **GS-16-02-03** Keyboard cheatsheet.
- [ ] **GS-16-02-04** 2D/3D correspondence explanations.
- [ ] **GS-16-02-05** Glossary and search.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

16.03 Developer docs

- [ ] **GS-16-03-01** Architecture.
- [ ] **GS-16-03-02** Schema and migrations.
- [ ] **GS-16-03-03** IPC APIs.
- [ ] **GS-16-03-04** Solver flags.
- [ ] **GS-16-03-05** Profile/repro commands.
- [ ] **GS-16-03-06** Accepted benchmarks.
- [ ] **GS-16-03-07** Contribution guidance.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

16.04 Support

- [ ] **GS-16-04-01** Diagnostic bundle.
- [ ] **GS-16-04-02** Redaction.
- [ ] **GS-16-04-03** Known issues.
- [ ] **GS-16-04-04** Reproducible steps.
- [ ] **GS-16-04-05** Screenshots.
- [ ] **GS-16-04-06** Crash reporting privacy.
- [ ] **GS-16-04-07** Feedback tracking.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

16.05 Product readiness

- [ ] **GS-16-05-01** Minimum useful feature tiers.
- [ ] **GS-16-05-02** MacOS support matrix.
- [ ] **GS-16-05-03** Offline operation.
- [ ] **GS-16-05-04** Accessibility.
- [ ] **GS-16-05-05** Install/update quality.
- [ ] **GS-16-05-06** License review.

Acceptance: Demonstrate this subsection using a real garment/project fixture, record intended data mutation and undo/redo/save/load, test precise geometry or physical metrics when relevant, and attach before/after evidence with an automated assertion.

4. Confirmed / recorded failures and diagnostic worklist

These are not speculative parity wishes; they come from the existing engineering audit and need current-build reproduction before a bug is closed.

- [ ] **BUG-001 Floating shirt:** reproduce body-to-neck/shoulder clearance and collar lift with current Thomas geometry; diagnose competing anchors and collider overlap; fix; verify all frames.
- [ ] **BUG-002 Excess strain:** prior 180 cm tee scenario recorded peak ~546% relative extension; reproduce with version/preset, locate source constraints, reduce to measured material tolerances, not merely a permissive regression cap.

- [ ] **BUG-003 Dynamic intersections:** prior 10-frame PPF tee recorded final 177 avatar intersection pairs and 56 self-intersection pairs; retest; implement CCD/coupled contact and certify continuous trajectory.
- [ ] **BUG-004 Sewing/contact conflict:** quantify each solver pass residual and resulting displacement; remove seam reopenings and interpenetrations.
- [ ] **BUG-005 PPF long-sew timeout:** profile pathological contact graph; enforce cancellability and convergence thresholds, valid failure reporting.
- [ ] **BUG-006 Fixture identity:** ensure 2D source pattern, 3D pose, avatar scale and collider mesh have exactly matching transform/units and source IDs across backends.
- [ ] **BUG-007 UI QA selector ambiguity:** prior `scripts/qa-clo-interface.mjs` matched multiple "Free sewing" buttons. Replace brittle exact-name assumptions with semantic selectors; re-run native/browser E2E.
- [ ] **BUG-008 Grading test coverage:** prior audit found missing `scripts/check-grading.mjs` command; add explicit grading golden tests, then register in consolidated QA.
- [ ] **BUG-009 Mesh/styleline partial tooling:** audit currently disabled/planned controls and incomplete mesh selection styles; verify actual state after latest patches; implement true persistent operations with 2D mapping.
- [ ] **BUG-010 DXF fidelity beyond existing golden samples:** verify rotated buttonholes, grainlines, notches, internal lines, curves, nested graded pieces, scaling and industrial vendor interchange.
- [ ] **BUG-011 Browser visual QA timeout under concurrent builds:** isolate stable dedicated QA run and save success/failure artifacts rather than counting timeout as pass.
- [ ] **BUG-012 Multiple-actor edit safety:** document ownership, avoid resets/cleans and fixture overwrites; snapshot before destructive migrations; verify simultaneous native build processes do not race.

5. Milestones and dependency gates

Milestone	Goal and deliverables	Blocking gates	Evidence required
M0: Document integrity	Canonical schema, action API, migrations, undo/save and source mapping	None	Open/modify/save/reopen/undo/redo across multiple fixtures
M1: Manufacturer-ready 2D	Exact CAD primitives/constraints, seam allowances, T-shirt/jeans base patterns	M0	Golden numeric patterns, fixture import/export tolerances
M2: Seam graph & meshing	Durable edge IDs, seam orientation, adaptive mesh, complete flat-to-3D mapping	M0–M1	Closed sewing graph and persistent remesh with stable mappings
M3: Physical fit truth	Unified avatar skin collider, CCD, coupled seam/contact, calibrated materials	M1–M2	Zero unsafe crossings across trajectory, acceptable fabric strain/gaps
M4: Reliable auto-fit	Inferred arrangement, reversible tweaks and scored candidate acceptance	M3	Tee fits at 160/180/200 cm across multiple poses
M5: Professional editing	3D selection/stylelines, hierarchical browser, property editing, trims and fabrics	M0–M2	Full multi-tool E2E CRUD + synchronized 2D/3D editing
M6: Factory outputs	Grading, BOM, POM, markers, standards-grounded export	M1,M5	Vendor fixture cross-import and production package completeness
M7: Complete visuals	Avatar poses, rendering, animation, library and Blender interchange	M3,M5	Consistent assets and approved rendered scenes
M8: Release quality	Native packaging, full regression, performance, security, docs	All previous	Release checklist, benchmark matrix, migration and recovery drills

6. Focused next-pass implementation order

1. **Do not lose existing source:** inspect working tree, confirm local backups and running solver processes; note other agents editing files.
2. **Fix structural reliability first:** canonical schema, piece/edge/seam IDs, explicit unit conversion, command transactions and undo/save tests.
3. **Complete the precision drafting golden shirt:** true four-piece pattern, seam graph, measured allowances, tolerance and grading fixtures.
4. **Add full-trajectory physical quality gates before more visual tweaks:** per-frame body/self contact, strain, seam p95/max and settling using the exact posed Thomas collider.
5. **Couple solver contact, stitching and material behavior:** maintain correct arrangement; never accept a fit only because a render looks good.

6. **Close 3D/editor workflow gaps:** editable stylelines and mesh selection, hierarchical Object Browser and numeric inspector, all reversible.
7. **Industrial export and calibration:** validate apparel factory files on external consumers and versioned benchmark fixtures.
8. **Advance rendering, animation and asset library only after verified data integrity and fitting milestones are stable.

7. Ownership and reporting template

For each engineering pass record: `ID | current evidence | exact files | fixture | before metrics | implementation | after metrics | automated test output | manual screenshot | native app build | status | next blocker`. Statuses: `unreviewed`, `confirmed absent`, `partial`, `implemented-unverified`, `verified`, `blocked`. Do not promote `partial` to `verified` on build-pass alone.

8. Source material within this repository

- `docs/CLO3D_NON_SIMULATION_FULL_BUILD_AUDIT_2026-10-13.md`
- `docs/CLO3D_CLOTHING_SIMULATION_GAP_AUDIT_2026-10-13.md`
- `docs/GARMENT_STUDIO_NATIVE_ROADMAP.md`
- `docs/ZOZO_PPF_ON_BODY_INTEGRATION_2026-10-13.md`
- `docs/GARMENT_SIMULATION_NEXT_GEN.md`
- `docs/ARCHITECTURE.md`
- `docs/CLO_UI_TOOL_INVENTORY.md`
- `docs/PPF_PHYSICAL_SEWING_IMPLEMENTATION_2026-10-13.md`
- Source, fixtures and tests under `src/`, `native_backend/`, `ppf_backend/`, `blender_bridge/`, `scripts/` and `native_app/`.

Inventory: 17 major categories, 124 functional subcategories and 924 atomic feature checklist items, plus 12 concrete recorded bug investigations and 14 universal done gates.

EXO_MASTER_ROADMAP.md

Audit interpretation: This annex preserves earlier engineering research for traceability. Read reported pass counts and statuses as of its original date only. The current 70-slot and 400-fabric inventories above supersede any older source totals.

EXO SAMPLE 3D — ENGINEERING MASTER ROADMAP

Status: proposed execution plan, 15 October 2026. **Product identity:** EXO Sample 3D. **Repository:** `mikristian/exo-sample-3d`; historical document filenames and internal identifiers are preserved for traceability. This roadmap is a contract for work and verification, not a declaration that the application matches CLO3D or produces certified factory deliverables.

Product contract and architectural boundaries

EXO aims to connect precise editable 2D patterns, sewn 3D geometry, realistic fabric responses, fit analysis, visual material design, and traceable manufacturing packages. A single versioned garment project must remain the source of truth for all views. An edit to a dart, seam allowance or grade rule should invalidate the relevant derived mesh, contact state, point-of-measure (POM) results, marker and export outputs. The application should never display an old simulation or export as if it reflected newer drafting geometry.

The source currently includes a React/Three.js browser workspace, a shipped macOS shell using WebKit, native C++ feature modules under `native_cpp/core/`, a developing Qt interface, and an optional Rust PPF sidecar. The active UI and the tested native CLI do not yet constitute a single independently validated fully native application. Existing C++ parity coverage is module-specific. JavaScript remains a reference behavior for many functions, and a close numerical match to JavaScript is not proof that the reference physics is physically correct. Preserve that distinction in status and release notes.

Read [architecture](ARCHITECTURE.md), [native conversion checkpoint](SAMPL3D_NATIVE_CPP_CONVERSION_LOOP_2026-10-15.md), [CLO production comparison](CLO_EXPORT_PRODUCTION_PARITY_2026-10-15.md), [simulation research](EXO_SIMULATION_RESEARCH.md) and [verification plan](EXO_VALIDATION_TEST_PLAN.md) together. The October production audit explicitly states that 90% CLO3D parity has **not** been demonstrated. Neither the existence of export files nor passing source parity tests is equivalent to acceptance by a cutter, grader, vendor or garment technician.

Work graph and dependency gates

The milestones below are numbered by dependency and verification effort, not by an asserted percentage complete. Parallel tasks are safe where they operate on separate data contracts and cannot silently alter shared geometry. Every GitHub milestone requires a named owner, evidence artifact, exact command, machine/OS/solver version, failure case, and acceptance reviewer. Do not promote an issue solely because a feature appears in a menu.

M0 — reproducibility, provenance and reference corpus

1. Capture immutable source revision, build flags, native runtime versions, material configuration and test seed with each fixture. Define schema revisions for garment projects, physics snapshots, export manifests and acceptance records. Store fixture metadata beside the fixture and avoid environment-dependent absolute paths.
2. Establish open, redistributable base garments: basic tee with conventional sleeves, sleeveless shell, trousers with inseam and fly, jacket with zipper and multi-layer cloth. Include explicit unit scales, grain angles, notches, panel winding, seams, symmetric references and annotated measurement landmarks. Use synthetic or freely licensed avatar geometries with documented redistribution rights.
3. Add a deterministic cross-backend harness that compares reference JavaScript outputs, native C++ transformations and Rust solver traces by operation and field. Declare tolerances per field and explain why they are physically meaningful. Every mismatch retains a minimal repro and observed numeric delta.
4. Set protected CI checks for format validity, build reproducibility, license inventory, data migrations, and sample exports. Run `npm run build`, `npm run check:non-simulation`, `scripts/check-native-cpp-conversion-coverage.mjs` and native test suites as applicable. The inventory is a set of covered exports, not a whole-product percentage.

Exit: on a clean checkout, another developer can run a documented subset without private assets; fixture licenses are declared; every baseline passes or its failures are visible. See [open-source validation](OPEN_SOURCE_VALIDATION_2026-10-14.md), [release audit](OPEN_SOURCE_RELEASE_READINESS_AUDIT_2026-10-13.md), and [licensing](LICENSING.md).

M1 — CAD accuracy and round-trip drafting

Keep all native drafting coordinates in declared physical units. Design a stable document model for panels, vertices, quadratic/curved edges, seam allowances, darts, internal lines, cutouts, grainlines, labels, linked mirror transforms, user-defined construction points and per-size grade rules. Editing must preserve constraint identity and undo/redo history. A notch tied to an edge parameter should stay attached after reasonable edits and should become a validation error if its parent edge is deleted. Imported patterns must be quarantined from simulation until winding, unit scale and topological checks pass.

Validate exact millimeter bounding geometry, edge sampling error, seam length, reversible transformations, nested undo groups, dangling seam references, nonmanifold contours and self-intersections. Cover keyboard and pointer tools, snapping in actual 2D viewport coordinates, scale/zoom behavior, high-DPI selection and redraw responsiveness. Existing relevant files include `src/core/dxf.js`, `src/core/grading.js`, `src/core/measurements.js`, and native tests such as `scripts/check-native-cpp-drafting-constructions.mjs`, `scripts/check-native-cpp-cad.mjs` and `scripts/check-native-cpp-project-geometry.mjs`. Read [2D drafting acceptance] (CLO_DRAFTING_WORKSPACE_MATERIAL_ACCEPTANCE_2026-10-14.md) before changing interaction contracts.

Exit: reference patterns survive save/open and export/import without lost construction semantics, and tolerances have been verified against independent geometry assertions. Persisted projects remain readable following schema upgrades.

M2 — sewn garment topology and mesh generation

Map seam segments to discretized mesh boundary edges using stable parent edge identifiers and arc-length parameters. Support opposing directions, seam-length imbalances, easing, pleats, darts, virtual zipper tracks, reversible seam assignments, layered seam junctions and incremental remeshing. Report sewable lengths, unmatched intervals and topology violations before assembling. A sewing error must remain visible even if a visual interpolation can hide the opening in a posed screenshot.

Generate UVs in grain-relative coordinates, preserve panel names/materials after subdivision, and propagate fabric orientation into anisotropic constitutive models. Verify triangle area and orientation, boundary invariants, panel spacing, and seam matching under changes to tessellation density. `native_cpp/core/pattern_mesh.hpp`, `scripts/check-native-cpp-pattern-mesh.mjs`, `scripts/check-native-cpp-seam-assembly-parity-bridge.mjs` and [native mesh audit] (NATIVE_MESH_QUALITY_PASS_2026-10-13.md) provide starting evidence. The advanced assembly tests must report contact and position agreement across tee and trousers rather than converting a one-frame pass into a complete fit claim.

Exit: deterministic, editable sewn rest state across multiple garment classes, with explicitly tracked invalid or incomplete seams.

M3 — physically defensible dynamics and fitting

Maintain a clear separation between real-time design preview and a slower acceptance-grade simulation. The browser solver may support immediate iteration; the Rust PPF contact solver may own physically demanding contact operations; C++ may own CAD, interchange, quality analysis and glue. A single orchestrator must expose backend identity, integration scheme, timestep, substeps, cloth thickness, material parameter units, initial pose, collision proxy source and solver stopping criteria. The viewport must not collapse diagnostics into an unexplained green status.

Implement and verify stretch, shear, bending, damping, mass, gravity, seams and boundary pins. Treat self-contact and continuous collision detection (CCD) as distinct requirements from body penetration correction. Tests must cover fast arm motion, cloth-on-cloth folds, crossing sleeve panels, layered jackets, narrow armholes, seam convergence and warm-start resets. Evaluate swept cloth/body and cloth/cloth interactions through complete trajectories. Require controlled comparisons against licensed or open physical benchmark data and physically measured drapes. See [simulation research] (EXO_SIMULATION_RESEARCH.md), [PPF body tee pipeline](PPF_BODY_TEE_PIPELINE.md), [contact safety] (PPF_CONTACT_SAFETY_NEXT_PASS_2026-10-13.md) and [actual fit audit] (PPF_ACTUAL_GARMENT_FIT_DEEP_AUDIT_2026-10-14.md).

Exit: recurring simulation fixtures meet numerical, geometric and trajectory acceptance limits recorded in the [test plan] (EXO_VALIDATION_TEST_PLAN.md). Until then, any status marked `fitCertified`, `factoryCertified` or CCD-complete remains false unless independently justified by the exact scenario.

M4 — fabric appearance and engineering parameters

Retain separate appearance and mechanical material models. Procedural albedo, normal, height and roughness do not establish bending stiffness or friction. Keep physical provenance for warp/weft elasticity, bending, density, damping, thickness and surface friction, with confidence intervals where measurements are uncertain. Support both isotropic and anisotropic profiles, grain rotation, different materials per panel, trims and hardware collision boundaries. Map UI parameters to defined units and solver-specific constitutive fields.

The current project describes 23 fabric presets and 13 procedural textile structures; inspect `scripts/check-native-cpp-textile-procedurals.mjs`, `scripts/check-native-cpp-fabric-pbr-pixels.mjs` and [fabric parity audit] (SAMPL3D_FABRIC_PARITY_AUDIT_2026-10-15.md). Validate continuity at UV seams, deterministic generated maps, texture-scale consistency in centimeters and visual behavior under changing lighting. Changing appearance should not

accidentally restart drape; changing mechanical parameters should invalidate physics caches and visibly request a new solve.

Exit: appearance reproducibility plus measured or explicitly estimated physical parameters, accompanied by uncluttered material provenance indicators.

M5 — fit analysis, avatars and animation

Formalize avatar import, mesh hygiene, collision normals, body landmarks, pose binding, shape scaling and motion interpolation. Create signed-distance or mesh contact proxies with documented error bounds and a way to compare proxy results with source triangles. Separate cloth strain from pressure estimates, local clearance from penetration depth, and posture feasibility from garment size recommendations. A color-coded strain map needs a legend, normalization rule and per-material thresholds. Expose analysis quality flags when motion sampling or contact convergence is inadequate.

Run baseline and perturbation tests on the tee, trousers and layered outfit, multiple sizes, and full motion sequences. Do not ship private scan assets or embedded proprietary avatars. Use open synthetic human meshes with correct notices. If a real reference body is used locally for testing, publish anonymized derived metrics only where distribution and privacy permissions permit.

Exit: trustworthy measurement landmarks, explainable heat maps and a repeatable on-body acceptance demonstration including failure cases.

M6 — manufacturing handoff

Use the [production pipeline](EXO_PRODUCTION_PIPELINE.md) as the exporter contract. Verify industrial 2D DXF interpretations, unit scaling, grade-rule preservation, marker orientation, lengths, material widths, physical plot scale, POM tolerances, and a complete BOM. Keep revision identifiers consistent across source project, drawings, grades, cut orders, costing and ZIP manifest. Test a complete handoff with a cooperating technician and at least one separate CAD/cutter system.

Current work already covers several source and native operations: see [CLO export comparison](CLO_EXPORT_PRODUCTION_PARITY_2026-10-15.md), `scripts/check-production-handoff.mjs`, `scripts/check-factory.mjs`, `scripts/check-native-cpp-factory-cli.mjs`, `scripts/check-native-cpp-markers.mjs`, and [native BOM/marker audit](NATIVE_CPP_BOM_AND_MARKER_PARITY_2026-10-15.md). Treat byte parity, unit parity and manufacturer compatibility as three different forms of evidence.

Exit: authorized pilot cutting from controlled reference patterns with signed dimensional QC. Factory certification requires documented supplier approval and an actual physical production review.

M7 — desktop consolidation and open-source release

Once contracts above stabilize, offer an independently built C++/Qt desktop with equivalent authoring, undo, 2D/3D navigation, project save/load and production actions. The user can continue working in the existing WebKit workspace until verified workflows move. Rust PPF can remain a separate process exposed over a stable versioned protocol; a Rust solver does not force a Rust UI. Confirm Qt distribution terms and upstream binary/source licenses before packaging. A public repository should include legitimate screenshots, reproducible demo assets and the V16 promo only if all media rights and asset licenses permit redistribution.

Exit: reproducible packaging, clean-machine smoke test, explicit bundled licenses, public issue board, vulnerability handling and reviewed release notes. See [Qt toolchain](CPP_QT6_TOOLCHAIN_READINESS_2026-10-15.md), [native roadmap](GARMENT_STUDIO_NATIVE_ROADMAP.md), [GitHub launch](OPEN_SOURCE_GITHUB_LAUNCH_PLAYBOOK.md) and [contribution guide](EXO_CONTRIBUTION_GUIDE.md).

Issue board taxonomy and prioritization

Create labels `cad`, `sewing`, `mesh`, `physics`, `ppf-rust`, `contact`, `material`, `avatar`, `fit`, `factory`, `interchange`, `desktop`, `docs`, `license`, `verification`, plus severity labels. An issue's first paragraph defines the user-visible failure or missing workflow; subsequent fields specify input fixture, expected behavior, observed behavior, candidate modules, acceptance gate, regression command and proof attachment. An umbrella tracking issue may link implementation, independent verification and external pilot tasks, but each subissue must close on its own artifact.

Prioritize defects that can misrepresent a garment's physical viability: silently inverted normals, body penetration, stale simulation shown after editing, wrong DXF units, altered grade dimensions, and false factory-ready flags. Next come fidelity improvements that influence purchase or manufacturing decisions; then rendering aesthetics, workspace refinement and optional interchange formats. A striking promo shot never substitutes for evidence of a solved collision or a measured cut dimension.

Decision record and release rule

For every migration, record a brief architecture decision: the data ownership change, why C++/Rust/JavaScript is appropriate, serialization boundary, numerical agreement conditions, backward compatibility, rollback route and test provenance. Preserve a testable reference path until coverage is sufficient to remove it. Avoid hardcoded fixture-specific output, test-only flags hidden from users or declaring coverage using only counted exports. A release can label modules `experimental`, `fixture-verified`, `pilot-validated` or `production-approved` only with attached proof of that particular claim. Maintain a visible unsupported-functions matrix rather than compressing the roadmap into a fabricated completion score.

CLO3D_NON_SIMULATION_FULL_BUILD_AUDIT_2026-10-13.md

Audit interpretation: This annex preserves earlier engineering research for traceability. Read reported pass counts and statuses as of its original date only. The current 70-slot and 400-fabric inventories above supersede any older source totals.

GARMENT STUDIO VS CLO3D: WHOLE-APPLICATION GAP AUDIT (EXCLUDING CLOTH SIMULATION)

Date: 2026-10-13. Scope: repository `src/`, `native_app/`, `scripts/`, `docs/`, production/export modules. Excludes solving/draping/cloth-body contact quality. This is a source/feature and bounded automated-test audit, **not** a certified parity review of every CLO version or an exhaustive interactive end-to-end test.

Evidence and test results

- `npm run build`: PASS.
- `node scripts/check-production.mjs`: PASS.
- `node scripts/check-factory.mjs`: PASS.
- `node scripts/qa-clo-interface.mjs`: FAIL, selector ambiguity (`getByRole('button',{name:'Free sewing'})` now matches two buttons). This is a UI test reliability issue, not proof that sewing is broken.
- `node scripts/check-grading.mjs`: unavailable / missing script; grading has an implementation in `src/core/grading.js`, but that test command did not verify it.
- Existing detailed implementation and inventory reports: `docs/CLO_UI_IMPLEMENTATION_2026_10_13.md`, `docs/CLO_UI_TOOL_INVENTORY.md`, `docs/CLO_ICON_VISUAL_AUDIT_2026-10-13.md`, `docs/CLO_DUAL_COLUMN_TOOLBAR_PLACEMENT.md`.

Status terms: **Foundation** = usable core with substantial missing workflows; **Partial** = limited subset or UI-to-inspector only; **Absent/unverified** = no validated complete implementation found in inspected code. None means CLO parity.

Category	Current evidence	Gap / next acceptance target	Priority
Precision 2D CAD	`src/components/PatternEditor.jsx`; select, edit points, split, curve, internal line, notches, measure; DXF import/export	Complete Bézier/arc precision, point/segment constraints, symmetry, true cut/spread/slash, dart manipulation, nested pieces, keyboard dimension input, multi-selection and precision snapping	P0
3D mesh/styleline editing	`src/App.jsx` 3D rail: Select Mesh and Edit Styleline are explicitly disabled/planned	Implement actual 3D mesh selection (box/lasso/brush) and editable stylelines, preserve mapping to 2D patterns and undo	P0
Sewing authoring (not physics)	Seam links, free sewing, edge correspondence and 2D/3D guides	Edit seam direction, partial/segment sewing, match lengths, seam allowances and seam walk interaction; robust editing and undo tests	P1
Toolbars and workspace	Two tool rails, object and property panes, menus, camera and view widgets	Correct CLO visual ordering and semantics, flyout variants, context-sensitive tool state, docking, persistence of workspace layouts; avoid proprietary icon copying	P1
Object Browser	Ten tabs: Scene, Fabric, Graphic, Button, Buttonhole, Topstitch, Puckering, Trim, Grading, POM (`src/components/CloUI.jsx`)	Real hierarchy, visibility/lock, drag ordering, search/filter across objects, multi-object editing; Puckering and several object types have shallow actions	P1
Property Editor	Pattern/fabric/avatar/seam/production inspectors	Complete selection-specific editors, per-object validation, batch edits, true CLO particle-distance-in-mm control instead of subdivision surrogate	P1
Fabrics and appearance	Fabric GSM, thickness, shrink, warp/weft controls, texture/graphics, rendering	Measured physical fabric library, texture transform UV workflow, PBR layers/normal/roughness, calibrated color management and swatch persistence	P1
Trims, graphics, stitching	`src/core/trims.js`, zippers, buttons/buttonholes, topstitch, graphic and trim object rows	True editable 3D hardware placement, realistic geometry, customizable stitch types, trim libraries, production metadata, puckering authoring	P1
Grading	`src/core/grading.js`, size offsets and size-run generation	Point-level rules and rule tables, linked/nested size runs, size-specific seam checks, industrial grading import/export, golden fixture tests	P1

Category	Current evidence	Gap / next acceptance target	Priority
Marker making	`src/core/marker.js` placement and marker reports	Grainline/nap constraints, fabric defects, stripe/plaid matching, multi-size and multi-ply marker strategy, verified material utilization	P2
Manufacturing / tech packs	`src/core/factoryPackage.js`, `src/core/measurements.js`, BOM/POM, graded DXF, marker reports, HTML and ZIP	More robust grading tolerances, annotations, construction detail, versioned approvals, vendor round-trip and verified export fidelity	P1
File interchange	Local garment project, DXF, OBJ/GLB export, factory ZIP	No validated native CLO `*.zprj` round-trip; assess AAMA/ASTM DXF interoperability, curve/notch/metadata fidelity, import warnings and round-trip fixture coverage	P1
Avatar tools	Avatar options, CLO Thomas asset and measurements, viewport transforms	Dedicated avatar editor, pose library, custom rig import, motion retargeting, skin/measurement precision, avatar accessory management	P1
Rendering	Still capture, environment and resolution settings	Studio-grade PBR lighting/camera presets, shadows and fabric microdetail, render queue and batch output, verified consistent color output	P2
Animation	Animation menu/workspace scaffold	Keyframe/timeline editing, motion import, clips, export, pose-to-animation workflows; no validated full implementation found	P2
Asset library	Garment/fabric/avatar/trims categories and local thumbnails	Complete reusable asset catalog, import tagging, folders, previews, filtering, versioning and project relinking	P2
Automation	`src/App.jsx` command parser implements narrow verbs (widen %, crop cm, seam allowance, fabric weight, simulate/reset)	Typed action API for *all* editing commands, schema validation, preview/rollback, agent-driven multi-step projects, regression tests	P1
Project reliability	Autosave/undo/redo and native wrapper; current production build passes	Comprehensive tool acceptance tests, migration/versioning, crash recovery, project round-trip, keyboard shortcuts, accessibility, QA on native bundle	P0

Recommended order (outside simulation)

1. **P0 CAD correctness:** establish editable topology, exact numeric geometry, reliable undo/redo, selection, snap and DXF round-trip. This is the foundation for every downstream feature.
2. **P0 real 3D tool implementation:** replace disabled mesh selection and styleline controls with real operations that propagate to 2D and persist.
3. **P1 editing completeness:** sewing graph editor, trims/topstitch/puckering, Object Browser and Property Editor semantics.
4. **P1 production reliability:** grading, marker, tech packs and DXF import/export acceptance fixtures; do not equate existence of a generator with factory-level certification.
5. **P1 automation and QA:** expose safe, reversible typed operations for AI and users; test every toolbar command against model mutation, undo, save/load and 2D/3D synchronization.
6. **P2 presentation ecosystem:** material library, avatar authoring, high-quality rendering, animation, asset management and customizable workspace.

Verification contract

For each CLO comparison tool, log `implemented`, `partial`, `planned` or `unsupported`, link source function and screenshot, then test: click -> intended model mutation -> undo -> redo -> save/load -> 2D/3D sync -> DXF/production export if applicable. Include numerical geometric checks, not only browser screenshot assertions. CLO version and toolbar personalization must be recorded for any 1:1 comparison. Do not silently activate placeholder buttons or imply a full feature from a tab or menu label.

CLO3D_PROPERTY_EDITOR_DEEP_AUDIT_20261016.md

Audit interpretation: This annex preserves earlier engineering research for traceability. Read reported pass counts and statuses as of its original date only. The current 70-slot and 400-fabric inventories above supersede any older source totals.

CLO PROPERTY EDITOR DEEP AUDIT AND EXO V2 PARITY SPECIFICATION

Date: 2026-10-16. **Deliverable:** official-reference inventory, local evidence boundaries, EXO V2 mapping, priority and acceptance criteria. **Scope:** right-side selection-sensitive Property Editor, including linked Object Browser property contexts. This is a design and source audit; it is not a claim that the installed CLO application's rendered UI was visually captured.

Evidence, version boundaries, and how to read this audit

- **O** means the control or behavior appears in an official CLO Help Center article. The linked pages are living documents and can describe newer releases than **CLO 7.0.242**. Therefore **O does not automatically mean verified in 7.0.242**.
- **L?** means exact label, order, defaults, ranges, visibility, and behavior must be checked in the *locally running* CLO 7.0.242 Property Editor. No screenshots or interaction with that application were available during this worker audit. Existing project audit records identify `/Applications/CLO_Standalone_OnlineAuth.app`, but a bundle's presence is not evidence of an active 7.0.242 session.
- **E** means directly supported by the current EXO V2 source examined on October 16, particularly `src/components/CloPropertyEditorAccordion.jsx`. **E-partial** means UI or data exists without verified matching physical semantics; **E-no** means no matching editor/solver binding was established in that inspected code. Other project work may be landing concurrently; verify after integration.
- **P** means proposed future EXO behavior. **Do not present P or L? as an observed CLO screenshot fact.**

Primary official reference: Property Editor. This confirms that the editor changes according to selected pattern, internal shape, sewing, stitch, sewing style, fabric, button, buttonhole and avatar, with simulation/wind/2D-window properties invoked via background context menus. Numeric fields support right-click Calculator and Convert, and items can carry an Item No. The specific properties below must be dispatched by **selection object kind**, not merely the active toolbar tab.

Important version caution: The Help Center includes older article families and pages updated in 2025–2026. For example Select Binding explicitly calls out a *Both (ver 7.0)* placement option, while newer seam-tape articles refer to 2024.2 migration. Record the local build and screenshot before calling an individual feature "exact CLO 7.0.242 parity."

Current EXO V2 code facts (read-only October 16 snapshot)

The editor at `src/components/CloPropertyEditorAccordion.jsx` exports eight IDs: `information`, `selected-line`, `simulation`, `fabric`, `bond`, `solidify`, `arrange`, and `miscellaneous`. A compact Property Editor remains docked below the Object Browser, with expandable sections. It distinguishes a Rust `selection.kind === 'line'` from `pattern`, draft line selection, some Object Browser kinds, and project fallback, but **does not yet have dedicated property forms for every CLO object kind**.

On a selected Rust line the editor displays **2D Line Length**, **3D Line Length**, **3D Stretch**, sewing-link IDs, and an editable **Target 2D length (cm)** specification (0.01–1000 cm, only with a Rust source and setter). Source text correctly states that a target length specification is *not* an elastic force. Draft-only selection can provide a 2D length while leaving 3D length unavailable. For Rust patterns, measured 2D/3D perimeters are displayed. Elastic and Seam Taping have disabled controls and explanatory readouts; Bond and Solidify are explicitly unconfigured. The simulation panel can issue simulate/pause/reset callbacks. Fabric panel displays assigned fabric and, in draft mode, swatch/GSM. Arrangement can report position X/Y/Z or draft staging offsets, but these do not establish CLO arrangement-point semantics. The rendering of children under the active workspace and the pattern-properties portal should remain intact during future upgrades.

EXO evidence pointers: `CloPropertyEditorAccordion.jsx:8–27` group mapping; `:38–50` selection/type resolution; `:85–110` measured line and unavailable forces; `:113–160` simulation, fabric, Bond, Solidify, staging and portal; `:166–185` accordion shell. See also `docs/CLO3D_PROPERTY_EDITOR_AUDIT_20261016.md` and `docs/EXO_V2_RIGHT_PROPERTY_EDITOR_20261016.md`. These documents reflect earlier checkpoints; when they disagree about V2 measured-line support, **the current source governs**.

Field notation

In each matrix, R = read-only computed/display, N = number, B = boolean, S = dropdown/enum, T = text, C = color/material picker, A = action. A range/default marked **?** is not confirmed by the cited source and requires a 7.0.242 capture. Length

defaults follow project unit settings (mm/cm/in) when applicable; do not hardwire centimeters as CLO's universal display unit. —— is unknown/unavailable, **not zero**.

Selection-context parity matrix

Context and trigger	Field / group (type; documented default, unit, or conditional)	CLO evidence	EXO V2 current	Implementation and version check
Pattern whole: Transform Pattern / Select	Information name/item number (T); assigned fabric (S); selected full-pattern 2D/3D perimeter (R length)	[Property Editor] (https://support.clo3d.com/hc/en-us/articles/115004877468-Property-Editor); [staff clarification of perimeter vs edge] (https://support.clo3d.com/hc/en-us/community/posts/4408171842969-3d-length-count-all-2d-lines)	E-partial: name/item, native perimeters, fabric readout	Keep perimeter separate from any selected segment. Test size and selection provenance; local category labels L?.
Pattern whole	Particle Distance (N mm); Layer (N index); fabric assignment (S); simulation/thickness/collision/pressure when present (N)	[Property Editor / select all same property] (https://support.clo3d.com/hc/en-us/articles/115004877468-Property-Editor); source inventory for exact field names in 7.0.242 **L?**	E-partial: generic simulation and fabric summaries	Back fields by actual mesh resolution, layer, collision and pressure state; don't infer unknown numerical defaults.
Pattern transform	Draft 2D translate/rotate and 3D placement (N position/angle), geometry-aware	[Select] (https://support.clo3d.com/hc/en-us/articles/115012227067-Select)	E-partial: staging X/Y/Z readout; broader EXO Pattern controls may be mounted	Distinguish editable transform command from simulated position and arrangement-point offset; inspect 7.0.242 selected pattern before locking exact field order.
Outer line/segment: Edit Pattern	2D Line Length and live 3D Line Length (R length)	[staff 2D/3D length explanation] (https://support.clo3d.com/hc/en-us/community/posts/4408171842969-3d-length-count-all-2d-lines)	**E** for selected Rust boundary; draft may be 2D only	Match stable pattern-edge ID; calculate 2D arclength from UV/draft contour and 3D arclength from mapped live vertices. Return unavailable on unmapped geometry.
Outer line/segment	Elastic enabled (B); Strength (N, official current guide default **10**); Ratio (N, **0–200%**, original **100%**); Segment Length (N length), Total Length for multiple segments (N length)	[Elastic] (https://support.clo3d.com/hc/en-us/articles/115012380228-Elastic)	E-partial: disabled Elastic, Strength 10 and Ratio 100% described as unsimulated	Implement edge rest/target length and solver force coupling before enabling; changing segment length and ratio must recompute consistently.
Outer line/segment	Seam Taping enabled (B), Width (N length), Preset (S); orange applied-edge highlight	[Seam Taping] (https://support.clo3d.com/hc/en-us/articles/115012381228-Seam-Taping)	E-no physical effect; disabled controls	Preserve edge-bound tape selection; preset must resolve to physical reinforcement, not merely orange stroke.
Seam tape instance	Extend Start/End (S: **None / To Outline / To Seam Taping**), provided intersecting geometry exists	[Seam Taping extension] (https://support.clo3d.com/hc/en-us/articles/115000574107-Seam-Taping)	E-partial: placeholder Extend None	Version-specific extension rules and 2024.2 migration require L?; test invalid nonintersecting target.
Internal line / shape: Edit Pattern	Length (R), elastic properties (B/N), Fold Angle (N degrees), Fold Strength (N, default ?); selected internal shape can receive Bond/Skive	[Elastic] (https://support.clo3d.com/hc/en-us/articles/115012380228-Elastic); [Fold Arrangement] (https://support.clo3d.com/hc/en-us/articles/115012226507-Fold-Arrangement); [Skive] (https://support.clo3d.com/hc/en-us/articles/115000616548-Skive)	E-partial line model; no dedicated internal shape editor established	Explicit `internalLine` / `internalShape` identities; Fold Angle 0 or 360 gives a sharp fold in the official fold-arrangement description.
Pin / fold / arrangement tool	Selected constraint, pin action/mode (A); Fold Angle on internal/sewing line after interactive fold (N degrees)	[Fold Arrangement] (https://support.clo3d.com/hc/en-us/articles/115012226507-Fold-Arrangement)	E-no dedicated property context established	Investigate local 7.0.242 Pin Property independently; do not invent a universal pin strength slider or assume every pin exposes a right-panel form.

Context and trigger	Field / group (type; documented default, unit, or conditional)	CLO evidence	EXO V2 current	Implementation and version check
Sewing line: Edit Sewing	Paired line ID, orientation and 2D seam lengths (R); Tension Strength (N, default ?); Tension Ratio (N %, official max **200%**)	[Sewing Tension] (https://support.clo3d.com/hc/en-us/articles/360001749347-Sewing-Tension)	E-partial: number/IDs of links, no actual paired seam inspector	Build a paired `sewingLine` selection; tension must update joining constraints. Do not confuse the elastic segment ratio with sewing tension ratio.
Sewing line	Turned (B) vs custom fold behavior; Fold Angle (N **0–360°**, 180° flat); Fold Strength (N, ?) disabled when incompatible with Turned	[Fold Seam Line] (https://support.clo3d.com/hc/en-us/articles/115000414567-Fold-Seam-Line); [CLO support discusses changed Turned UI] (https://support.clo3d.com/hc/en-us/community/posts/40547802776985-Need-help-in-property-editor-sewing-line-type)	E-no dedicated toggles verified	Make conditional enablement explicit; 7.0.242 label `Sewing Line Type` vs `Turned` L?.
Sewing line	3D Seamline Intensity/Thickness (N, visual dimensions), possibly sewing style assignment (S)	[CLO community support on 3D seamline] (https://support.clo3d.com/hc/en-us/community/posts/360014571514-Noticeable-Seams)	E-no dedicated scene seamline visual property established	Separate visible seam rendering from physical seam strength. Verify current version and ranges locally.
Topstitch asset: Object Browser Topstitch	Name (T), Type (S: **OBJ / Texture**), Preset/Shape (S: Custom, Bartack, Buttonhole, Overlock, Pickstitch, Single, Single_Offset_0, Zigzag), Color/material (C), Offset (N length)	[Topstitch Item/Property] (https://support.clo3d.com/hc/en-us/articles/115013590827-Topstitch-Item-Property)	E-partial: browser kind dispatch to generic information; no contextual editor	Create style asset schema with render representation and assignable ID; changing a preset field can switch preset to Custom.
Topstitch asset	Stitch Length in **SPI** (N); Space (N positive/negative); Thread Thickness (N with **Tex, Ticket/Tkt, Metric/Nm, Denier, mm** units); face (S Front/Back/Both); Number of Lines (N), Distance conditional on multiple lines (N length)	[Topstitch Item/Property] (https://support.clo3d.com/hc/en-us/articles/115013590827-Topstitch-Item-Property); [line count] (https://support.clo3d.com/hc/en-us/articles/115000560528--Style-Property-Number-of-Topstitch-Line-Setting)	E-no functional style fields established	Preserve SPI semantics (higher SPI = shorter stitch); unit conversions must use proper thread count relations, not blanket millimeter conversions.
Topstitch asset	Up to **5** stitch layers (A/B), specification/material sharing toggles (B), OBJ polygon optimization conditional on OBJ (N)	[Edit Topstitch] (https://support.clo3d.com/hc/en-us/articles/115002752908-Edit-Topstitch)	E-no	Conditional layer tabs/rows and render path; proof of effective geometry/texture.
Topstitch instance: select on pattern	Style assignment (S); Extend Start/End (S); Corner (S Curved/Right Angled/Smooth); Z Offset (N length); Flip (B) only where direction can change	[Edit Topstitch] (https://support.clo3d.com/hc/en-us/articles/115002752908-Edit-Topstitch); [Flip] (https://support.clo3d.com/hc/en-us/articles/115000497067--Property-Flip-Topstitch)	E-no	Keep instance position and offset separate from shared style; Flip disabled on cutting line per CLO source.
Binding: select binding	Name (T), Line Length (R), Type (S Under / Over / **Both in 7.0**), Width (N **1–50 mm**), Length (% **50–200%** or explicit length), Particle Distance (N **4–20 mm**), collision thickness (N), Fabric/Topstitch/Sewing Line Type (S), Grain Bias/Warp/Custom and Angle (S/N), Extend (B)	[Select Binding] (https://support.clo3d.com/hc/en-us/articles/360000628128-Select-Binding)	E-no dedicated binding context established	Useful 7.0-specific reference; selected binding should be a first-class item with separate physics and fabric assignment. Legacy article lists slightly different particle-distance range; use version capture.
Fabric asset: Object Browser Fabric	Name (T), physical-property **Preset** (S) and Detail (N fields), assigned usages (R)	[Property Editor] (https://support.clo3d.com/hc/en-us/articles/115004877468-Property-Editor); [CLO designers on changing physical preset] (https://support.clo3d.com/hc/en-us/community/posts/4404012517657-Issue-of-puffiness)	E-partial: swatch/name/GSM, native fabric library linked through workspace	Separate a selected **fabric asset** from a selected **pattern wearing fabric**. Saving should retain physical preset and axes.

Context and trigger	Field / group (type; documented default, unit, or conditional)	CLO evidence	EXO V2 current	Implementation and version check
Fabric physics	Density/weight, thickness, stretch/weft/warp/bias, bending, shear, friction, damping and related detail fields (N; exact 7.0.242 labels, units, preset defaults **L?**)	[physical preset discussion] (https://support.clo3d.com/hc/en-us/community/posts/900002284526-Problem-with-garment-collission?page=1); [fabric digitization limits] (https://support.clo3d.com/hc/en-us/articles/40040576374425-Non-testable-Fabrics-for-CLO-Fabric-Digitization-Service)	E-partial: material catalog and physics research exist; no complete exact Property Editor field mapping established	Use measured/calibrated anisotropic fabric properties; no claim of exact CLO coefficients without test data. Distinguish grams per square meter from solver mass units.
**Fabric appearance **	Color (C), texture/material maps and material-type options (S/C), optical preview	[Color Window] (https://support.clo3d.com/hc/en-us/articles/360047316974-Color-Window-How-do-I-change-colors-in-my-project); [material effects] (https://support.clo3d.com/hc/en-us/articles/40040576374425-Non-testable-Fabrics-for-CLO-Fabric-Digitization-Service)	E-partial: swatches, material preview/surface code exists	Implement base color, opacity, normal/bump, roughness, metallic, displacement only as supported; physical and rendering properties must remain distinct. Exact CLO labels L?.
Graphic instance / asset	Graphic placement, physical dimensions, rotation, visibility, material/color/texture (T/N/B/C; exact label/default L?)	[Color Window] (https://support.clo3d.com/hc/en-us/articles/360047316974-Color-Window-How-do-I-change-colors-in-my-project); [Property Editor] (https://support.clo3d.com/hc/en-us/articles/115004877468-Property-Editor)	E-partial: generic object kind via browser	Capture selected 2D/3D graphic in 7.0.242 to verify transform coordinate space and size-lock behavior before claiming precise label parity.
Button / buttonhole *	Selected Button or Buttonhole context; style assignment (S) and asset material/color (C), transform/size/rotation as version applicable (N, L?)	[Apply Buttonhole Style] (https://support.clo3d.com/hc/en-us/articles/115002382348-Apply-Buttonhole-Style); [Color Window] (https://support.clo3d.com/hc/en-us/articles/360047316974-Color-Window-How-do-I-change-colors-in-my-project)	E-partial: Object Browser kind acknowledged only	Build distinct button instance, button style, buttonhole instance, and buttonhole style. Verify 7.0.242 sewing/fastening interaction and defaults in app.
Avatar / scene accessory	Avatar selection displays avatar properties; selectable props, shoe, hair, OBJ/FBX objects may expose color (C)	[Property Editor] (https://support.clo3d.com/hc/en-us/articles/115004877468-Property-Editor); [Color Window] (https://support.clo3d.com/hc/en-us/articles/360047316974-Color-Window-How-do-I-change-colors-in-my-project)	E-partial: Avatar workspace mapping to generic miscellaneous	Separate avatar transform, dimensions, pose/skin material/arrangement bounding volumes as distinct capabilities; exact 7.0.242 controls L?.
Graded piece / grading	Size-aware pattern data and graded measurement displays (S/N/R; labels/defaults L?)	[Create POM] (https://support.clo3d.com/hc/en-us/articles/360025716774-Create-POM)	E-partial: browser recognizes Grading, no dedicated contextual fields	Maintain base size, size ID, graded point rules and measurements; show rows by size without overwriting base geometry.
POM: ruler selected / POM tab	Single/group measurement (R length); **3D Length** in Property Editor; graded lengths in POM tab when graded (R)	[Create POM] (https://support.clo3d.com/hc/en-us/articles/360025716774-Create-POM)	E-partial: browser recognizes POM, no evidenced POM inspector	Track individual ruler vertices, traced outline segments and cross-pattern connectors (connectors excluded from grouped length). 2D vs live 3D distinctly labeled.
Bond: pattern/internal shape	Bond (B), physical property Preset/Detail (S/N), local stiffening	[Bond] (https://support.clo3d.com/hc/en-us/articles/115000574207-Bond-How-to-Add-Interlining-Fusible-Reinforcement)	E-no physics, explanatory readout	Implement a local bonded area and material composite; test drape changes quantitatively before enabling toggles.

Context and trigger	Field / group (type; documented default, unit, or conditional)	CLO evidence	EXO V2 current	Implementation and version check
Skive: pattern/internal shape	Skive percentage (N **1–100%** per official guide), softer region when simulating	[Skive] (https://support.clo3d.com/hc/en-us/articles/115000616548-Skive)	E-no	Separate subtractive effective thickness/stiffness from Bond; scope selection to valid pattern/internal shape.
Solidify: selected 3D pattern	Solidify action (A/B), Strength (N, higher preserves current drape more strongly)	[Solidify] (https://support.clo3d.com/hc/en-us/articles/115000306568-Solidify)	E-no physical strength, read-only text	Store reference 3D shape and solver constraint target; do not treat as Freeze/Strengthen.
Arrange pattern via arrangement point	Point Name (S/T), Shape Style (S Flat/Curved), Position X/Y, Offset, Orientation (N/S) conditional on applied arrangement point	[Arrangement Property Setting] (https://support.clo3d.com/hc/en-us/articles/115002101968-Arrangement-Property-Setting)	E-partial: staging XYZ only	Assign point ID, avatar bounding volume, wrap and orientation; hide point-specific fields for freely moved patterns.
Edit arrangement point on avatar	Name, Arrangement BV, X, Y, Offset, Wrap Direction (T/S/N)	[Add Arrangement Point] (https://support.clo3d.com/hc/en-us/articles/115000430768-Add-Arrangement-Point)	E-no dedicated context	Split point geometry/ownership from arranged garment transform.
Simulation settings / wind: right-click 3D background	Simulation Property or Wind Property (context selection; parameter fields L?)	[Property Editor] (https://support.clo3d.com/hc/en-us/articles/115004877468-Property-Editor)	E-partial: buttons & status in generic Simulation accordion	Background click must clear object selection before opening settings. Validate field-level native behavior in live runtime.
**2D window background **: right-click 2D background	2D Window Property context; rendering/environment or background controls depend on version, labels L?	[Property Editor] (https://support.clo3d.com/hc/en-us/articles/115004877468-Property-Editor)	E-partial: Render route, not verified CLO window context	Record 2D display settings, background image/color, grids and units in local screenshot; do not conflate 2D window background with 3D render studio settings.

Conditional behavior and exactness rules

1. **Selection drives schema.** `pattern`, `outerEdge`, `internalLine`, `internalShape`, `sewingPair`, `seamTape`, `topstitchStyle`, `topstitchInstance`, `binding`, `fabricAsset`, `graphic`, `button`, `buttonhole`, `avatar`, `gradedPiece`, `POM`, `arrangementPoint`, `simulationSettings`, `windSettings`, `window2D`, or `none`. The current generalized eight-group accordion can stay as shell, but only context-valid sections may appear. No stale pattern name/measurement after deselection.
2. **Every numeric field declares dimensionality:** length (mm/cm/in), area, ratio %, angle degrees, SPI, thread thickness unit (Tex/Tkt/Nm/Den/mm), per-area mass (GSM), or dimensionless scalar. UI conversions never rewrite physical meaning. Null/unmapped values are `Unavailable`.
3. **Do not invent defaults.** Documented values are Elastic Strength 10, Elastic Ratio 100%, and ratio range 0–200%; Skive 1–100%; topstitch max five layers; Binding Width 1–50 mm, Length 50–200%, Particle Distance 4–20 mm as described in the linked current binding article; Fold Angle 0–360°. The remainder are L? until version-specific capture.
4. **CLO instance versus asset:** a Topstitch or Fabric style entry has shared specifications; its instance on a garment has spatial placement/assignment. Editing the selected garment line must not silently change the shared global style.
5. **Conditional disabling:** topstitch Flip inactive on pattern cutting lines; stitch-line distance only for multiple lines; stitch OBJ optimization only in OBJ mode; binding Extend must have valid surrounding geometry; sewing Fold fields depend on Turned state; pattern arrangement-point fields require an arrangement point; native physics toggles require a solver-capability response.
6. **Transactions:** mutation includes selected object ID, revision and validated units, and must reject stale updates or missing geometry. Document model update, render update, undo/redo and saved reload must agree.
7. **Multi-select:** heterogeneous contexts show common compatible fields and mixed values; homogeneous edges may expose Total Length. Never silently choose the first target.
8. **Version comparability:** capture CLO 7.0.242 (exact About dialog/build) with selected pattern, outer edge, internal line/shape, sewing line, taped segment, topstitch asset/instance, Fabric asset, graphic, button/hole, avatar, graded piece, POM, Bond/Skive, arrangement point, and blank 2D/3D canvas. Capture expanded accordions, units, tool, disabled states, tooltip, and one benign field change per context. Record screenshot names and observations in a follow-on evidence appendix.

Prioritized implementation roadmap

Priority	Concrete deliverable	Acceptance gate
P0-A	Add typed selection dispatch and per-object capability model across 2D, 3D, Object Browser. Keep existing eight accordion visual shell and hosted advanced property controls working.	Clicking each available object shows its identity and correct context; gray background clears selection; no stale values; header and visible groups stable.
P0-B	Complete trustworthy measured geometry: individual outer edges, entire pattern perimeters, source 2D UV vs current mapped simulated 3D path, group totals, units. Preserve current Rust line measurement implementation.	Known 10 cm straight edge reads 10 cm 2D; pulled geometry can read >10 cm 3D; unmapped line returns --; change cm to mm/in without mutating stored length.
P0-C	Central field registry with `kind,type,unit,min,max,step,default,readOnly,availability,tooltip,comm and` and source-specific verification.	Unknown/range-invalid numbers rejected; disabled fields cannot dispatch; numeric calculator converts dimensions accurately.
P1-A	Outer/internal Elastic and Seam Tape schemas, native edge constraints and reinforcement material transfer to Rust; separate sewing-pair tension/fold commands.	Solver parameter actually updates; monitored force/rest target changes and simulated cloth behavior responds. UI refuses unsupported source/selection.
P1-B	Fabric asset physics and visual appearance inspector, with material calibration provenance; related style/instance editors for topstitch, graphics, button/hole and binding.	Assign/change asset, undo/redo, save/reload, verify only intended instances and physics/render outputs change.
P1-C	Bond, Skive and Solidify solver semantics; pattern-relative arrangement/point placement.	Bond raises locally measured resistance; Skive lowers it; Solidify holds a recorded drape with strength-controlled response; arrangement links survive round trip.
P2-A	Grading/POM and avatar contexts; right-click Simulation/Wind/2D Window property windows; mesh/Pin advanced contexts.	Graded POM reports correct per-size lengths; background routes open settings context; scene objects and pinned geometry never steal pattern properties.
P2-B	CLO 7.0.242 visual comparison, capturing every conditional accordion and exact width, typography, ordering, default and selection interaction.	Screenshot pairs annotated by app version and test project; parity score reported per context, with unknowns explicitly unresolved.

Acceptance test suite: scenarios to automate and visually inspect

ID	Setup / action	Required result
PE-01	Select pattern then edge then blank 2D gray.	Context changes `pattern → outerEdge → none`; stale lengths and fabric details disappear.
PE-02	Select 10 cm draft edge with no 3D correspondence.	2D length = 10.00 cm; 3D = unavailable; no copied 10.00 value.
PE-03	Select mapped edge, drag fabric with live simulation, compare 2D/3D arclengths.	Independent readings and positive/negative elongation update with solver snapshot revision.
PE-04	Change display units cm → mm → in.	Round-trip values equivalent within display precision; underlying geometry unchanged.
PE-05	Edit target 2D specification 0, -2, 1000.01, valid 25.5, then stale revision.	Invalid values and stale request rejected; valid edit only changes specification; no false elastic reaction claim.
PE-06	Enable Elastic ratio 80% with backed native edge constraint, simulate.	Rest target and solver constraints change; actual observable contraction under relevant conditions; undo restores. Gate test until implemented.
PE-07	Add tape to edge, adjust width/preset/extension, save/reload.	Tape identity and thickness material survive and influence drape; missing intersection disables extension. Gate test until implemented.
PE-08	Select sewing pair, adjust tension/fold; toggle Turned.	Matching paired seam gets updated; conditional Fold fields disable; all edits persist and undo. Gate test until implemented.
PE-09	Edit topstitch asset vs one stitch instance.	Style change affects assigned instances; placement/flip change affects chosen instance; cutting-line Flip disabled.
PE-10	Select Fabric asset, change physical preset, then optical texture.	Physics change influences simulated drape; optical change updates surface without silently altering solver mass or stiffness.
PE-11	Select internal shape, apply Bond, Skive or Solidify sequentially.	Each action exhibits its distinct physical meaning; unsupported control disabled and explained.
PE-12	Grade two sizes; create group POM across two patterns.	Length per grade matches traced pattern portions; bridging gap excluded, live 3D length independent.

ID	Setup / action	Required result
PE-13	Select pattern with no arrangement point, then assign one.	Arrangement-point fields appear only after assignment; staging position and point offset never alias.
PE-14	Right-click empty 3D and 2D areas.	Proper Simulation/Wind vs 2D Window context offered without pattern selection carryover.
PE-15	Keyboard navigation, multi-select, reload and undo.	Accordion state accessible; mixed values accurate; repeated selection and persisted properties remain consistent.

Remaining local CLO 7.0.242 inspection required

To finish **version-specific** parity, a reviewer with access to the live installed application should record the selected tool and exact Property Editor screenshot for every row, especially pattern thickness/collision, graphics, detailed physical material coefficients, button/buttonhole defaults, grading, Pin, Avatar, Wind and 2D background. Copy each label, min/max/step, initial value, unit and dependent visibility into the matrix. No such screenshot inspection was completed in this audit, so these rows intentionally retain **L?** rather than claiming exact CLO layouts or control coverage.

Official source index: Property Editor, Elastic, Seam Taping, Seam Tape Extend, Sewing Tension, Fold Seam Line, Topstitch, Binding 7.0, Skive, Solidify, POM, Arrangement, Buttonhole Style.

CLO7_LIVE_PROPERTY_EDITOR_AUDIT_20261016.md

Audit interpretation: This annex preserves earlier engineering research for traceability. Read reported pass counts and statuses as of its original date only. The current 70-slot and 400-fabric inventories above supersede any older source totals.

CLO 7.0.242 LIVE PROPERTY EDITOR AUDIT — 2026-10-16

Evidence and observed selection state

This audit uses a **real native CLO window capture**, not an EXO simulation. The signed `EXO CLO Inspector.app` wrote `~/Library/Application Support/EXO-CLO-Inspector/clo-property-editor-reference.png` and `latest-status.json` at **2026-10-16 21:32:22 UTC**. A copy of the screenshot is `docs/clo7-live-property-reference-20261016.png`. Its bottom-left status reads **Version: 7.0.242 (r39887)**. The window title is **CLO Standalone OnlineAuth Academic**; the open file is `Default_Modelist.zprj`. The helper reports `accessibility: true`, `screenRecording: true`, CLO PID 22394, one AX window and no warnings. The AX capture contains named toolbar buttons, but **no Property Editor descendants or individual field labels**. A successful accessibility permission check does not imply complete AX exposure of the game's custom-rendered editor.

Initial direct visual observations (LIVE, before selection): The right-hand dock has **Object Browser above Property Editor**, with a distinct separator/header. Object Browser shows one checked, white-swatch **FABRIC 1** row. The visible 2D Pattern Window is a blank grid and the 3D viewport contains no garment. In this initial screenshot the Property Editor is **blank**. A checked fabric row alone does not establish an active property selection. **Subsequent approved live captures now demonstrate a successfully selected fabric and populated Property Editor**, detailed below. The earlier empty-state evidence remains valid for that earlier state only.

Subsequent live Fabric selection: exact observed rows

Native evidence files: `integrations/clothcraft/output/clo3d-reference/clo-after-fabric-click.png` (selected FABRIC 1, top of inspector) and `integrations/clothcraft/output/clo3d-reference/clo-property-fabric-scroll-deep-4.png` (same selected fabric, scrolled into Material). Both screenshots display CLO **7.0.242 (r39887)**. The Object Browser row is visibly highlighted blue and the right Property Editor is populated with a **Fabric** context. This is **LIVE evidence of a fabric asset selection** despite the blank garment views; it supersedes the earlier statement that no fabric-property selection had been seen.

Screen/order	Exact CLO 7.0.242 label	Observed value / control	Evidence
Top context	Fabric	Header with **Open** and **Save** actions	`clo-after-fabric-click.png`
Information group	Information	Expanded disclosure	`clo-after-fabric-click.png`
Information 1	Name	**FABRIC 1**	`clo-after-fabric-click.png`
Information 2	Item No.	**None**	`clo-after-fabric-click.png`
Information 3	Construction	**None**, expandable row	`clo-after-fabric-click.png`
Information 4	Supplier	**None**	`clo-after-fabric-click.png`
Information 5	Owner	**None**	`clo-after-fabric-click.png`
Material group	Material	Expanded; mode **PBR**	`clo-after-fabric-click.png`
Material face selectors	Front / Back / Side	Three distinct visible tabs; independence/editability **not tested**	`clo-after-fabric-click.png`
Material mapping	Texture Mapping	**Repeat**	Both selected-fabric captures
Material type	Type	**Fabric_Matte**	`clo-property-fabric-scroll-deep-4.png`
Material subgroup	Basic Parameters	Expanded	`clo-property-fabric-scroll-deep-4.png`
Basic Parameters	Texture	Expandable asset/map row	`clo-property-fabric-scroll-deep-4.png`

Screen/order	Exact CLO 7.0.242 label	Observed value / control	Evidence
Basic Parameters	Normal Map	Expandable asset/map row	`clo-property-fabric-scroll-deep-4.png`
Basic Parameters	Displacement Map	Expandable asset/map row	`clo-property-fabric-scroll-deep-4.png`
Basic Parameters	Color	**(None)** alongside a white swatch	`clo-property-fabric-scroll-deep-4.png`
Basic Parameters	Opacity	**100**, slider shown	`clo-property-fabric-scroll-deep-4.png`
Basic Parameters	Opacity Map	Expandable asset/map row	`clo-property-fabric-scroll-deep-4.png`
Material subgroup	Reflection	Expanded/visible near bottom; child values cut off	`clo-property-fabric-scroll-deep-4.png`

These captures verify labels, row sequence and *displayed state for this selected FABRIC 1*. They do **not** verify other PBR material types, dropdown option lists, save effects, mapping dimensions, face-specific changes, physical-property coefficient values or later scrolled sections. `None`, `100`, `Repeat` and `Fabric_Matte` are observed values for this sample, not universal defaults.

AX-confirmed adjacent controls (AX): The window AX children expose `Select/Move`, `Edit Sewing`, `Segment Sewing`, `Free Sewing`, `Select Binding`, `Binding`, `Select/Move Button`, `Button`, `Transform Graphic`, `Fold Arrangement`, `Edit Texture (3D)`, among other toolbar controls. These are **tool names, not Property Editor fields**, and cannot establish which contextual fields would appear in the right panel.

Context-by-context field audit

Legend: **LIVE** directly seen in screenshot; **AX** named by helper; **OFFICIAL** published CLO behavior (version-specific order/defaults are unverified until a 7.0.242 selected-object capture); **EXO** inspected `src/components/CloPropertyEditorAccordion.jsx` and related existing project implementation notes. A blank live column is a visibility limitation, not evidence that CLO lacks the control.

Selection / UI trigger	CLO fields or conditional rules supported by published references (**OFFICIAL**, not visible LIVE)	LIVE observation	EXO comparison / actionable gap
None, blank 2D/3D views	Property Editor follows active selection; background context menus may expose Simulation, Wind or 2D Window properties. [Property Editor](https://support.clo3d.com/hc/en-us/articles/115004877468-Property-Editor)	**LIVE: empty right Property Editor, no fields.**	EXO currently resolves a `project` fallback into eight accordion groups and a `PROPERTY EDITOR` subject header (source lines 60–69, 239–269); **gap P0:** clear the context-specific field body when truly unselected, while retaining any deliberately separate project-settings UI.
Whole pattern, selected	Name/item identity; fabric assignment; 2D/3D perimeter; particle distance, layer and simulation-dependent properties; placement/arrangement fields where applicable. [Property Editor](https://support.clo3d.com/hc/en-us/articles/115004877468-Property-Editor)	Not shown	**EXO partial:** pattern/native readouts, selected fabric and draft property edits; confirm precise CLO 7.0.242 group names, order, defaults, thresholds and unavailable states with a selected sample pattern.
Outer pattern edge / segment	2D versus measured 3D line length; Elastic with Strength **10** and Ratio **0–200%, initial 100%** as documented; Seam Taping width/preset. [Elastic](https://support.clo3d.com/hc/en-us/articles/115012380228-Elastic), [Seam Taping](https://support.clo3d.com/hc/en-us/articles/115012381228-Seam-Taping)	Not shown	**EXO partial:** actual native edge lengths, stretch and a separate editable drafting target; Elastic and seam tape are explanatory/disabled pending physical solver commands. Verify single vs multi-edge Total Length and group visibility.

Selection / UI trigger	CLO fields or conditional rules supported by published references (**OFFICIAL**, not visible LIVE)	LIVE observation	EXO comparison / actionable gap
Seam tape / selected taped edge	Extend Start / End: **None**, **To Outline**, **To Seam Taping**, subject to intersecting geometry. [Seam Taping] (https://support.clo3d.com/hc/en-us/articles/115000574107-Seam-Taping)	Not shown	**Gap:** typed seam-tape instance, geometry-gated extension and physically applied tape width/material. Version-specific UI may differ from latest help pages.
Internal line or shape	Line length, fold angle/strength as tool-dependent; possible Elastic and Bond/Skive context. [Fold Arrangement] (https://support.clo3d.com/hc/en-us/articles/115012226507-Fold-Arrangement), [Skive] (https://support.clo3d.com/hc/en-us/articles/115000616548-Skive)	Not shown	**Gap:** first-class internal-line/shape stable IDs and dedicated controls, with physically backed fold behavior.
Sewing relationship / seam	Paired segment lengths, sewing tension strength/ratio (up to **200%** per guide), Turned/fold angle **0–360°**, Fold Strength with conditional availability. [Sewing Tension] (https://support.clo3d.com/hc/en-us/articles/360001749347-Sewing-Tension), [Fold Seam Line] (https://support.clo3d.com/hc/en-us/articles/115000414567-Fold-Seam-Line)	Not shown; **AX** does expose Edit Sewing/Segment Sewing tools	**EXO partial:** native sewing pair IDs, UV/3D partial lengths, gap, direction and native stiffness readouts; missing solver-backed tension and Turned/fold editing. Record actual 7.0.242 conditional UI in a live seam selection.
Fabric asset selected in Object Browser	Asset name; physical preset/details, color and surface material settings, with warp/weft/bending/shear and other physical attributes in applicable detail views. [Property Editor] (https://support.clo3d.com/hc/en-us/articles/115004877468-Property-Editor)	**LIVE VERIFIED:** selected **FABRIC 1** opens **Fabric** editor; **Information** fields Name, Item No., Construction, Supplier, Owner; **Material** mode PBR with Front/Back/Side, Texture Mapping Repeat, Type Fabric_Matte, Basic Parameters and Reflection. See exact-row table above. **Physical Property values were not visible in these two captures.**	**EXO implementation update (reported by prime):** Fabric now mirrors **Information / Material / Physical Property** sections and displays independent face controls as **disabled** pending support. Follow up with EXO runtime comparison of field order and editable behavior; maintain disabled face controls until actual independent material channels exist. Physical calibration and complete CLO coefficients remain unverified.
Topstitch style asset	Type **OBJ/Texture**, preset/shape, color, offset, SPI, space, thread thickness with thread-count units, lines/distance, up to five layers and conditional OBJ controls. [Topstitch Item/Property] (https://support.clo3d.com/hc/en-us/articles/115013590827-Topstitch-Item-Property), [Edit Topstitch] (https://support.clo3d.com/hc/en-us/articles/115002752908-Edit-Topstitch)	Not shown	**EXO partial:** topstitch *instance* group with stitch length/SPI/offset/line count; **gap:** separate shared style asset, multilayers, thread specs.
Topstitch instance	Applied style, Extend Start/End, Corner, Z Offset; Flip conditional on valid geometry. [Edit Topstitch] (https://support.clo3d.com/hc/en-us/articles/115002752908-Edit-Topstitch), [Flip Topstitch] (https://support.clo3d.com/hc/en-us/articles/115000497067--Property-Flip-Topstitch)	Not shown	**Gap:** stable instance selection with per-placement controls and disabled Flip where forbidden.
Binding	Type **Under/Over/Both (Both introduced in version 7.0)**, width **1–50 mm**, length **50–200%** or length units, particle distance **4–20 mm**, fabric/topstitch/grain and related properties. [Select Binding] (https://support.clo3d.com/hc/en-us/articles/360000628128-Select-Binding)	Not shown; **AX** exposes Select Binding and Binding	**Gap:** dedicated binding asset/instance context and constraints, typed units and persisted physical changes.
Graphic / button / buttonhole	Graphics have placement and appearance; buttons/buttonholes have distinct instances and styles with version-dependent size, placement,	Not shown; **AX** exposes Button and Transform Graphic tools	**EXO partial:** separate typed draft editors for size, placement, appearance and annotations; missing CLO-equivalent

Selection / UI trigger	CLO fields or conditional rules supported by published references (**OFFICIAL**, not visible LIVE)	LIVE observation	EXO comparison / actionable gap
	rotation and material. [Property Editor] (https://support.clo3d.com/hc/en-us/articles/115004877468-Property-Editor), [Buttonhole Style] (https://support.clo3d.com/hc/en-us/articles/115002382348-Apply-Buttonhole-Style)		native physics/fastening and style asset contexts.
Bond / Skive / Solidify	Bond physical reinforcement, Skive **1–100%**, Solidify shape-retention strength. [Bond] (https://support.clo3d.com/hc/en-us/articles/115000574207-Bond-How-to-Add-Interlining-Fusible-Reinforcement), [Skive] (https://support.clo3d.com/hc/en-us/articles/115000616548-Skive), [Solidify] (https://support.clo3d.com/hc/en-us/articles/115000306568-Solidify)	Not shown	**EXO:** explanatory groups, without equivalent material/solver transformations; gate enablement on native support.
Avatar / arrangement point	Avatar's relevant item properties or pattern-specific point name, style, offset/position/orientation after an arrangement point exists. [Property Editor] (https://support.clo3d.com/hc/en-us/articles/115004877468-Property-Editor), [Arrangement] (https://support.clo3d.com/hc/en-us/articles/115002101968-Arrangement-Property-Setting)	Not shown	**EXO partial:** avatar generic context and position readouts, without a validated CLO arrangement-point schema.
POM / grading	Measurement length and graded per-size data as applicable; 3D POM length distinguished from 2D. [Create POM] (https://support.clo3d.com/hc/en-us/articles/360025716774-Create-POM)	Not shown	**EXO partial:** selected POM/grading readouts and Production operators, with limited native 3D grade-aware measurements.

Source mapping and implementation roadmap

`src/components/CloPropertyEditorAccordion.jsx` provides the selection-context shell; `docs/EXO_CLO3D_V2_PROPERTY_EDITOR_IMPLEMENTATION_20261016.md` documents its earlier implementation stage. The prime reports a newer EXO Fabric editor mirroring **Information / Material / Physical Property** and intentionally disabled independent face controls; this audit records that update without claiming a new EXO screenshot test. `docs/CLO3D_PROPERTY_EDITOR_DEEP_AUDIT_20261016.md` supplies the official-reference matrix and proposed acceptance cases. Earlier documents asserting no native visual evidence are superseded for **both** the blank context and the **selected Fabric** context described here.

1. **P0: empty-state parity.** In a V2 browser/native app QA session, select no object and compare the right property body against the new CLO capture; ensure no pattern-specific accordions or stale values leak into an empty selection. Distinguish the currently visible Fabric browser listing from active Fabric *property selection*. Preserve other project controls in their proper location.
2. **P0: complete live selected-context evidence. Fabric asset is now captured and partially mapped.** Continue scrolling its **Physical Property** group and inspect material Reflection subfields, face tab behavior and dropdown choices. Separately select whole pattern, edge, internal line, sewing pair and topstitch style/instance in an expendable CLO test project. Save screenshot and AX JSON *per state* and record exact labels, order, values, units and disabled conditions. Confirm build remains 7.0.242.
3. **P1: schema and parity.** Tie each selection kind to a typed field registry with `name`, `type`, `unit`, `range`, `readOnly`, `condition`, `capability` and native/draft command. Compare observed screenshots with rendered EXO selection states; prioritize pattern/edge/sewing/fabric before minor style contexts.
4. **P1: actual physical behavior.** Add solver-backed edge elastic, seam tape and fold/tension; then Bond/Skive/Solidify; verify native changes alter cloth behavior rather than only editor text. Add style-vs-instance separation for topstitch and hardware. Existing official reference ranges are candidates for validation, not proof of identical 7.0.242 defaults.
5. **P2: repeatable evidence.** Capture a side-by-side comparison per context with scale, platform, selection ID, active tool and accordion expansion; test edit, undo/redo, reload and unit conversions, including mixed selections and unavailable 3D mappings. Only then assign a version-specific parity score.

Evidence limit: A signed helper successfully selected the live **FABRIC 1** asset and captured the two approved screenshots; the audit reads those screenshots. CLO pattern, edge, sewing, topstitch and other selection contexts remain unobserved. No dropdown was expanded in the evidence, no independent face setting was changed, and the visible screenshots do not establish Physical Property coefficients. This documentation update did not alter any CLO garment or EXO application source.

EXO_400_FABRIC_LIBRARY_AND_WASH_SYSTEM.md

Audit interpretation: This annex preserves earlier engineering research for traceability. Read reported pass counts and statuses as of its original date only. The current 70-slot and 400-fabric inventories above supersede any older source totals.

EXO 400+ FABRIC LIBRARY AND DIGITAL WASH SYSTEM

Research, implementation and engineering acceptance specification — 16 October 2026. The implemented module references below are verified by a programmatic catalog audit; future high-fidelity features in this document remain proposals. It does not certify that EXO currently has 400 experimentally characterized/photorealistic fabrics or that its implementation matches CLO.

Implemented scope (validated October 16, 2026)

Subsequent real-scan appearance pass on the same 400

All 400 presets were re-audited against actual installed and SHA-256-verified CC0 photographs; the result is in `docs/reports/EXO_400_FABRICS_COVERAGE.csv` and `.json` with exactly 400 rows.

Evidence state	Actual count
EXO garment fabric presets	**400**
Unique installed CC0 photographic PBR source materials	**67** (37 Poly Haven, 30 ambientCG)
Unique CC0 sources actually used by presets	**44**
Verified installed images (1k JPG albedo, OpenGL normal, roughness)	**201**
Fabric presets referencing an installed photographed material	**385**
Construction-reference assignments	**220**
Related-fabric proxy assignments (NOT an exact fabric scan)	**165**
Textile-specific generated appearance fallback (mostly fine/open mesh)	**15**
New per-specimen mechanically measured/calibrated materials	**0**

Real sources were obtained from official Poly Haven and ambientCG APIs under CC0 licenses (see `https://polyhaven.com/license` and `https://docs.ambientcg.com/books/website-licensing/page/license-information`), with SHA-256 checksums, author/source and provider metadata preserved in `src/data/polyhavenScans.generated.json` and `src/data/ambientcgScans.generated.json`. No proprietary CLO textures are used. Resumable installers are `scripts/install-polyhaven-fabric-scans.py` and `scripts/install-ambientcg-fabric-scans.py`; the combined strict asset registry is `docs/reports/EXO_400_LICENSED_SCAN_REGISTRY.json`. `src/data/polyhavenFabricAssignments.js` documents assignment to the 80 base archetypes, inherited by four additional variants; no extra material specimen is created by editing color or GSM.

Photo maps and provider-declared repeat scaling are installed via `src/data/polyhavenScans.js` and rendered in both browser and native swatches. Some ambientCG assets do not publish physical repeat measurements, so their default repeat scale is marked **unknown/illustrative** rather than verified. Colorways use non-destructive photographic tonal retargeting: source yarn detail is retained while the user adjusts color and wash strength. The 15 unmatched materials retain procedural textures suited to fine meshes/tulle and unsupported physical constructions (neither substitutes for yarn-level cloth simulation).

Clarification of evidence grade: The strict photo registry may grade a fabric "A" for complete **color + normal + roughness maps**. That is optical map coverage only. It is not a claim of color-spectrophotometric accuracy, a scan of the exact EXO fabric variation, or laboratory-determined drape/stress data. Related-fabric proxies are clearly separated in the audit. The 1k source maps suit interactive previews; higher-resolution originals may be available for later selective re-import. The two source caches occupy approximately **191 MB** (90 MB Poly Haven and 101 MB ambientCG); additional bulk downloading was paused because the Mac had approximately 19 GiB free.

The physical importer in `src/core/fabricEvidenceCalibration.js` supports traceable mass/thickness measurements and directional laboratory curves with provenance and unit checks. In the editor, use **Physical Fabric > Import Lab Specimen Data (JSON)** to add *user-provided* specimen data. Areal mass and thickness can be converted directly to the solver's SI quantities; uncalibrated tensile, bending and shear fits remain advisory. A user's submitted file is marked

`imported-unverified` until instrument and specimen traceability are independently confirmed. This is intentionally not an automated 400-material calibration claim.

The visual preview contact sheet generated from all current preset colors and photo references is `screenshots/EXO_400_FABRIC_CONTACT_SHEET.jpg`. Its JPG thumbnails are graphic previsualizations, not WebGL screenshots or measured color targets. The main and native PBR rendering use the source image, normal and roughness maps.

- **Exactly 400** named, unique, selectable React material records: 80 original archetypes plus 320 physical/finish variants, assembled from `src/data/fabricCatalog400.js` and `src/data/defaultProject.js`. Four variants per base change weight, thickness, bending, directional stretch, friction and optical controls—not only color. They are parametric estimates, **not 400 unique fabric specimens**.
- **405** Rust fabric choices in `src/core/clothcraftFabrics.js`: 400 aligned construction IDs plus five extra original instrument/reference specimens retained for backward compatibility. The same underlying material models are **not** used by the React and Rust solvers; conversion remains approximate.
- **16** selectable, parameterized digital appearance effects, including modeled enzyme, stone, acid-marble, ozone, bleaching, localized PP-style fading, garment dye, whiskers, seam-edge fading and distress. `src/core/clothWashProcedurals.js` generates deterministic periodic masks; `src/core/clothWashImage.js` provides browser sRGB image grading. Project wash metadata is stored on fabric and changes remain optical-only in the main React workflow.
- `src/core/textileProcedurals.js` applies color-graded wash to generated yarn/base color maps, while `GarmentViewport.jsx` and `FabricMaterialPreview.jsx` also apply visual grading to loaded photographic color maps. The source image itself is not overwritten. Original procedural normal/roughness and uploaded maps remain independent.
- `src/core/clothcraftFabricSurface.js` renders matching preconfigured washes on saved native fabric preset names, but **the native workspace does not yet offer a persisted arbitrary wash layer editor**. Native custom finish stack persistence is a future bridge.
- Automated tests `scripts/check-apparel-catalog.mjs`, `scripts/check-cloth-washes.mjs`, and `scripts/check-cloth-wash-image.mjs` verify exact catalog size, finite parameters, uniqueness, deterministic tiling and reversible unmodified source RGB. `npm run build` and `npm run check:materials` passed. Physical drape certification, full-page visual regression, manufacturer wash QA and per-panel seam masks are not established by these checks.
- Native exhaustive simulation fixture `integrations/clothcraft/check-v2-fabrics.mjs` is time-intensive at 405 materials. Run `EXO_FABRIC_QA_LIMIT=20 node integrations/clothcraft/check-v2-fabrics.mjs` for a deterministic distributed sample plus high-priority denim/terry/leather/suede variants (27 samples), which passed material assignment, saved reopen, undo/redo and finite native steps. Pure consistency tests validate **all 405 material descriptors**, but the 405-case full native simulation exercise was terminated for runtime cost and must not be reported as passed.

Quality classification of the new 320 variants: Grade C/D *estimated construction descriptors and generated visuals*, zero newly instrument-measured textile specimens. A/B measured-fabric targets below are future quality gates; showing a count of 400 must never imply calibrated fabrics or scanned material photos.

Verified reference and scope

The official CLO Express Denim Wet Wash guide (updated 19 August 2026; feature introduced in CLO 2025.1) says to select **Material Type: Denim**, supply a denim **Base Color Map**, then choose a wash effect. CLO describes the effects as changes to the fabric's **Base Color Map**. The eight published presets are **Unwashed, Rinse, Enzyme, Stone, Acid, Bleach, Ozone, Custom**. CLO's stated controls are: **Wash Map concentration/intensity**, **Transformation** (scale, rotation, translation), **Blend** (with Overlay recommended for many denim washes), **Absorbency**, **Cycle Duration**, and **Seam Abrasion** (modifies appearance associated with pattern puckering). The article documents categories and qualitative behavior, not precise numerical control ranges, pixel algorithms, fabric test measurements, or a reproducible CLO rendering baseline. Accordingly, EXO should expose independently designed functionality and use its own names and algorithms rather than infer unpublished CLO internal formulas.

The related official CLO Material Type article describes Denim alongside fabric reflection/material types. Source reference is for comparative research only; no CLO screenshots, source images, procedural shader code, proprietary assets, or wash maps are to be copied.

Proposed EXO system architecture

Maintain three linked but independently versioned records:

1. **Fabric master**: unique stable ID, human-readable name, construction/fiber composition, nominal GSM, measured or estimated thickness, warp/weft grain, mechanical calibration with evidence level, optical PBR family, base color/texture, yarn-scale texture repeat, licenses and source provenance. A master is a real fabric category/specimen variant, not a colorway or wash alias.
2. **Fabric appearance variant**: parent fabric ID, colorway, UV repeat, base PBR maps and an ordered, editable **digital finish stack**. Store original map references and parameters; regenerate derivative textures without overwriting originals. Share finish templates between fabrics but keep per-variant inputs independent.
3. **Panel assignment**: pattern panel ID, parent fabric/variant ID, stable grainline, UV transform, panel-specific masking, seam references, optional overrides. Persist alignment to the 2D pattern as well as the simulated 3D garment; avoid binding wear marks to arbitrary world-space coordinates.

Suggested asset graph:

```
FabricMaster(id, mechanics, PBR, evidence)
  -> AppearanceVariant(id, base maps, finishStack[])
  -> PanelAssignment(panelId, uvLayout, grainline)
  -> SeamTopology(pairings, stitch lines, allowances, puckering)
  -> TextureBake(albedo, roughness, normal?, height?, AO?)
  -> Browser/Native render + swatch + BOM/techpack export
```

An illustrative finish-layer schema (a *design target*, not a claim of existing EXO API) is:

```
{
  "id": "finish-unique-id",
  "kind": "denim-wash",
  "preset": "stone",
  "enabled": true,
  "seed": 31415,
  "strength": 0.45,
  "durationVisual": 0.5,
  "spread": 0.35,
  "blendMode": "overlay",
  "transform": {"offsetU": 0, "offsetV": 0, "scaleU": 1, "scaleV": 1, "rotationDeg": 0},
  "maskAssetId": null,
  "seamAbrasion": 0.4,
  "edgeWear": 0.3,
  "reversible": true
}
```

Normalize every numeric field to a published EXO range and reject nonfinite values. The 0–1 values above are proposed UI conventions, **not official CLO ranges**. Preserve seed, implementation version, color space, inputs and checksums so exports can be reproduced. Finish edits must participate in undo/redo, versions, named colorways, manufacturer comments and export metadata.

Denim wash preset targets: appearance only

Digital preset	Visible result to target	Proposed masks and texture response
Unwashed / raw	Preserved dark, saturated dye, sharp twill relief	Identity base color; no wash mask
Rinse	Subtle evenness and small reduction in apparent saturation	Low-amplitude broad mask, minimal localized contrast
Enzyme-inspired	Gentle broken-in fading with soft transitions	Low-frequency attenuation plus mild raised-area wear
Stone-inspired	Uneven highlights, pronounced seam/edge wear	Multi-scale wear noise, fold-ridge and edge masks
Acid-inspired	High-contrast irregular cloud/marble pattern	Seeded clustered mask with smooth/fragmented boundaries
Bleach-inspired	Localized to broad color lightening	Controlled pigment-removal color blend, preserve yarn detail
Ozone-inspired	Gentle fairly uniform fade	Broad low-contrast fading mask
Custom	User-owned photography or authored masks	Editable map transforms, channel blending and intensity

These labels describe **visual approximations**. A digital filter does not chemically change dyestuff, predict process chemistry, validate environmental impact, soften a real fabric, or prove garment longevity. Even when the appearance suggests a softer hand, solver properties stay fixed until separately provided with measured post-treatment mechanical values and explicit user approval to change the simulated fabric.

Extend finishing beyond denim through the same stack: pigment or garment-dye look, vintage wash, sunfade, localized whiskers, honeycombs, knee/seat wear, edge abrasion, soft overdyed patina, uneven dye uptake, suede nap direction variation, leather patina and highlight wear, cotton/fleece garment aging, and print crackle. Use applicable masks per material; do not imply that every wet process is valid for leather, silk, wool or treated composites. Names such as 'potassium wash' should be presented as visual looks only, with no chemical formulations, process recipes, concentrations, times, temperatures or manufacturing operating instructions.

Correct placement on UVs, patterns, and stitches

UV invariants. Generate primary repeat UVs from each 2D panel in centimeters. Keep fabric twill/knit orientation registered to grainline. On a stitched 3D garment, the mask must follow simulated vertex UVs during motion and deformation. Pattern resizing and grading must regenerate appropriate UV coordinates and remap authored local mask landmarks. Preserve authored offsets and seam continuity when panels are split or re-sewn.

Construction-aware masks. Derive edge-distance maps from **valid pattern boundary edges**, separately tagging fold, hem, cut, bound, zipper, pocket, waistband, seam allowance and internal stitch lines. Use signed or unsigned distance in **physical centimeters**, accounting for texel density. A fade near a hem should remain similar physical width on differently sized panels. Use user controls for seam bias and localized exclusion masks; avoid automatic artificial bright seams on raw or taped, sealed, reversible or non-denim materials.

Seam registration. Sewn edges refer to paired pattern edge IDs and correspondence/arclength. Edge-wear and puckering patterns should interpolate along those paths, with orientation and stitch density attached to the seam. Avoid accidental discontinuities when a wash crosses front/back joins; intentional panel discontinuities are allowed as a design choice. For whiskering, offer user-placed anchor areas at hip/upper thigh with directional fold curves and displacement from avatar pose, then bake on the panel UVs. Seat/knee worn regions should use labeled user-adjustable pattern zones, not hardcoded pixel rectangles that shift with size.

Map pipeline. Base color in sRGB; scalar/normal/height masks in non-color data space. Perform blending in a documented linear-light workflow and convert for storage/viewing as needed. Preserve weave and twill contrast by altering dye/pigment appearance without deleting the entire yarn normal map. Anti-alias masks, generate mip levels, check UV seams at render distance, and use a deterministic fallback when a map is absent. Roughness or normal response may be altered in an independent optical layer where appropriate; no direct change to collision thickness, mass, bending stiffness, tensile stiffness or friction from a color-only wash.

Recommended generation dependency graph: `finish params -> UV/edge/zone masks -> dye/fade color composite -> optional optical map composite -> mipmapped render assets`. Cache by fabric variant, panel geometry/UV revision, seam topology revision, finish version, input image hash and deterministic seed. A color-only operation must not reset simulated particle positions or rewrite garment pattern geometry.

Measured physics versus visual plausibility

Current project reference paths include `src/core/clothcraftFabrics.js`, `src/data/apparelMaterialExpansion.js`, `src/core/fabricSurfaceModel.js`, `src/core/fabricCalibration.js`, `src/core/textileProcedurals.js`, `src/core/fabricPbrGenerator.js`, `src/components/ClothCraftFabricLibrary.jsx`, and `src/components/FabricMaterialPreview.jsx`. Existing source already distinguishes approximate material appearances from mechanical inputs; a source comment in `clothcraftFabrics.js` notes measured mass/thickness for some specimens but **estimated** elasticity and bending. `fabricCalibration.js` fits elementary measured tensile responses, not a complete fabric constitutive model. The [existing EXO materials reference](EXO_FABRIC_MATERIALS_AND_TRIMS.md) documents normal-map and shader limitations.

Physical fidelity requires specimen-level evidence for areal density (g/m²), thickness (mm with defined pressure), warp/weft extension versus applied load, shear, bend hysteresis, damping, surface friction, compression, shrinkage and behavior after real finishing when relevant. Record test method, specimen conditioning, force/displacement curve, uncertainty, and solver-specific fit/error. For denim, compare raw and physically finished *measured* samples before offering 'wash changed hand feel' simulations. An artist-made appearance may still be visually convincing without these measurements: label that honestly as **optically authored / mechanics estimated**.

Practical simulation gate: identical garment, poses, timestep and initial conditions must show unchanged cloth positions after an appearance-only edit within stated floating-point tolerance; independently applied measured mechanical updates should alter corresponding cloth observations when re-simulated. Test UV stability on a moving avatar and cloth self-contact; shader displacement and painted creases do not prove actual folded topology, abrasion damage or additional collision geometry.

Swatch photography and texture capture protocol

1. Obtain a physical reference swatch with recorded consent/license, manufacturer/lot, fiber content, weave/knit, nominal GSM, finishing history, dimensions and measurements. Capture raw and finished variants when available. Do not assign a chemical-process label solely from visual similarity.

2. Condition and flatten a representative area consistently; capture a **measured centimeter scale**, neutral gray reference, calibrated color target and fiber/grain direction indicator. For PBR, obtain high-resolution macro photographs and garment-scale images under stable diffuse illumination. Include raking light and cross/polarized lighting when equipment allows; document the lighting and lens profile.
3. Capture controlled front/back albedo, weave/nap relief, low-angle directional reflection and transmitted-light reference when relevant (especially sheer cloth). For pile materials, photograph multiple fiber orientations because velvet, suede and brushed fleece change substantially with viewing direction.
4. Correct distortion and perspective against the scale marker, white-balance/color-manage with chart profile, document RGB working space, remove illumination gradients without erasing material texture and create edge-compatible tile crops. Do not mistake shaded wrinkles for diffuse pigmentation.
5. Derive roughness, normal and displacement maps through separately validated imaging or artist calibration. A single diffuse photo cannot uniquely recover physical roughness, normal microgeometry or BRDF. Mark inferred maps **estimated**. Set physical repeats in millimeters/centimeters and use matching UV axes.
6. Compare the rendered flat swatch, sphere/macro sample and full draped garment under matching reference studio lights at near and far distances. Save baseline captures with color target, input image hashes, source rights, calibration notes and compare dates. Use a calibrated display for judgment; no screen preview promises production dye match.

Capture minimum for priority families: rigid/stretch denim (face/back), cotton jersey, French terry (loop/back), fleece, poplin, canvas, twill, leather, suede, silk satin, wool suiting, linen, nylon ripstop, rib knit, velvet, mesh and sheer organza. Build family-specific test variants: black/white/high-chroma dyes, fine/coarse weave, shallow/deep relief, heavy/light weights, brushed and coated finishes, cross-grain orientation.

400+ catalog acceptance

Count definition: target >= `400` **unique meaningful fabric masters** identified by stable IDs and differing in material composition, construction, weight, finish substrate or other defensible physical/material identity. Multiple dye colors, repeated presets, slight noise seeds, wash look variants, or near-identical parameter clones do **not** increase the unique-master count. Keep appearance variants separate; they may number in the thousands but must never inflate the underlying 400 claim.

Coverage plan (targets, not verified current counts):

Material family	Unique masters target	Examples / coverage
Cotton woven, canvas, twill, shirting	68	poplin, oxford, broadcloth, sateen, duck, drill, voile
Cotton and blended knit	62	jersey, interlock, piqué, rib, ponte, French terry, fleece
Denim	52	rigid/stretch, varying weight, warp/weft blend, slub, black, raw, lightweight
Wool and animal-fiber fabrics	45	suitings, flannel, melton, felt, worsted, cashmere blends
Linen, hemp and bast blends	31	plain weave, canvas, slub, twill, knit, mixes
Silk, satin and lightweight dress fabrics	35	charmeuse, crepe, georgette, chiffon, organza, taffeta
Synthetic technical fabrics	59	nylon/polyester woven, ripstop, mesh, shell, coated, spacer
Leather, suede and alternatives	27	nappa, split suede, nubuck, bonded, PU and microfiber
Pile, specialty and composite	33	velvet, corduroy, faux fur, jacquard, lace, quilted laminates
Total planned	**412**	All category counts require machine-audited distinct source entries

Run a full automated catalog audit on the **actual assembled runtime catalog**, not source-file text counts. Assert 400+ stable unique IDs and names; valid, normalized numeric optics and finite physical data with units; correct family assignment; fabric master ≠ colorway ≠ wash variant; deterministic import/export; editable controls that persist after reopening; successful fabric assignment to an actual 2D panel and visible 3D mesh. Deduplicate case/spacing aliases and compare fingerprints (fiber composition, knit/weave, GSM band, thickness, stretch profile, optic family), flag suspicious near-identical records for manual review.

For **each** master, validate 1) source/provenance and license metadata, 2) useful swatch/fallback, 3) color map plus calibrated or explicitly estimated roughness/normal/relief, 4) consistent cm UV scale and grain direction, 5) nonzero credible physical metadata and its evidence status, 6) serialized save/load and production data export, 7) viewport application and 8) accessible search/filter/grouping. Quality is graded separately: **A** specimen-measured mechanical +

color-calibrated optical reference; **B** part-measured with optical reference; **C** literature-estimated and artist-generated; **D** placeholder. Report A/B/C/D counts; a 400 total with 390 placeholders fails the intended quality bar.

For denim finishes, minimum golden scenes: raw indigo, sulfur black, light washed blue, heavyweight/slubby twill, stretch denim, jean front with fly/pockets/hem, jean back with seat/pockets, jacket with yoke and cuffs. Snapshot all eight presets across representative scenes and compare a physical/photo reference where licensed. Include intensity endpoints, UV transform rotations and scales, independent seed determinism, blend selection, area masks, seam abrasion at three levels, extreme panel aspect ratios, resizes/grading, symmetry, undo/redo, round-trip save/load, 3D pose motion, lighting changes and distance mipmapping. In particular assert finish edits do not modify recorded physical solver state.

Visual QA must catch: faded areas that drift in world space during cloth motion; identical wear printed onto both legs despite different panel mappings; seams glowing uniformly despite absence of puckering; twill detail disappearing under strong bleach looks; unlicensed photographs shipped with default assets; wash texture repeats exposing obvious tile seams; albedo treated as data or normal maps treated as sRGB; progress UI announcing '400 realistic fabrics' with no verified runtime total. Preserve separate measured optical similarity scores and solver fitting errors where evidence exists.

Suggested release gates: catalog count and integrity tests green; every entry can be assigned and reopened; each family has a documented manual lighting/visual sample; denim golden scenes reviewed; no catastrophic UV seams in sampled garments; wash stack exports are deterministic and reversible; no unlicensed reference maps; calibrated-vs-estimated badges visible; measured physics claims limited to measured specimens. Establish tolerances with baseline image differences and panel displacement fixtures rather than inventing unverified universal pixel/perceptual thresholds.

Product semantics, rights, and safety

Label interface controls **Digital Wash / Finish Appearance**, with a separate optional **Measured Post-Finish Mechanics** asset only when actual tests support it. Store wash settings and reference image provenance in editable techpacks: finished-look target, source fabric, colorway, layers, reference photographs, panel mask placements, revision notes and manufacturer comments. Export swatches and a visual spec with a clear disclaimer that manufacturing results require separate factory sampling and approval.

Accept user-owned or appropriately licensed swatches and original procedural generators; track source URL, creator, license, attribution restrictions and whether redistribution/derivatives are allowed. Avoid bundling CLO media, logos, proprietary textures and reverse-engineered code. Never offer operational chemical wash formulas or infer worker/environmental safety from a rendered wash effect. Physical production instructions should be written only from qualified manufacturing documentation and applicable safety standards outside this digital appearance specification.

Sources

- CLO, Express Denim Wet Wash, official support guide, updated 19 August 2026.
- CLO, Material Type, official support guide.
- EXO code reviewed read-only: `src/core/clothcraftFabrics.js`, `src/core/fabricSurfaceModel.js`, `src/core/fabricCalibration.js`; background architecture: [EXO Fabric Materials and Trims](EXO_FABRIC_MATERIALS_AND_TRIMS.md). All proposed fields and targets remain unimplemented unless separately verified by code and tests.

FABRIC_PHYSICS_AND_CLO_RESEARCH_2026-10-16.md

Audit interpretation: This annex preserves earlier engineering research for traceability. Read reported pass counts and statuses as of its original date only. The current 70-slot and 400-fabric inventories above supersede any older source totals.

EXO CLOTHCRAFT V2 — FABRIC SCIENCE, CLO FEATURE REFERENCE, CALIBRATION AND RENDERING

Research date: 2026-10-16. **Scope:** public CLO documentation, textile measurement standards, graphics practice and a read-only EXO source inspection. **Evidence convention:** “documented CLO behavior” means CLO's published UI/test behavior; “EXO observation” means the inspected repository; “proposed” means engineering guidance; “estimate” means an unverified starting point. Neither published CLO physics presets nor calibration internals are copied here.

1. Summary of findings

Fabric realism has **two coupled but independently testable systems**: (1) physical dynamics, whose key inputs are areal mass, anisotropic tensile extension, shear, bending, thickness/collision, friction and energy dissipation, and (2) optical response, which depends on yarn-scale pattern, normal/height, roughness, reflectance, fibers, translucency, side-specific appearance and lights. A physically correct drape can look synthetic if its fiber surface is flat; a photorealistic swatch can still behave like cardboard.

CLO provides configurable fabric physical properties and material maps. Its CLOFAB interface (current published guidance dated September 2026) supports a fabric type, composition and density workflow, adjustable advanced physical properties, generated/scanned PBR maps, preview on cylinder or dummy, and saving fabric with textures. Its legacy Fabric Kit documents weighing, thickness gauging, warp/weft/bias bending and stretch testing. These are **measurement and workflow references**, not evidence that EXO uses CLO's proprietary model or its calibration data.

References: https://support.clo3d.com/hc/en-us/articles/49343947711769-CLOFAB ; https://support.clo3d.com/hc/en-us/articles/360041074334-Fabric-Kit-Manual ; https://support.clo3d.com/hc/en-us/articles/115000483087-Adjust-Stretch-Weft-Warp-Shear

2. CLO: what is confirmed publicly

- **Physical axis directions:** CLO names Stretch-Weft, Stretch-Warp and Shear in Fabric > Property Editor > Physical Property > Detail. CLO's guidance says increasing shear resistance changes how easily cotton/denim crease, while softer silk/jersey can use lower resistance. Do not map those slider numbers directly to EXO N/m without an identified calibration experiment. Source: https://support.clo3d.com/hc/en-us/articles/115000483087-Adjust-Stretch-Weft-Warp-Shear
- **Legacy Fabric Kit:** public instructions describe cutting prescribed swatches, weighing multiple samples, thickness using a gauge, three bending orientations (weft, warp, bias) with geometry readings and a stretch force device. Follow the exact instrument manual for swatch sizes/loads rather than improvising its calibration formula. Source: https://support.clo3d.com/hc/en-us/articles/360041074334-Fabric-Kit-Manual
- **CLOFAB (2026):** catalog type/composition/density create an initial behavior, with Advanced Details to inspect/edit. Image maps may be captured or scanned, imported in coordinated filename sets, and saved to a fabric file with available textures. The workflow previews drape as cylinder circle/square or dummy cape. CLO describes integration with its zFab Kit; this does not imply universal interchangeability of measurements from other kits. Source: https://support.clo3d.com/hc/en-us/articles/49343947711769-CLOFAB
- **Material channels:** CLO lists base color, normal, displacement, opacity and reflection-related controls; material appearance can be configured for front/back/side, with front/back different when same-material sharing is disabled. Its PBR map generation covers normal, displacement, opacity, roughness and metalness. Source: https://support.clo3d.com/hc/en-us/articles/360047317454-Set-Fabric-Materials ; https://support.clo3d.com/hc/en-us/articles/33852438561049-PBR-Map-Generator-ver-2024-1
- **Height vs simulator geometry:** CLO's displacement support documents render-only effects in the cited workflow. EXO should likewise explicitly distinguish render displacement from actual collision thickness, unless a physics collision implementation is verified. Source: https://support.clo3d.com/hc/en-us/articles/360025439933-Displacement-Map-ver5-1-0
- **Accuracy claims:** CLO's public FAQ states approximately “90% accuracy” for its own library/Fabric Kit fabrics and cautions arbitrary changes can affect it. That statement has no publicly defined error metric or universal transfer claim; EXO must report its own test-specific measured errors. Source: https://support.clo3d.com/hc/en-us/articles/115013660747-How-accurate-are-virtual-fabrics

- **CLO Fabric Guide:** official downloadable fabric guide is linked from CLO support and discusses physical properties/textures. It is useful for UI terminology, not a license to redistribute private presets. Source: https://support.clo3d.com/hc/en-us/articles/360001436227-CLO-Fabric-Guide

3. Canonical EXO SI schema and governing relationships

Use measured material records with **unit-tagged** parameters and full provenance. Store scalar values in SI and convert solely in UI. Recommended representation:

Quantity	Canonical storage	Display	Method / model meaning
Areal mass `rhoA`	kg/m²	g/m² (GSM)	dry conditioned measured swatch mass divided by area
Physical thickness `t`	m	mm	measured at **specified compression pressure**; physical collision thickness separate
Extensional stiffness `Kw,Kf`	N/m	N/m	tangent/secant membrane force per width versus engineering strain, at recorded strain/loading
Shear stiffness `G`	N/m	N/m	for dimensionless engineering shear strain; state convention/angle
Bending rigidity `Dw,Df`	N·m	μN·m	shell plate/membrane energy per area convention; unit depends on exact discretization
Fiber direction `theta`	rad	degrees	warp/wale local rotation with UV/pattern pieces
Damping	model-dependent	unit-tagged	viscous, Rayleigh, or dimensionless per-step ratio must be distinguished
Surface friction `mu`	dimensionless	dimensionless	static/kinetic, cloth-cloth and cloth-avatar tested separately
Rest/locked strains	ratio	%	shrinkage and imposed residual strain must not be confused
Optical roughness, metalness, sheen	dimensionless	0–1	shader parameters, **not** textile mechanical roughness

Conversion: `rhoA[kg/m²] = GSM/1000`, `t[m]=thicknessMm/1000`, `D[N·m]=valueInMicroNewtonMeter × 1e−6`. Example: 460 GSM = 0.460 kg/m²; 1.2 mm = 0.0012 m. Gravitation load per actual fabric area: `q = rhoA × g` N/m², with `g ≈ 9.81 m/s²`.

Strain and energy, illustrative continuum model: engineering strain `epsilon=(L−L0)/L0` is dimensionless; surface force per width `N=K epsilon` (N/m) only for a locally linear tensile region; energy per undeformed surface area `U_stretch≈0.5 K epsilon²` (J/m²). For anisotropic fabrics use `U≈1/2 [Kw epsilon_w² + Kf epsilon_f² + G gamma²]`, with extra coupling/nonlinear recovery terms when calibrated. For bending with curvature `kappa` (1/m), `U_bend≈1/2 D (kappa−kappa0)²` (J/m²), hence `D` dimensions N·m. The discrete solver's spring stiffness must be derived from mesh spacing and its specific energy; inserting continuum values blindly changes behavior with tessellation.

Dynamics: `M x_ddot + C x_dot + grad(E_stretch+E_shear+E_bend+E_contact+E_seam)=f_external`. Cloth is often nonlinear, especially knit, bias and leather. Compression buckles instead of sharing tensile stiffness; use one-sided tensile response or calibrated buckling. XPBD compliance has dimensions/solver scaling different from stiffness: convert using the implementation's constraint energy and time step, then verify time step and mesh convergence. Friction coefficients belong in the collision/contact model and are distinct from a fabric PBR roughness map.

Caution: `stretchWarp:0.12` in defaultProject.js is a ratio-like authoring field, while `warp` in ClothCraft's solver record is N/m. They cannot be assigned to each other by name. Bending coefficients in the default project's `bend:0.68` are authoring/normalized controls and are not N·m. Mechanical simulation thickness should be tested for how collisions actually use the parameter.

4. EXO implementation audit as of inspection

Inspected path	Evidence in code	Gap / action
`src/data/defaultProject.js`	Initial presets jersey, heavy French terry, rigid denim, silk and interfacing plus poplin/canvas/wool/leather/suede/nylon/linen/knits/velvet/organza/corduroy variants; fields include GSM, `stretchWarp`, `bend`, `thickness`, PBR controls. Variants are explicitly PPF-inspired **estimates**.	Preserve provenance; convert parameter meaning through a named adapter with unit assertions. Do not call authoring values laboratory measurements.
`src/core/clothcraftFabrics.js`	Ten material examples with `gsm`, `thicknessMm`, extension %, `shear`, two `bending` directions, source/evidence; `presetProperties` populates `density`, `thickness`, `warp`,	Some specimen mass/thickness from MIT research dataset, but stiffness and many optical values are **estimates**. Treat derived stiffness as model-

Inspected path	Evidence in code	Gap / action
	`weft`, `shear`, `bend_warp`, `bend_weft`. `FABRIC_FIELDS` lists display conversions.	dependent; verify `stiffnessForExtension` assumptions and test cases before labeling physically calibrated.
`src/core/fabricSurfaceModel.js`	clamps opacity, optical displacement, roughness, metalness, sheen, anisotropy, clearcoat, procedural weave color variation; comments say optical displacement is not collision geometry.	Add calibrated asset metadata: texels per cm, capture illumination, color space, map channel conventions, tangent sign and scale.
`src/components/FabricMaterialPreview.jsx`	Three.js `MeshPhysicalMaterial` preview, generated/file texture channels including base/normal/roughness/metalness/opacity/displacement.	Verify UV orientation and equal physical repeat across pieces, proper linear-vs-sRGB decoding, normal strength, physically plausible fiber sheen under multiple light setups.
`src/core/clothcraftFabricSurface.js`	separate ClothCraft material initializes roughness/sheen from `appearance` and grain angle.	Compare resulting viewport appearance to richer preview: material parity and import behavior need visual regression.
`native_cpp/core/fabric_surface.cpp`	native PBR normalization/naming support exists.	Validate both native and browser paths agree for every map and unit conversion.

No source inspection by itself proves the cloth solver reproduces laboratory tensile/shear curves, geometric pile, leather hysteresis, fabric self-contact under load, or all maps in live garment rendering. Those require the tests below.

5. Measurement and calibration protocol

1. **Identify each physical sample:** manufacturer, SKU/lot, fiber %, blend, weave/knit type, finish, washed/unwashed, actual dye shade, fabric face/back, sample orientation and date. Condition swatches under a specified measured humidity and temperature; record them for repeatability.
2. **Measure areal mass and thickness:** cut multiple known-area swatches, weigh with calibrated scale, measure under stated compressive force/foot area. Save mean, standard deviation and raw samples; thickness is pressure dependent particularly for terry, fleece and suede.
3. **Tensile and recovery:** measure warp/weft (or wale/course) force-versus-extension at *several* low-strain loads, then unload/reload to measure hysteresis and residual strain. A single “stretch %” without force and recovery is insufficient.
4. **Bias and shear:** conduct bias-extension or dedicated shear rig with specimen geometry, pretension and angle recorded. Calibrate against observed folds, not the texture image. Distinguish bias extension from a pure-shear modulus.
5. **Bending:** separate warp/weft/bias cantilever or low-force tester; capture bend length, fold direction and recovery. ASTM D1388 measures bending length/flexural rigidity. For napped/coated textiles, compare face-in/face-out if asymmetric.
6. **Contact:** inclined-plane or pull tests for fabric/fabric and fabric/skin surrogate under known load; record static, kinetic, grain-direction and fabric-back differences. Collar/hood contact is sensitive to friction.
7. **Validation geometry:** circular drape over pedestal, 20–30 cm hanging strip, controlled bias-square, fold compression test and a standard T-shirt/sweatshirt on a fixed avatar. Record camera, lighting, scale, swatch orientation, garment size and sewing.
8. **Fit parameters:** optimize mass/thickness directly from measurement; fit directional tensile/shear curves before bending and damping, then tune contact friction. Use held-out drape tests to check against overfitting; never optimize appearance parameters to disguise mechanical errors.
9. **Quantify accuracy:** report silhouette mean boundary distance (mm), static hem depth (mm), curvature distribution, number and wavelength of folds, shoulder/chest strain, dynamic oscillation decay (s), penetration counts/depth (mm), and runtime. Compare same avatar, gravity, mesh resolution, seam pattern and camera.
10. **Convergence & reproducibility:** test mesh spacing (e.g. 20, 10, 5 mm), time step and solver iterations; record build commit, machine, physics backend, parameter file and random seed. Values should remain within selected tolerances after refinement.

Standards/references: https://store.astm.org/standards/d1388 ; https://textiles.ncsu.edu/tpacc/comfort-performance/kawabata-evaluation-system/ ; https://research.hva.nl/en/publications/white-paper-ieee-industry-connections-ieee-ic-the-measurement-of-/ ; https://support.clo3d.com/hc/en-us/articles/360041074334-Fabric-Kit-Manual

Proposed acceptance targets (engineering goals, NOT achieved results)

- Schema round-trip: no unit loss; mechanically identical after save/load to within 1e-6 relative for numeric SI values.

- Uniform-cloth swatch gravity: measured areal mass within scale uncertainty; static silhouettes target ≤ 10 mm mean deviation on controlled mid-size swatches **after calibration**.
- Directional tests: reproduce force-vs-extension within 15% at defined strain points where instrument uncertainty permits, and report outliers; at least three repeats per orientation.
- Bend and folds: target 15% error in mean bend length and fold wavelength; explicitly compare warp and weft.
- Mesh convergence: selected observable drift ≤ 5% between the two finest practical discretizations; no explosive motion or systematic penetration.
- Render: check scale accuracy to 1 mm over 100 mm of fabric detail, no base-color or normal map seams under standardized grazing light. Visual review remains necessary.

Targets can be tightened after collecting actual cloth and experimental evidence. Relative thresholds are not claims about achieved commercial accuracy.

6. Material-specific physical and rendering requirements

Cotton plain weave / poplin: relatively low extension on yarn axis, larger diagonal shear; visible spun fiber irregularity and balanced fine plain weave. Use actual yarn count/repeat scale; natural matte diffuse reflectance and modest fiber sheen. Typical use cases: poplin shirts, canvas garments, oxford. Canvas generally requires greater areal mass and bending stiffness than poplin, but values depend strongly on finishing and weight.

Denim (rigid and stretch): anisotropic warp-faced twill; indigo warp / pale weft color separation is critical in closeups. Warp/weft/bias stiffness and directional fold response; rigid variants generally extend less than elastane stretch variants. Render diagonal ribs, subtle warp slubs, seam abrasion, fading concentrated around creases and garment stitching. Texture wear maps should be aligned to the pattern grain, never generated randomly per triangle. Stretch denim must have measured recovery and time-dependent behavior; do not assume any denim preset describes every wash.

French terry / fleece: knit jersey face plus loop pile backside (or brushed fleece); higher loft and compressibility, directional stretch and recovery, surface friction dependent on side. Represent face and backside separately, back loop normals/height, thick hemlines, ribbed cuffs and realistic compression around seams. A flat single normal on both sides will misrepresent this fabric. Measure thickness under load; pile optical displacement cannot replace bulk contact mechanics.

Jersey / rib knit / interlock: more anisotropic extension and nonlinear elastic recovery than typical woven fabrics. Jersey stitch and rib wales need real physical repeat. Ribbed cuffs recover elastically and contract under extension; add calibrated stitch/cuff rest dimensions, not shader-only ridges. Simple membrane shell does **not resolve individual knit loops**; microstructure is approximated through effective anisotropic constitutive behavior.

Wool suiting / melton: wool suiting has variable bending and recovery due to weave and pressing. Melton is densely felted/fuller with soft diffuse surface and thick edge. Highlight fiber flyaway and subtle broad sheen; avoid aggressive large-scale normal noise. Coats may need laminated/interfacing zones.

Leather (nappa, lambskin, cowhide): tanned hide is not a woven cloth. It may have substantial nonlinear stiffness, anisotropy, plastic set, grain-dependent folding, limited rebound and material-specific friction. Treat top grain, finish, pores, creases, natural variation and cut direction as independent data; separate smooth finished top face from flesh side. Shader clearcoat describes coatings, not a universal "leather gloss." A generic thin-shell solver approximates leather response but does not simulate damage, cracking, delamination or permanent creasing until explicitly implemented.

Suede / nubuck: napped fiber response, soft broad specular lobes, highly directional visible nap and color change with brushing. Use tangent-aligned fiber anisotropy, micro-normal structure, fine fuzz/sheen, direction-sensitive reflectance and a face/back distinction. Artificially raising roughness to 1 without nap direction is insufficient. Suede thickness and friction depend on surface/finish and conditioning.

Silk satin / twill / chiffon / organza: satin highlights require correct low-frequency anisotropic reflectance orientation and small-scale yarn maps. Chiffon/organza need appropriate porosity/opacity, two-sided shading, thin silhouette treatment and collision separate from transparency. Organza can be optically sheer yet mechanically comparatively crisp. Do not infer bending stiffness from transparency.

Nylon, ripstop, softshell: capture ripstop reinforcement cell pitch, coated-surface specular and dye variation. Lightweight mass does not imply a universal low bending value (coatings alter folds). Softshell may be multi-layer; effective bulk properties should be measured for the stack. Rain beading/wetness is optional optical behavior, independent from PBR dry fabric and cannot prove waterproofness.

Velvet / corduroy: fiber pile anisotropic lobe and fold-dependent shade changes. Corduroy directional wales should have physical pitch; velvet piles need view-angle-dependent fiber shading. Thick pile collision is an approximation unless contact geometry includes loft.

Interfacing / padding / bonded composites: use explicit layered laminate material properties (areal mass total, directional rigidities, adhesive coupling, thickness), bonding location and seam constraints. For trapunto padded hoodie segments, panel-scale quilt stitches and internal loft controls are required; optical bump alone cannot reproduce puff volume or compression.

Hardware: metal buttons, snaps, eyelets and rivets need real object geometry, attachment constraints, collision and correct metal/roughness workflows. Zippers need tape stiffness, tooth/slider geometry, seam coupling and state (open vs closed). Plastic button transmissivity and brushed metal anisotropy are appearance controls. Hardware mass and pendulum motion must be modeled separately from the cloth. Distinguish rendered trims from verified functional physics.

7. Optical map pipeline and QA

Capture: photograph/scans with a ruler, diffuse cross-polarized albedo capture where possible, and multi-directional directional lighting for height/normal reconstruction. Include face/back swatches, color checker and neutral white balance. Store ICC/color-space provenance, sample dimensions, texture repeats and rights/source. A photo under directional light already includes shadows and highlights, so directly using it as base color tends to double-light the fabric.

Channel semantics:

- **Base color/albedo:** sRGB texture transformed into linear working space; remove baked specular/shadow and retain authentic yarn color variation.
- **Normal:** tangent-space linear map, document OpenGL/DirectX green channel polarity, ensure grain direction transforms correctly with 2D panel rotations/mirror/UV.
- **Height/displacement:** linear meters or millimeters per authored amplitude, visual geometry only unless solver explicitly uses it; verify edges and silhouette.
- **Roughness / metalness / ambient occlusion / opacity:** linear scalar textures, calibrated roughness distribution, metalness ~0 for uncoated textiles. Metallic threads and metal studs are separate materials. AO shouldn't darken base color twice; opacity can cause depth sorting issues.
- **Sheen / anisotropy / clearcoat:** artist controls with material-specific justification; physically sensible fiber orientation varies over cut pieces and backfaces.
- **Maps at scale:** store real-world `texelsPerMeter` or `repeatWidthM`/`repeatHeightM`. Clothing pattern UVs represent centimeters, not arbitrary model-scale repetition. Add seam-consistent UV orientation and per-panel grain transform.

Rendering tests: calibrated neutral HDRI and raking side light, daylight plus tungsten, rotating turntable, light/dark swatches, flat vs draped images, 1 cm and 10 cm closeups; white diffuse sphere plus gray reference; normal-only and roughness-only debug. Compare live viewport, fabric preview, export (Blender/glTF, if supported) and any native renderer for parity. Check glTF channel conventions before exporting. Avoid using random tiled high-frequency normals that shimmer at distance: mipmapping, anisotropic filtering and correct LOD are essential.

Official optical sources: https://support.clo3d.com/hc/en-us/articles/360047317454-Set-Fabric-Materials ; https://support.clo3d.com/hc/en-us/articles/33852438561049-PBR-Map-Generator-ver-2024-1 ; https://support.clo3d.com/hc/en-us/articles/360025439933-Displacement-Map-ver5-1-0

8. Recommended engineering milestones

M1 — provenance and units. Define `MaterialSpec v1`: unique IDs, family, face/back materials, areal density, measured thickness and pressure, warp/weft/shear force curves, directional bending, static/dynamic friction, grain angle, source license, measured/estimated flag and uncertainty. Keep a versioned adapter from editor presets and ClothCraft native fabrics. No silent unit coercion.

M2 — sample-driven physics. Acquire/populate 2–3 specimens each of jersey, heavyweight French terry, rigid/stretch denim, garment nappa leather, suede, canvas, wool and nylon. Produce measured low-strain curves. Extend solver constitutive support only after baseline comparison; check mesh dependence and fabric hand interaction against test strips.

M3 — optical texture library. Capture 4K+ (where justified by physical repeat, not resolution alone) tileable face/back swatches; author true-scale PBR and nap-direction fields. Maintain distinct geometry, physics and render asset references. Add glTF and Blender material conversion tests.

M4 — trims and multilayers. Validate attachment, fabric-hand dragging, buttons and zipper tape on garments. Add lining, interfacing and padding behavior, then quantify collision/contact. Record unresolved constraints on the project progress hub.

M5 — garment reference benchmark. Photograph physical T-shirt, French terry hoodie, denim jacket and leather panel garment on matched form; simulate same authored patterns and published measurement records. Compare drape and render independently, then run controlled user visual reviews.

9. Documentation/source and licensing cautions

A KES system measures bending, tensile, compression and shear, but its result units and acquisition conventions are not automatically compatible with a CLO Fabric Kit, other commercial solvers or EXO. An IEEE/HvA comparison reviews the

shortcomings of cross-system fabric measurement and highlights the need for standardization. See: https://textiles.ncsu.edu/tpacc/comfort-performance/kawabata-evaluation-system/ ; https://research.hva.nl/en/publications/white-paper-ieee-industry-connections-ieee-ic-the-measurement-of-/ .

For research specimens and sample metadata, EXO cites the MIT material dataset (https://people.csail.mit.edu/klbouman/pw/projects/materialproperties/dataset.html). Read its reuse terms and actual measurement rows before republishing data. EXO's `clothcraftFabrics.js` states clearly that a number of mass/thickness values have a measured source but elasticity/bending and optics are estimated; retaining that distinction is mandatory.

CLO's UI, article descriptions and public fabric science inform functional requirements; proprietary presets, kit conversion algorithms, textures and other restricted assets are not assumed accessible or reusable. No numerical EXO simulation error or fabric match is claimed here without sample tests and repeatable outputs.

10. Primary reference index

1. CLO, *CLOFAB* (2026): https://support.clo3d.com/hc/en-us/articles/49343947711769-CLOFAB
2. CLO, *Fabric Kit Manual*: https://support.clo3d.com/hc/en-us/articles/360041074334-Fabric-Kit-Manual
3. CLO, *Adjust Stretch Weft/Warp, Shear*: https://support.clo3d.com/hc/en-us/articles/115000483087-Adjust-Stretch-Weft-Warp-Shear
4. CLO, *Set Fabric Materials*: https://support.clo3d.com/hc/en-us/articles/360047317454-Set-Fabric-Materials
5. CLO, *PBR Map Generator*: https://support.clo3d.com/hc/en-us/articles/33852438561049-PBR-Map-Generator-ver-2024-1
6. CLO, *Displacement Map*: https://support.clo3d.com/hc/en-us/articles/360025439933-Displacement-Map-ver5-1-0
7. CLO, *CLO Fabric Guide*: https://support.clo3d.com/hc/en-us/articles/360001436227-CLO-Fabric-Guide
8. ASTM D1388, *Stiffness of Fabrics*: https://store.astm.org/standards/d1388
9. NC State T-PACC, *Kawabata Evaluation System*: https://textiles.ncsu.edu/tpacc/comfort-performance/kawabata-evaluation-system/
10. IEEE/HvA, *Measurement of Fabric Properties for Virtual Simulation — Critical Review*: https://research.hva.nl/en/publications/white-paper-ieee-industry-connections-ieee-ic-the-measurement-of-/
11. MIT, *Fabric Material Properties Dataset*: https://people.csail.mit.edu/klbouman/pw/projects/materialproperties/dataset.html
12. Ji, Li, Qiu (2006), *Dynamic Draping Behavior of Woven and Knitted Fabrics*: https://journals.sagepub.com/doi/10.1177/1528083706055753

11. Implementation completed in this pass

The expanded physical/optical library is present in the running application source, not only on a roadmap:

Component	Working integration	Measurement/realism limit
`src/data/apparelMaterialExpansion.js`, `defaultProject.js`	Added 57 named construction-based examples to existing preset IDs, bringing the React fabric menu to **80 presets**. App fabric library is searchable/grouped and swatches can be assigned to 2D pieces for use by the 3D editor.	New GSM, thickness, extension, bend and finish values are explicitly **unmeasured estimates**; a fabric name is not a universal constitutive law.
`src/core/clothcraftFabrics.js`, `ClothCraftFabricLibrary.jsx`	Rust workspace selector contains **66 examples** with membrane stiffness, shear, two bending directions, mass, thickness, and grain-relative appearance. Fabric edits retain the Rust project's save/undo path.	Browser `bend` is dimensionless while native stiffness/bending values are model-specific SI-like parameters. Conversion between solvers is heuristic, not physically established or CLO-compatible.
`src/core/textileProceduralS.js`, `clothcraftFabricSurface.js`	Differentiated procedural surface relief for common woven/knit/pile constructions, denim twill, leather grain, suede nap, synthetic mesh, bouclé and more. Native materials now get a neutral pigment-variation tile alongside bump with controlled material roughness, sheen and anisotropy.	These are artistic/surface mathematical approximations. They are **not** photographic scan assets, individual loop/yarn simulation, measured BRDF/BTDF or calibrated 3D pile volumes.
`src/core/fabricCalibration.js` and `ClothCraftFabricLibrary.jsx`	An interactive strip test calculator converts actual input width/gauge length, warp/weft force and elongation to a small-strain effective tension modulus in N/m; result may be assigned to native material. Pure helper also computes mass, shear estimate, cantilever bending estimate and incline friction.	Native nonlinear solver parameter meanings require fixture calibration; one-point force tests cannot capture hysteresis, anisotropic coupling or nonlinear plasticity.

Component	Working integration	Measurement/realism limit
`src/data/hardwarePresets.js`, `src/App.jsx`, `GarmentViewport.jsx`	**18** buttons/buttonholes/zippers/snaps/rivets/buckles reference objects. Three supported annotation families are selectable in Sewing and asset library; buttons have disc/hole geometry, face-aligned placement and shader finish.	Snap/rivet/buckle placement and rigid physics, zipper interlocking teeth and button-thread behavior remain to be implemented. Dimensions are generic estimates, not supplier specifications.
`scripts/check-apparel-catalog.mjs`, existing checks	Catalog uniqueness, finite physics values, rendering map objects, SI conversion examples and hardware annotations checked; build, material, buttonhole and factory checks can be run independently.	Programmatic tests cannot substitute for comparative physical drape, spectrophotometer/color management or 3D scan fidelity.

Next calibration priority: obtain physical samples for heavyweight cotton French terry, 12–14 oz rigid denim, 2% elastane stretch denim, garment nappa leather and sanded suede. Record specimen manufacturer/fiber content, areal mass, thickness under specified test pressure, 3–5 warp/weft/bias force-extension readings per sample, independent bending/shear experiments, static sliding surface pair, neutral light PBR captures and seam/crease recovery after cycles. Tie each experimental dataset to one immutable preset ID and display a calibrated/estimated indicator. Nothing in this pass establishes full CLO library equivalence.

EXO_PPF_MATERIALS_IMMERSIVE_STUDIOS_FUTURE_ARCHITECTURE_2026-10-16.md

Audit interpretation: This annex preserves earlier engineering research for traceability. Read reported pass counts and statuses as of its original date only. The current 70-slot and 400-fabric inventories above supersede any older source totals.

EXO: PPF PHYSICS, MEASURED TEXTILES, PBR MATERIALS, AND IMMERSIVE STUDIOS

Engineering research extension | 2026-10-16 | FUTURE PLANNING ONLY

Program navigation: [Integrated ecosystem implementation program](EXO_ECOSYSTEM_MASTER_IMPLEMENTATION_PROGRAM_2026-10-16.md) and [PPH/MAT/STU/COM candidate engineering issues](EXO_ECOSYSTEM_CANDIDATE_ENGINEERING_BACKLOG_2026-10-16.md).

This document develops the next EXO Studio concept: fast garment editing, offline PPF reference contact, real material calibration, realistic textile surfaces, photography studios, and an explorable room of the user's clothing collection. **None is requested to be implemented in this turn.** Do not alter the running V2 app, original garment projects, native solver, or local PPF jobs.

Companions: [Spatial master research](EXO_SPATIAL_PROJECTION_GARMENT_STUDIO_AND_AR_CONCEPTS_2026-10-16.md), [ten-page plan](EXO_SPATIAL_STUDIO_TEN_PAGE_IMPLEMENTATION_BLUEPRINT_2026-10-16.md), [PPF fit audit](PPF_ACTUAL_GARMENT_FIT_DEEP_AUDIT_2026-10-14.md), [upstream PPF frame forensics](PPF_UPSTREAM_PIN_MOTION_FORENSICS_2026-10-14.md), [fabric library](../integrations/clothcraft/FABRIC_LIBRARY_V2.md), and [EXO fabric optics](EXO_FABRIC_MATERIALS_AND_TRIMS.md).

1. One clothing design, multiple separately verified representations

1. **Authoring source:** pattern polygons/curves in measured local coordinates, seam graph, stitch assignments, internal lines, trims, grain direction, measurement spec and graded sizes. Immutable revisions + permitted edits are the source of truth.
2. **Physics input:** triangulated rest meshes, masses, local warp/weft orientations, calibrated constitutive material and collision parameters, avatar collider, staging plan, sewing constraints and chosen solver.
3. **Physics output:** timed drape frames with the exact source/solver/material hashes, seam/contact/strain QA report, simulation cache status and warning badges.
4. **Presentation:** mesh and UVs, appropriate textures, garments on mannequins, shaders, poses, stage/HDRI, camera, video and optimized gallery GLB, all labeled for their actual physical status.
5. **Publishing:** a separately authorized visual product page and optionally downloadable pattern/manufacturing data, material assets and studio presets, each with rights and visibility distinct from the original.

A good-looking PBR shader cannot repair an invalid cloth fit. A dense collider does not guarantee reliable seam closure. A published gallery look is not permission to redistribute the original source project.

2. Current implementation audit: what actually works

ClothCraft V2: local native Rust VBD/GPU VBD, three-vertex barycentric Hand spring with continuous target updates, panel selection/movement and material assignment. Its nine native exemplar materials are cotton woven, cotton/poly interlock jersey, wool, denim, nappa leather, light silk, linen, nylon ripstop and stretch knit. The preset mass and some thickness inputs are based on specimens; stretch/shear/bend and look are approximations. The V2 surface generator uses small procedurally repeated bumps and basic MeshPhysicalMaterial properties. There is **no experimentally validated complete friction, plasticity, knit-loop or certified textile fit** in this pipeline. Relevant files: src/core/clothcraftFabrics.js, clothcraftFabricSurface.js, integrations/clothcraft/exo-session.rs, workspace.mjs and FABRIC_LIBRARY_V2.md.

Broader original Garment Studio: src/core/fabricSurfaceModel.js, textileProcedurals.js, fabricPbrGenerator.js, components/FabricMaterialPreview.jsx and GarmentViewport.jsx provide richer visual texture and environment authoring. Its source model is **not automatically the same** as the native ClothCraft V2 source model. Migrate through a typed compatibility adapter and comparison fixtures, not by reusing raw numeric labels blindly.

PPF Contact Solver: the installed macOS arm64 Metal runtime has been executed from the EXO PPF bridge. It produced real native cloth trajectories, not a visual fake, but the October 14 historical tee with 98 stitches failed final production fit: mean seam gap near 1.01 cm, maximum about 9.98 cm, cloth/body penetrations beginning frame 3, and swept collision checks failed. A separate near-assembled test failed the required maximum 3 cm initial sewing gap. The upstream 241-frame dress example also had body intersections at frames 87, 90 and 174 under the independent audit. Smoothing a

pinned-body keyframe sequence reduced crossings but did not eliminate them. These are **hard blockers** to calling PPF results certified for fit on this project. See the linked local audit before making a replacement-engine decision.

Official PPF upstream: https://github.com/st-tech/ppf-contact-solver. The Apache-2.0 project expressly describes the solver as an offline/non-production-ready backend and notes slow CPU/Metal performance. Its claim about internal contact safety does not override EXO's separately observed full-trajectory penetration tests. Our goal is to test facts rather than adopt marketing guarantees.

3. The hybrid simulation architecture

Live track: responsive VBD/XPBD

Editor mouse/hand input → raycast to the visible cloth triangle → barycentric weights + bounded target in world meters → native temporary grab → step VBD/GPU VBD → valid drape snapshot → Three.js projection and studio preview. User release, camera loss, app close, mode switch or timeout removes the spring. Maintain revision ownership and a last valid solver state. Do not let an unsupported or unauthenticated website invoke raw local commands.

Offline reference track: PPF

Take a **frozen** garment source revision and material profile revision. Validate finite geometry, UV/grain topology, 2D-to-3D seam correspondence and initial assembly (no initial disallowed crossings, correct orientation, stitch distances below a chosen gate). Build PPF shell/body groups with explicit pinned-shell vs static policy, thickness and collision settings. Run isolated stages with bounded CPU/GPU quotas and log exact backend/mesh/params. Collect all native time samples and independent tests of intersections, seam closure, stretch, body clearance and swept motion.

States: pending → input validation → staging → native run → full-frame audit → **accepted, rejected, failed, cancelled**. A solver process exit of 0 means only completed, never automatically accepted. Rejected outputs remain private diagnostics; they cannot silently overwrite an accepted drape, visually approved showroom look or production specification.

Initial research experiments

- Reproduce the unchanged upstream fitting scene on current hardware with commit and asset hashes.
- Compare STATIC Thomas vs fully pinned SHELL Thomas using the same starting mesh/sewing/material/gravity/time; one policy variable changed.
- Fix actual garment 2D→3D armhole, shoulder and side seam correspondence, orientation, physical sewing lengths and nonintersecting initial layout.
- Independently test every saved frame with triangle intersection and swept CCD checks, including avatar keyframe boundaries; compare native contact diagnostics with independent geometry checks.
- Benchmark CPU/GPU runtime and memory at multiple triangle counts. Do not extrapolate full interactive speed from a ten-frame 500-vertex test.
- Measure material parameter convergence separately for jersey, French terry, denim and leather; don't use arbitrary visual presets as physical proof.

Release gate for any PPF-accepted fitting

At minimum require fully enumerated source triangles and stitches; valid material, body, pose and settings; full contiguous frame series and correct frame mapping; zero forbidden body/self penetrations in the tested trajectory; independently checked temporal/swept contacts; bounded maximum/mean seam gaps and regional body clearances; capped nonphysical principal strain; and comparison against actual clothing where users claim true fit. Existing example PPF thresholds from the audit (engineering targets, not certifications) are final seam mean ≤0.2 cm, maximum ≤1 cm, shoulder mean clearance ≤1.5 cm and p90 ≤3 cm, torso mean ≤2 cm. Log failures in red, not merely warning-colored footnotes.

System integration

A future SimulationJob references garmentRevision, triangulationHash, materialProfileVersion, bodyColliderVersion, backendName/Version, runConfigHash, progress, statistics, finalMeshRef, frameCacheRef, acceptanceReport and retention. This fits alongside the creator platform's upload permissions but should **not** run automatically on public uploads: compute/GPU expenditure and potentially hostile meshes require isolation, quotas and explicit authorization.

4. What a textile profile actually needs

The authoring UI should expose user-friendly physical values but the backend must store canonical units and source measurements.

Field	Physics meaning	Canonical value	Input/calibration note
GSM	mass per surface area	kg/m² = GSM/1000	Weigh known sample area; not garment total mass
Thickness	actual textile compressed/uncompressed thickness	m	Test pressure/conditioning matters
Warp extension curve	tension/strain along grain	load and strain samples	Many fabrics nonlinear and direction-dependent
Weft extension curve	cross grain tension/strain	load and strain samples	Jersey/rib knit differ substantially
Bias/shear response	resistance to rhombus deformation	calibrated force/angle	Drape and folds heavily depend on this
Warp/weft bending	directional moment/curvature	solver-model documented	Bending should be tested separately from stretching
Damping/hysteresis	loading/unloading loss and recovery	model-specific	Fabric may remain deformed or relax
Friction	cloth/cloth and cloth/body resistance	contact-model-specific	Need skin, underlying fabric and surface variants
Compression/loft	piled/knit/leather thickness under contact	nonlinear measured curve	Important for French terry, fleece, padded garments
Plasticity	permanent shaping/creases	strain history law	Denim/leather may need irreversible deformation
Environment	humidity/temp conditioning	specimen metadata	Can alter results
Uncertainty	provenance and confidence	sample/testing status	Default estimates are not scientific certificates

A **220 GSM** sample has 0.220 kg/m² nominal sheet mass. At 0.4 m² of cut material the surface-mass-only estimate is 0.088 kg before seam allowances, doubled/folded parts, trims, lining and adhesives. This arithmetic is not fabric hand, stiffness, thread weight or fitted garment mass certification.

Test kit and protocol: controlled-size specimen and scale → conditioned GSM/thickness measurement → repeated warp/weft/bias stretch at more than one load → three-point bending/cantilever strips → friction/sliding against representative body/fabric → drape photo with measured folds → repeat after release and use least-squares/model fitting to infer solver coefficients. Keep raw observation, curve, specimen batch and confidence distinct from fitted backend coefficients.

Important mapping rule: raw user-facing warp/weft stretch in percent at a named force is **not** automatically interchangeable with PPF normalized Young's modulus, Poisson ratio, thickness and bending parameter; a VBD N/m or N·m coefficient is not numerically the same as a PPF FEM parameter. Create separate calibrated maps for each solver and compare with the same physical fixture. Build backend-specific parameter dictionaries from the same canonical measurement set.

Material priority

Tier 1: cotton jersey, French terry, fleece, heavyweight cotton, canvas, raw and stretch denim, polyester/cotton knit, ribbing and wool suiting. Tier 2: nappa leather, suede, silk/satin, nylon ripstop, fleece/pile, technical shell/coated fabrics, linen, viscose and velvet. Tier 3: multi-layer padded/trapunto, bonded constructions, fusible interfacing, quilting, down/synthetic fill, zippers/hardware with rigid interaction, anisotropic directional trims. Specimen provenance matters more than an impressive count of 300 unmeasured presets.

Suggested MaterialPhysicalProfile fields: materialId, specimenId, description, composition, construction, weaveDirection, surfaceMassKgM2, thicknessM, extensionCurvesByDirection, shearCurve, bendingCurvesByDirection, frictionPairs, damping, compression, permanentSet, uncertainties, environmentalCondition, dataSource, license, validationStatus and derivedBackendCoefficients.

5. Optical appearance is a separate PBR/material problem

A cloth can look like velvet while simulating with a denim preset, so **do not assume appearance identifies physics**. Every MaterialAsset should link independently to MaterialPhysicalProfile and MaterialVisualProfile.

MaterialVisualProfile: baseColor map/linear color, normalMap, height/displacement, roughness map, ambient occlusion map, weave/nap/loop scale (millimeters/cm), fiber tangent direction, sheen, sheen roughness, anisotropy, clearcoat (only treated/coated surface), thickness/transmission/opacity where relevant, surface front/back, alpha handling, UV scale and

source rights. For leather/suede, use grain/nap direction, microrelief and controlled grazing highlights; for denim, repeatable twill angle and dyed warp/weft; for French terry, face knit and terry loop reverse, using separately assigned front/back materials if render backend supports it.

Color pipeline: base-color texture interpreted as sRGB then converted to renderer working color space; scalar roughness/metalness, height and normal are data textures and should not receive sRGB correction. Keep real-world repeat scale and correct directional UV across panel movement. Do not stretch a 3 mm stitch texture to cover a whole sleeve as an arbitrary repeating square. Use texture mipmapping and anisotropic filtering, plus a macro/micro view.

Relevant glTF extensions include KHR_materials_sheen for textile grazing highlights and KHR_materials_anisotropy for directed highlights; renderer and exporter feature support must be checked. Display normal/roughness/albedo maps independently for debugging, and compare shader output under the **same** lighting, camera, color management, exposure and monitor profile.

Procedural vs measured scans: procedural weave can fill placeholders cheaply; calibrated, tileable captured materials with multiscale normals and reflectance should carry a measured-scan flag. A photo without scale bar, lighting calibration, texture rights and direction annotation is not a physically correct textile source.

Hardware: zipper pullers, metal buttons, buckles and studs need their own real geometry, metallic/dielectric shading and physically meaningful BOM item; they should not simply be embedded in a flat albedo texture. Rigid-body contact for metal trims requires separately validated solver support.

Sample materials gallery tests

A sample scene contains flat swatch macro, bending strip, draped hanging square, hoodie torso and close-up seam at identical calibrated camera/light positions. Repeat for jersey, French terry, denim, leather and suede. Report **physics test status** beside **visual material fidelity**; do not silently award a single composite 90% realism score.

6. Photography Studio: build the stage, not a filter

A virtual studio is a **scene asset** with 3D stage geometry, backdrop/floor, camera rigs, light rigs, HDR environment, world scale, shadow/contact settings, color management, render engine, stage props and asset rights. Keep its original .blend master editable and export runtime derivatives for web. Blender GLB export does not preserve Blender area lights and complete World lighting as losslessly interchangeable scene lighting; EXO web scene recipes must reinstantiate suitable light rigs.

Suggested source scene presets:

- **White Infinity E-commerce:** white cyclorama and three-point softboxes, front/back/profile and detail photography camera; same frame crop across products.
- **Neutral Gray Fabric Lab:** measured white/gray card, neutral broad illumination plus optional grazing light, swatch spheres, material comparison.
- **Concrete Editorial:** original concrete studio, warm key/cool rim, directional floor shadows and fashion still framing.
- **Architectural Gallery:** original stone/concrete galleries with columns/vault/tracery and controlled materials, not copied proprietary architectural meshes.
- **Muslin Origins:** draped cream muslin backdrop, soft documentary lighting and macro stitch shots.
- **Black Stage:** rim-lit garments and hardware, shadow-preserving exposure.
- **Runway Theater:** time-controlled dressed avatars, path markers, backstage and audience cameras.
- **Outdoor Light:** authorized HDRI and an original ground surface.

Photography panel UX: select Look → Stage thumbnail → Avatar(s) and Pose → Camera/Shot → Lights/Exposure → Render Preview → Queue High-Quality Render → Review/Compare/Re-render → Add to collection or publish. Editing an exposure shouldn't rewrite source textile reflectance; an editorial renderer must not overwrite physically validated neutral swatch captures.

ShotManifest: lookId/version, mannequinIds/poses, garmentRevision/acceptedDrapeCache or visual-only mesh, studioId, camera position/target/lens/crop, lighting rig and HDRI rights, tone map/exposure, resolution, renderEngine/version, outputs and resulting file hashes. Include base PBR visual profile IDs and warn on missing maps.

7. An immersive walk-through 'Wardrobe Museum'

The user described a **cool room they can walk through**, showing mannequins dressed in all saved clothing. This is a separate view over the same Look and Studio objects, not a second set of duplicate garments.

Spatial UX and room design

Lobby: a 3D architectural hall with collection/season portals, search, favorites, new uploads and 'continue editing'. **Collection rooms:** runway-like rows of mannequin plinths, each representing a Look; hover on an outfit for floating panel info and thumbnails. **Material Room:** fabrics, swatches, denim/leather close-up and physical vs visual badges. **Pattern Archive:** framed technical drawings and 2D panels that open in controlled edit mode; unauthorized viewers can see only permitted redacted previews. **Photo Studio Room:** an active cyc/set the viewer enters to change cameras/lights and take images. **Runway Gallery:** a walkable fashion presentation, with saved animation/video and look details. **Marketplace doorway:** discover permitted public creator looks; no auto-download of their private sources.

Interaction: desktop WASD + mouse (with optional pointer-lock using Three.js PointerLockControls), joystick/gamepad and accessible point-and-click navigation, click-to-teleport, home/reset viewpoint, room mini-map and 'flat accessible browse' alternative. Smooth walking must have collision against walls, floor and mannequins, bounds/navmesh, camera near/far limits and motion-sickness controls. Do not force free-moving first-person on mobile; offer touch orbit and scene hotspot navigation. Render an outline around the chosen mannequin, not heavy emissive fill on the clothing itself.

Privacy: by default only the user's own garments appear. A public showroom is an opt-in published scene with separately licensed textures, mannequins and photos. Team-only rooms respect project ACLs and should not leak hidden pattern measurements via scene files, network responses, embedded GLB extras or developer tools.

Performance and scalability

Stage scene construction in Blender: master .blend → verify licenses, UV, rig, scale and texture dependencies → bake/optimize runtime GLB with LOD variants → compress textures to KTX2/Basis where renderer/device supports → GPU instancing for repeated room props/mannequin plinths → lazy-load rooms, high-resolution fabrics only for nearby outfits, texture memory budgets and recycled cached shaders → preserve a low-performance fallback gallery view.

Provisional performance gate: aim for usable 60 FPS on the chosen current Mac at a specified 1080p window and scene with ~20 mannequins, but benchmark hardware rather than claim this as achieved. Record frame-time median/p95, GPU memory, draw calls, load time and low-detail fallback. Navigation collision should be independent of clothing's full FEM triangles; retain authored garment mapping for inspection but use optimized collision proxies.

No need to simulate every garment live: store accepted drape cache or representative visual look geometry. Full real-time PPF solves while walking an entire room would be a poor use of GPU and is not supported by the current solver.

Example user journey

Open EXO Library → click 'Enter My Room' → walk through the Archive and choose a hoodie mannequin → review front/back, fabric close-up and PPF status → switch colorway → enter its photography studio → render a 3/4 shot → add image to a visual-only creator profile. Selecting 'Open in EXO' returns to the local editable file only when the user is the owner and source is locally available.

8. Engineering phases and acceptance tests

Phase	Future implementation unit	Must prove before proceeding
PH-01	Unified physical/visual material schema, provenance and unit adapters	No 1000× errors; edited material assignment matches known native fixture
PH-02	Specimen measurements and backend parameter fit	Same swatch curve and drape repeatably map to VBD and PPF separately
PH-03	PPF job adapter and failing-fixture regression	Historical crossings correctly rejected; no accepted-cache overwrite
PH-04	Nonintersecting seam assembly and body policy experiments	Actual seam and full-frame contact criteria pass on reference garment
PH-05	Shared realistic fabric/trim PBR assets	Stable texel scale, color space and maps across swatch, garment and scene
PH-06	Three Blender studio masters + web runtime derivatives	Camera/lighting compare, no missing textures, source .blend preserved
PH-07	Walkable single-user wardrobe lobby and Look plinths	Verified first-person navigation, click-to-view, private asset non-leakage
PH-08	Multiple collection rooms, accessibility, performance, public room mode	Permission tests, LOD/budget tests, flat-list fallback, URL sharing only of authorized assets

PPF verification harness should run input validation, staging metrics, fixed/pinned body comparison, all-frame mesh crossing, swept CCD, seam convergence, strain, clearance and disabled acceptance when tests are missing. A screenshot or beautiful final render is not an acceptance fixture.

Materials verification harness should compare fabric specimen/provenance, measurement samples, warp/weft directional response, unit conversions, renderer PBR channels, physical pitch, PPF coefficient mapping and deterministic source-to-render export. Show an 'unmeasured' flag whenever applicable.

Room verification harness should inspect GLB packet for hidden private pattern data, camera navigation collision, load-time RAM/GPU, access control to every asset endpoint, output reproducibility and outside-account denial.

9. Rough dependency/effort bands, not commitments

- Material schema + unit conversions and tests: 1–3 engineer-weeks.
- Four core measured textile specimens and calibration pipeline: 4–8+ engineer-weeks plus real specimen/professional testing access.
- PPF reference solver experimentation: 4–10+ engineer-weeks; significantly more if sewing/contact remains unstable.
- PBR material library and visual previews: 3–6 engineer-weeks plus look-dev artist time.
- First three Blender studios and still photo queue: 3–6 engineer-weeks with a Blender artist.
- Walkable local wardrobe prototype, one room and 10–20 original garments: 3–6 engineer-weeks, plus content preparation. A polished multiroom online experience and multiplayer are larger subsequent programs.

These are order-of-magnitude exploratory bands, **not promises or a budget approved for spending**. The public platform below should begin with read-only visual previews and static assets before adding remote heavy compute.

10. Source references and rights

- Original PPF solver (Apache-2.0): https://github.com/st-tech/ppf-contact-solver ; upstream itself warns that Metal/CPU can be slow and solver is experimental.
- Cloth physics reference (MIT): https://github.com/InteractiveComputerGraphics/PositionBasedDynamics
- Khronos glTF sheen: https://github.com/KhronosGroup/glTF/blob/main/extensions/2.0/Khronos/KHR_materials_sheen/README.md
- Khronos glTF anisotropy: https://github.com/KhronosGroup/glTF/blob/main/extensions/2.0/Khronos/KHR_materials_anisotropy/README.md
- Blender Asset Browser: https://docs.blender.org/manual/en/5.2/files/asset_libraries/introduction.html
- Three.js first-person control: https://threejs.org/docs/pages/PointerLockControls.html
- Local EXO reference: [PPF Metal cloth test](PPF_METAL_CLOTH_VALIDATION_2026-10-13.md), [fabric research](FABRIC_PHYSICS_AND_CLO_RESEARCH_2026-10-16.md), [Garment Studio actual PPF fit audit](PPF_ACTUAL_GARMENT_FIT_DEEP_AUDIT_2026-10-14.md).

Decision: the right architecture is a hybrid physical engine plus independently validated material and photo assets, with a walkable gallery sharing the same Look library. It does **not** require rewriting EXO V2 or shipping PPF as its real-time Hand backend.

EXO_ECOSYSTEM_MASTER_IMPLEMENTATION_PROGRAM_2026-10-16.md

Audit interpretation: This annex preserves earlier engineering research for traceability. Read reported pass counts and statuses as of its original date only. The current 70-slot and 400-fabric inventories above supersede any older source totals.

EXO ECOSYSTEM — INTEGRATED RESEARCH AND EXECUTION PROGRAM

Version: concept planning v1 | **Date:** 2026-10-16 | **Status:** FUTURE WORK ONLY

New UI research and clickable concept: [EXO Community + Pattern Lab premium UI/UX master](EXO_COMMUNITY_PATTERN_LAB_PREMIUM_UI_UX_MASTER_2026-10-16.md), [production design specification](EXO_COMMUNITY_PATTERN_LAB_PREMIUM_UI_UX_SYSTEM_2026-10-16.md) and [isolated HTML prototype](../concepts/exo-community-ui/index.html). This is not integrated with the native app or a deployed website.

New premium UX research and working isolated UI prototype: [EXO Community + Pattern Lab Design System and Build Plan](EXO_COMMUNITY_PATTERN_LAB_PREMIUM_UI_UX_MASTER_2026-10-16.md). Prototype source: [EXO Community/UI concept](../concepts/exo-community-ui/index.html), with public website, sample asset detail, embedded Pattern Lab mock, and sample Web Tech Pack revision comparison. The live EXO app and real community website are **not** modified/deployed.

Issue-level planning: [39 additional PPF/material/studio/community task cards](EXO_ECOSYSTEM_CANDIDATE_ENGINEERING_BACKLOG_2026-10-16.md), plus [48 spatial tasks](EXO_SPATIAL_STUDIO_EXECUTION_BACKLOG_2026-10-16.md) and [14 Web Tech Pack tasks](EXO_INTERACTIVE_WEB_TECHPACK_PLANNING_2026-10-16.md). None are filed or approved for development.

This is the **decision and execution layer** tying together four independent, technical research documents:

- [PPF physics, measured materials, lighting and immersive wardrobe studios](EXO_PPF_MATERIALS_IMMERSIVE_STUDIOS_FUTURE_ARCHITECTURE_2026-10-16.md).
- [Community creator profiles, pattern pool, remix marketplace and collaboration](EXO_COMMUNITY_MARKETPLACE_SOCIAL_COLLABORATION_ARCHITECTURE_2026-10-16.md).
- [Interactive 2D/3D Web Tech Pack and designer/factory revision approvals](EXO_INTERACTIVE_WEB_TECHPACK_PLANNING_2026-10-16.md).
- [Spatial/projected garment workshop and original ten-page implementation plan](EXO_SPATIAL_STUDIO_TEN_PAGE_IMPLEMENTATION_BLUEPRINT_2026-10-16.md) plus [the original future-concepts master](EXO_SPATIAL_PROJECTION_GARMENT_STUDIO_AND_AR_CONCEPTS_2026-10-16.md).

Scope lock: Nothing in this plan authorizes building or deploying a site, connecting accounts, publishing patterns, charging buyers, uploading work, enabling camera capture, modifying garments, running PPF jobs or altering the installed EXO app. All task IDs below are candidate issues, not filed or assigned issues.

A. Product map and central principle

EXO Studio remains the desktop/native authoring and simulation app. It owns editable garments, original patterns, 2D/3D source, live ClothCraft VBD, optional audited PPF high-fidelity jobs, physical textile test data, procedural/scanned PBR materials, and Blender-created studio presentation. Users retain local ownership of design files.

EXO Community is a separate social and marketplace website. Creators have profiles and private brain-like garment libraries; publish selected Looks as visual-only; optionally grant or sell a specific pattern, garment display mesh, material/PBR texture, manufacturing template, trim, studio scene, avatar or runway asset. Other users follow, curate, download, remix with explicit rights and enter walkable rooms of permitted Looks.

EXO Teams / Tech Packs is a secured workspace on the platform for people developing real garments with other designers, suppliers and manufacturers. The interactive spec includes CAD-linked POM tables, graded sizes, fabrics, trim/BOM, construction and sampling notes, reference photos, pinned comments and immutable manufacturer-specific releases. Factories may propose edits; they cannot silently overwrite a designer's approved source.

Three unequal levels of access must always remain separate:

1. **Can see** a public or team-approved sanitized rendered outfit.
2. **Can download** particular asset files under exact free or purchased license.
3. **Can edit/approve** private source geometry, manufacturing data or a team's working release.

One right never implies the other two. An authorized manufacturer may have full pack data while the creator's followers see only a fashion film or orbitable, reduced 3D view.

Example day-in-the-life

A user opens their private **Wardrobe Room**, sees all their owned or imported hoodie collections on mannequins, and clicks a garment. The inspector shows fabric weight and measured/estimated reliability, material textures, 2D/3D previews and two simulation badges (interactive preview vs PPF full-frame verified/rejected). From there they can enter a self-authored Concrete Studio, choose a camera/light and render photos. For a chosen garment the **Tech Pack** shows the approved size/spec, comments from Manufacturer A on sleeve construction and designer responses. Once ready, the user may publish **only a visual-only public Look** on their profile. A different item may publish a legally licensed **free or paid pattern** and receive remix citations. The hidden source, client list and factory chat remain private.

B. System decomposition and integration rules

\\\`mermaid

flowchart TB

subgraph Local["EXO Studio / local device"]

CAD["Editable pattern, seams, materials & revisions"]

LIVE["Interactive native VBD cloth + Hand"]

PPF["Isolated offline PPF contact jobs"]

RENDER["Blender masters / Three.js previews / local render"]

CAD --> LIVE

CAD --> PPF

CAD --> RENDER

PPF -->|"accepted or rejected + exact fixture"| RENDER

end

CAD -->|"explicit selected release/upload only"| PUB["Publisher + rights/permission gate"]

RENDER -->|"sanitized visual derivatives only"| PUB

PUB --> OBJ["Private source storage + safe public preview derivatives"]

subgraph Web["Separate EXO Community / Teams platform"]

ID["Identity / profiles / organizations / ACL"]

REG["Immutable versioned asset registry"]

SOCIAL["Explore / follow / collections / creator rooms"]

LIC["Free / paid entitlement + remix licenses"]

PACK["Interactive versioned Web Tech Pack"]

JOB["Quarantined conversion / thumbnail jobs"]

OBJ --> JOB

OBJ --> REG

ID --> REG

REG --> SOCIAL

REG --> LIC

REG --> PACK

end

\\\`

Boundary: the website never mounts the user's entire EXO workspace or raw native localhost Rust commands. EXO Studio initiates a narrow, authenticated, explicit artifact publication job, lists every file to be sent and the permissions, and receives a publish receipt with exact version/hash. A remote viewer cannot start native PPF runs or read local pattern vertices unless a separately authorized future service explicitly offers those operations through reviewed capabilities.

Four catalogs, one identity graph

- **Garment catalog:** source garment + exact variants/sizes + editable and display derivatives + fabrication readiness.
- **Materials catalog:** physical specimen profile separate from PBR visual texture/profile; each can carry independent licensing.
- **Studio catalog:** original editable Blender masters, downloadable studio recipes, optimized web scenes, cameras and lighting rigs.
- **Community catalog:** publication packages, author/team, free/paid rights, dependency/remix lineage, gallery rooms, reviews and collections.

Use stable UUIDs with immutable version IDs, origin and content hashes. The same garment source can support 15 different Looks without copying the original. A pattern listing can point to an authorized immutable production package while the creator's private working Tech Pack remains hidden.

C. Technical data contracts worth specifying before code

Domain	Required entity & IDs	Immutable versus mutable	Forbidden assumption
Design	Garment, PatternPiece, Edge, Seam, Revision	Release immutable; working branch revisioned	3D mesh implies editable CAD
Physics	MaterialPhysicalProfile, SolverRun, ColliderVersion, AcceptanceReport	Solver run snapshot + QA immutable	Solver completed means safe fit
Optics	MaterialVisualProfile, TextureSet, MapProvenance	Versioned texture source and render profile	Beautiful texture implies measured physics
Studio	StudioMaster, StudioRuntime, Look, RenderJob	Master protected, published Look frozen	Blender GLB retains entire lighting rig
Platform	Account, Profile, Organization, AssetVersion, PublicationPackage	Published asset version and license receipt immutable	Public preview means source download
Collaboration	ProjectACL, Comment, Proposal, Revision, FactoryRelease, Acknowledgment	Proposal audited; factory release immutable	Factory can edit accepted spec directly
Marketplace	Offer, Entitlement, Purchase, Settlement, Refund, LicenseGrant	Purchase receipt and delivered version immutable	Payment frontend flag grants download

Concept API examples (not active):

- \`POST /v1/publish/preview\` — preview selected files, metadata and rights that *would* be uploaded; no side effect.
- \`POST /v1/publish/commit\` — explicit creator confirmation; upload approved source/derivative with expected version and idempotency key.
- \`GET /v1/assets/{id}/versions/{v}/preview\` — sanitized permitted derivative, never underlying private source.
- \`POST /v1/remixes\` — validate immutable origin, all parent licenses, selected allowed derivative files and creator attribution.
- \`POST /v1/teams/{id}/techpacks/{pack}/proposals\` — propose structured field patch referencing parent release and stable comment anchor.
- \`POST /v1/techpacks/{pack}/releases\` — approver-only, revision-guarded immutable snapshot, generate exact export manifest.
- \`POST /v1/techpacks/{pack}/releases/{release}/acknowledgments\` — factory scoped to received release.
- \`POST /v1/orders/checkout\` and webhook settlement — price/entitlement calculated and verified server-side; no direct client-chosen prices.

Database/caching split: relational database for access, comments, product, asset versions, orders, factory releases and immutable audit; private object storage for source, public CDN for specifically generated display derivatives; explicit safe

upload/conversion workers; search index of only public metadata. The local app should not depend on an internet round trip to continue using its existing 2D CAD/simulator.

Capability model for visual-only publishing

The owner selects what is visible: rendered still, orbitable simplified GLB, optional branded studio and selected public fabric descriptions. The publish processor strips source patterns/grades, measurement tables, internal comments, PPF solver coefficients where private, texture EXIF/hidden paths, glTF extras and unpublished other people's IP. Private source files get independent storage keys never transmitted in public responses. Server ACL and signed asset access controls are tested per request. 'Hide Download' UI alone does not secure a file already sent to the browser.

D. Hybrid physics and trustworthy fabric status

Interactive engine: native ClothCraft VBD/GPU VBD supports hand dragging and fast visual response in user workflow. It is appropriate for temporary constraints, stop/reset and room look previews once cached. Its nine native materials are not a completed, lab-calibrated apparel physics library.

High-fidelity contact engine: PPF is a separate asynchronous Rust/Metal batch backend with independently checked trajectory. Its existing EXO run history contains real body crossings and excessive seam gaps; baseline acceptance explicitly failed. Next work must prioritize sound initial sewing assembly, static-vs-pinned body A/B, material conversion and independent full-frame plus swept contact tests. A green 'Fit Verified' badge must not appear on a solver merely finishing with finite output.

Material model: a cloth's **physical** sheet mass, thickness, warp/weft/bias stretch, shear, bending, friction pairs, hysteresis and compression are a measured specimen or a flagged estimate. **Visual** albedo, normal, roughness, textile sheen, anisotropy, UV repeat/grain and clearcoat are independent rendering assets. Show both:

- Physical: \`Measured\` / \`Calibrated to solver X\` / \`Estimated preset\` / \`Unknown\`.
- Visual: \`Scanned, scale-checked\` / \`Procedural\` / \`Uploaded, unverified\`.
- Fit: \`Visual preview\` / \`Solver ran, rejected\` / \`Accepted for reference fixture\` / \`External garment validation\`.

User can select a material under a local hoodie's panels and instantly see woven/knit optics, but to claim credible drape, material coefficients must be fit from the actual sample using known experimental loading, documented solver mappings and same reference garment. Tie source specimen to PBR texture rights only if creator licensed both.

Solving and cost: production site starts with uploaded previews and already cached drape. No anonymous, user-uploaded mesh automatically triggers PPF Metal jobs. Future compute processing needs queuing, per-user quotas, memory/time constraints, cancellation, malware/file type checks and acceptance reports; schedule and charges must be explicitly approved.

E. Virtual wardrobe room, photography and web viewer

Local EXO Studio provides the primary private **walkable closet/gallery**, with original Blender stages such as White Infinity, Gray Fabric Lab, Concrete Gallery, Architectural Archive, Black Void, Muslin, Runway and Outdoor. The user can navigate WASD/mouse or teleport, choose mannequins and cameras, swap garments, walk into a photography studio and export a local Look. Multiple sizes/colorways and saved avatar poses should feel like an organized fashion collection rather than a filesystem.

Web EXO Community reuses **only licensed/sanitized published Look derivatives**, optimizes GLB and KTX2 textures, lazy-loads each showroom, restricts private content, and offers accessible 2D galleries. A public room should never reference unapproved private materials/patterns. Users may download or license original scene assets only when that package is explicitly offered separately.

Performance plan: 1 room/10 avatars first, then 20 and several rooms; scene LOD, texture streaming, renderer disposal, shader cache, instanced decor and navigation colliders. Measured FPS and memory on target desktop/mobile matter more than a scene screenshot. Never simulate all outfits in PPF while browsing; use cached drape/visual meshes.

Photography: three original Blender masters before dozens of presets. One neutral e-commerce and one fabric lab for truthful textile checks; a creative Concrete/Gallery set for campaigns. Each Look records garment and asset hashes, avatar rig, pose, lighting, camera, tone mapping and license. A WebGL scene must recreate its own light rig because Blender World and area lights are not faithfully carried by a generic GLB export. Premium offline renders remain a separate local Blender job.

F. Interactive tech packs and manufacturer collaboration as independent product

The web Tech Pack should feel like a **living garment specification website**, not just an embedded PDF. It has tabbed 2D/3D technical views, size POM tables with method/tolerance, material BOM, cost-restricted sections, construction lines,

hardware and zipper references, fit photos, sampling rounds, per-component comments and revision timeline.

The designer owns a Draft branch with private formula, cost and sourcing information. An internal reviewer may annotate actual panel, seam, 3D triangle, measurement row or image. The designer explicitly makes immutable **Release R01** with linked EXO source revision and export hashes. Manufacturer A receives only the approved sections/files and can annotate, upload sample measurements/photos, request alternates and acknowledge. A request to change the cuff width becomes **a proposal**, not a silent edit. The designer creates Draft 02, resolves comments, tests pattern/tech pack, approves **Release R02** and the factory sees exactly what changed. The platform remembers R01 as superseded evidence.

Existing EXO source already generates `\tech-pack.json\`, `\factory-spec.html\`, `\pom-spec.csv\`, material/BOM, trims/hardware and DXF export in original Garment Studio; this is a schema/asset source, **not an interactive online techpack feature yet**. The website should ingest typed metadata, not derive production specs from screenshot OCR. CLO-SET and Techpacker currently provide relevant 3D tech packs, 3D annotations, revisions and factory comments; EXO's differentiator could be linking every item to native pattern/sewing IDs, formally approved immutable manufacturing releases and a community view-only boundary.

No cross-tenant leakage: Manufacturer B may not see A's unit prices, quotes, comments or approved private patterns, even if both collaborate with the same brand. Public followers and marketplace buyers should not receive draft manufacturing notes unless the creator intentionally packages them under a separate license.

G. Creator marketplace and remix network

Public profiles: username/cover/bio, favorite Looks, authored garments, studio rooms, followers/following, free patterns, shop items, licensed remixes, collections and optional verified brand/team account. Private 'My Brain Library' remains separate: drafts, all pattern iterations, measurements, hidden notes, visual references, unfinished design and purchased assets.

Publishing classes: private, unlisted specifically granted, public visual-only, free licensed download, remixable project, paid downloadable asset, team workspace or approved factory release. A listing may have multiple *different* permissioned packages (public still and purchasable DXF; downloadable scene but private original .blend).

Free mass pattern pool: structured size metadata, format validation, author/source license, production-readiness flag, description/preview and download receipts; filters by garment type, style, size, file format and permitted remix/commercial usage. Creators can upload free work without losing ownership. Paid pattern downloads add entitlement, seller onboarding and refund/chargeback handling.

Attribution: every fork tracks exact parent asset revision and rights. Terms distinguish personal use, commercial sewing/production, derivative remixes, source redistribution and promotional use. A studio may contain third-party HDRIs whose original owners do not permit redistributing texture binary files. A contributor royalty graph is a future legally reviewed capability, not something safely inferred merely from asset dependencies.

Social: follow, like, save, comment, remix request, personalized/chronological feed, creator room visit, shared collections, mute/block/report, and moderation. A view-only author can opt out of comments and downloadable derivatives. Keep chronological option for discovery and audit recommendation quality before popularity loops.

Team working: organization roles and per-project ACL, direct garment sharing, invitations/revocations, pinned comments and review queue. A future CRDT may safely merge text/cursor presence, but simultaneous CAD point edits, critical BOM quantities and approved versions need revision guard and conflict review.

Commerce/security: quarantine untrusted uploaded files; never execute arbitrary .blend, model scripts or ZIP payloads on the server unsandboxed. Payments must be verified via provider webhooks, not client flags. Follow seller identity verification, taxes/reporting, platform liability and licensing advice with counsel before launching paid commerce. U.S. copyright and other laws differ by jurisdiction and asset type; don't assume all patterns or designs have identical rights.

H. Program sequence with specific exit gates

P0 — Plan, rights and canonical geometry

Owner: product/EXO architect, security, apparel technologist. **Deliverables:** sample garments and hashed source fixtures, source-of-truth decision, units, license/consent policy, cloud/compute threat model. **Gate:** user confirms intended prototype and publishing rights; no existing private project automatically uploaded.

P1 — Local material provenance + PPF research

Owner: native/physics + textiles. **Deliverables:** unified material measurements/version scheme, typed solver adapters, failing historic PPF fixture, controlled contact A/B test plan, identical-swatch render test. **Gate:** invalid physical simulation remains rejected, no claimed full fit until tests pass.

P2 — Studio library and first walkable room

Owner: Blender/look-dev, graphics. **Deliverables:** three original studios, versioned Look metadata, 10 mannequins/looks in private walkable hall, accessible list fallback, material inspector. **Gate:** original garment hashes unchanged, no licensing violations, target FPS/LOD measured.

P3 — Community core, creator profiles and safe read-only previews

Owner: web/data/security. **Deliverables:** accounts, profiles, private/public asset boundary, source-safe upload, sanitized preview and creator libraries. **Gate:** cross-account negative ACL tests and malicious file rejection; published preview has no private CAD/techpack bytes.

P4 — Free downloads, remix licensing and collection graph

Owner: platform/rights. **Deliverables:** free asset pages, immutable versions, licenses/receipts, search/follows, source dependency graphs and format readout. **Gate:** unlicensed remix blocked and granted download package hashed.

P5 — Team workflow + read-only interactive Web Tech Pack

Owner: EXO interoperability, web full-stack. **Deliverables:** garment/POM/BOM/technical view linked to typed native exports, private factory role and annotated comments. **Gate:** Manufacturer A cannot see B/private fields; measurement units and size variants match known EXO fixture.

P6 — Draft/release workflow + sampling

Owner: apparel specialist, backend/security. **Deliverables:** structured change requests, sample photos/measurements, version diff, approver-signed R01/R02 and acknowledgment, historical export hashes. **Gate:** released specs immutable and diff reproducible, access revocable.

P7 — Paid marketplace and seller operations

Owner: commerce/legal/security. **Deliverables:** seller accounts, payment provider integration, order entitlements, refunds, payout/tax controls, moderation and support operations. **Gate:** unauthorized download denied, chargeback handled according to defined rights, financial audit trail retained.

P8 — Public creator rooms and advanced collaboration

Owner: realtime/graphics/privacy. **Deliverables:** public walkable creator showcases with only licensed derivatives, optional shared view/presence, optimized 3D rooms and accessibility. **Gate:** room performance, permissions and moderation at real expected traffic.

P9 — Optional advanced PPF, royalties and AR

Owner: research/physics/commercial. **Deliverables:** validated PPF high-quality processing only after physical contact acceptance, legally reviewed multi-author revenue shares, separately scoped AR try-on/fit research. **Gate:** never present unvalidated output or noncommercial research model as factory-grade or unrestricted commercial product.

Prioritization: working slices, not dependency lock-in

Do **not** require PPF to pass to launch a local visual studio. Do **not** require the paid store to provide a free community pool. Do **not** require expensive 3D rooms before simple creator galleries and download pages. Tech Pack collaboration may ship for a private small brand/workshop before a public social launch, if permissions/version contracts are ready. Keep authorization for these as separate decisions.

I. Early integrated test matrix

Acceptance fixture	Required evidence	Failure behavior
220 GSM jersey	0.220 kg/m² canonical mass; real swatch provenance; texture not treated as physics	Reject inconsistent units and clearly label estimates
Hoodie PPF result	Full-frame seam gaps, strain, independent body/self crossings and swept report	Rejected output remains visible as diagnostic, not a verified fit
White/Concrete/Black studios	Same accepted Look, original .blend scene and camera/light manifest	Missing maps/lighting flag; sources not overwritten
Wardrobe Hall	Ten original private Looks, walk/teleport, click-to-inspect, list fallback	Do not leak other accounts' assets

Acceptance fixture	Required evidence	Failure behavior
Visual-only portfolio page	Sanitized GLB; private POM, pattern rules and tech pack excluded from all network bytes	Privacy test fails release
Free remixable pattern	Explicit license, source hash, size/format metadata, parent credit	Deny fork if licence doesn't allow
Paid pattern	Buyer entitled to exact immutable paid package, no unauthorized direct object-store access	Reject and audit unsigned/non-entitled request
Manufacturer review	A sees only approved R01 fields, comments pinned to cuff POM	Cross-factory access denied
Revised sample R02	Immutable R01+R02 with field-level diff, actor, acknowledgment	Old release cannot be overwritten
Security scanning	Untrusted ZIP/GLB/.blend processed with sandbox and quotas	Reject/hold without exposing filesystem

J. Architecture and research references

- Existing [EXO source-to-factory artifacts](EXO_PRODUCTION_PIPELINE.md).
- [PPF actual garment collision failure audit](PPF_ACTUAL_GARMENT_FIT_DEEP_AUDIT_2026-10-14.md).
- [Fabric optics, physical and PBR authoring source](EXO_FABRIC_MATERIALS_AND_TRIMS.md).
- [Interactive Web Tech Pack details](EXO_INTERACTIVE_WEB_TECHPACK_PLANNING_2026-10-16.md).
- CLO-SET Web Tech Pack Construction; 3D annotations with Web Tech Pack.
- Techpacker maker/factory versioned sharing.
- CLO-SET CONNECT free/paid store; component creator design-source policy.
- Khronos glTF sheen; material anisotropy.
- Yjs collaborative shared types; OWASP upload safety.
- Stripe Connect marketplace settlements; Three.js PointerLockControls.

No-build decision: This ecosystem program is a concept and implementation plan only. No EXO source code, system configuration, public account, browser session, payment connection or garment files were changed by this document.

EXO_INTERACTIVE_WEB_TECHPACK_PLANNING_2026-10-16.md

Audit interpretation: This annex preserves earlier engineering research for traceability. Read reported pass counts and statuses as of its original date only. The current 70-slot and 400-fabric inventories above supersede any older source totals.

EXO WEB TECH PACK — LIVING PRODUCT SPECIFICATION, REVISIONS & FACTORY COLLABORATION

Future concept research and implementation plan | 16 October 2026 | No build authorized

Program navigation: [Integrated ecosystem master architecture](EXO_ECOSYSTEM_MASTER_IMPLEMENTATION_PROGRAM_2026-10-16.md), [shared platform/physics/material candidate backlog](EXO_ECOSYSTEM_CANDIDATE_ENGINEERING_BACKLOG_2026-10-16.md) and [community creator marketplace](EXO_COMMUNITY_MARKETPLACE_SOCIAL_COLLABORATION_ARCHITECTURE_2026-10-16.md).

Vision: transform the static 2D paper/PDF tech pack into a secure **interactive garment product website**. A designer and manufacturer inspect an approved 2D/3D garment, measurements, size grading, trims, fabrics, construction instructions, reference images and sample review threads in one linked workspace. Each update is documented, compared, accepted or rejected, and attributed to the correct person. The original factory export remains available as a **versioned fallback**, but the living web specification becomes the collaborative source for future product development.

Cross-references: [EXO marketplace and collaboration architecture](EXO_COMMUNITY_MARKETPLACE_SOCIAL_COLLABORATION_ARCHITECTURE_2026-10-16.md), [EXO physical/visual garment research](EXO_PPF_MATERIALS_IMMERSIVE_STUDIOS_FUTURE_ARCHITECTURE_2026-10-16.md), [EXO production pipeline](EXO_PRODUCTION_PIPELINE.md), [technical pattern/manufacturing parity plan](GARMENT_STUDIO_COMPLETE_CLO3D_PARITY_MASTER_PLAN_2026-10-13.md).

1. Target experiences and views

Designer Workspace: full editable product source, technical drawings, formulas, grading, fabrics and suppliers, costs, confidential sample notes, design inspirations, internal targets, restricted assets, manufacturing release tools, private team tasks and proposal merges.

Manufacturer Workspace: the specific approved release and selected fields explicitly shared with this factory: measurement/size charts and tolerances, pattern package when authorized, construction callouts, BOM material identifiers, approved colors, trim placement, QA expectations, reference photos and sample comments. Manufacturer can flag manufacturing concerns, attach production photos, propose alternate stitch/materials and request clarification. Changes are **proposed**, not immediately authoritative.

Public/Portfolio View: only a separately published visual outfit presentation and any intentionally public facts. No internal POM rows, proprietary measurement formulas, construction recipes, raw pattern coordinates, vendor costs or factory messages should be sent to browser clients from this view.

Buyer/License View: if a designer sells a downloadable garment pattern or tech pack, show only the purchased licensed package/version; a marketplace purchase does not automatically grant access to the internal collaborative factory workflow.

User flow: hoodie sample review

1. Designer creates a hoodie in EXO Studio, links fit avatar, fabric profile, variant/size run and a visual garment preview.
2. Designer opens **Web Tech Pack > Draft 04**, adds chest/waist/sleeve length tables, tolerance notes, grading, construction stitches, padded-hood details, hardware BOM and reference images.
3. Internal patternmaker and brand lead review with annotations pinned to 2D panel edges and 3D shoulder/hood geometry.
4. Designer creates immutable **Factory Release R01** from a specified garment source revision, POM, BOM and construction snapshot. Release shows creation time, author, status and full hash.
5. Assigned Manufacturer A receives a secure project-scoped link to R01, can see permitted drawings and download only approved pattern/export attachments.
6. Factory attaches a photo of Sample 1 and comments on the sleeve cuff row: “The opening is 1.2 cm too wide relative to R01 tolerance.” Designer gets an assigned issue linked to the specific size and POM ID.

7. Designer edits a new Draft 05, adjusts the point of measure, updates tech notes, checks the pattern source and asks for review.
8. Diff screen compares R01 and proposed R02: updated cuff dimensions across sizes, fabric and stitch spec unchanged, revised reference photo added, unresolved comments flagged.
9. Approved R02 is issued; Manufacturer A must acknowledge receipt. R01 remains accessible as superseded evidence rather than silently changing.
10. Subsequent Sample 2 feedback attaches to R02, not to an undefined 'latest' floating version.

Result: creators and factories can reconstruct exactly what was approved, when, by whom, with which garment/measurement files. A screenshot and PDF of each release can be generated for offline production.

2. Browser layout and content panels

Proposed route /teams/{team}/styles/{style}/techpacks/{pack}/revisions/{revision}; public routes never mirror this private path unless specifically approved.

Top: style name, development season, SKU, colorway, current status (Draft / In Review / Changes Requested / Approved / Released / Superseded / Archived), active source document and release version, factory partner, language and updated timestamp.

Left contents: Overview, Moodboard/References, Technical Flat, 3D Garment, POM & Sizes, Grade Rules, Fabric/Material, Trims/Hardware, Sewing & Construction, Colorways/Artwork, Labels & Packaging, BOM & Costing, Sampling, QC/Approvals, Files, Revision History.

Center: selectable 2D flat diagrams and 3D garment with annotations. Manufacturer can click zipper, seam, printed graphic or edge and read only permitted technical callouts. Images can be zoomed, layered and commented upon. Side-by-side visual for original vs revised garment at matched camera and pose. Simple mobile read-only layout with anchored comment cards when canvas is too small.

Right: properties/measurements and contextual comments, task assignee, linked sample photos, approval state, discussion history, change proposals, mentions and files. Pinned comments should have resolved/open filters and mention notification controls.

Footer / sidebar: status changes, upcoming sampling dates, references to approved PDF/export ZIP and audit history; keyboard-accessible comment navigation.

Version comparison: per-field colored diff at category level, including centimeters/millimeters with tolerance, size chart modifications, color/material changes, added/removed pattern pieces, stitch/fastener changes, images, and approval notes. Some 3D geometry differences cannot be represented as simple numeric values; provide 2D contour/3D ghost overlays in matching coordinate frames plus a changed-panel list.

3. Technical source data reused from EXO

Existing EXO already generates useful manufacturing artifacts: src/core/measurements.js builds a structured Tech Pack object; src/core/factoryPackage.js generates tech-pack.json, pom-spec.csv, cost/factory data and export ZIP; native_cpp/core/tech_pack_export.cpp and native_cpp/core/factory_package_files.cpp offer a parallel native path including factory-spec.html, materials-bom.csv, construction-seams.csv and trims-hardware.json. These are **existing export paths**, not evidence of an operational collaborative Tech Pack website.

The future web schema should import stable EXO IDs and original source revision; it should not parse PDF text to recover manufacturing facts. Accept a semantic schema version, SI canonical measures, display cm/mm/in conversions, sizes and grading metadata, and links to visual read-only source previews. If some EXO fields are missing, show **not supplied** rather than fabricating a value.

Version data types (proposal)

- TechPack: packId, garmentId, teamId, ownerId, currentWorkingDraft, releasedVersionIds, factoryScopes, productStatus.
- TechPackRevision: revisionId, parentRevisionId, sourceGarmentRevision, contentHash, actor, createdAt, status, schemaVersion, reviewPolicy, fieldChangeSet.
- SectionSnapshot: refImage entries, drawings, POM table, fabric/profile references, BOM, trims, artwork, grading, construction, labeling, packaging, qcNotes, supplier-facing instructions.
- SpecLine: stable lineId, category, pointOfMeasure, description, sizeDimensionValues, baseSize, toleranceUpper/Lower, methodOfMeasurement, units, validationStatus.
- Attachment: fileId, source type, uploader, version, access classification, malware scan state, external provenance/license, linked entity ID.

- Sample: factory, sampleRound, size/colorway, shipped/received dates, measurement report, photo set, problems and resolution state.
- Proposal: proposed patch with expectedRevision and target IDs, factory notes, priority, affected variants, evidence and review decision.
- Release: immutable snapshot hash, approver IDs and time, approved buyer/factory audience, signature/acknowledgment, supersedes, export manifest.
- Comment: threadId, stable anchor object, author, mentions, createdAt, visibility scope, status, assignedTo, edit history and evidence.

Sample state graph: Draft → Internal Review → Approved → Released; Released may become Superseded only when another approved release is issued. Rejected proposals and unresolved comments remain historical records. Edits to a released version must never occur in-place; create a new draft based on its snapshot.

4. Designer vs manufacturer view: security is mandatory

Access is controlled server-side per team, project, release, actor role and exact file, not just CSS tabs. Designer can see internal cost assumptions and factory quotes separately; Manufacturer A sees only their explicitly approved release and related discussion, not Manufacturer B's pricing, drafts or internal creative comments. A guest link may be time-limited and restricted to a specific released revision; “unlisted” alone is not sufficiently safe.

Review roles: Owner / Designer / Patternmaker / Approver / Manufacturer / QC / Guest. Permissions include view_public_preview, read_private_working_draft, modify_design_spec, propose_factory_change, comment, upload_sample_evidence, approve_internal, release_to_factory, factory_acknowledge, download_pattern_package and access_costs. Assign grants at project and release scope, and explicitly deny grants by default.

Private mode: a creator can still publish the outfit visually to the public social site, while hiding full measurements, pattern geometry, PPF parameters and the whole tech pack. This requires a sanitized visual-only package and no unapproved technical metadata in the GLB, network response or public image alt text.

5. Comments tied to stable 2D and 3D anchors

2D anchor: reference garment sourceVersion, panelId, curveId/edgeId/pointId, local (u,v)/distance along edge plus fallback world-space XY in canonical units, visibility, and viewport transform version. Best anchor is a stable logical edge or measurement ID rather than raw pixel location.

3D anchor: mesh asset version, triangle ID + barycentric weights + optional surface normal and garment panel ID. A physically simulated mesh can be remeshed/reindexed: prefer logical source panel/feature identifier; if remapping fails, mark **anchor needs relinking** and keep the original screenshot so the remark does not jump to an unrelated vertex.

Measurement anchor: pointOfMeasure stable ID + size/colorway and field path. **BOM anchor:** line item ID and vendor/material row. **Image anchor:** stable image ID and normalized [0..1] XY. **Workflow anchor:** sampling stage, date, release version or request ID.

Each thread should support reply, mention, assign, resolve/reopen, attachment, notify participants and immutable edit/audit records. A factory cannot rewrite earlier acknowledged statements or approved specs without an audit trail. A comment edit may be allowed with separate original and new text records under retention rules.

Collaboration technology choice

Yjs/CRDT can handle concurrent rich-text notes and ephemeral presence, but measurement tables, BOM quantities, source pattern geometry and release approval must use validated server transactions and expectedRevision locking. Two people entering different numbers for the same cuff width cannot safely “auto-merge” into a new production value. Show a conflict with both proposals, not silent last-write-wins. Offline comments sync only after server ACL and released revision are revalidated.

6. Data correctness: BOM, size runs and reference images

POM changes need cross-size constraints, e.g. allowed tolerance by size and method of measurement, not simply free-form paragraphs. The draft editor shows cm/in preview conversion but stores one canonical unit. Compare original design measurements with *actual pattern or 3D derived measured values* only if those measurements are robustly extracted. Display separate columns for target spec, tested sample actual, deviation and approve/rework decision.

BOM must preserve supplier, material ID, supplier SKU, composition, estimated usage/quantity by garment/size, unit price, effective date, colorway restrictions and substitutes. Public garment pages never reveal confidential supplier/unit costs unless approved. Production export source XML/CSV/JSON fields must correspond to displayed release, with hashes to prevent ambiguity.

Reference imagery is not merely a moodboard decoration. Store target detail (e.g. cuff shape, topstitch, lining, wash outcome), uploader, rights, size-specific scope, review status, and linkage to the relevant panel, POM, BOM or construction line. Present images alongside the actual accepted spec; a factory sees whether an image is “inspiration,” “illustrative non-binding” or “approved manufacturing reference.”

A Technical Flat is distinct from a beauty render. It must show authorized seam construction, grain/curve references and technical dimensions, not claim production correctness based on the photo scene. An external downloaded pattern's license may permit viewing but not factory mass production; attach license terms to each approved package.

7. Manufacturing workflow and sampling iteration

Dev Stage: concept → first pattern → proto sample → fit sample → size set → preproduction sample → production release → shipment QC. These are configurable labels per manufacturer; the product must not invent a globally mandatory supply-chain process.

At each sample round the factory can upload sample photos, POM actuals, workmanship findings, production feasibility, material alternates and shipping evidence. The designer reviews deviations and either accepts-with-reason, requests rework, or drafts a formal spec change. A sample approval must name size/colorway, fabric lot where relevant, inspector and exact release hash. It cannot blanket-approve later unreviewed size changes.

Change Request records target revision, rationale, proposed field diff, affected BOM/POM/pattern list, cost/schedule impact estimate, manufacturer discussion and final decision. Approved requests create or update a **new draft**; the designer publishes a new immutable release after review. Manufacturers acknowledge R01→R02 and the site shows that R01 is superseded. For factories offline, distribute a release-tagged PDF/ZIP with a QR/link/reference ID; that export never becomes a mutable “latest.pdf” with no snapshot.

8. Feature-level engineering work packages

Task	Dependencies	Build specification	Completion proof
TP-001 Define TechPack/Revision schema	EXO source inventory	Stable IDs, variant/size, line-level units, release hashes	Semantic version migration and fixture round-trip
TP-002 Authoring-source adapter	TP-001	Import existing tech-pack.json, BOM/POM and native patterns read-only	Same known garment and sizes in web view
TP-003 Visual 2D/3D viewer	TP-002	Sanitized geometry, technical flat and optional 3D garment panels	Public vs factory rights validated
TP-004 POM & measurement editor	TP-001,002	Canonical mm/cm, tolerances, grade rules and sample actuals	Unit/size invariants; conflicting changes rejected
TP-005 BOM & materials editor	TP-001,002	Variant-specific fabric/trim item refs, supplier annotations, costing ACL	No price/private data leaked to external view
TP-006 Reference image and sample round	TP-001	Secure image uploader, linked callouts, sample metadata and status	No unapproved attachments or leaks
TP-007 Anchored comments & tasks	TP-003–006	Comments on stable POM, panel, vertex, BOM, image and stage IDs	Persist across revisions or become explicit orphan
TP-008 Role-based factory portal	TP-001,007	Secure manufacturer accounts, scoped release viewer and proposal rights	Manufacturer A cannot access B or design private notes
TP-009 Version diff & immutable release	TP-004–008	Snapshot, compare, approve, supersede and acknowledge	Two releases reconstructable bit-for-bit
TP-010 PDF/ZIP fallback export	TP-009	Numbered snapshot outputs, downloadable under ACL, hash + timestamp	Output matches approved fields, historical export unchanged
TP-011 Notifications/mentions	TP-007,008	Team/factory mentions, review reminders, digests and mute	Scope-correct notifications without confidential excerpts
TP-012 Mobile & accessibility QA	TP-003–011	Read-only mobile, keyboard comments, color contrast and zoom	Independent actor/capability tests
TP-013 Integration with social portfolio	TP-008,009	Publish only explicit visual-only derivative from pack	Public cannot fetch POM/BOM/private file bytes
TP-014 Audit, rollback & security	all	Append-only event trail, ACL regression, restore and import conflict	Every approved change attributed; no silent overwrite

Proposed sequencing: TP-001–003 (read-only spec page) → TP-004–007 (working review room) → TP-008–010 (factory-controlled release) → TP-011–014 (safe notifications/public integration and reliability). Do not start public factory

invitation until the ACL and immutable snapshot engine passes repeated tests.

Qualitative staffing and effort

Dedicated backend/security, EXO CAD schema expert, UI/3D specialist and production-apparel reviewer. A basic read-only Tech Pack linked to an existing design may be a several-week prototype; manufacturer-grade permissions, concurrent reviews, versioned releases, audit and robust exported factory files are a multi-month independent product effort, dependent on schema and review cycles. **No approved delivery date or spend is implied.**

9. Proposed web screen sketches (textual)

Designer: header "HOODIE 001 · Draft 05 · Awaiting Pattern Review"; subnav "3D / 2D / Measurements / Fabrics / Construction / Samples / Versions"; left technical drawing; right open issue "Cuff POM #C17 Sample1 +1.2cm" assigned to designer; inline side-by-side R01→Draft05, private cost table and button "Request approval" (not automatically release).

Manufacturer: header "HOODIE 001 · Approved R01 · Acknowledge Revision"; center construction drawing and approved size chart; right sections "Factory Questions / Upload Sample / Propose Change"; R02 notice only after approved; no source formula edit controls, private design notes, or internal cost fields.

Public: header "HOODIE 001 by Creator"; 3D orbit, stage photo, fabric description and allowed public notes; optional Buy/Download Pattern if separately licensed. No techpack version/internal notes or factory assets transmitted.

10. Prior art: what existing products prove

Techpacker publicly advertises digital card-based tech packs, real-time comments, teamwork and version comparison. Its help pages document versioned sharing to manufacturers and note read-only factory functions including commenting and attaching files. Backbone documents product feedback revisions and pulling fit/sample reports into tech packs. This validates **specific feature categories**, not a claim that all competitors have the exact EXO 3D-native data/approval model.

EXO should differentiate by a first-class **native garment-to-TechPack link** (stable 2D/3D garment IDs), comment anchoring in the actual cloth scene, exact version/size diff, independently approved factory snapshots, 3D rooms and creator-public storefront privacy separation.

Additional directly relevant CLO-SET benchmark (verified 2026-10-16): CLO-SET's Web Tech Pack Construction documentation describes an online tech pack populated from a garment's 3D content, data synchronized from CLO3D BOM, editable construction markers and annotation of materials, trims and sewing methods. It notes that the Construction pane's support differs between its legacy and newer content-page views; do not infer all accounts/products expose the same menu. Its 2025 annotation update describes selecting pattern, fabric or trim components in a 3D Tech Pack pane and creating linked 3D annotations. **This proves the 3D data-linked approach is commercially relevant, not that EXO can copy CLO proprietary source or inherits CLO cloud features automatically.**

Differentiation gates for EXO: (1) the browser never uses PDF extraction as the authoritative POM/BOM source; (2) comment anchors carry stable EXO pattern feature IDs and a mesh revision rather than only pixels; (3) factories read exact approved release Rnn and must acknowledge changes; (4) the system can publish a visual-only social Look without delivering private techpack/manufacturing records; (5) permitted downloads and pattern licensing are separately recorded from manufacturer collaboration grants. Competitors may already provide portions of these; validate actual release/version before making comparison claims.

Research:

- Techpacker current feature overview: https://techpacker.io/
- Techpacker secure factory sharing and versions: https://helpcenter.techpacker.com/hc/en-us/articles/360022720553-How-to-share-a-Tech-pack-with-Manufacturers
- Techpacker comments: https://helpcenter.techpacker.com/hc/en-us/articles/360017617214-How-to-communicate-with-team-members-and-manufacturers
- Techpacker factory role behavior: https://documentation.techpacker.com/techpack/communicate/share.html
- Backbone product revisions: https://intercom.help/backbone-help-center/en/articles/5803263-product-feedback-revisions
- Yjs shared text for concurrent notes: https://docs.yjs.dev/getting-started/working-with-shared-types
- CLO-SET Web Tech Pack Construction and BOM integration: https://support.clo-set.com/hc/en-us/articles/45303219780377-Web-Tech-Pack-Construction
- CLO-SET Annotation and 3D Tech Pack: https://support.clo-set.com/hc/en-us/articles/45303128139673--2025-04-Updates-in-CLO-SET-Annotation-work-with-3D-Tech-Pack

Decision: Treat the Web Tech Pack as a collaborative, permissioned product development room with machine-readable source data, immutable release approvals and linked manufacturing evidence; static PDF becomes one export of a frozen

version, not the sole collaboration interface. Nothing is built or published by this document.

EXO_COMMUNITY_MARKETPLACE_SOCIAL_COLLABORATION_ARCHITECTURE_2026-10-16.md

Audit interpretation: This annex preserves earlier engineering research for traceability. Read reported pass counts and statuses as of its original date only. The current 70-slot and 400-fabric inventories above supersede any older source totals.

EXO COMMUNITY — PROFILES, GARMENT PATTERN POOL, CREATOR MARKETPLACE & COLLABORATION

Product + platform architecture research | 2026-10-16 | Concept only, no implementation

User experience extension: [Premium web Community, in-editor Pattern Lab, and sample Web Tech Pack design system](EXO_COMMUNITY_PATTERN_LAB_PREMIUM_UI_UX_MASTER_2026-10-16.md). Isolated clickable prototype: [Website and Pattern Lab](../concepts/exo-community-ui/index.html). Demo-only rights and import states; no production service is enabled.

Program navigation: [Integrated ecosystem implementation program](EXO_ECOSYSTEM_MASTER_IMPLEMENTATION_PROGRAM_2026-10-16.md) and [creator marketplace / team candidate engineering issues](EXO_ECOSYSTEM_CANDIDATE_ENGINEERING_BACKLOG_2026-10-16.md).

Product direction. Build an independent consumer-facing EXO Community website, connected to EXO Studio via explicit user-controlled publish and download workflows. The site is a Sketchfab-like fashion network: people have maker profiles, outfits, patterns, fabrics, sewing assets, rigged avatars, Blender studios and fashion show environments. Others can browse a huge free pool, download legally allowed assets, pay for others, remix with correct attribution, collaborate with teams, follow designers and visit walkable virtual fashion spaces.

Protect creator agency: Public visibility of a beautiful garment is not consent to share editable CAD patterns, grading rules, a manufacturing tech pack, raw measured fabric data, private team discussions or full-resolution meshes. Uploaded source, presentation derivatives, approved downloads, team-only working versions and public profile entries must be different storage/security concepts.

Related: [EXO PPF/materials/rooms research](EXO_PPF_MATERIALS_IMMERSIVE_STUDIOS_FUTURE_ARCHITECTURE_2026-10-16.md), [spatial blueprint](EXO_SPATIAL_STUDIO_TEN_PAGE_IMPLEMENTATION_BLUEPRINT_2026-10-16.md), [execution backlog](EXO_SPATIAL_STUDIO_EXECUTION_BACKLOG_2026-10-16.md), [interactive web tech pack future](EXO_INTERACTIVE_WEB_TECHPACK_PLANNING_2026-10-16.md).

1. Two products, one creator identity

EXO Studio is the local native/browser garment editor: editable 2D patterns, dimensions, native/PPF simulations, material authoring, 3D avatars, local pattern manufacturing export, staged photography and immersive wardrobe. Production data stays local until expressly uploaded or connected.

EXO Community is the website: public/private creator pages, a 3D viewer, free pattern pool, commercial marketplace, remix tree, likes/follows/collections, creator rooms and collaborative team workrooms. It only receives artifacts the creator chooses to publish or share, never blanket access to local disk.

EXO Teams / Product Development adds organization projects, separate permissions, comments, review/approval and interactive web tech packs. A brand may invite manufacturers to review an approved specification without offering that proprietary source to random followers.

Suggested website navigation:

- Discover: personalized but controllable chronological/search feed, categories, latest public garment looks, pattern pool and trending studios.
- Patterns: free/open and paid CAD/pattern listings with formats, sizes, number of pieces, seam-allowance status and compatibility notes.
- Marketplace: Garments, Patterns, Fabrics/Materials, Trims/Hardware, Avatars, Studios/Scenes, Runways, Collections, Tools/Templates.
- Profiles: creator's bio, collections, visual garment wall, followers/following, remix credits, stores, work published, external links and badges.
- Rooms: immersive private/public outfit museums from verified published Look assets.
- Teams: workspaces, review queues, assigned tasks, secure files, comments and live Tech Packs.

- My Library: owned assets, downloaded licenses, saved patterns, purchased items, remixes and subscriptions.
- Upload/Publish: guided wizard with private content selection, license, copyright, preview, scan and final publish confirmation.

2. Models of publication: never treat public as downloadable

A product/creator entry can expose different packages:

Publish intent	Public can see	Public receives	Ownership constraint
Private project	Nothing	No network response	Owner/team only
Unlisted proof/review	Selected preview only, gated link	No download unless granted	Token scoped and revocable
Public visual portfolio	Thumbnails, filtered 3D display, permitted notes	No editable pattern/mesh source	Viewer-only derivative with no confidential metadata
Free pattern	Preview + selected distributable pattern files	ZIP/DXF/SVG/AAMA only if legitimately exported and documented	Explicit free license and creator permission
Open remix project	View + specified source, version and dependencies	Source files under clear permission to fork	Preserve upstream credits, derivative terms
Paid digital product	Teaser and license tiers	Entitled signed download after confirmed transaction	Seller rights and platform settlement
Team project	Role-filtered metadata, previews and annotations	Files based on project role and explicit grant	Organization/workroom ACL
Factory tech pack	Approved snapshot and revisions	Factory-scoped approved specifications and authorized packages	Manufacturer cannot silently edit designer's master

View-only is not perfect DRM: users may screenshot visible artwork or reverse-engineer some approximate geometry. Security means **never transmit restricted source bytes**, not simply hiding a download button. Generate genuinely reduced/rebaked display derivatives that omit proprietary 2D coordinates, formulas, grade rules, source filenames, metadata, author comments and precise manufacturing seam labels. Scrubbing glTF extras and metadata is necessary but not sufficient; inspect network payloads. Even an accurate 3D view may expose commercially relevant visual design, which the creator accepts when publishing.

3. Rich catalog taxonomy and universal asset records

AssetType: garment pattern, garment display/look, fabric physical profile, fabric visual/PBR scans, hardware trim, graphic/print, embroidery/seam template, avatar/pose/rig, studio environment, runway set, collection/lookbook, manufacturing template and tutorial.

Asset record: assetId UUID, type, creator/team owner, title, slug, summary, searchable tags, garment category, applicable sizes/region standards, software/format compatibility, asset source reference, rights and license, base price and currency, preview references, thumbnails, approved downloadable package refs, dependency IDs/versions, content hash, publishing state, editability badge, external source provenance, maturity/validation status and moderation status.

Pattern specifics: paper/reference pattern vs verified production-grade document, dimension units, grading coverage, seam allowance type, notches, drill marks, grainline, internal construction lines, number of pieces, fabric width/layout advice, optional PDF/SVG/DXF/other legally interoperable formats. Public listing must not claim factory-ready accuracy from unverified SVG screenshots.

Fabric specifics: sample source, measured GSM and thickness, optional loading curves/solver coefficients, physical calibration status, weave/knit description, UV repeat scale, separately licensed texture maps, author and colorway. A material can be downloaded as visual-only (PBR textures) without giving the commercial lab measurement set, or vice versa where terms allow.

Studio specifics: source .blend allowed only when creator authorizes, portable GLB scene when compatible, original HDRIs and props with their own rights, cameras/lights, environment preview, performance budgets and scene license. A studio's purchased license does not magically include permission to redistribute bundled third-party material packs.

Derivatives: each derived publication links sourceAssetId and exact sourceRevision, the base material/trim/studio dependency manifest, declared transformations, original creator credit, evidence of license compatibility and a graph of upstream distribution terms.

Creator profile layout

Cover image and username; studio alias/handle; public biography and optional verified business profile; featured outfits in 3D, public room entrance, Pattern Pool submissions, fabrics/studios tabs, followers/following, saved collections, remix lineage and seller store. A private dashboard holds draft and team projects, unshared studies and paid sales reports, never visible merely because the public visits the profile.

Suggested creator types: independent designer, clothing brand, patternmaker, textile artist, 3D artist, factory account and educator. No occupation claim is verified until the platform has a responsible verification process.

4. Remix graph and licenses are the product, not an afterthought

Creators want to build on a shared sweatshirt base, modify panels and publish a new hoodie or reuse an original fabric/studio. A naive download button will break attribution and potentially infringe rights.

Core principles: the upload author must attest that they hold redistribution and derivative rights; remixable requires an explicit license that permits modification; paid/downloadable is *not* automatically remixable or resellable; each derived item records exact upstream versions and original authors. Store a license receipt at each download and purchase.

User-facing options: visual display only; free personal use; free with attribution; open commercial remix under a specified compatible license; paid personal/commercial usage with optional redistribution rights; separate extended physical-production usage. Do not invent ambiguous 'free to use however you want' permissions. Creative Commons licenses can cover some art assets; source code uses software licenses; garment patterns and manufacturing terms may need carefully drafted industry-specific license terms and local legal review.

Composite remix licensing: a derivative garment may incorporate an original base pattern, a paid denim visual material, a button asset and a licensed runway stage. Its public metadata can credit all contributors, but the right to package or resell component files depends on every original license. Some assets are *reference in project* rather than redistributable. Automatically show conflicts and block a publication containing incompatible rights.

Royalty split (long-term optional): map dependencies to an entitlement graph, choose explicit percentages and order-of-operations for each paid sale, cap complexity/cycles, handle refunds/chargebacks, and store settlement evidence. This is a commercial/legal project of its own, not something a runtime asset hash alone can solve. CLO-SET CONNECT's design-source policy offers a concrete reference for distributing proceeds to component creators; do not simply copy its fee schedule or assume the same terms work for EXO.

Marketplace economics research

CLO-SET CONNECT already lists Garment, Fabric, Trim, Avatar and Scene categories with free/paid publishing and basic/extended licensing. Its help center currently says it charges 30% commission; this is a competitive observation, **not** a recommended EXO fee. Sketchfab demonstrates 3D annotations, profiles, embedding and private/unlisted controls. Differentiate EXO by genuine pattern source/size metadata, author-approved remix rights, material mechanics, interactive factory tech packs and local EXO app round trips.

Stripe Connect is a candidate for seller onboarding, marketplace settlements, payouts, refunds and fee splitting. Its pricing, tax collection roles, supported geographies and liability must be reviewed when implementing; do not hardcode a nonexistent fee or claim it handles every seller's tax obligations automatically. Paid sales should require a real entitlements state and signed short-lived download, not a frontend purchase flag.

5. Collaboration: independent editing, review, and permissions

Team workspace: organization brand, teams, project boards, roles, shared asset registry, milestones, assigned tasks, review notes, garment version history and manufacturing partners. Invite by verified user/account and explicit project scope, with invitation expiry/revocation.

Proposed roles (least privilege):

- Owner: invite/revoke admins, modify rights, publish and initiate sales.
- Admin/Manager: manage team work and assign reviews; cannot transfer ownership without owner approval.
- Designer/Patternmaker: edit authorized garment sources/measurements and propose a revision.
- Material specialist: edit approved fabric/trim records, not unrestricted all-project assets.
- Reviewer: annotate and approve only when designated; no general edit.
- Manufacturer: read approved release snapshot, propose questions/change requests and attach sample evidence; **cannot silently overwrite canonical spec**.
- Client/Guest: restricted preview/feedback, expiration and watermark optional.
- Public user: only sanctioned released files and public derivative.

Real-time collaboration: rich-text descriptions, comments, checklists and annotations can use a CRDT such as Yjs for concurrent text; structured garment CAD editing should initially use transaction locks/optimistic revisions, deterministic

operations and conflict resolution, **not** merge arbitrary point edits by last-write-wins. An edit to the same curve control point at two devices needs a visible conflict rather than hidden data loss. Live viewers may follow a coworker's camera, inspect a diff or comment anchored to an object stable ID.

Branching and approvals: Draft branch → internal review → technical/pattern review → manufacturer request → approved immutable Release, with ability to fork a new draft. One release version is always identifiable for a purchase/factory run; a previous approval must not be implicitly inherited after measurement or material changes. Events have actor, clock time, old/new revision, project/garment, before/after data paths and reason.

Comments: anchor to pattern panel ID and point/edge stable ID, 3D triangle barycentric surface with mesh revision, BOM line, size/measurement row, image pixel normalized coordinates or document page/section. When geometry changes, try remap anchor with known stable IDs; otherwise mark **unresolved / orphaned** rather than silently pinning to an unrelated point. Thread, mentions, resolution, assigned owner, due date and evidence image belong to the comment record. See the [web tech pack plan](EXO_INTERACTIVE_WEB_TECHPACK_PLANNING_2026-10-16.md).

6. Web architecture and systems

Frontend: separate EXO Community app (e.g., modern React framework, browser 3D renderer built atop Three.js); public SEO pages, profile and catalog. The existing EXO desktop app should integrate only through a user-controlled publisher/importer with scoped account authentication, not by opening local ports to the internet.

Backend: authenticated API, relational database for profiles/permissions/assets/orders/version metadata, private blob/object storage for originals, derivative/CDN storage for public preview, entitlement service for purchased assets, search index with optional semantic tags, background job queue for thumbnails/virus scans/conversions. Organization/project isolation enforced on server, not just hidden frontend tabs. Audit logs are append-only from the application perspective.

File pipeline: signed upload destination → quarantine → MIME/magic/type scan and strict allowed formats → decompression bomb / path traversal / executable-content checks → size, file count, triangle, texture and archive limits → provenance/license review → safe source storage → sandboxed parser/render conversion → public derivative preview → user approves exact publish bundle → CDN. Never directly execute code embedded in a model asset or Blender .blend; studio scene processing is a sandboxed offline job. Follow the OWASP upload guidance.

Private boundaries: raw original patterns, manufacturing specs, personally identifying account data, factory notes and all purchased sources live in private storage. Every download checks entitlement and exact asset version at request time. Public search only indexes public sanitized metadata. A profile viewer should never receive hidden measurements or private object-store URLs. Avoid 'unlisted' as the sole protection for confidential factory specs.

Conceptual normalized database entities

users / profiles / organizations / org_members / teams / projects / project_members / garments / source_revisions / assets / asset_versions / asset_files / licenses / entitlements / imports / derivative_edges / collections / collection_items / follows / reactions / comments / threads / moderation_reports / studio_rooms / looks / reviews / releases / manufacturer_partners / orders / order_items / payment_events / settlements / notifications / immutable_audit_events.

Relational invariants: asset_versions immutable when published; orders reference stable offered asset version/license price at purchase time; derivation graph is acyclic (unless explicitly represented as non-derivative reference); creator can remove future sales while retaining necessary proof for prior buyers under policy; hard deletion/erasure honors privacy law and retained transaction obligations. Sales or asset counts on a profile are derived from authorized queries, not editable arbitrary fields.

Example asset manifest (illustrative only)

asset: "uuid"

type: "garment-pattern"

owner: "creator-uuid"

edition: "2026.1"

visibility: "public-preview-only"

previews: ["approved-preview-id"]

sourceFiles: ["private-source-object-id"]

downloads: []

grants: [{ principal: "team", role: "viewer" }]

sourceDependencies: []

rights: { display: true, download: false, remix: false }

provenance: { uploader: "...", rightsAttested: true }

validation: { unitsChecked: false, productionReady: false }

This is a **schema sketch**, not a functioning security policy or final licence legal text. All grants require backend verification; a public JSON snippet should never expose private source object IDs to a different account.

7. Trust, moderation, creator protection and community safety

Large public libraries require active safety design: copyrighted works, counterfeit brand logos, misleading 'certified fit' listings, spyware disguised as downloadable tools, plagiarized patterns, unauthorized human scans/photos, explicit or harmful imagery, harassment in comments and spam. Design reporting, appeal and action workflows, a repeat-infringer/copyright notice process, seller verification where needed, rate limits, bot protection, account recovery and moderation audit records.

A 'verified garment' badge must mean a documented technical process, not paid status. Separate **3D preview checked**, **pattern scale checked**, **measured textile source**, **simulation contact-checked**, **manufacturer approved**, and **community reviewed**, each with its own evidence. Never display a generic 'physically accurate' stamp based on a beauty render.

Protect against scraping only to the extent possible; a public view naturally exposes visual design. The material promise is that proprietary source data stays private unless the owner grants it. Each buyer/download receives receipts of license terms, file hashes and seller identity; sellers can see usage summaries without automatically learning buyers' private project files.

Follow controls: chronological feed available, mute/block creators, disable comments, disable remix requests, private bookmarks and opt-out of personalized recommendations. Give creators analytics without exposing individual user behavior beyond what the terms and privacy notice permit.

8. Product phases and realistic build order

Gate	Future release	Key deliverable and acceptance
C0	Rights, schema and privacy strategy	License vocabulary, creator protections, upload threat model, public/private manifests
C1	Public visual gallery + profiles	Account, creator portfolio, authorized thumbnails and 3D viewer; private data never transmitted
C2	Free pool + downloads	Searchable licensed pattern files, verifiable hashes, format metadata, free entitlements and download receipts
C3	Remix lineage	Fork explicit remixable source only, attribution and dependency tree; incompatible terms block publish
C4	Teams and secure project review	Role-based assets, comments, release snapshots and correct revocation/permission QA
C5	Paid marketplace	Seller onboarding, order entitlements, payout/refund accounting, buyer rights and tax/compliance review
C6	Virtual wardrobe rooms and studio assets	Walkable user rooms, scene template downloads, source-safe optimized GLB
C7	Advanced multiplayer and revenue-share derivatives	Presence, optional shared showroom, robust rights splitting and moderation at scale

Do **not** implement payments before correct entitlements/licensing or a safe uploader. Do **not** implement collaborative direct CAD edits before reliable version/revision management. Do **not** start public uploaded PPF processing before job isolation, cost controls and physical validity checks.

Suggested relative effort: profile/gallery prototype 4–8 engineer-weeks once the platform foundation exists; free pool and safe importer 4–8; teams/comments 6–12; paid marketplace 6–12+ plus legal/accounting; remix royalties and large 3D rooms substantially more. These are nonbinding exploratory bands, not calendar estimates or a promise that this is a small side project.

QA success journeys

Creator: signup → private draft hoodie → publish sanitized view-only showcase → another account views geometry but receives no editable size/measurements/edge data → remove it → confirm access revoked. **Patternmaker:** upload explicitly

licensed free pattern with sizing → format/scale validations → stranger downloads with receipt → forks only where remix license allows → original creator linked. **Brand team:** invite a reviewer and manufacturer into a specific garment release → guest sees approved spec, comments, no confidential formulas → approval history cannot be rewritten. **Seller:** post paid texture pack → customer receives exact entitled version after charge → revoked link prevents fresh unauthorized download; refund and rights behavior follows explicit store policy. **Gallery:** 20 mannequin looks in a walkable original room, optimized with LOD, and a list fallback for accessibility.

9. Platform business model: options to evaluate, not decided

Potential revenue streams: low commission on paid digital item sales; optional paid extra storage for large libraries; brand/team seats; optional advanced offline PPF batch processing with transparent compute credits; premium showroom hosting; custom branded pages or manufacturer review seats. **Keep a meaningful free tier** for public patterns and educational collaboration. Core private drawing rights and safety features should not be withheld for monetization.

Do not assume 'free download' licenses can be flipped to paid without respecting prior rights. Publish versioned product/licensing changes for new customers and preserve historical entitlements. Test economics: median storage per garment, upload conversion costs, image CDN, GPU rendering, payment fees, refund exposure and support/moderation before setting platform pricing.

10. External reference documentation

- CLO-SET CONNECT Store and uploading/licensing options: marketplace categories and free/paid tiers.
- CONNECT design-source and component royalties: derivative revenue attribution precedent, not EXO terms.
- Sketchfab plans: profile previews, embed/annotation and private/unlisted capabilities as documented by vendor.
- Stripe Connect: marketplace onboarding and payouts.
- Yjs shared types: candidate for collaborative rich text/comments, not a reason to merge physical CAD geometry without review.
- OWASP secure file upload: public upload threat model.
- GitHub license guidance: public viewing alone does not authorize use and derivatives.

Decision: The site should be an independent creator platform over the same intentional asset/garment identity model, with publicly visible fashion work, independently licensed downloadable sources, a huge free pattern pool, optional paid products, accessible walkthrough rooms and robust team/project ACLs. **Nothing is uploaded, sold, published or deployed by this research.**